Atomic Data and Nuclear Data Tables 96 (2010) 586–644

Contents lists available at ScienceDirect

Atomic Data and Nuclear Data Tables

journal homepage: www.elsevier.com/locate/adt

A critical compilation of experimental data on spectral lines and energy levels of hydrogen, deuterium, and tritium

A.E. Kramida *

Atomic Physics Division, National Institute of Standards and Technology, 100 Bureau Dr., MS 8422, Gaithersburg, MD 20899, USA

A R T I C L E I N F O

Article history:
Available online 9 June 2010

A B S T R A C T

For more than 50 years, Charlotte Moore's compilation of atomic energy levels and its subsequent revisions have been the standard source of reference data for the spectra of hydrogen and its isotopes. In those publications, theoretical data based on quantum-electrodynamic calculations have been given. This reflects the fact that the theory of the hydrogen spectrum has been perfected to an extent far exceeding the capabilities of the best measurements. However, rapid advances in the techniques of laser spectroscopy and optical frequency metrology have recently put experiments on a par with theory in terms of precision. This calls for construction of new comprehensive data sets for H, D, and T that summarize the latest experimental work and can be directly compared with the modern theoretical reference data. The present work compiles several tens of recent measurements of the hydrogen, deuterium, and tritium fine and hyperfine structure intervals and presents sets of energy levels and Ritz wavelengths derived from those measurements. Data exist for the fine structure of energy levels of hydrogen and deuterium up to principal quantum number $n = 12$. For higher lying levels, there are many observed lines with unresolved fine structure. From those observations, level centers (centers of the fine structure) are derived by a least-squares optimization, and Ritz wavelengths of series with upper levels up to $n = 40$ are obtained. For tritium, the $n = 2$ and 3 energy level intervals are derived from experimental observations.

* Fax: +1 301 975 3038.
E-mail address: Alexander.kramida@nist.gov

0092-640X/$ - see front matter © 2010 Elsevier Inc. All rights reserved.
doi:10.1016/j.adt.2010.05.001

Contents

1. Introduction 587
2. Brief historical review 589
3. Hyperfine structure 591
   3.1. Calculations of hfs splittings 592
      3.1.1. hfs Splittings of *n*S levels in hydrogen 593
      3.1.2. hfs Splittings of non-S levels in hydrogen 593
      3.1.3. hfs Splittings in deuterium 594
      3.1.4. hfs Splittings in tritium 595
4. Fine structure measurements in hydrogen 596
5. Level optimization for hydrogen 600
6. Fine structure measurements in deuterium 603
7. Level optimization for deuterium 605
8. Fine structure measurements and derived energy levels in tritium 606
9. Comparison with QED calculations 608
10. Line series of hydrogen 609
   10.1. Laboratory observations 609
   10.2. Solar observations 610
   10.3. Radio recombination lines 611
   10.4. The concept of distinctive energies 611
   10.5. Level optimization for experimental data on line series 613
   10.6. Intensities of the series lines 615
   10.7. Use of the hydrogen and deuterium series lines for calibration 616
11. Conclusions 617
   Appendix A. Supplementary data 618
   References 618
   Explanation of Tables 621
   Tables
      1. Hfs splittings and off-diagonal shifts in hydrogen (Hz) 624
      2. Hfs splittings and shifts in deuterium (Hz) 625
      3. Hfs splittings and off-diagonal shifts in tritium (Hz) 626
      4. Optimized energy levels of hydrogen derived from available experimental transition frequencies compared with QED calculations of Jentschura et al. [10] 627
      5. Frequencies of fine-structure transitions in hydrogen derived from the level-optimization procedure compared with QED calculations of Jentschura et al. [10] 628
      6. Optimized energy levels of deuterium derived from available experimental transition frequencies compared with QED calculations of Jentschura et al. [10] 631
      7. Frequencies of fine-structure transitions in deuterium derived from the level-optimization procedure compared with QED calculations of Jentschura et al. [10] 632
      8. Experimental and theoretical energy levels of tritium 636
      9. Optimized distinctive energy levels of the hydrogen series (see Section 10.4 for the definition of distinctive energy levels) 637
      10. Observed and Ritz wavelengths and wave numbers of the hydrogen series lines 638
      11. Wave numbers of the Lyman series lines of hydrogen and deuterium derived from experimental fine-structure intervals weighted by calculated line strengths 641
      12. Designations of fine-structure components and their groups, used by different authors for the Balmer-α lines, and their relative intensities according to different models 642
      13. Wave numbers of component groups and their separations for the Balmer series lines of hydrogen and deuterium derived from experimental fine-structure intervals weighted by different models 643

1. Introduction

The hydrogen spectrum has fundamental importance for modern science. Many books have been written about it. Excellent reviews of experimental and theoretical work on hydrogen before 1990 were given in the collection of articles edited by Series [1] and in the articles by Beyer [2] and by Pipkin [3]. The most important of the later experiments are reviewed in the publications of the Task Group on Fundamental Constants of the Committee on Data for Science and Technology (CODATA) [4–6].

Hydrogen is the simplest atomic system and thus has always attracted the attention of experimentalists and theorists in their attempts to understand the physical nature of light radiated by atoms. More than 500 scientific papers have been published on its spectrum. A reasonable question is: Why do we need one more? First of all, there are many formulas evolving in their complexity as experimental measurements of increasing precision revealed inconsistencies in them, sometimes leading to major revisions in our understanding of nature. The applicability of various theoretical approximations strongly depends on the accuracy with which it is necessary to know the spectrum. Indeed, in a large number of practical applications the simplest Bohr description based on the planetary model of an electron rotating around a heavy nucleus is sufficient (see, for example, Woodgate [7]). However, this simple model fails to explain the fine structure of the series lines, which is quite easily observed in laboratory and astrophysical conditions. More elaborate relativistic theory also provides a set of formulas describing the spectrum [8]. These formulas, although more complex than that of the Bohr model expression, can still be evaluated using a calculator. They describe the fine-structure splitting of the series lines. However, highly accurate measurements revealed that there are not only small numerical differences between those formulas and observed spectral intervals, but there is a qualitative difference between observations and the relativistic model; namely, the observed spectrum contains more fine-structure components than the relativistic theory predicts [9]. This fact led to the discovery of the zero point fluctuations of electromagnetic field permeating the vacuum. It turned out that these fluctuations are responsible for the phenomenon of spontaneous emission of radiation by atoms. A small discrepancy between the old theory and experiment once again led to a major revision of our understanding of atoms and fields. This discrepancy has been well described by modern quantum electrodynamics (QED). However, at this time there is no simple formula that can be evaluated with a calculator to obtain the spectrum. QED calculations require deep understanding of the physics of particles and fields and are highly technically involved. Several tens of theoretical investigations on this subject have been published in the last decade. Highly accurate numerical results obtained by Jentschura et al. [10] can be found on the Internet and used as reference data.

However, even those high-precision results do not give a full description of the observed spectrum. First of all, they do not include the hyperfine structure (hfs). The hfs results from the magnetic interaction between the nucleus and the bound electron. It is manifested as a small splitting of the fine-structure energy levels. Although it is small compared to the breadth of the fine structure, it has some important consequences. The most famous of them is the existence of the hydrogen maser. The ground state of hydrogen and its isotopes is split into two hfs components. One may arrange experimental conditions under which there is a population inversion between these two hfs states leading to stimulated emission in the microwave range. For hydrogen, the frequency of the hfs maser has been measured to an extraordinary precision of seven parts in 10^{13} [11–14] and is equal to 1 420 405 751.768(1) Hz [14]. For deuterium and tritium, the corresponding frequencies are 327 384 352.522 2(17) Hz [15] and 1 516 701 470.775(7) Hz [16]. The theory of the hfs is not as well developed as the theory of the fine structure. It is appropriate here to give a quotation from the recent paper of Jentschura and Yerokhin [17]:

> "Unfortunately, our theoretical understanding of the ground-state hfs is limited by the insufficient knowledge of the nuclear charge and magnetization distributions, whose contribution of about −50 kHz (30 ppm) cannot be accurately calculated at present."

The recent work of Carlson et al. [18] has reduced the theoretical uncertainty of the hydrogen 1S hyperfine splitting to about 1 ppm, but even those advanced calculations still cannot compete with the best experimental measurements of this quantity.

Approximate formulas for the hfs splittings in hydrogen, deuterium, and tritium have been given by several authors [19–23]. An outline of methods of more accurate numerical calculations of hfs for arbitrary excited states of hydrogen and deuterium was given by Brodsky and Parsons [20]. Numerical calculations for several excited states of hydrogen and deuterium have been carried out by several authors, but we could not find any general formulas beyond the simple approximation of Bethe and Salpeter [19] in the literature.

Another limitation of theory is related to the use of hydrogen and deuterium lines for wavelength calibration of spectra emitted by laboratory and astrophysical plasmas. For example, sealed deuterium discharge lamps are widely used for radiometric calibration of laboratory spectra, and their wavelength scale is often determined from the Balmer series lines of deuterium. In astrophysics, various parameters of emitting plasmas are often derived from observed positions and shapes of hydrogen series lines. In such plasmas, the observed fine structure intervals depend on the excitation conditions. The reason for this dependence is that the shapes of spectral lines emitted by hydrogen or deuterium atoms are always broadened and distorted by effects of motion of atoms, as well as by interactions between emitting atoms and other particles. For example, the Balmer series lines are usually observed in laboratory discharge spectra as doublets with separations between the components of the order of 0.3 (cm^{-1}) [1] (or 9 GHz) [25], while the widths of the components are typically about 0.1 cm^{-1} (or 3 GHz). We now know that each of the doublet components is actually comprised of several individual fine and hyperfine components, which blend together into unresolved profiles whose exact shape depends on the populations of excited states and various broadening mechanisms in the gaseous medium. This dependence on the excitation conditions becomes especially important for the spectra of astrophysical objects. While a laboratory experiment can be arranged so that various line-broadening and shifting factors are minimized, astrophysical observations cannot be optimized in this way. There we can observe only the results of the physical processes occurring in solar and stellar atmospheres and other astronomical objects. These processes involve radiation trapping and transfer, as well as collisional kinetics. Exact calculation of these processes from first principles is currently impossible. In fact, astronomers compare the observed wavelengths of the hydrogen series lines with the calculated values to derive various physical parameters of the emitting media, such as the macroscopic velocities of movements of stars, electron densities, and temperatures. Thus, it is important to know the reference values of these wavelengths pertinent to the astrophysical conditions. These values do not necessarily coincide with the average weighted line centers computed assuming Boltzmann populations of excited states.

The purposes of the present compilation are as follows: (1) to provide a reference set of essentially experimental values of the energy levels and wavelengths for the fine structure of hydrogen, deuterium, and tritium, and compare them with the most precise modern calculations; (2) to provide a reference set of the most precise data available for hyperfine structure of hydrogen, deuterium, and tritium, and outline the areas where more theoretical work is needed; (3) to provide a reference set of wavelengths for various series of hydrogen, derived from solar observations, for use in diagnostics and modeling of solar spectra; and (4) to provide practical guidance regarding the accuracy of reference wavelengths of the H and D Balmer series lines for wavelength calibration of various laboratory discharge lamps.

The data presented in this paper is organized in tables arranged in the following way. The tables containing illustrative material necessary for understanding of the text are given within the text and are labeled with letters A through M. The tables containing the main results (i.e., reference data such as recommended values of hyperfine intervals, optimized energy levels, and transition wavelengths) are numbered with Arabic numerals and placed in the end of the article.

The units used in the tables and corresponding text sections are dictated by the context in which the data are used. For example, the high-precision data for the hyperfine and fine structure intervals (Tables 1 through 8 and Sections 3 through 9) are given in the units of Hertz, kiloHertz, or megaHertz (depending on the precision of the data), since most of those data are obtained by direct frequency measurements. The less precise energy intervals in H and D line series, pertinent to solar observations (Table 9 through Table 13 and Section 10) are given in the units of reciprocal centimeters, which is more convenient to represent their precision. The wavelengths of the series lines discussed in Section 10 are given in angstroms (1 Å = 0.1 nm) for the visible and ultraviolet (UV) spectral ranges and in micrometers for the infrared, because most of

[1] The customary unit cm^{-1} for energy levels and intervals, used here, is related to the SI unit for energy (Joule) and frequency (Hertz) as follows: 1 cm^{-1} is equivalent to $1.986\,445\,501(99) \times 10^{-23}$ J [24] or exactly 29 979 245 800 Hz.

those lines were measured using traditional techniques in the time when the angstrom and the micrometer were standard units of wavelength for these spectral ranges.

In the following section we will give a brief historical review of experiments concerning the hydrogen spectra. In Section 3, we give an account of existing measurements and calculations of the hyperfine structure of H, D, and T. Section 4 contains a critical review of measurements of the fine structure intervals in H. Section 5 describes the level optimization procedure, which we used to derive the energy levels of hydrogen and optimized experimental values (the so-called Ritz values) of fine-structure transition frequencies. Section 6 reviews measurements of the fine-structure intervals in deuterium. Section 7 provides details of the level optimization procedure for deuterium. In Section 8, we describe the existing measurements of the fine structure of tritium and our derivation of the energy levels of this spectrum. In Section 9 we compare the experimental values of energy levels obtained with results of modern QED calculations. Section 10 is devoted to observations of the series lines of hydrogen and deuterium. It is focused on practical recommendations for the use of those spectral lines to calibrate astrophysical and laboratory spectra of moderate resolution, in which individual fine- and hyperfine-structure components of the series lines are blended into partially resolved profiles. The final section draws conclusions and gives a brief summary of the main results. The main numerical results, which are the tables of recommended values of hyperfine splittings, energy levels, and fine-structure intervals in H, D, and T, as well as derived wavelengths of series lines of H and D, are given in the tables.

2. Brief historical review

Our current understanding of the structure and properties of atoms stems from the quantum theory first proposed by Bohr [26]. This theory explained the Balmer and Paschen series of lines in the spectrum of hydrogen and helium as observed to that time and predicted several new line series, which were successfully observed later by Lyman [27], Brackett [28], Pfund [29], and Humphreys [30]. According to Bohr's theory, the energy levels of atoms are degenerate in the sense that they are fully described by a single principal quantum number n by means of a simple formula

$$E(n) = -RZ^2\big/n^2, \tag{1}$$

where R is a constant (subsequently called the Rydberg constant) related to the electron's mass and charge and Planck's constant h, and Z is the number of protons in the nucleus ($Z = 1$ for hydrogen, but we keep it in all formulas valid for other hydrogen-like systems). Bohr's theory assumed that the nucleus, positioned in the center of the reference frame, has infinite mass. Taking into account the finiteness of the nuclear mass, Bohr's formula can be approximately corrected for an arbitrary hydrogen-like atom with atomic number Z by scaling the Rydberg constant according to the ratio of masses of the electron and the nucleus,

$$R = R_\infty/(1 + m_e/M_N), \tag{2}$$

where the constant R_∞ is now applicable to any hydrogenic ion.

It was soon discovered that the lines of the Balmer series appear as closely separated doublets. Separations within these doublets were measured to high precision by many observers. This called for revision of Bohr's theory that was first developed by Sommerfeld, Unsöld, and Dirac. In this new theory, the effects of special relativity were taken into account, and a new approximate, but more accurate, formula was obtained for the energy levels of hydrogen and hydrogen-like ions. Specifically, it was found that the energy spectrum of an electron in a static Coulomb field of an infinitely heavy point nucleus can be represented by the Dirac formula [8]

$$E_D(n,j) = 2\left(R_\infty/\alpha^2\right)[f(n,j) - 1], \tag{3}$$

where $f(n,j) = [1 + (Z\alpha/(n-\delta))^2]^{-1/2}$, $\delta = (j+\frac{1}{2})(1-[1-(Z\alpha/(j+\frac{1}{2}))^2]^{1/2})$, α is a dimensionless constant called the fine-structure constant, and j is an additional quantum number corresponding to the total angular momentum of the electron. The latter results from combining the orbital motion of the electron with its intrinsic angular momentum called spin.

An approximate account for the finiteness of the nuclear mass was first made in the same way as for the nonrelativistic Bohr formula, namely, by replacing R_∞ with $R_\infty m_r/m_e$, where m_r is the reduced mass of electron, $m_r = m_e/(1 + m_e/M_N)$. However, it was soon realized that a general solution of a relativistic quantum mechanical problem of two interacting particles is very difficult to obtain. Garcia and Mack [31] gave a formula describing the largest of the additional corrections. A more thorough treatment by Grotch and Yennie [32,33] gave an expression for small terms not accounted for by Garcia and Mack [31]. The result of Grotch and Yennie was later reproduced by Sapirstein and Yennie [34] (see also Eides et al. [35] and references therein) who gave it in a form similar to the following,

$$E_{rel} = E_D\frac{m_r}{m_e} - \frac{R_\infty[f(n,j)-1]^2 m_r^2}{\alpha^2 m_e(m_e + M_N)} + (1-\delta_{l0})\frac{R_\infty(Z\alpha)^4 m_r^3}{\alpha^2\kappa(2l+1)n^3 m_e M_N^2} + \cdots \tag{4}$$

where l is the nonrelativistic orbital angular momentum quantum number, κ is the angular-momentum-parity quantum number, $\kappa = (-1)^{j+l+1/2}(j+1/2)$, and δ_{l0} is a Kronecker delta, $\delta_{l0} = 1$ ($l = 0$), $\delta_{l0} = 0$ ($l > 0$). Sapirstein and Yennie [34] noted that the complete mass dependence through order α^4, including the small term that breaks the degeneracy in l, was obtained originally by Barker and Glover [36]. However, these mass correction terms were given in the latter paper in a less general form.

In the relativistic treatment, the motion of the nucleus introduces further small contributions to the energy levels, which are called relativistic recoil corrections. Expressions for these were given by Erickson [37] and Sapirstein and Yennie [34]. Since they are considerably more complex, we do not give them here. These and other corrections to the hydrogen energy levels are systematically reviewed in the article accompanying the CODATA 2006 set of fundamental constants [6,24].

In the Dirac theory, the levels with the same principal quantum number n, which were merged together in Bohr's model, become separated if they have different values of j. For the Balmer-α line, the consequence of the Dirac theory is that, instead of just one line corresponding to the $n = 2 \rightarrow n = 3$ transition, there appear five lines with various relative intensities. This development is illustrated in Fig. 1. In this figure, the components of the Balmer-α line are numbered in the order of decreasing predicted intensity, as suggested by Williams [38] with further breakdown of components 2 and 3 into sub-components *a* and *b* suggested by Kibble et al. [39].

The two strongest components, 1 and 2, are responsible for the fact that the Balmer-α line appears to be a doublet. In deuterium, due to its heavier mass, the Doppler broadening is smaller, which allowed Williams [38] to partially resolve the smaller component 3.

Qualitatively, the Dirac theory explained the observations of the Balmer series very nicely. However, the measured doublet separations were consistently smaller than those predicted by the Dirac theory (see, for example, Houston and Hsieh [40] and Houston [25]). Pasternack [41] analyzed those deviations between observations and the theory and showed they could be explained by an upward shift of the $2S_{1/2}$ level by about 0.03 cm^{-1} (equivalent to ≈1 GHz). The reality of this shift was later convincingly proven

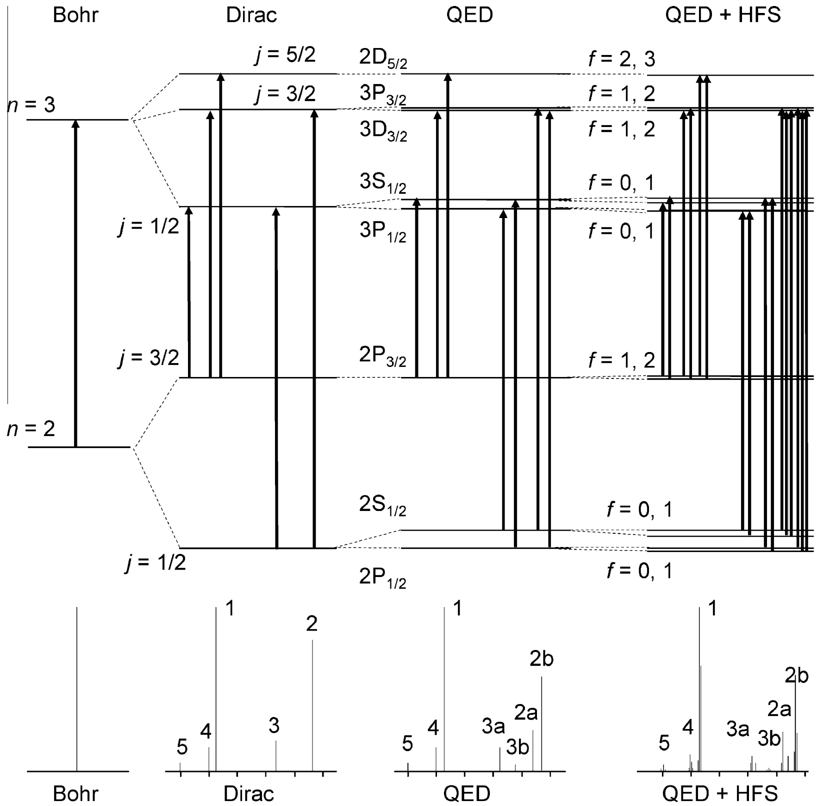

Fig. 1. The structure of the Balmer-α line in various theoretical approximations (not to scale). The heights of the vertical lines in the bottom part of the figure represent calculated relative intensities of transitions assuming Boltzmann populations of the upper levels.

by the series of experiments of Lamb and co-workers [9,42–47]. It was explained by the new QED theory developed first by Bethe [48,49]. Although some approximate formulas were put forward in the early QED calculations, it was made clear that the approximating expansions in terms of powers of αZ and $\ln(\alpha Z)$ converge very slowly, and it becomes exceedingly difficult to make calculations with increasing precision. Examples of QED calculations for hydrogen-like systems can be found in the work of Garcia and Mack [31], Erickson [37], and Johnson and Soff [50]. Important corrections to Erickson's expressions [51,37] for high-order self-energy contributions to Lamb shifts were made by Mohr [52] and Sapirstein [53]. The most recent developments are described by Kotochigova et al. [54] and Jentschura et al. [55].

It should be noted that the relativistic corrections given by formula (4) remove the degeneracy in l and hence also introduce a small shift between $2S_{1/2}$ and $2P_{1/2}$. However, this relativistic shift is much smaller than the shift observed by Lamb et al. The l-dependent term in formula (4) contributes about 2 kHz to the frequency of the $2P_{1/2}$–$2S_{1/2}$ transition in hydrogen, while the size of the Lamb shift is about 1 GHz. Historically, because these relativistic corrections were calculated theoretically somewhat later than the experimental discovery of the Lamb shift, they are not considered to be part of the latter.

At the level of resolution achieved in the experiments of Lamb et al. [42–47], it became important to account for the hyperfine structure in the hydrogen and deuterium spectra. The currently accepted definition of the hfs can be found in Woodgate's textbook [7]. It is the energy structure caused by the properties of the atomic nucleus other than its electric charge. If the nucleus has a non-zero spin I (which is true for all hydrogen isotopes), the dominant contribution to the hfs is due to the magnetic dipolar interaction between the nuclear magnetic moment and the magnetic field produced at the nucleus by the electron. For electronic states with $l > 0$ (p, d, etc.), the nucleus is subjected to a magnetic field due to the orbital moment of the electron. For all electronic states, there is also a magnetic field produced by electron's intrinsic magnetic moment. Nuclei with $I \geqslant 1$, such as deuterium ($I = 1$) but not hydrogen and tritium ($I = 1/2$), have an electric quadrupole moment. This interacts with electron's electric field gradient and gives rise to a second contribution to the hfs.

The hyperfine interactions split each fine-structure energy level into a hyperfine multiplet. The magnitudes of the splittings can be approximately calculated within the framework of perturbation theory. The relative intensities of the hyperfine transitions depicted in Fig. 1 were taken from White and Eliason [56], where they were calculated using sum rules for oscillator strengths. Description of the relevant Ornstein–Burger–Dorgelo sum rule can be found, for example, in chapter 7.4 of Woodgate [7] (rephrased here for hfs multiplets): the sum of the intensities of all the transitions from an initial level or to a final level of a multiplet is proportional to the statistical weight, $2F + 1$, of that level; the constant of proportionality is common to all levels of a given multiplet.

The hyperfine splitting of the $2S_{1/2}$ level of hydrogen was observed in Lamb's experiments, as well as in the later saturated absorption experiments of Petley et al. [57]. The observed spectrum of the Balmer-α line given in Fig. 2 of the latter paper is in remarkable agreement with the theoretical picture in Fig. 1. It demonstrates well-resolved components 1, 2a, 2b, 3a, 3b, and 4, as well as partially resolved hfs splitting in components 2a, 3a, and 3b. A full account of experimental measurements and calculations of the hfs in hydrogen and its isotopes will be given in the following section. The magnitude of the hfs splittings decreases with increasing principal quantum number n approximately as $1/n^3$, while the natural widths of the levels decrease approximately as $1/n^2$. Thus, it becomes exceedingly difficult to observe and measure the hfs in highly excited states. The highest non-S state for which it was measured is $2P_{1/2}$ in hydrogen [58]. This measurement, having the relative uncertainty of only 0.2%, was quite an achievement,

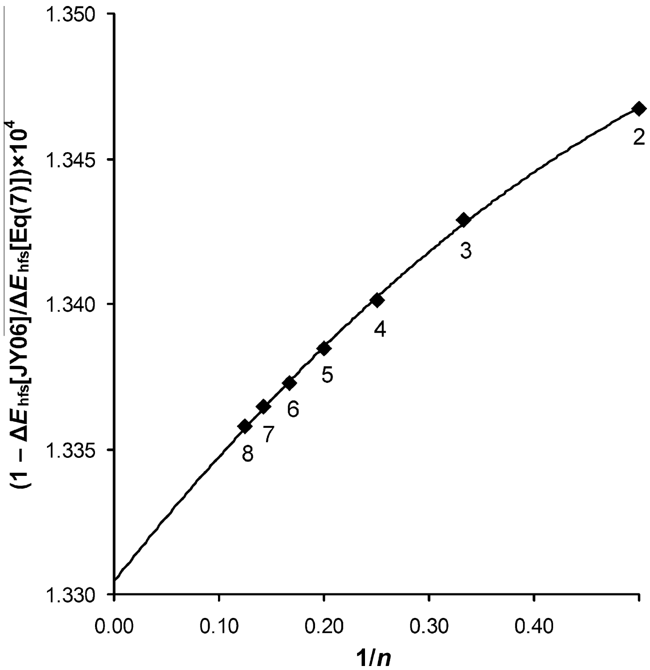

Fig. 2. The difference from unity of the ratio of precisely calculated nS hfs splittings [17] to those calculated by formulas (5–7 and 9), magnified by a factor of 10^4. The solid curve is a parabola obtained by a least-squares fitting of the data points from Jentschura and Yerokhin [17] for n = 2–8. The values of n are given next to the data points.

since the hfs interval for this energy level is only about half of the natural linewidth. As demonstrated by Lundeen et al. [58], using the technique of separated oscillatory radio-frequency fields it is possible to decrease the width of observed resonances to about 1/3 of the natural linewidth.[2] Thus, despite the fact that the hfs splittings of the hydrogen nL levels with $n \leqslant 12$, $L = 1, 2$ constitute only 12–60% of the natural linewidths, it does not mean that they are impossible to measure. The hfs splittings of the nS levels are greater by two orders of magnitude, and they exceed the natural linewidths for $n \leqslant 150$. The hfs intervals in deuterium are smaller than those in hydrogen by approximately a factor of 8, which makes their measurement even harder. The relative sizes of the hfs in H, D, and T are determined mainly by their nuclear magnetic moments, 2.792 847 356(23)[3], 0.857 438 230 8(72), and 2.978 962 448(38) [6,24], respectively (in units of the nuclear magneton) and by the twofold greater nuclear spin of D compared to H and T.

Rapid development and refinement of microwave, radio-frequency, and laser-spectroscopy measurement techniques beginning in the early 1970s resulted in several dozen high-precision measurements of various fine and hyperfine intervals in hydrogen and its isotopes. The most significant results of those measurements will be reviewed in the following sections.

3. Hyperfine structure

To our knowledge, the first measurement of the hyperfine structure of the ground state of hydrogen and deuterium was accomplished by Nafe et al. [61]. Their results, 1 421.3(2) MHz for hydrogen and 327.37(3) MHz for deuterium, were obtained by means of the atomic beam magnetic resonance method. The method was later refined by Nafe and Nelson [62], resulting in improved values, 1 420.410(6) and 327.384(3) MHz, respectively. With the same method, Nelson and Nafe [63] measured the ground-state hfs interval in tritium to be 1 516.702(10) MHz. Further refinement of the method allowed Prodell and Kusch [64–66] and Kusch [67] to obtain the improved results 1 420.405 73(5), 327.384 24(8), and 1 516.701 70(7) MHz for H, D, and T, respectively. Using an alternative microwave absorption technique, Wittke and Dicke [68] obtained a similar-precision value 1 420.405 80(6) MHz for hydrogen. Anderson et al. [69] using an optical polarization method obtained 1 420.405 726(30), 327.384 349(5), and 1 516.701 396(30) MHz for H, D, and T, respectively. Using a similar method, Pipkin and Lambert [70] obtained 1 420.405 749(6) MHz for hydrogen and 1 516.701 477(6) MHz for tritium. The latter result for tritium was in large disagreement with Anderson et al. [69].

The invention of the hydrogen maser by Kleppner et al. [71] led to a major improvement in the precision of the measurements of the ground-state hfs. Their original result for hydrogen, 1 420.405 762(4) MHz [71], was greatly improved by Crampton et al. [72], who obtained 1 420.405 751 800(28) MHz. The same team measured the ground-state hfs interval in deuterium [73] and tritium [16] masers. Their results were 327.384 352 30(25) MHz for deuterium and 1 516.701 470 808 7(71) MHz for tritium. These values were obtained using their previously measured value for hydrogen [72] as a frequency reference. Several high-precision measurements of the ground-state separation were made later using masers in hydrogen [74–78] and deuterium [79,15]. The measurements on hydrogen were critically analyzed by Karshenboim [14], who recommended the final value 1 420.405 751 768 MHz with a conservative estimate of uncertainty as ±1 mHz. The values actually measured by Mathur et al. [16] and Wineland and Ramsey [15] were the ratios of the maser frequencies in tritium and deuterium to the one in hydrogen. Thus, adjustment of the value of the hydrogen maser frequency requires a corresponding adjustment of the tritium and deuterium values. Using this adopted value for the hydrogen maser frequency, we re-calculated the value for tritium from Mathur et al. [16] to be 1 516.701 470 775(7) MHz. It turns out that the adopted value for the hydrogen maser frequency coincides with the reference value used by Wineland and Ramsey [15] in their deuterium measurement, so their result for deuterium, 327.384 352 522 2(17) MHz, does not require any correction.

The hfs of the metastable excited state $2S_{1/2}$ in hydrogen and deuterium was first measured by Heberle et al. [80] and Reich et al. [81] using an atomic beam magnetic resonance method. They obtained 177.556 86(5) MHz for hydrogen and 40.924 439(20) MHz for deuterium. Rothery and Hessels [82] refined the value for hydrogen by using a similar method (i.e., driving the magnetic-dipole radio-frequency transition in a thermal beam). Their result was 177.556 785(29) MHz. The reduction of the uncertainty was primarily due to a lower temperature of the atomic beam (410 K compared to 2 800 K in Ref. [80]), which resulted in much narrower line shapes. Kolachevsky et al. [83] developed a more accurate optical differential method based on two-photon Doppler-free spectroscopy on a cold (5–6.5 K) atomic beam. Their results, 177.556 860(16) MHz for hydrogen and 40.924 454(7) MHz for deuterium, are the best currently available experimental values. (See Note added in proof).

The only reported measurement of the $2P_{1/2}$ hfs interval for hydrogen is that by Lundeen et al. [58], which yielded 59.22(14) MHz. There are no reported measured values for deuterium. However, as early as in 1953 Triebwasser et al. [46], by modeling the observed line profiles in their radio-frequency measurements, were able to confirm theoretically calculated hyperfine

[2] The principle of obtaining sub-natural line widths in a separated oscillatory fields experiment was described by Fabjan and Pipkin [59] with a reference to an earlier paper by Hughes [60] who suggested it. Fabjan and Pipkin noted that this technique was also independently suggested by R.H. Dicke (no specific reference was given).

[3] In this work, the number in parentheses given after a numerical value means the uncertainty in the units of the last decimal place. The uncertainties are always meant to be in terms of one standard deviation, although it is often difficult to obtain the exact meaning of uncertainties quoted from other publications.

splittings of $2P_{1/2}$ and $2P_{3/2}$ in both hydrogen and deuterium to within ±0.20 MHz.

For other excited levels of hydrogen and its isotopes, there are no known measurements of the hfs splittings. Since their values are required for calculation of corrections to the measured fine-structure intervals, they must be derived theoretically. Calculations of hfs are reviewed in the following sections.

3.1. Calculations of hfs splittings

The first-approximation theory of hfs was well-developed in the first half of the 20th century. Analytical expressions for the hfs energy shifts can be found in several textbooks (see, e.g., [1,7,19,21,84]). Assuming that the hfs is much smaller than the fine structure, an approximate formula for the magnetic-dipole hfs shifts of hydrogenic energy levels relative to formula (4) can be written as

$$\Delta E_{nljIF} = \frac{A_{nlj}}{2}[F(F+1) - I(I+1) - j(j+1)] + \Delta E^{\text{off-diag}}_{nljIF}, \tag{5}$$

where n is the principal quantum number, l is the orbital quantum number of the electron, j is the quantum number describing the intermediate angular momentum (obtained by adding the electron spin to its orbital angular momentum), F is the total angular momentum of the atom (obtained by adding the angular momentum I of the nucleus to j), and A_{nlj} is the magnetic-dipole hfs constant expressed by the formula

$$A_{nlj} = 2\alpha^2 Z^3 R_\infty c \frac{\mu_e \mu_{\text{nucl}}}{\mu_B^2 \left(1 + \frac{m_e}{M_N}\right)^3 n^3 j(j+1)(2l+1) I} F^{\text{rel}}_{nlj}, \tag{6}$$

where μ_e and μ_{nucl} are the absolute values of magnetic moments of the electron and nucleus, respectively, μ_B is the Bohr magneton, and I is the spin of the nucleus ($I = 1/2$ for hydrogen and tritium, and 1 for deuterium). The product $R_\infty c$, as well as A_{nlj} and ΔE_{nljIF}, are in units of Hertz ($R_\infty c = 3.289\,841\,960\,361(22) \times 10^{15}$ Hz [24]). F^{rel}_{nlj} is a relativistic correction factor close to unity. The quantity $\Delta E^{\text{off-diag}}_{nljIF}$ in formula (5) is a small correction due to off-diagonal terms in the Hamiltonian (introduced by Brodsky and Parsons [20]; it will be discussed further below).

From formula (5), the interval $\Delta E_{hfs}(F-1, F)$ between two adjacent levels in a hyperfine multiplet can be written as

$$\Delta E_{\text{hfs}}(F-1, F) = A_{nlj} F + \Delta E^{\text{off-diag}}_{nljI}(F-1, F). \tag{7}$$

This formula is an analog of the Landé interval rule for the fine-structure intervals in *LS* coupling (see, e.g., Woodgate [7], Chapter 7.3).

In Dirac's theory, the ratio μ_e/μ_B is exactly equal to 1, as well as the ratios $\mu_p/(e\hbar/2M_p)$, $\mu_t/(e\hbar/2M_t)$, and $\mu_d/(e\hbar/2M_d)$. However, precise experimental measurements yield $\mu_e/\mu_B = 1.001\,159\,652\,181\,11(74)$, $\mu_p/(e\hbar/2M_p) = 2.792\,847\,356(23)$, $\mu_t/(e\hbar/2M_t) = 8.918\,170\,61(12)$, and $\mu_d/(e\hbar/M_d) = 0.857\,012\,728(7)$ [24]. The deviations of these ratios from unity (called magnetic moment anomalies) are due to interactions of the particles with quantum fields existing in vacuum. For electrons, these interactions are described by the QED theory, and for nuclei they are described by quantum chromodynamics. Thus, formula (6) partially accounts for the quantum field effects in an empirical way.

The relativistic correction factor F^{rel}_{nlj} was shown by Sobel'man [21] (see formula 27.29 on p. 253) to be proportional to the integral coefficient[4]

$$C^{-2} = \int (gfr^{-2}) r^2 dr, \tag{8}$$

where f and g are functions defined by Eqs. 26.66 and 26.67 in Ref. [21]. Shabaev [86] found an exact dependence of C^{-2} on the quantum numbers and Z. Using his expression for C^{-2}, it can be shown that

$$F^{\text{rel}}_{nlj} = \frac{1 + \dfrac{1 - N - \dfrac{2\kappa|\kappa|}{n} q}{2\kappa - 1}}{(1-q)N^4 \left[1 - \dfrac{4(\alpha Z)^2}{4\kappa^2 - 1}\right]}, \tag{9}$$

where $\kappa = (-1)^{j+l+1/2}(j + 1/2)$, $q = 1 - [1 - (\alpha Z/\kappa)^2]^{1/2}$, and $N = [1 - 2|\kappa|\,(n - |\kappa|)q/n^2]^{1/2}$.

Jentschura and Yerokhin [17] have derived a similar expression for F^{rel}_{nlj} given by their formula (8)[5]. When comparing formulas (7) and (8) from Ref. [17] and formula (27.29) from Ref. [21] with our formulas (5–7, 9), it is useful to note that the following relations can be easily derived

$$\frac{\kappa}{|\kappa|(2\kappa + 1)} \equiv \frac{1}{|2\kappa + 1|} \equiv \frac{1}{2l + 1}. \tag{10}$$

Using these relations, it can be shown that formula (8) from Jentschura and Yerokhin [17] is identical to our formula (9).

For approximate calculations, formula (9) can be expanded as a series of powers of a small parameter αZ. Brodsky and Parsons [20] found the leading term in this expansion for the $2P_{1/2}$ and $2P_{3/2}$ levels. From their Eq. (8)

$$F^{\text{rel}}_{2p_{1/2}} \approx \frac{47}{24}(\alpha Z)^2, \quad F^{\text{rel}}_{2p_{3/2}} \approx \frac{7}{24}(\alpha Z)^2. \tag{11}$$

Numerically, the relative difference of factors calculated with the above approximate formulas and with formula (9) is 9×10^{-9} and 2×10^{-10} for the $2P_{1/2}$ and $2P_{3/2}$ levels, respectively.

The contribution of the off-diagonal Hamiltonian $\Delta E^{\text{off-diag}}_{nljF}$ was first discussed by Brodsky and Parsons [20], who found approximate expressions for the corresponding terms in the Hamiltonian for hydrogen and deuterium (see their Eqs. 10a and 10b). This contribution is proportional to the product $\langle \boldsymbol{I} \cdot \boldsymbol{L} \rangle$ and therefore is non-zero only for non-S states with non-zero F. For non-S states, the effect of this term on the energy levels can be understood as mutual repulsion of the levels with the same F but different j. Thus, it leads not only to a change in the hyperfine splittings, but also to a shift of the centers of gravity of hyperfine multiplets. Lundeen and Pipkin [88] calculated the value of the off-diagonal contribution of the $2P_{1/2}$ and $2P_{3/2}$ hydrogen levels with $F = 1$ to be 2.5 kHz. Thus, due to this effect, the hydrogen level $2P_{1/2,F=1}$ shifts downwards by 2.5 kHz, and the level $2P_{3/2,F=1}$ shifts upwards by the same amount. It should be noted that the value of this shift is correctly given in Fig. 11 of Lundeen and Pipkin [88], but it is misprinted as 0.0025 kHz in the text. This is confirmed by independent calculations of Latvamaa et al. [89]. The value 2.5 kHz was also given in Lundeen's thesis [90]. The off-diagonal corrections for other hydrogenic levels, to our knowledge, have never been published. They are discussed in the following section.

For the $2P_{1/2}$ level, due to the off-diagonal hfs interaction the center of gravity of the unresolved hfs doublet is shifted downwards by 1.9 kHz from the value computed without account for the hfs interaction (i.e., using the relativistic formula (4)). It is worthwhile to note that the theoretical reference values of energy levels of hydrogen and deuterium, given in the database [10], result from high-precision QED calculations that do not include the

[4] This result was originally obtained by Breit [85] who used a somewhat different notation.

[5] A similar result was obtained earlier by Nefiodov et al. [87]; see also references therein.

Table A
Comparison of hfs intervals of non-S states in hydrogen obtained by different authors.

H Level		hfs splitting[a] (MHz)	Reference
$2P_{1/2}$	Theory	(59.1501)(1)[b]	[58][b]
		(59.172)	[58][c]
		(59.169)(2)	[94,97][d]
		(59.1696)(6)	[95,88]
		(59.1695)(6)	This work
	Experiment	59.22(14)	[58]
$2P_{3/2}$	Theory	(23.6521)(8)	[97,98][d]
		(23.6516)(2)	[88][e]
		(23.65157)(24)	This work
$3P_{1/2}$	Theory	(17.537)	[99]
		(17.5268)(4)	[59][d,f]
		(17.528)(4)	[100][d]
		(17.53180)(18)	This work
$3P_{3/2}$	Theory	(7.0080)(16)	[100][d]
		(7.00792)(7)	This work
$3D_{3/2}$	Theory	(4.205)(3)	[100][d]
		(4.2050)(25)	This work
$3D_{5/2}$	Theory	(2.7034)(17)	[100][d]
		(2.701)(3)	This work
$4D_{5/2}$	Theory	(1.1414)(9)	[101][g]
		(1.1394)(11)	This work
$8D_{3/2}$	Theory	(0.222)	[102]
		(0.22177)(13)	This work
$8D_{5/2}$	Theory	(0.14268)	[103][h]
		(0.14243)(14)	This work
$8F_{5/2}$	Theory	(0.10191)	[103][h]
		(0.10162)(6)	This work

[a] Calculated (theoretical) values are given in parentheses. Uncertainty in the last decimal place is given in parentheses after the value, where available.
[b] From the text on p. 380 of Lundeen et al. [58] (the value is quoted from Lundeen's thesis [90]; the uncertainty is greatly underestimated; see text).
[c] From Fig. 1 of Lundeen et al. [58] (calculated without the off-diagonal correction).
[d] The given value of the hfs interval is inferred from the total hfs correction to the measured fine-structure interval given in the quoted paper.
[e] From Fig. 11 of Lundeen and Pipkin [88].
[f] From Table V of Fabjan and Pipkin [59] (whose method of calculation was not explained; it is not clear if this value includes the off-diagonal correction).
[g] The uncertainty ±100 Hz given by Weitz et al. [101] was greatly underestimated (see text).
[h] From Fig. 12 of de Beauvior et al. [103] (calculated with the Fermi formula from Bethe and Salpeter [19]).

hfs effects, as is indicated there. In this database, excitation energies of hydrogen are given with estimated uncertainties ranging from 40 Hz (for $2S_{1/2}$) to 2.4 kHz (for $2P_{1/2}$). While the above-mentioned off-diagonal hfs shift is smaller than the indicated uncertainty, it is comparable in size.

3.1.1. hfs Splittings of nS levels in hydrogen

From the experimental point of view, the *n*S levels of hydrogenic systems are especially important because of their small natural width. The metastable 2S level of hydrogen has a natural lifetime of ≈1/8 s [91] and therefore the two-photon 1S–2S transition is highly attractive for high-precision measurements. Although the other *n*S levels ($n > 2$) can decay to the 2P levels via electric dipole transitions, their natural linewidths (⩽1 MHz) are much smaller than those of the non-S states. Therefore, they are also attractive to both experimentalists and theorists.

For the *n*S levels of hydrogen, a number of theoretical calculations have been published. Those calculations consider higher-order QED corrections beyond formula (6). However, as mentioned in the Introduction, calculations from first principles are limited by the lack of knowledge of certain parameters of the internal structure of the nucleus. On the other hand, the existence of high-precision experimental data on the 1S and 2S hfs splittings makes it possible to derive empirical corrections accounting for these nuclear parameters. These corrections can be scaled to higher values of *n*. This was done by Jentschura and Yerokhin [17] for the hydrogen *n*S levels with $n \leqslant 8$, for which they calculated the hfs splittings with estimated relative uncertainties increasing from 1.7×10^{-9} for $n = 2$ to 1.1×10^{-8} for $n = 8$. To obtain the values for the *n*S hfs splittings for $n \geqslant 9$, we have plotted the ratio of the precise hfs values from Jentschura and Yerokhin [17] to those obtained from formulas (5–7, 9) against $1/n$ (see Fig. 2). A least-squares fitting then yields the following approximate formula for the correction to formulas (5, 6, 9)

$$A_{ns} \approx A_{ns}(\mathrm{Eq.6})[1 - F_{\mathrm{corr}}(\mathrm{H}, ns)], \tag{12}$$

$$F_{\mathrm{corr}}(\mathrm{H}, ns) = 1.33045 \times 10^{-4} + 4.5583 \times 10^{-6}/n - 2.5917 \times 10^{-6}/n^2. \tag{13}$$

For $n = 2$–8, the values of hfs splittings calculated with the interpolation formulas (12), (13) differ from those given by Jentschura and Yerokhin [17] by a fraction of 7×10^{-9}. Thus, we expect that extrapolation to higher *n* values should be valid within a fractional uncertainty of at most 10^{-7}. Since the values of the hfs splittings of highly-excited levels are small and decrease rapidly with increasing *n*, this uncertainty should be sufficiently small for any practical application.

3.1.2. hfs Splittings of non-S levels in hydrogen

For the $2P_{1/2}$ and $2P_{3/2}$ levels of hydrogen, the calculation of Brodsky and Parsons [20] remains the main source of accurate hfs data. The hfs splittings of $2P_{1/2,3/2}$ are especially important since they affect the high-precision measurements of the $2P_{1/2}$–$2S_{1/2}$ Lamb shift [92–96,88] and the $2S_{1/2}$–$2P_{3/2}$ fine-structure interval [97,98]. Brodsky and Parsons [20] gave explicit formulas for these hfs intervals (formulas (8, 9), and (10a) of their paper). Most authors investigating hydrogen and deuterium use those formulas in different approximations. The values of hfs splittings obtained by different authors are compared in Table A (note that in all tables in the present paper, purely theoretical values are enclosed in parentheses, and semi-empirical ones are enclosed in square brackets).

Lundeen et al. [58] give two different calculated values for hfs($2P_{1/2}$). In the text on page 380 they give a value 59.150 1(1) MHz with a reference to Brodsky and Parsons [20]. In fact, this value is quoted from Lundeen's thesis [90], where it was calculated using a truncated version of formula (8) of Brodsky and Parsons without account for relativistic correction and anomalous magnetic moments of electron and proton. This value was also used by Lundeen and Pipkin [92] to calculate the hfs correction necessary to obtain the $2P_{1/2}$–$2S_{1/2}$ Lamb shift from their measurement of the $2P_{1/2,F=1}$–$2S_{1/2,F=0}$ hyperfine interval. In Fig. 1 of Lundeen et al. [58], the value 59.172 MHz is given. As follows from our calculations, this value was obtained by using the complete version of Brodsky and Parsons' formula (8) but without the off-diagonal corrections (formula (10a) of Ref. [20]). Such values should not be used if uncertainty smaller than ±20 kHz is required. The off-diagonal corrections to the $2P_{1/2}$ and $2P_{3/2}$ hfs splittings were calculated by Lundeen [90], Newton et al. [95] and Lundeen and Pipkin [88] by diagonalizing the matrix resulting from formulas (10a) and (16) of Brodsky and Parsons [20]. These corrections result in a downward shift of the $2P_{1/2,F=1}$ level by 2.5 kHz and an upward shift of the $2P_{3/2,F=1}$ by the same amount. The other levels are not affected. The same result was independently obtained by Latvamaa et al. [89].

The final result for the hfs splitting of $2P_{1/2}$ obtained by Lundeen and Pipkin [88] is 59.169 6(6) MHz, in perfect agreement with Newton et al. [95]. The value 59.169 MHz, inferred from the hfs correction to Lamb shift given by Andrews and Newton [94] and Safinya et al. [97], is consistent with 59.169 6 MHz [95,88] within rounding errors. Our calculation using formula (8) from Brodsky and Parsons [20] with the current values of the fundamental constants [24] without account of the off-diagonal correction yields 59.172 0 MHz, only 100 Hz away from the value 59.172 1 MHz in Fig. 11 of Lundeen and Pipkin [88]. Subtracting the off-diagonal correction 2.5 kHz, we obtain 59.169 5 MHz, in fair agreement with Newton et al. [95] and Lundeen and Pipkin [88].

For the $n \geqslant 3$ levels, there are only a few published data. The value for the hfs of $3P_{1/2}$ given in Fig. 1 of the early work of Fabjan and Pipkin [99] should be ignored, as it disagrees both with the values obtained later by the same authors [59] and by Van Baak et al. [100]. The latter two values agree well with each other. Fabjan and Pipkin [59] did not specify whether or not they included the off-diagonal correction in their calculation. Van Baak et al. [100] specifically note that they applied the off-diagonal correction. Unfortunately, they gave only the values of hfs corrections to the measured frequency intervals rounded to three decimal places (in units of megahertz) and indicated that the uncertainties are roughly on the level of 1 kHz. These corrections are fractions of the hfs intervals, so this uncertainty translates to the larger values given in Table A. Our calculations described below yield a value of the $3P_{1/2}$ hfs splitting that is higher than Fabjan and Pipkin's [59] by about 10 combined uncertainties. Given the estimated uncertainty of our value is 10 ppm, the 23 ppm uncertainty quoted by Fabjan and Pipkin [59] appears to be too small; the 10 times greater uncertainty given by Van Baak et al. [100] is more realistic.

Weitz et al. [101] calculated the hfs splitting of $4D_{5/2}$ by scaling from the hfs of 1S. This procedure cannot provide an uncertainty smaller than 800 ppm. Therefore, we increased their 100 Hz estimated uncertainty ninefold to 900 Hz. Their value agrees with our calculation within the combined uncertainties.

To calculate the hfs splittings of the $8D_{5/2}$ and $8F_{5/2}$ levels, de Beauvoir et al. [103] used a modified Fermi formula similar to our formulas (5) and (6) without the off-diagonal and relativistic corrections [104]. According to a private communication from F. Nez, the value of the hfs shift of the $8D_{5/2,F=2}$ level was misprinted in their Table 5. The correct and more precise values of hfs shifts are given in Fig. 12 of their paper.

Lundeen and Pipkin [88] stated that the uncertainty of the calculations of Brodsky and Parsons [20] for the hfs of $2P_{1/2}$ and $2P_{3/2}$ is 10 ppm. Although the effects of the distribution of magnetization within the nucleus, which are as large as 30 ppm for S-levels, are not accounted for by Brodsky and Parsons [20], these effects are much smaller for non-S levels. The same argument applies to higher-order radiative corrections. The relativistic corrections, as mentioned above, are accounted for by Brodsky and Parsons [20] quite well. Thus we conclude that the 10 ppm uncertainty estimate is justified and apply the same estimate to the calculations of the other nP levels taking into account the off-diagonal corrections.

The formulas (8) and (9) of Brodsky and Parsons [20] differ from our formulas (5–7) in the following way. Their formulas (8), in addition to the factor $g_s/2$ (which corresponds to the factor μ_e/μ_B in our formula (5)), contain terms proportional to $(g_s - 2)$ and m_e/M_N, the largest of which is proportional to $(g_s - 2)$. The numerical coefficients in this term are different for the $2P_{1/2}$ and $2P_{3/2}$ levels. This implies that they should depend on j and probably on l. We calculated the hfs splittings of nP, nD, and nF levels with $n \leqslant 12$ using formulas (8), (9), (10), and (16) from Brodsky and Parsons [20] having replaced the relativistic correction factor with that from our formula (9) and using the $1/n^3$ scaling for the Fermi splitting E_F. For the nD and nF levels, we neglected the possible dependence of the additional terms in formulas (8) of Brodsky and Parsons [20] on j and l and assumed that these coefficients for nl $(j = l - 1/2)$ and nl $(j = l + 1/2)$ are the same as for $nP_{1/2}$ and $nP_{3/2}$, respectively. The relative uncertainty introduced by this assumption is of order $(g_s - 2)/2 \approx 10^{-3}$.

The off-diagonal hfs interaction contributes about 100 ppm to the value of the hfs splitting of $2P_{3/2}$. Despite its smallness, it is conceptually important, since it introduces a shift of centers of gravity of fine-structure levels split by hyperfine interactions. The off-diagonal corrections to the hfs can be calculated using formulas (10) and (16) of Brodsky and Parsons [20] for arbitrary n and l. The character of these corrections is similar for all nl $(l > 0)$. Namely, there are only two mutually repulsed hyperfine levels for each nl with $F = l$ (e.g., $2P_{1/2,F=1}$ and $2P_{3/2,F=1}$, or $3D_{3/2,F=2}$ and $3D_{5/2,F=2}$). The magnitude of this repulsion can be approximately calculated by using perturbation theory

$$\Delta E^{\text{off-diag}}_{nljIF} = \left\langle H^{\text{off-diag}}_{hfs,nljj'IF} \right\rangle^2 \Big/ \Delta E_{njj'}, \tag{14}$$

where the matrix elements of the off-diagonal Hamiltonian are given by formulas (10) and (16) of Brodsky and Parsons [20], and $\Delta E_{njj'}$ is the difference of relativistic energies of levels with quantum numbers j and j' computed with formula (4). We estimate the relative uncertainty of this calculation to be 10^{-3}. It is interesting to note that the off-diagonal hfs shifts of nl levels with the maximum possible $l = n - 1$ decrease very slowly with increasing n. Thus, for the levels $3D_{3/2,5/2}(F = 2)$, the off-diagonal shift is 2.4 kHz, which is only slightly smaller than for $2P_{1/2,3/2}(F = 1)$, 2.5 kHz, and for $12N(F = 11)$ it is 0.9 kHz.

The values for the hydrogen hfs splittings quoted from the literature or calculated as described above (with account for the off-diagonal shifts) are collected in Table 1. In this table, we also give the values of the shifts of centers of gravity of the hyperfine multiplets caused by the off-diagonal hfs.

3.1.3. hfs Splittings in deuterium

The measurements of the 1S and 2S hfs splittings in deuterium were reviewed in the beginning of Section 3. Besides these, there are no known measurements of hfs in deuterium. Thus, the values of hfs splittings for other deuterium levels must be obtained from theoretical calculations.

By calculating the ratios of measured hfs splittings for 1S and 2S to those obtained from formulas (6, 7, 9), we derived correction factors for the higher nS levels using a formula similar to formula (13)

$$F_{corr}(\mathrm{D}, ns) = 4.592\,377 \times 10^{-4} + 2.338\,3 \times 10^{-6}/n. \quad (15)$$

Since there are only two experimental points (n = 1 and 2), this formula has only a linear term proportional to $1/n$. To estimate the systematic error in the extrapolation of formula (15) to $n > 2$, caused by truncation of the expansion in powers of $1/n$ as compared to the more precise formula (13) for hydrogen, we made a similar fitting for hydrogen. Namely, we truncated the extrapolation formula (13) to the same number of terms as in formula (15) and fitted it to the measured hfs splittings for n = 1 and 2 in hydrogen. The approximate values of the hfs splittings of the hydrogen nS levels ($3 \leqslant n \leqslant 12$) obtained are systematically lower than the more accurate ones obtained in Section 3.1.1 with the complete formula (13), the relative error being $\leqslant 1.8 \times 10^{-7}$. We assume that the systematic errors for deuterium, caused by truncation of the expansion in formula (15), are of the same size. Combining this estimate with the measurement uncertainty of the hfs splitting of the deuterium 2S level (±7 Hz [83]), we estimate that the relative error in the extrapolation of formula (15) to the nS levels of deuterium with $n \leqslant 12$ is within 2.4×10^{-7}. Assuming that the dependence of the correction factor on n is similar to that depicted for hydrogen in Fig. 2, we can expect that the deuterium hfs splittings extrapolated using formula (15) (see Table 2) are systematically slightly underestimated (within the given error limit).

Drake et al. [22,23] suggested an approximate formula for calculating the hfs energy shifts in hydrogen-like atoms (see footnote to Table V in Ref. [22] and footnote to Table I in Ref. [23])

$$\Delta E_{nljIF} = \frac{\alpha_{\mathrm{HFS}}}{n^3}\left(\frac{F(F+1) - j(j+1) - I(I+1)}{j(j+1)(l+0.5)}\right). \quad (16)$$

This formula is similar to our formulas (5, 6) without the relativistic and off-diagonal corrections. We note that the value of α_{HFS} given by Drake et al. [22] for deuterium (32.74 MHz) was too small by a factor of 1.25. In a subsequent paper Drake et al. [23] corrected this value to 40.923 MHz. Our calculations yield α_{HFS} = 40.918 36 MHz for deuterium, 266.341 02 MHz for hydrogen, and 284.399 13 MHz for tritium.

The diagonal magnetic hfs splittings of the nP levels can be calculated using formulas (8) and (9) from Brodsky and Parsons [20]. The off-diagonal corrections for arbitrary nl can be calculated using formulas (10b) and (16) of Brodsky and Parsons [20]. In these calculations, one should note that the definition of the "anomalous magnetic moment" of the deuteron κ_{D} used by Brodsky and Parsons [20] differs from the definition of "magnetic moment anomaly" used in the currently adopted set of fundamental constants [24]. According to Brodsky and Parsons [20]

$$\kappa_{\mathrm{D}} = g_{\mathrm{d}}(M_{\mathrm{d}}/M_{\mathrm{p}}) - 1 \approx 0.714, \quad (17)$$

where g_{d} is the g-factor of the deuteron ($g_{\mathrm{d}} \equiv \mu_{\mathrm{d}}/(e\hbar/2M_{\mathrm{p}})$ = 0.857 438 230 8(72) [24]) and M_{d} and M_{p} are masses of the deuteron and the proton. The magnetic moment anomaly of the deuteron a_{d} is defined in the CODATA-2006 set of constants [24] as follows

$$a_{\mathrm{d}} = g_{\mathrm{d}}(M_{\mathrm{d}}/2M_{\mathrm{p}}) - 1 \approx -0.143. \quad (18)$$

For the values of the nP magnetic hfs splittings calculated with the formulas from Brodsky and Parsons [20] (accounting for the off-diagonal correction), we adopted the same relative uncertainty as for hydrogen, which is estimated by Lundeen and Pipkin [88] to be 10 ppm (see the previous section).

The character of the off-diagonal magnetic hfs shifts in deuterium is similar to that in hydrogen. For each nlF ($l > 0$, $l - 1/2 \leqslant F \leqslant l + 1/2$), there are two levels with $j = F$ and $j = F + 1$ that are shifted by the same amount in opposite directions (all levels with $l = 0$ and levels with $l > 0$, $F = l \pm 3/2$ are not shifted). The largest shifts are 0.6 kHz for the levels $2\mathrm{P}_{1/2,3/2}$ ($F = 3/2$).

To calculate the magnetic hfs for the nD and nF levels, we used the same approach based on formulas (8), (9), (10), and (16) of Brodsky and Parsons [20], as we did for hydrogen. The relative uncertainties of these calculations are estimated to be 10^{-3}.

Unlike the proton and the triton, the deuteron has a non-zero electric quadrupole moment Q_{d} = 0.2860(15) fm^2 [105]. This leads to appearance of quadrupole hyperfine shifts in deuterium. Brodsky and Parsons [20], quoting Lamb [44], in their Appendix B stated that the electric quadrupole moment can only affect the $2\mathrm{P}_{3/2}$ level (and not $2\mathrm{P}_{1/2}$), and the additional energy of this level is of the order 6 kHz. We calculated the quadrupole hfs by using formula (27.45) from Sobel'man [21] for the quadrupole hfs constant B_{nlj}. Neglecting relativistic effects

$$B_{nlj} \approx \frac{3(Q_{\mathrm{d}}a_0^{-2})Z^3 R_\infty c}{8n^3 I(2I-1)j(j+1)(l+1)(l+0.5)l} \approx \frac{126000(700)}{n^3 j(j+1)(l+1)(l+0.5)l}[\,\mathrm{Hz}], \quad (19)$$

and the quadrupole hfs shifts for individual hyperfine levels are

$$\Delta E^{\mathrm{quad}}_{nljF} = B_{nlj}C(C+1), \quad (20)$$

where $C = F(F+1) - j(j+1) - I(I+1)$.

For the $2\mathrm{P}_{1/2}$ levels, both hfs levels with $F = 1/2$ and $F = 3/2$ are shifted upwards by the same amount, 14 kHz, while for $2\mathrm{P}_{3/2}$ the shifts vary between 2.8 kHz and 28 kHz. Thus, Lamb's estimates of electric-quadrupole shifts of 2P quoted by Brodsky and Parsons [20] appear to be too low. We calculated the electric-quadrupole hfs shifts for all nP, nD, and nF levels with $n \leqslant 12$ using formulas (19) and (20). Relative uncertainties of this calculation are mostly due to the measurement uncertainty of Q_{d}, which is 5×10^{-3}.

Calculated values of hfs splittings and shifts in deuterium, along with the 1S and 2S measurements [15,83], are collected in Table 2.

There are only a few published theoretical values of the hfs intervals for excited states of deuterium. They are compared with our calculations in Table B.

3.1.4. hfs Splittings in tritium

There are no experimental data on tritium hfs splittings besides those for the 1S ground level described in the beginning of Section 3. This is because tritium is radioactive and hard to handle.

Table B
Comparison of hfs intervals in deuterium calculated by different authors.

D level	Interval	hfs splitting[a] (MHz)	Reference
$4\mathrm{S}_{1/2}$	$F = 1/2–3/2$	(5.115 54)(3)	[101]
		[5.115 551 2](12)	This work
$4\mathrm{D}_{5/2}$	$F = 3/2–5/2$	(0.146 17)	[101][b]
		(0.145 43)(15)	This work
	$F = 5/2–7/2$	(0.204 64)	[101][b]
		(0.204 52)(21)	This work
$8\mathrm{D}_{5/2}$	$F = 3/2–5/2$	(0.183)	[103][c]
		(0.181 78)(18)	This work
	$F = 5/2–7/2$	(0.256)	[103][c]
		(0.255 7)(3)	This work

[a] Calculated (theoretical) values are given in parentheses. The uncertainty in the last decimal place is given in parentheses after the value, where available.
[b] Calculated by scaling from the hfs of 1S.
[c] From Table 5 of de Beauvior et al. [103] (calculated with the Fermi formula from Bethe and Salpeter [19]).

Thus, the values of hfs splittings for other tritium levels must be obtained from theoretical calculations.

By calculating the ratio of the measured hfs splitting for 1S to that obtained from formulas (5–7), we derived the correction factor for the higher nS levels using a formula similar to formulas (13) and (15)

$$F_{\mathrm{corr}}(\mathrm{T}, ns) = 1.417\,870 \times 10^{-4}. \tag{21}$$

Since there is only one experimental point available for $n = 1$, this formula has only a constant term. Using a similar procedure to that applied to deuterium (i.e., truncating the correction formulas (13) and (15) to leave only a constant term), we estimate that the relative error in extrapolation of formula (21) to nS levels with $n \leqslant 12$ is within 2 ppm. Assuming that the dependence of the correction factor on n is similar to that depicted for hydrogen in Fig. 2, we can expect that the tritium hfs splittings extrapolated using formula (21) (see Table 3) are systematically underestimated within the given error limit.

To calculate the hfs splittings for non-S levels, we used the same method as for hydrogen. We replaced the proton mass M_{p}, magnetic moment μ_{p}, and magnetic moment anomaly κ_P in formulas (8), (9), and (10a) of Brodsky and Parsons [20] with the corresponding values for the triton from Mohr et al. [24], assuming

$$\kappa_{\mathrm{T}} \equiv a_{\mathrm{t}} \equiv (\mu_{\mathrm{t}}/\mu_{\mathrm{N}})(M_{\mathrm{t}}/M_{\mathrm{p}}) - 1 \approx 7.91817. \tag{22}$$

This calculation should be accurate within 10 ppm for the nP levels and 1000 ppm for all other levels.

Calculated values of hfs splittings and shifts in tritium, along with the experimental value for the 1S level [16], are collected in Table 3.

4. Fine structure measurements in hydrogen

Table C lists the most precise measurements of fine structure intervals in hydrogen selected for the present compilation. Modern high-precision measurements are very complex experiments in which care is required to take into account many physical effects. In most cases, the actually measured quantities are energy separations between completely or partially resolved hfs or Zeeman sublevels. Especially in the latter case, derivation of the fine-structure data from the measured signals is possible only by detailed quantum–mechanical modeling of signal shapes. To simplify the discussion, we list the hfs intervals as quantities measured in those experiments, based on modeling described, for example, by Fabjan et al. [106] and Brown and Pipkin [107]. For each experiment, where possible, we indicate the size and uncertainty of the largest purely theoretical correction required to obtain the fine-structure interval from the actually measured quantity. It can be seen from Table C that in most cases the largest theoretical correction is due to the hyperfine splitting of one or both fine-structure levels involved in the measured transition. Besides that, there are usually several other effects causing systematic shifts, such as dc or ac Stark effects, linear and quadratic Zeeman effects, photon recoil, Doppler shift, etc. Although these effects also require theoretical modeling, in most cases this modeling is semi-empirical, that is, based on additional experimental information. Thus, the hyperfine structure is the main factor that is considered purely theoretically in the vast majority of the selected experiments.

In several experiments where the hyperfine structure was not completely resolved (e.g., Refs. [106–110]) theoretical modeling of line shapes and shifts required the use of calculated hfs intervals along with some assumptions about the population distribution and transition strengths among hfs sublevels. Although it is not always possible to extract information about the total size of contribution of these effects to the measured intervals, this size can be estimated using the data from other experiments where similar fine-structure intervals were measured by different methods. The size of these effects is usually a fraction of the hfs splitting of the levels involved. Examples of the methods of calculation of such fractions for single-photon and two-photon transitions are given below.

According to White and Eliason [56], one can calculate the relative intensities of single-photon electric-dipole hyperfine transitions (n_i,l_i,j_i,F_i)–(n_k,l_k,j_k,F_k) using the corresponding formula for the fine-structure multiplet (see, for example, Sobel'man [21], formula (31.40); this formula follows from the sum rules described in Section 2) by replacing S with I, L with j, and j with F

$$I_{\mathrm{rel}} = (2F_i + 1)(2F_k + 1)\begin{Bmatrix} j_i & F_i & I \\ F_k & j_k & 1 \end{Bmatrix}^2, \tag{23}$$

where $I = 1/2$ for hydrogen and tritium and 1 for deuterium. Thus, in experiments involving excitation from a single hyperfine component of the metastable 2S level to nP levels (e.g., Zhao et al. [111]) the unresolved hyperfine components of nP are weighted with factors that differ from the statistical weights $(2F_k + 1)$. To derive the position of the center of gravity of nP from such measurements, a small correction is required. The size of this correction can be easily calculated using the formula above.

A corresponding formula for the relative intensities of two-photon hyperfine transitions was first given by Cagnac et al. [112]. Extensive theoretical investigation of multiphoton transitions was made by Grynberg in his thesis [113] and published in Grynberg and Cagnac [114]. For two-photon hyperfine transitions, one finds

$$I_{\mathrm{rel}} \propto (2F_i + 1)(2F_k + 1)\begin{Bmatrix} j_k & j_i & 2 \\ F_i & F_k & I \end{Bmatrix}^2. \tag{24}$$

From the above discussion it is clear that, in order to derive the fine-structure energy levels from the available experimental data, it is necessary to assume the validity of theoretical values of hfs splittings reviewed in Section 3. This assumption, although well justified by available experimental data on hydrogen, deuterium, and tritium, as well as on other atomic ions, still leaves some room for criticism and underlines a necessity of further experimental research on the hfs of hydrogen and its isotopes. However, these purely theoretical corrections constitute only a small fraction of the measured quantities, in most cases smaller than 0.15%. Thus, even if a measured quantity cannot be considered as purely experimental, it is essentially such. This was an important criterion for selecting the data for the present compilation.

For example, the $2\mathrm{P}_{1/2} - 2\mathrm{S}$ Lamb shift measurement of Pal'chikov et al. [96] was not included in Table C because the actually measured quantity in this experiment was the ratio of the frequency of the $2\mathrm{P}_{1/2,F=1} - 2\mathrm{S}_{1/2,F=0}$ hyperfine interval to the purely theoretical decay rate of its lower level. Both quantities in this ratio are of the same order of magnitude. Although this method indeed provides a good independent test of internal consistency of quantum theory, it cannot be considered as an essentially experimental measurement of the Lamb shift.

A few of the measurements listed in Table C stand out as having none or very small theoretical corrections to the measured quantities. They result from the experiments of Hänsch et al. [115], de Beauvoir et al. [103], and Weitz et al. [101]. Those and other experiments are discussed below.

Hänsch et al. [115] measured the 1S–2S two-photon transition frequency by phase coherent comparison with a microwave cesium fountain clock using a laser frequency comb. Additional details about this experiment can be found in the works of Fischer et al. [122] and Niering et al. [123]. The value for the 1S–2S frequency is obtained from the measured $1\mathrm{S}_{F=1} - 2\mathrm{S}_{F=1}$ hfs interval

Table C
The best available measurements of fine-structure intervals in hydrogen (MHz).

Fine structure interval	Measured value (MHz)	Actually measured interval	Largest theoretical correction[a]	Ref.
$1S_{1/2}-2S_{1/2}$	2 466 061 413.187 074(34)	$1S_{F=1}-2S_{F=1}$	None	[115]
$2P_{1/2}-2S_{1/2}$	1 057.847(9)	$2P_{1/2\,F=1}-2S_{F=0}$	(14.792 4)(2) = 1/4 hfs($2P_{1/2}$)	[88][b]
$2P_{1/2}-2P_{3/2}$	10 969.13(10)	$2P_{1/2\,F=0\,1}-2P_{3/2\,F=1\,2}$	hfs($2P_{1/2}$,$2P_{3/2}$)[c]	[108][c]
$2S_{1/2}-2P_{3/2}$	9 911.201(12)	$2S_{F=0}-2P_{3/2\,F=1}$	(14.782 3)(2) = 5/8 hfs($2P_{3/2}$)	[98][d]
$2P_{1/2}-3D_{3/2}$	456 685 852.8(17) + x	$2P_{1/2\,F=0\,1}-3D_{3/2\,F=1\,2}$	hfs($2P_{1/2}$,$3D_{3/2}$)[c]	[116][c,e]
$2S_{1/2}-3P_{1/2}$	456 681 549.9(3) + x	$2S_{F=1}-3P_{1/2\,F=0\,1}$	(1.460 98)(2) = 1/12 hfs($3P_{1/2}$)	[111][f]
$2S_{1/2}-3P_{3/2}$	456 684 800.1(3) + x	$2S_{F=1}-3P_{3/2\,F=1\,2}$	(−1.459 98)(2) = −5/24 hfs($3P_{3/2}$)	[111][f]
$2P_{3/2}-3D_{5/2}$	456 675 968.3(34) + x	$2P_{3/2\,F=1\,2}-3D_{5/2\,F=2\,3}$	hfs($2P_{3/2}$,$3D_{5/2}$)[c]	[116][c,e]
$2S_{1/2}-4P_{1/2}$	616 520 017.568(15)	$(2S_{F=1}-4P_{1/2\,F=0\,1})-1/4(1S_{F=1}-2S_{F=1})$	(−0.841) 2S–4P photon recoil	[117][g]
$2S_{1/2}-4S_{1/2}$	616 520 150.636(10)	$(2S_{F=1}-4S_{F=1})-1/4(1S_{F=1}-2S_{F=1})$	(0.005)(3) 2nd-order Zeeman shift	[101][h]
$2S_{1/2}-4P_{3/2}$	616 521 388.672(10)	$(2S_{F=1}-4P_{3/2\,F=1\,2})-1/4(1S_{F=1}-2S_{F=1})$	(−0.841) 2S–4P photon recoil	[117][g]
$2S_{1/2}-4D_{5/2}$	616 521 843.441(24)	$(2S_{F=1}-4D_{5/2\,F=2\,3})-1/4(1S_{F=1}-2S_{F=1})$	(−0.221 6)(21) = −7/36 hfs($4D_{5/2}$)	[101][i]
$2S_{1/2}-6S_{1/2}$	730 690 017.097(21) + $x/4$	$(2S_{F=1}-6S_{F=1})-1/4(1S_{F=1}-3S_{F=1})$	None	[103][j]
$2S_{1/2}-6D_{5/2}$	730 690 518.592(11) + $x/4$	$(2S_{F=1}-6D_{5/2,F=2\,3})-1/4(1S_{F=1}-3S_{F=1})$	(−0.065 64)(6) = −7/36 hfs($6D_{5/2}$)	[103][k]
$2S_{1/2}-8S_{1/2}$	770 649 350.012(9)	$2S_{F=1}-8S_{F=1}$	None	[103]
$2S_{1/2}-8D_{3/2}$	770 649 504.450(8)	$2S_{F=1}-8D_{3/2\,F=1\,2}$	(0.027 72)(3) = 1/8 hfs($8D_{3/2}$)	[103]
$2S_{1/2}-8D_{5/2}$	770 649 561.584(6)	$2S_{F=1}-8D_{5/2\,F=2\,3}$	(−0.027 69)(3) = −7/36 hfs($8D_{5/2}$)	[103]
$2S_{1/2}-10D_{5/2}$	789 144 886.411(39)	$2S_{F=1}-10D_{5/2\,F=2\,3}$	(−0.014 179)(14) = −7/36 hfs($10D_{5/2}$)	[118][l]
$2S_{1/2}-12D_{3/2}$	799 191 710.473(9)	$2S_{F=1}-12D_{3/2\,F=1\,2}$	(0.008 213)(9) = 1/8 hfs($12D_{3/2}$)	[103]
$2S_{1/2}-12D_{5/2}$	799 191 727.404(7)	$2S_{F=1}-12D_{5/2\,F=2\,3}$	(−0.008 206)(8) = −7/36 hfs($12D_{5/2}$)	[103]
$3P_{1/2}-3S_{1/2}$	314.818(48)	$3P_{1/2\,F=1}-3S_{F=0}$	(4.382 95)(5) = 1/4 hfs($3P_{1/2}$)	[59][m]
$3P_{1/2}-3D_{3/2}$	3 244.9(31)	$3P_{1/2\,F=0\,1}-3D_{3/2\,F=1\,2}$	hfs($3P_{1/2}$,$3D_{3/2}$)[c]	[110][c]
$3S_{1/2}-3P_{3/2}$	2 933.5(12)	$3S_{F=0\,1}-3P_{3/2\,F=1\,2}$	hfs($3S_{1/2}$,$3P_{3/2}$)[c]	[106][c]
$3S_{1/2}-3D_{3/2}$	2 929.9(8)	$3S_{F=0\,1}-3D_{3/2\,F=1\,2}$	hfs($3S_{1/2}$,$3D_{3/2}$)[c]	[109][c]
$3S_{1/2}-3D_{5/2}$	4 013.155(48)	$3S_{F=0}-3D_{5/2\,F=2}$	(1.576)(2) = 7/12 hfs($3D_{5/2}$)	[119]
$3D_{3/2}-3P_{3/2}$	5.5(9)	$3D_{3/2\,F=1\,2}-3P_{3/2\,F=1\,2}$	(0.3)(3) Unresolved, partly decoupled hfs	[120]
$3D_{3/2}-3D_{5/2}$	1 083.0(29)	$3D_{3/2\,F=1\,2}-3D_{5/2\,F=2\,3}$	hfs($3D_{3/2}$,$3D_{5/2}$)[c]	[110][c]
$3P_{3/2}-3D_{5/2}$	1 078.0(11)	$3P_{3/2\,F=1\,2}-3D_{5/2\,F=2\,3}$	hfs($3P_{3/2}$,$3D_{5/2}$)[c]	[110][c]
$4P_{1/2}-4S_{1/2}$	133.2(6)	$4P_{1/2\,F=0\,1}-4S_{F=0\,1}$	hfs($4S_{1/2}$,$4P_{3/2}$)[c]	[106][c]
$4P_{1/2}-4P_{3/2}$	1 370.85(22)	$4P_{1/2\,F=0\,1}-4P_{3/2\,F=1\,2}$	(0.00)(4) hfs($4P_{1/2}$,$4P_{3/2}$)	[121]
$4P_{1/2}-4D_{3/2}$	1 371.1(12)	$4P_{1/2\,F=0\,1}-4D_{3/2\,F=1\,2}$	hfs($4P_{1/2}$,$4D_{3/2}$)[c]	[106][c]
$4S_{1/2}-4D_{3/2}$	1 235.0(21)	$4S_{F=0\,1}-4D_{3/2\,F=1\,2}$	hfs($4S_{1/2}$,$4D_{3/2}$)[c]	[110][c]
$4S_{1/2}-4P_{3/2}$	1 237.79(29)	$4S_{F=0\,1}-4P_{3/2\,F=1\,2}$	hfs($4S_{1/2}$,$4P_{3/2}$)[c]	[107][c]
$4S_{1/2}-4D_{5/2}$	1 693.0(4)	$4S_{F=0\,1}-4D_{5/2\,F=2\,3}$	hfs($4S_{1/2}$,$4D_{5/2}$)[c]	[110][c]
$4D_{3/2}-4F_{5/2}$	456.8(16)	$4D_{3/2\,F=1\,2}-4F_{5/2\,F=2\,3}$	hfs($4D_{3/2}$,$4F_{5/2}$)[c]	[106][c]
$4D_{3/2}-4D_{5/2}$	458.0(22)	$4D_{3/2\,F=1\,2}-4D_{5/2\,F=2\,3}$	hfs($4D_{3/2}$,$4D_{5/2}$)[c]	[110][c]
$4P_{3/2}-4D_{5/2}$	455.7(16)	$4P_{3/2\,F=1\,2}-4D_{5/2\,F=2\,3}$	hfs($4P_{3/2}$,$4D_{5/2}$)[c]	[106][c]
$4D_{5/2}-4F_{7/2}$	227.96(41)	$4D_{5/2\,F=2\,3}-4F_{7/2\,F=3\,4}$	hfs($4D_{3/2}$,$4F_{5/2}$)[c]	[106][c]
$5P_{1/2}-5S_{1/2}$	64.6(50)	$5P_{1/2\,F=0\,1}-5S_{F=0\,1}$	hfs($5S_{1/2}$,$5P_{3/2}$)[c]	[106][c]
$5P_{1/2}-5D_{3/2}$	704(7)	$5P_{1/2\,F=0\,1}-5D_{3/2\,F=1\,2}$	hfs($5P_{1/2}$,$5D_{3/2}$)[c]	[106][c]
$5S_{1/2}-5P_{3/2}$	622(10)	$5S_{F=0\,1}-5P_{3/2\,F=1\,2}$	hfs($5S_{1/2}$,$5P_{3/2}$)[c]	[106][c]
$5P_{3/2}-5D_{5/2}$	232.2(29)	$5P_{3/2\,F=1\,2}-5D_{5/2\,F=2\,3}$	hfs($5P_{3/2}$,$5D_{5/2}$)[c]	[106][c]
$5D_{5/2}-5F_{7/2}$	117.2(15)	$5D_{5/2\,F=2\,3}-5F_{7/2\,F=3\,4}$	hfs($5D_{3/2}$,$5F_{5/2}$)[c]	[106][c]

[a] The uncertainty given for corrections accounting for non-statistical excitation of hfs sub-levels includes only the uncertainty of the adopted hfs splittings. Numerical values of fractions of hfs intervals in the corrections were calculated using formulas (23), (24).

[b] The value for the frequency of the $2P_{1/2}$–2S transition is obtained by adding 0.002 MHz to the "Lamb shift" measured by Lundeen and Pipkin [88] to account for the off-diagonal hfs correction to the $2P_{1/2}$ hfs splitting (see text).

[c] In the measurements of Fabjan et al. [106], Brown and Pipkin [107], Baird et al. [108], Glass-Maujean [109], Glass-Maujean et al. [110], and Hänsch et al. [116], theoretical values of hfs were used along with some assumptions about populations of hfs components. It is not possible to estimate the numerical values of corrections resulting from these assumptions.

[d] The value for the frequency of the 2S–$2P_{3/2}$ transition is obtained by adding 1 kHz to the interval given by Hagley and Pipkin [98] to account for the off-diagonal hfs correction to the $2P_{3/2}$ hfs splitting (see text).

[e] Recalibrated using the $2S_{1/2}$–$3P_{3/2}$ measurement of Zhao et al. [111] (see text).

[f] Re-calibrated using the correction factor determined from the 2S–$4P_{1/2\,3/2}$ frequencies measured by Zhao et al. [121] and Berkeland et al. [117] (see text).

[g] We derived the $2S_{1/2}$–$4P_J$ frequencies from the original measured quantities, which are the frequency differences $\nu(2S_{F=1}-4P_J)-1/4\nu(1S_{F=1}-2S_{F=1})$, by adding the hfs correction obtained from the adopted value of the 2S hfs splittings (see text) and 1/4 of the $1S_{F=1}$–$2S_{F=1}$ frequency measured by Hänsch et al. [115].

[h] We derived the $2S_{1/2}$–$4S_{1/2}$ frequency from the original measured quantity, which is the frequency difference $\nu(2S_{F=1}-4S_{F=1})-1/4\nu(1S_{F=1}-2S_{F=1})$, by adding the hfs correction obtained from the adopted values of the 2S and 4S hfs splittings (see text) and 1/4 of the $1S_{F=1}$–$2S_{F=1}$ frequency measured by Hänsch et al. [115].

[i] We derived the $2S_{1/2}$–$4D_{5/2}$ frequency from the original measured quantity, which is the frequency difference $\nu(2S_{F=1}-4D_{5/2})-1/4\nu(1S_{F=1}-2S_{F=1})$, by adding the hfs correction obtained from the adopted values of the 2S hfs splitting (see text) and 1/4 of the $1S_{F=1}$–$2S_{F=1}$ frequency measured by Hänsch et al. [115].

[j] We derived the $2S_{1/2}$–$6S_{1/2}$ frequency from the original measured quantity, which is the frequency difference $\nu(2S_{F=1}-6S_{F=1})-1/4\nu(1S_{F=1}-3S_{F=1})$, by adding the hfs correction obtained from the adopted values of the 1S, 2S, 3S, and 6S hfs splitting to 1/4 of the fine-structure interval $1S_{1/2}$–$3S_{1/2}$ obtained from the level optimization (see text). The additive quantity "+1/4 x" in the second column reflects the systematic uncertainty equal to 1/4 of the uncertainty of the Ritz value of the $1S_{1/2}$–$3S_{1/2}$ interval based on the 2S–3P measurements of Zhao et al. [111].

[k] We derived the $2S_{1/2}$–$6D_{5/2}$ frequency from the original measured quantity, which is the frequency difference $\nu(2S_{F=1}-6D_{5/2\,F=2\,3})-1/4\nu(1S_{F=1}-3S_{F=1})$, by adding the hfs correction obtained from the adopted values of the 1S, 2S, 3S, and 6D hfs splitting to 1/4 of the fine-structure interval $1S_{1/2}$–$3S_{1/2}$ obtained from the level optimization (see the previous footnote).

[l] Recalibrated using the revised value of the reference laser frequency (see text).

[m] Corrected for the adopted value of the $3S_{1/2}$ hfs splitting (see text).

246 606 001 102 474 851(34) Hz by adding the hfs correction [hfs(1S)–hfs(2S)]/4 = 310 712 223(4) Hz. The uncertainty of the latter value is determined by the measurement uncertainty of the 2S hfs splitting [83].

The article of de Beauvoir et al. [103] contains a detailed review of the series of experiments made by that team (see Refs. [118,124–127] and references therein). Those experiments used Doppler-free two-photon laser spectroscopy with a metastable atomic beam. The optical frequencies of the 2S–8S, 2S–8D, 2S–10D, and 2S–12D two-photon transitions were measured by direct comparison with a reference laser frequency standard. For the 2S–6S and 2S–6D intervals, the measured quantity was the difference between the 1S–3S frequency and four times the 2S–6S,D frequencies. A more detailed description of this frequency comparison is given by Bourzeix et al. [125]. To determine the values of the 2S–6S,D frequencies from those frequency-difference values, we used the Ritz value of the 1S–3S frequency, 2 922 743 277.97(22) MHz, resulting from the level optimization procedure (see subsequent sections). This value is mainly based on the 2S–$3P_{1/2}$ measurement of Zhao et al. [111] and the $3P_{1/2}$–3S measurement of Fabjan and Pipkin [59] and the highly accurate 1S–2S measurement [115]. Other chains of measured frequencies, such as ν(1S–2S)–ν($2P_{1/2}$–2S) + ν($2P_{1/2}$–$3D_{3/2}$)–ν(3S–$3D_{3/2}$) and ν(1S–2S) + ν(2S–$2P_{3/2}$) + ν($2P_{3/2}$– $3D_{5/2}$)–ν(3S–$3D_{5/2}$), also contribute to the optimized Ritz value of the 1S–3S frequency. Since 1/4 of this frequency is added to the measured intervals to obtain both 2S–6S and 2S–$6D_{5/2}$ intervals, the resulting values contain the same systematic uncertainty equal to one quarter of the uncertainty of the 1S–3S Ritz value (i.e., ±0.06 MHz). This systematic uncertainty is expressed as a "+1/4x" additive to the 2S–6S and 2S–$6D_{5/2}$ values given in Table C. It will further be shown that the actual value of x may be much greater, since the systematic uncertainty of the Zhao et al. [111] measurements is probably greatly underestimated. The absolute measurement of the 1S–3S transition frequency, which is currently underway [128–130], should eliminate this systematic uncertainty.

Lundeen and Pipkin [88] measured the hyperfine interval $2P_{1/2,F=1}$–$2S_{1/2,F=0}$. From this measured interval they deduced the value of the n = 2 Lamb shift, 1 057.845(9) MHz. This quantity is not the same as the $2P_{1/2}$–$2S_{1/2}$ fine-structure interval. As illustrated by their Fig. 11, the Lamb shift is the QED correction to the center of gravity of the $2P_{1/2}$ hyperfine doublet, calculated without account for the off-diagonal hfs correction. In contrast, the $2P_{1/2}$–$2S_{1/2}$ fine-structure interval given in Table C and subsequent tables of the present work corresponds to the difference of frequencies of the centers of gravity of "true" hyperfine doublets (i.e., it includes the shift of the center of gravity of $2P_{1/2}$ caused by the off-diagonal hfs; see Section 3.1).

In the experiment of Hagley and Pipkin [98], the measured quantity was the $2S_{1/2,F=0}$–$2P_{3/2,F=1}$ hyperfine interval. To obtain the fine-structure interval, we added the hfs correction equal to (5/8) × hfs($2P_{3/2}$)–(1/4) × hfs(2S). The resulting value is higher by 1 kHz than the value given by Hagley and Pipkin [98]. This does not imply any correction to their value but reflects a difference in terminology, similar to the Lamb shift transition discussed above. Hagley and Pipkin give a value for the "interval in the absence of hyperfine structure", while we give a value for the interval between the centers of gravity of the $2S_{1/2}$ and $2P_{3/2}$ levels. The difference is caused by the off-diagonal hfs interaction that shifts the center of gravity of $2P_{3/2}$ upwards (see Section 3.1).

In an earlier experiment, Kaufman et al. [131] measured the $2S_{1/2}$–$2P_{3/2}$ fine-structure interval using a radiofrequency method similar to that of Hagley and Pipkin [98]. They obtained 9 911.377(26) MHz, which strongly disagrees with the results of Hagley and Pipkin [98], 9 911.201(12) MHz, Cosens and Vorburger [132], 9 911.173(42) MHz, and Shyn et al. [133], 9 911.250(63) MHz. Consequently, we made no use of it.

We obtained the value for the $2P_{1/2}$–$3D_{3/2}$ interval by adding the separation between $2S_{1/2}$–$3P_{3/2}$ and $2P_{1/2}$–$3D_{3/2}$, 1 052.7(17) MHz, measured by Hänsch et al. [116] to the $2S_{1/2}$–$3P_{3/2}$ interval measured by Zhao et al. [111]. Subtracting the ($2P_{1/2}$–$3D_{3/2}$)–($2P_{3/2}$–$3D_{5/2}$) separation, 9 884.5(29) MHz [116], yields the value for the $2P_{3/2}$–$3D_{5/2}$ interval. Although in a later experiment using a similar saturated absorption technique Petley et al. [57] achieved better spectral resolution, their quoted uncertainties for various estimated corrections are much greater than those given by Hänsch et al. [116]. The values of these two intervals obtained from line separations measured by Petley et al. [57], 456 685 854.3(39) MHz for $2P_{1/2}$–$3D_{3/2}$ and 456 675 967.9(36) MHz for $2P_{3/2}$–$3D_{5/2}$, are in good agreement with those given in Table C. Since the adopted values of the 2P–3D intervals are based on the 2S–3P measurement of Zhao et al. [111], they possess the same systematic shift "+ x."

Berkeland et al. [117] measured the frequency differences between the ($2S_{F=1}$–$4P_{1/2,F=0,1}$, $4P_{3/2,F=1,2}$) and one quarter of the ($1S_{F=1}$–$2S_{F=1}$) transition. The latter transition was later precisely measured by Hänsch et al. [115]. Thus, to obtain the 2S–4P intervals, the only model-dependent corrections needed are the hfs shifts caused by non-statistical populations of the unresolved hyperfine sublevels of the $4P_{1/2}$ and $4P_{3/2}$ levels. These corrections, involving theoretical values of the 4P hfs intervals and transition rates, were calculated by Berkeland et al. [117] by solving density-matrix equations. These corrections were found to be 33(8) kHz and −17(4) kHz for the $4P_{1/2}$ and $4P_{3/2}$ levels, respectively. The 2S–4P photon recoil correction (absent for two-photon transitions) has a much larger contribution of −841 kHz. However, its calculation is straightforward and does not involve any assumptions other than the law of momentum conservation. The other hfs-related quantity required to obtain the fine-structure intervals is the 1S–2S hfs correction equal to (5/16) × hfs(2S)–(1/16) × hfs(1S). Using the precise experimental values for the 1S and 2S hfs intervals (see Section 3), we obtained the value −33 289 kHz for this correction. The value used by Berkeland et al. [117] was −33 291 kHz. Thus, the fine-structure intervals given in Table C are greater by 2 kHz than those obtained by summing the relevant entries in Table 1 of Berkeland et al. [117]. This correction is well within the uncertainties quoted in their paper.

Using a similar method, Weitz et al. [101] measured the frequency differences between the ($2S_{F=1}$–$4S_{1/2,F=1}$, $4D_{5/2,F=2,3}$) and one quarter of the ($1S_{F=1}$–$2S_{F=1}$) transition. Since the 2S–4S and 2S–4D two-photon transitions involve simultaneous absorption of two counterpropagating photons, there is no photon recoil. Thus, there are only very small theoretical corrections to the measured intervals, especially in the case of the 2S–4S transition, where the large hfs intervals allowed excitation of selected F = 1 hyperfine sublevels. In the case of $2S_{F=1}$–$4D_{5/2,F=2,3}$, however, the intensities of the unresolved hfs components were assumed to be proportional to their calculated oscillator strengths. The smallness of the resulting correction (−0.222 MHz) and the increased estimated uncertainty of the final result (±0.024 MHz, as compared to the ±0.010 MHz for 2S–4S), seem to justify the use of this assumption.

Garreau et al. [118] measured the frequencies of the $2S_{1/2,F=1}$–$nD_{5/2}$ (n = 8, 10, 12) two-photon transitions relative to the known frequency ν_f of a He-Ne laser stabilized on the hyperfine component f of the R(127) 11-5 line in $^{127}I_2$. This laser was calibrated against the standard He-Ne laser of the Institut National de Métrologie (INM12) and its frequency was found to be shifted by +29 kHz with respect to the INM12 laser. In a subsequent work by the same group [103], the $2S_{1/2,F=1}$–$8D_{5/2}$ and $2S_{1/2,F=1}$–$12D_{5/2}$ transitions were re-measured with much lower uncertainties, and it was noted that the frequency ν_f of the INM12 laser had been re-measured in 1992 to a greater precision of ±3.4 kHz (see Eq. (32) of de Beauvoir et al. [103]). Biraben [134] noted that the quantities actually measured by Garreau et al. [118] are ratios of frequencies of the H and D transitions to the frequency of the reference laser. Thus, he multiplied the H and D frequency values by a correction factor equal to the ratio of the new and old values of the frequency ν_f of the reference laser used by Garreau et al. [118], 473 612 353 616(10) kHz and 473 612 353 725 kHz, respectively. The resulting corrected value for the hydrogen $2S_{1/2}$–$10D_{5/2}$ transition is given in

Table C. The similarly corrected $2S_{1/2,F=1}$–$8D_{5/2}$ and $2S_{1/2,F=1}$–$12D_{5/2}$ frequencies from Garreau et al. agree with those from de Beauvoir et al. [103] (corrected as described above) within a few kilohertz. However, the various other corrections (such as second-order Doppler shift, Stark shifts, etc.) were calculated by Garreau et al. much less accurately than in the work of de Beauvoir et al. [103]. The differences of the corrected final values of the 2S–$8D_{5/2}$ and 2S–$12D_{5/2}$ fine-structure frequencies of Garreau et al. [118] from those of de Beauvoir et al. [103] are −24 kHz and −15 kHz, respectively. These differences lie well within the uncertainty limits given by Garreau et al. [118].

Zhao et al. [111] used Fabry–Perot interferometry to measure the frequencies of the $2S_{F=1}$–$3P_{1/2}$ and $2S_{F=1}$–$3P_{3/2}$ transitions excited by a laser perpendicular to a collimated atomic beam. They used two different methods of frequency measurement, the virtual mirrors and the phase shift methods. The latter method was deemed by Zhao et al. [111] to be less dependable. Therefore, we quote here only the results obtained by the virtual mirrors method. The measured wave numbers of the 2S–$3P_{1/2}$ and 2S–$3P_{3/2}$ transitions in hydrogen were 15 233.256 807(10) cm^{-1} and 15 233.365 224(10) cm^{-1}, corresponding to the frequencies 456 681 550.2(3) MHz and 456 684 800.4(3) MHz, respectively. As a frequency standard, Zhao et al. [111] used a He-Ne laser stabilized on the hyperfine component *g* of the *R*(127) 11-5 line in $^{127}I_2$. Their laser was identical in design and operating conditions to the one used by Jennings et al. [135]. The latter authors made the first direct measurement of the frequency of an iodine-stabilized He-Ne laser and reported the frequency 473 612 340.492(74) MHz corresponding to the *g* component. The newer, more accurate, value of this reference laser frequency recommended by the Bureau International des Poids et Mesures (BIPM) [136] is 473 612 340.406(11) MHz, which is 0.086 MHz lower than the value from Jennings et al. [135], used by Zhao et al. [111].

However, the operating conditions of the He-Ne laser in the experiments of Jennings et al. [135] and Zhao et al. [111] were somewhat different from those recommended by BIPM at that time. According to Jennings et al. [135], the frequency of the laser operating under their conditions was lower than the frequency under standard operating conditions by 41 kHz. Furthermore, the currently recommended frequency given above corresponds to new BIPM recommendations regarding the operating conditions of the laser [136]. These conditions are somewhat different from the ones referred to by Jennings et al. [135]. In particular, they specify an additional constraint on the cell-wall temperature, (25 ± 5) °C, and require a particular method of frequency stabilization, namely, the third harmonic detection technique. This reflects the fact that, as found by many investigators (see, e.g. Lea et al. [137], Madej et al. [138], and references therein), a good reproducibility of the He-Ne-I_2 frequency standard can be achieved only if the laser is operating under conditions that are defined and controlled more strictly than it was known at the time of the experiments of Jennings et al. [135] and Zhao et al. [111]. The frequency stabilization method of Zhao et al. [111], as described in their later paper [121] (using a third-derivative signal) is different from the current BIPM recommendations. It is not clear whether this method is identical to the one used by Jennings et al. [135]. Therefore, the assertion of Zhao et al. [111,121] that the frequency of their reference laser was the same as measured by Jennings et al. [135] cannot be verified. However, it is possible to re-calibrate the measurements of Zhao et al. [111,121] using the hydrogen 2S–$4P_{1/2,3/2}$ transition frequencies accurately measured by Berkeland et al. [117]. This re-calibration procedure is described below.

Using a method similar to that for the Balmer-α components, Zhao et al. [121] measured the frequencies of the $2S_{F=1}$–$4P_{1/2}$ and $2S_{F=1}$–$4P_{3/2}$ transitions. Applying corrections for the hfs of 2S and 4P, they obtained the fine-structure frequencies 616 520 018.02(7) MHz and 616 521 388.87(16) MHz for the 2S–$4P_{1/2}$ and 2S–$4P_{3/2}$ transitions, respectively (the indicated uncertainties are statistical only, as quoted from Table II of Zhao et al. [121]). We note that the correction for the hfs of $4P_{1/2}$ and $4P_{3/2}$ was estimated by Zhao et al. [121] (see their Table I) to be +0.561(30) MHz and −0.561(30) MHz, respectively. From our calculations of the hfs intervals (see Section 3.1.2), these corrections are $(1/12) \times \mathrm{hfs}(4P_{1/2}) =$ 0.616 MHz and $(-5/24) \times \mathrm{hfs}(4P_{3/2}) = -0.616$ MHz, respectively. Applying the revised hfs correction, we obtain 616 520 018.08(7) MHz and 616 521 388.82(16) MHz for 2S–$4P_{1/2}$ and 2S–$4P_{3/2}$, respectively. Combining the statistical uncertainties with the uncertainties of various corrections specified in Table I of Zhao et al. [121] (Stark and light shifts, second-order Doppler effect, and hfs of 4P) yields the total measurement uncertainties (excluding the reference laser) of 0.11 MHz and 0.18 MHz, respectively.

The adopted frequencies of these transitions were determined from the measurements of Berkeland et al. [117] and Hänsch et al. [115] as described above. They are 616 520 017.568(15) MHz and 616 521 388.672(10) MHz (for 2S–$4P_{1/2}$ and 2S–$4P_{3/2}$, respectively). The correction factor for the frequency measurements of Zhao et al. [121] is determined as a mean of the ratios of the adopted values to the frequencies measured by Zhao et al. [121] weighted by the inverse squares of the measurement uncertainties. This correction factor is $1 - (6.6 \pm 1.6) \times 10^{-10}$. The absolute value of the correction exceeds the original estimate of the uncertainty of the standard laser frequency [111,121] by more than a factor of four. This clearly indicates that the laser used by Zhao et al. [111] (or at least its operating conditions) were different from those of Jennings et al. [135]. We assume that both the laser and its operating conditions (including the method of frequency stabilization) were the same in all hydrogen and deuterium measurements of Zhao et al. [111,121]. Applying the above correction factor to the 2S–$3P_{1/2}$ and 2S–$3P_{3/2}$ frequencies measured by Zhao et al. [111], we obtain 456 681 549.9(3) MHz and 456 684 800.1(3) MHz, respectively. We note that for these transitions our calculations of the hfs(3P) corrections coincide with those of Zhao et al. [111]. Since there are some doubts about the validity of the re-calibration procedure, we retain in the 2S–3P intervals the additive quantity *x* denoting the possible systematic shift.

Fabjan and Pipkin [59] measured the frequency of the $3P_{1/2,F=1}$–$3S_{1/2,F=0}$ hyperfine transition using a radio-frequency separated oscillatory fields technique. From this measured frequency, they deduced the $3P_{1/2}$–$3S_{1/2}$ fine-structure interval by adding a correction accounting for the $3P_{1/2}$ and $3S_{1/2}$ hfs intervals. In this derivation they used a theoretical value for the $3S_{1/2}$ hfs interval, 52.6111(11) MHz. The currently adopted value is 52.609 473 2(3) MHz, as determined by Jentschura and Yerokhin [17] using semi-empirically corrected QED calculations (see Section 3). Using this value for the hfs of $3S_{1/2}$ results in a small adjustment (by −1 kHz) of the $3P_{1/2}$–$3S_{1/2}$ fine-structure interval given by Fabjan and Pipkin [59]. This adjustment is well below their quoted uncertainty, ±48 kHz. The subtle difference between the notions of the $n = 3$ Lamb shift and the $3P_{1/2}$–$3S_{1/2}$ fine-structure interval ("interval in the absence of hyperfine structure" vs. "interval between the centers of gravity of the hfs doublets", similar to the $n = 2$ case discussed above) amounts to a negligibly small difference of about 500 Hz and so was neglected in our consideration of this transition.

A large number of fine structure intervals between excited levels measured in experiments of Fabjan and Pipkin [59], Fabjan et al. [106], Glass-Maujean [109], Glass-Maujean et al. [110], Weber and Goldsmith [120], Van Baak et al. [119], Zhao et al. [121], and other experiments included in Table C provide numerous interconnections between the excited levels. To derive the energy level values from those measured intervals, it is necessary to use a least-squares level optimization procedure, which is described in the following section.

5. Level optimization for hydrogen

To derive optimized energy levels from the measured fine structure intervals listed in Table C, we used the least-squares optimization (LOPT) code [139]. In the list of measured transitions there are several isolated groups of levels that are not connected to the ground level by any observed transition. Besides that, there are some levels for which no transitions were observed (e.g., $6D_{3/2}$, $9S_{1/2}$, $9D_{3/2,5/2}$, $10S_{1/2}$, $10D_{3/2}$), and determination of the $6S_{1/2}$ and $6D_{5/2}$ levels requires a knowledge of the $3S_{1/2}$ excitation energy that was not measured directly. Therefore, the level optimization procedure involved several iterations.

The first iteration consisted of two steps. In the first, the $6S_{1/2}$ and $6D_{5/2}$ levels were not included. The $3S_{1/2}$ energy was determined from four observed transitions connecting it to the ground state via different sets of intermediate transitions. Some examples were given in the previous section. Then the frequencies of the 2S–6S and 2S–$6D_{5/2}$ transitions were determined by adding 1/4 of the 1S–3S frequency to the beat frequencies measured by de Beauvoir et al. [103]. In a second step, from these indirectly determined frequencies, we obtained the $6S_{1/2}$ and $6D_{5/2}$ energies.

To determine the separation of the $n = 5$ levels from the ground level and to find several unobserved fine-structure levels, we used various interpolation and extrapolation procedures. The first of these is the fitting of the Ritz series formulas for the $nD_{5/2}$ and $nS_{1/2}$ series. Several types of such formulas were suggested in the early 20$^{\text{th}}$ century to describe regularities in the observed energy intervals of series of lines of hydrogen and other elements (see, for example, works by Ritz [140], Fowler [141,142], and Curtis [143,144]). The particular formula applied here is the so-called extended Ritz formula

$$\delta_n = c_0 + c_1/(n-\delta_n)^2 + c_2/(n-\delta_n)^4 + \ldots, \tag{25}$$

where δ_n is the quantum defect describing an empirical correction to formula (1) for the excitation energy E_n of the level with the principal quantum number n

$$E_I - E_n = RZ^2/(n-\delta_n)^2, \tag{26}$$

where E_I is the ionization energy.

Theoretical quantum–mechanical grounds for the validity of formulas (25), (26) were laid out by Hartree [145] and extended by Drake [146]. The constants c_0, c_1, c_2, ... can be determined by a non-linear least-squares fitting of a few known energy levels in the series. To carry out this fitting, we used the computer code RITZPL [147].

From the first iteration of the least-squares fitting of transition energies, the $3D_{5/2}$, $4D_{5/2}$, $6D_{5/2}$, $8D_{5/2}$, $10D_{5/2}$, and $12D_{5/2}$ energy levels were determined with uncertainties varying between 6 kHz (for $8D_{5/2}$) and 300 kHz (for $3D_{5/2}$). To determine the ionization energy and coefficients c_1, c_2, ... in formulas (25), (26), we used several fittings with varying numbers of terms in formula (25). *A priori*, it is clear that the number of available experimental data points is insufficient to capture the high precision of measurements with such fitting. For example, Weber and Sansonetti [148] in their fitting of 30 members of the $nD_{5/2}$ series of Cs I (which is similar in character to the hydrogen series) had to use the six-parameter Ritz formula in order to fit the levels with uncertainties of about 0.9 MHz.

With six available data points, the maximum number of terms in formula (25) that can be fitted is five, if the ionization energy is used as a free variable. Thus, the fitting of the five-term Ritz formula should produce the best possible results for the $nD_{5/2}$ series of hydrogen. However, this is an exact fit, since the number of free parameters is equal to the number of data points. Therefore, the uncertainties of this fit have to be estimated from some other considerations rather than from the usual standard deviations of the fitted quantities. To obtain these estimates, we compared the fitted values of ionization energy and $n = 5$, 7, 9, and 11 levels obtained with the five-parameter Ritz formula with those of the four- and three-parameter fittings, and the experimental values of $n = 3$, 4, 6, 10, and 12 levels with those fitted by the four- and three-parameter Ritz formulas. This gave us 22 values of differences between the various level values, which constitutes sufficiently large statistics. Thus estimated, the standard deviation of the fitting is 0.7 MHz. The resulting value of the ionization limit is 3 288 086 856.8(7) MHz, which is lower than the result of the precise QED calculation of Jentschura et al. [10] by 0.3 MHz. We consider this a good agreement. From this fitting, the energies of the $nD_{5/2}$ ($n = 5$, 7, 9, and 11) levels are determined with uncertainties of ±0.7 MHz. The parameters of the various fittings of the $nD_{5/2}$ series are summarized in Table D.

The next step is to determine the energies of the $nS_{1/2}$ levels with $n = 5$, 7, and 9–12. For this series, experimental values of

Table D
Ionization energy and quantum defect expansion coefficients for the $nD_{5/2}$ series of H obtained by fitting different versions of the extended Ritz formula (25).

	3-term Ritz formula	4-term Ritz formula	5-term Ritz formula (exact fit)
E_I	3 288 086 854.5(6) MHz	3 288 086 856.06(22) MHz	3 288 086 856.8 MHz
$c_0 \times 10^6$	7.03(7)	7.51(6)	7.91
$c_1 \times 10^6$	−60.4(16)	−86(3)	−127
$c_2 \times 10^6$	153(9)	580(50)	2 040
$c_3 \times 10^6$	–	−2 090(250)	−22 500
$c_4 \times 10^6$	–	–	92 160
Std. dev. of levels	0.5 MHz	0.09 MHz	–

Table E
Quantum defect expansion coefficients for the $nS_{1/2}$ series of H obtained by fitting different versions of the extended Ritz formula (25).

	4-term Ritz formula	5-term Ritz formula	6-term Ritz formula (exact fit)
E_I (fixed)	3 288 086 856.8 MHz	3 288 086 856.8 MHz	3 288 086 856.8 MHz
$c_0 \times 10^6$	23.11(18)	23.80(8)	24.14
$c_1 \times 10^6$	−48(3)	−72(3)	−96
$c_2 \times 10^6$	82(10)	311(26)	829
$c_3 \times 10^6$	−52(7)	−760(80)	−4 920
$c_4 \times 10^6$	–	500(60)	12 380
$c_5 \times 10^6$	–	–	−8 210
Std. dev. of levels	7 MHz	1.1 MHz	–

energy levels are available for n = 1, 2, 3, 4, 6, and 8. The uncertainties vary between 30 Hz (n = 2) and 300 kHz (n = 3). As in the case of the $n\mathrm{D}_{5/2}$ series, the number of available data points is insufficient to reproduce this high experimental precision. Unlike the $n\mathrm{D}_{5/2}$ series, in this case the fittings with 3- or 4-term Ritz formulas produce considerably larger deviations from experiment than the fitting with the 5-term formula. Theoretically, it is clear that this is due to much larger contributions of the QED effects in the nS levels. For this reason, when fitting the nS series we fixed the value of the ionization energy at the result obtained from the fitting of the $n\mathrm{D}_{5/2}$ series and used only two different fittings with 6- and 5-term Ritz formulas. As with the $n\mathrm{D}_{5/2}$ series, the best fit is obtained with the largest number of terms in formula (25), in this case 6. This is an exact fit with no free parameters. To estimate the uncertainties, we compare the fitted values of the nS (n = 5, 7, 9–12) levels resulting from the 6-term Ritz formula with those from the 5-term formula, and the experimental levels for nS (n = 1–4, 6, 8) levels with those from the 5-term formula. This gives us 12 values for deviations of the n = 1–12 levels, the average deviation being 0.9 MHz. Combining the average deviation in quadrature with the estimated uncertainty of the ionization energy (0.7 MHz, see above), we obtain an estimated uncertainty of the fitted level values of 1.2 MHz. The parameters of the various fittings of the $n\mathrm{S}_{1/2}$ series are summarized in Table E.

The $n\mathrm{D}_{3/2}$–$n\mathrm{D}_{5/2}$ splittings have been measured for n = 3 and 4 by Glass-Maujean et al. [110] with uncertainties 2.9 MHz and 2.2 MHz, respectively. The level-optimization procedure reduces the uncertainties of the Ritz values of these intervals to 0.6 MHz and 1.0 MHz, respectively. For n = 8 and 12, the nD fine-structure splittings are determined by the 2S–$n\mathrm{D}_{3/2,5/2}$ frequency measurements of de Beauvoir et al. [103] with uncertainties ±10 kHz to ±12 kHz. The 5D fine-structure splitting is determined from the n = 5 measurements of Fabjan et al. [106] with a relatively large uncertainty of 14 MHz. We can use an interpolation along the series of n to find this interval with a much lower uncertainty and to find the values of the $n\mathrm{D}_{3/2}$–$n\mathrm{D}_{5/2}$ splittings for n = 6, 7, and 9–11, for which there are no experimental data available. As illustrated in Fig. 3, the $n\mathrm{D}_{3/2}$–$n\mathrm{D}_{5/2}$ frequency intervals ΔE_{nD} multiplied by n^3 are almost constant along the series of n. Thus, a simple fitting with a parabola yields accurate interpolated values for the entire interval n = 3 to n = 12. These interpolated values agree well with precise QED calculations of Jentschura et al. [10].

The $n\mathrm{P}_{1/2}$–$n\mathrm{S}_{1/2}$ fine-structure intervals have been measured for n = 2 and 3 by Lundeen and Pipkin [88] and Fabjan and Pipkin [59] with uncertainties of ±9 kHz and ±50 kHz, respectively. For n = 4, although the direct measurement by Fabjan et al. [106] has a relatively large uncertainty of ±0.6 MHz, a much more accurate value can be obtained from the frequencies of the $2\mathrm{S}_{1/2}$–$4\mathrm{P}_{1/2}$ and $2\mathrm{S}_{1/2}$–$4\mathrm{S}_{1/2}$ transitions measured by Berkeland et al. [117] and Weitz et al. [101]. The uncertainty of this indirectly determined value is 18 kHz. For n = 5, the most accurate value available was measured by Fabjan et al. [106] with an uncertainty of ±5 MHz. A much more accurate value can be obtained by an extrapolation along the series of n.

As seen from Fig. 4, the frequencies of the $n\mathrm{P}_{1/2}$–$n\mathrm{S}_{1/2}$ fine-structure intervals $\Delta E_{n\mathrm{SP}}$ multiplied by a factor n^3 are nearly constant along the series. Therefore, an extrapolation of $\Delta E_{n\mathrm{SP}} \times n^3$ to $n > 4$ provides reliable estimates of $\Delta E_{n\mathrm{SP}}$. These extrapolated intervals are compared in Table F with precise QED calculations of Jentschura et al. [10]. The agreement is quite satisfactory.

The $n\mathrm{P}_{1/2}$–$n\mathrm{P}_{3/2}$ splittings $\Delta E_{n\mathrm{P}}$ were measured directly for n = 2 and n = 4 by Baird et al. [108] and Zhao et al. [121]. The uncertainties of those measurements are ±100 kHz and ±170 kHz, respectively. The values obtained for these two splittings in the first iteration of the least-squares level optimization procedure have much smaller uncertainties, ±15 kHz and ±18 kHz. The optimized values are mainly based on the $2\mathrm{P}_{1/2}$–$2\mathrm{S}_{1/2}$, $2\mathrm{S}_{1/2}$–$2\mathrm{P}_{3/2}$, and $2\mathrm{S}_{1/2}$–$4\mathrm{P}_{1/2,3/2}$ measurements of Lundeen and Pipkin [88], Hagley and Pipkin [98], and Berkeland et al. [117], respectively. The optimized values of $\Delta E_{n\mathrm{P}}$ for n = 3 and n = 5 have considerably larger uncertainties, ±0.4 MHz and ±11 MHz, respectively. Much more accurate values can be obtained from an interpolation along the series of n. We again can use the empirical fact that the quantity $\Delta E_{n\mathrm{P}} \times n^3$ is almost constant along the series. This is illustrated in Fig. 5.

The experimental data for the $n\mathrm{P}_{1/2}$–$n\mathrm{P}_{3/2}$ splittings are consistent with a constant value of $\Delta E_{n\mathrm{P}} \times n^3$ = 87 752.4 MHz, which is a weighted average of four experimental values. The uncertainties of interpolated and extrapolated values are estimated as being entirely due to the experimental uncertainty of the n = 4 value scaled as $1/n^3$. The resulting interpolated and extrapolated splittings are

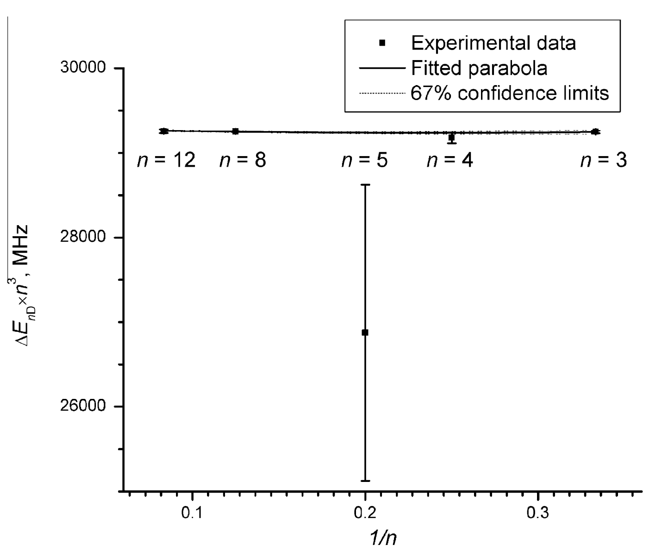

Fig. 3. Frequencies of the $n\mathrm{D}_{3/2}$–$n\mathrm{D}_{5/2}$ fine-structure intervals ΔE_{nD} in hydrogen scaled by n^3. The experimental error bars for n = 3, 8, and 12 are too small to be seen. The solid curve is a parabola that fits the experimental data points for n = 3, 4, 8, and 12. The dashes (visible only for $n < 5$) are estimates of the limits of the interpolation error.

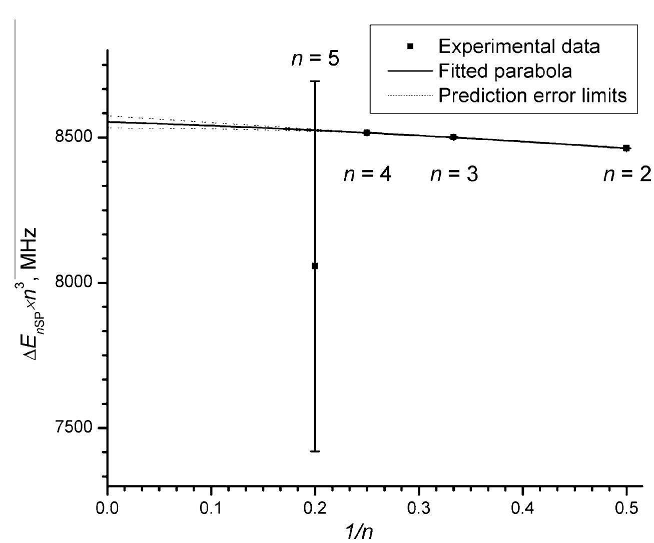

Fig. 4. Frequencies of the $n\mathrm{P}_{1/2}$–$n\mathrm{S}_{1/2}$ fine-structure intervals ΔE_{nSP} in hydrogen scaled by n^3. The experimental error bars for n = 2, 3, and 4 are too small to be seen. The solid curve is a parabola that fits the experimental data points, and the dashes are estimates of the limits of the extrapolation error.

Table F
Comparison of extrapolated $n\mathrm{P}_{1/2}$–$n\mathrm{S}_{1/2}$ fine-structure intervals $\Delta E_{n\mathrm{SP}}$ with the QED calculation of Jentschura et al. [10].

n	$\Delta E_{n\mathrm{SP}}^{\mathrm{exp}}$ (MHz)	Ref.	$\Delta E_{n\mathrm{SP}}^{\mathrm{extrap}}$ (MHz)	$\Delta E_{n\mathrm{SP}}^{\mathrm{QED}}$ [10] (MHz)	$\Delta E_{n\mathrm{SP}}^{\mathrm{extrap}} - \Delta E_{n\mathrm{SP}}^{\mathrm{sssQED}}$ (MHz)
2	1057.847(9)	[88]		(1057.8440)(24)	
3	314.82(5)	[59]		(314.8779)(7)	
4	133.066(18)	[117,101]		(133.0775)(3)	
5	65(5)	[106]	[68.201](19)	(68.19543)(16)	0.006(19)
6			[39.494](23)	(39.48435)(9)	0.010(23)
7			[24.882](20)	(24.87231)(6)	0.010(20)
8			[16.674](16)	(16.66582)(4)	0.009(16)
9			[11.714](13)	(11.70655)(3)	0.007(13)
10			[8.541](11)	(8.534922)(21)	0.006(11)
11			[6.418](9)	(6.412886)(16)	0.005(9)
12			[4.944](7)	(4.939834)(12)	0.004(7)

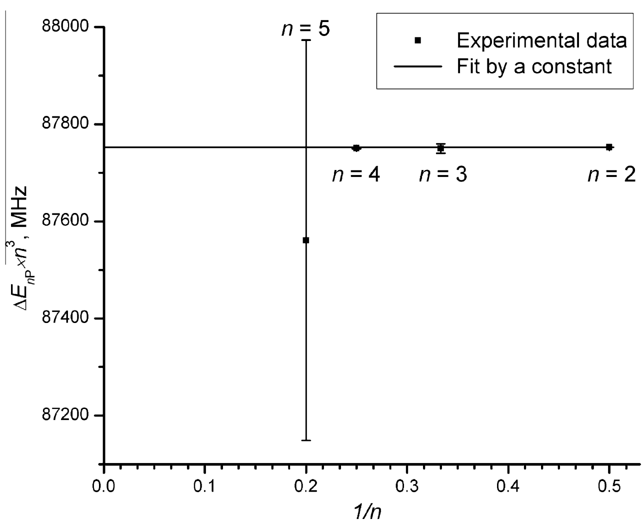

Fig. 5. Frequencies of the $n\mathrm{P}_{1/2}$–$n\mathrm{P}_{3/2}$ fine-structure intervals ΔE_{nP} in hydrogen scaled by n^3. The experimental error bars for $n = 2$ and 4 are too small to be seen. The solid line is a constant fitting the experimental data points. The experimental data results from the second step of the least-squares level optimization procedure (see text).

in excellent agreement with the QED calculations of Jentschura et al. [10], all deviations being smaller than 1.5 kHz. The interpolated $3\mathrm{P}_{1/2}$–$3\mathrm{P}_{3/2}$ interval, 3250.09(3) MHz, is an order of magnitude more precise than the value obtained in the initial least-squares level optimization.

In total, we obtained 34 interpolated and extrapolated fine-structure intervals. These intervals were added to the input of the least-squares level optimization program on equal footing with 40 measured ones to make the final iteration of the level optimization procedure. The resulting optimized energy levels are listed in Table 4. For rounding the level values, the general "rule of 24" was used, according to which the given values are accurate to at least 24 units of the last given decimal place. In several cases, extra digits are necessary in order to reproduce exactly known frequencies of transitions connecting the level with some other levels.

The overall agreement of the optimized experimental energy levels with the QED calculations of Jentschura et al. [10] is good. Out of 57 experimental level values in Table 4, only 14 deviate from the calculated values by more than one standard deviation. However, 10 levels deviate from the QED calculations by more than two standard deviations. This indicates some systematic effect. The largest deviations are observed for the $n = 3$ levels. The entire $n = 3$ complex of levels is based on the 2S–3P frequencies measured by Zhao et al. [111]. These five levels systematically deviate from the QED calculations by −0.68 MHz on average. This should be compared with the original estimate of the uncertainty of the reference laser frequency in the Zhao et al. measurements [111], ±0.07 MHz. The most probable explanation is that our assumption that the standard laser was handled by Zhao et al. [111] in the same way as in their later work [121] is incorrect. If the operational procedures of the standard laser were different, it may have significantly changed the laser frequency. The determination of the $n = 6$ levels is based on one quarter of the 1S–3S transition frequency, which is in turn based on the same measurements of Zhao et al. [111]. Their optimized energies are systematically lower than the calculated ones by −0.17 MHz on average, which is equal to one quarter of the deviation observed for the $n = 3$ levels. This supports the assumption that the Zhao et al. [111] measurements have a systematic shift. However, since the details of their experiment are not available, the only way to confirm it is to repeat those measurements.

Table 5 lists the input frequency intervals (measured and derived semi-empirically from Ritz-formula fittings, interpolations, and extrapolations, as described above), along with the Ritz values of these intervals (differences between the corresponding energy levels) resulting from the least-squares level optimization procedure. In this table, we also give a comparison with theoretical values of these intervals resulting from the precise QED calculations of Jentschura et al. [10]. The LOPT code, along with the optimized values of energy levels, also calculates the Ritz values of transition energies and their uncertainties derived from the covariance matrix of the input system of linear equations. This makes it possible to calculate uncertainties of any transition between the derived energy levels. Thus, we included in Table 5 several transitions that were not part of the level optimization procedure either because their energies were not measured or the measurement uncertainty was too large to have a noticeable effect on the optimized energies. In particular, we included the 1S–2P transition measured by Sansonetti et al. [149], 1S–3P and 1S–5P transitions measured by Wilkinson [150], and 1S–nP ($n = 4$, 6–12) transitions measured by Curdt et al. [151]. Transitions that were not measured are denoted with a symbol "P" (meaning "predicted interval") in the column of references.

In Tables 4 and 5 we also include the calculated shifts of the fine-structure levels and transitions caused by the off-diagonal magnetic dipole hfs interaction. These shifts are very small (compared to experimental uncertainties) and thus are only of theoretical interest at present. From Table 4, it can be seen that for the 2P, 3D, and 4F levels the magnitudes of the off-diagonal hfs shifts have about the same value or exceed the uncertainties of the high-precision QED calculations [10], which are carried out in an approximation that does not describe the hfs interactions. Table 5 shows that the effect of the hfs is more pronounced for fine-structure intervals between highly excited states. For example, the off-diagonal hfs correction for the $4\mathrm{D}_{3/2}$–$4\mathrm{D}_{5/2}$ interval is 1.052 kHz, which should be compared with the uncertainty of 21 Hz for the fine-structure QED calculation [10]. Thus, in calculations of the fine-structure intervals the hfs should not be neglected.

Table G
The best available measurements of fine-structure intervals in deuterium (MHz).

Fine structure interval	Measured value (MHz)	Actually measured interval	Largest theoretical correction[a]	Ref.
$1S_{1/2}$–$2S_{1/2}$	2466732407.52171(15)	H($1S_{F=1}$–$2S_{F=1}$)–D($1S_{F=3/2}$–$2S_{F=3/2}$)	None	[152]
$2P_{1/2}$–$2S_{1/2}$	1059.28(6)	$2P_{1/2\,F=3/2}$–$2S_{F=1/2}$, $2P_{1/2\,F=3/2}$–$2S_{F=3/2}$	(4.54446)(5) = 1/3 hfs($2P_{1/2}$)	[153,155][b]
$2S_{1/2}$–$2P_{3/2}$	9912.61(30)	$2S_{F=3/2}$–$2P_{3/2\,F=5/2}$	(−2.72694)(5) = −1/6 hfs($2P_{3/2\,F=1/2-F=3/2}$) −1/2 hfs($2P_{3/2\,F=3/2-F=5/2}$)	[47,156][d]
$2P_{1/2}$–$3D_{3/2}$	456810113.8(19) + *x*	H($2P_{1/2}$–$3D_{3/2}$)–D($2P_{1/2}$–$3D_{3/2}$)	None (Boltzmann population of hfs assumed)	[157][c]
$2S_{1/2}$–$3P_{1/2}$	456805811.7(3) + *y*	$2S_{F=3/2}$–$3P_{1/2\,F=1/2\,3/2}$	(0.448839)(6) = 1/9 hfs($3P_{1/2}$)	[111][e]
$2S_{1/2}$–$3P_{3/2}$	456809062.6(3) + *y*	$2S_{F=3/2}$–$3P_{3/2\,F=1/2\,3/2\,5/2}$	(−0.448429)(8) = −5/36 hfs($3P_{3/2\,F=1/2-3/2}$)−1/4 hfs($3P_{3/2\,F=3/2-5/2}$)	[111][e]
$2P_{3/2}$–$3S_{1/2}$	456796251(30)	D($2P_{3/2}$–$3S_{1/2}$) − D($2P_{3/2}$–$3D_{5/2}$)	hfs($2P_{3/2}$,$3S_{1/2}$,$3D_{5/2}$)[f]	[158][g]
$2P_{3/2}$–$3D_{5/2}$	456800225.9(16) + *x*	H($2P_{3/2}$–$3D_{5/2}$)–D($2P_{3/2}$–$3D_{5/2}$)	hfs($2P_{3/2}$,$3D_{5/2}$)[f]	[157][c]
$2P_{1/2}$–$4D_{3/2}$	616690180(40)	2SP–4SPD (2a + 2b + 3a + 3b)	(Boltzmann population of fine structure)	[162][i]
$2S_{1/2}$–$4P_{1/2}$	616687769.99(19)	$2S_{F=3/2}$–$4P_{1/2\,F=1/2\,3/2}$	(0.18935)(3) = 1/9 hfs($4P_{1/2}$)	[121][e]
$2S_{1/2}$–$4S_{1/2}$	616687903.573(20)	($2S_{F=3/2}$–$4S_{F=3/2}$)−1/4($1S_{F=3/2}$–$2S_{F=3/2}$)	(0.029)(15) Second-order Zeeman shift	[101]
$2S_{1/2}$–$4P_{3/2}$	616689141.73(17)	$2S_{F=3/2}$–$4P_{3/2\,F=1/2\,3/2\,5/2}$	(−0.189182)(3) = −5/36 hfs($4P_{3/2\,F=1/2-3/2}$)−1/4 hfs($4P_{3/2\,F=3/2-5/2}$)	[121][e]
$2S_{1/2}$–$4D_{5/2}$	616689596.72(4)	($2S_{F=3/2}$–$4D_{5/2\,F=3/2\,5/2\,7/2}$) − 1/4 ($1S_{F=3/2}$–$2S_{F=3/2}$)	(−0.06807)(7) = −7/45 hfs($4D_{5/2\,F=3/2-5/2}$) −2/9 hfs($4D_{5/2\,F=5/2-7/2}$)	[101]
$2P_{3/2}$–$4D_{5/2}$	616679760(50)	2P–4SD (1 + 4 + 5)	(Boltzmann population of fine structure)	[162][i]
$2P_{1/2}$–$5D_{3/2}$	690691810(50)	2SP–5SPD (2a + 2b + 3a + 3b)	(Boltzmann population of fine structure)	[162][i]
$2P_{3/2}$–$5D_{5/2}$	616681100(70)	2P–5SD (1 + 4 + 5)	(Boltzmann population of fine structure)	[162][i]
$2P_{1/2}$–$6D_{3/2}$	730890320(60)	2SP–6SPD (2a + 2b + 3a + 3b)	(Boltzmann population of fine structure)	[162][i]
$2P_{3/2}$–$6D_{5/2}$	730879480(80)	2P–6SD (1 + 4 + 5)	(Boltzmann population of fine structure)	[162][i]
$2P_{1/2}$–$7D_{3/2}$	755128600(60)	2SP–7SPD (2a + 2b + 3a + 3b)	(Boltzmann population of fine structure)	[162][i]
$2P_{3/2}$–$7D_{5/2}$	755117710(50)	2P–7SD (1 + 4 + 5)	(Boltzmann population of fine structure)	[162][i]
$2P_{1/2}$–$8D_{3/2}$	770860360(210)	2SP–8SPD (2a + 2b + 3a + 3b)	(Boltzmann population of fine structure)	[162][i]
$2S_{1/2}$–$8S_{1/2}$	770859041.246(7)	$2S_{F=3/2}$–$8S_{F=3/2}$	None	[103]
$2S_{1/2}$–$8D_{3/2}$	770859195.702(6)	$2S_{F=3/2}$–$8D_{3/2\,F=1/2\,3/2\,5/2}$	(0.008518)(7) = 1/12 hfs($8D_{3/2\,F=1/2-3/2}$) + 3/20 hfs($8D_{3/2\,F=3/2-5/2}$)	[103]
$2S_{1/2}$–$8D_{5/2}$	770859252.850(6)	$2S_{F=3/2}$–$8D_{5/2\,F=3/2\,5/2\,7/2}$	(−0.008510)(7) = −7/45 hfs($8D_{5/2\,F=3/2-5/2}$) −2/9 hfs($8D_{5/2\,F=5/2-7/2}$)	[103]
$2P_{3/2}$–$8D_{5/2}$	770849570(210)	2P–8SD (1 + 4 + 5)	(Boltzmann population of fine structure)	[162][i]
$2P_{1/2}$–$9D_{3/2}$	781645760(300)	2SP–9SPD (2a + 2b + 3a + 3b)	(Boltzmann population of fine structure)	[162][i]
$2P_{3/2}$–$9D_{5/2}$	781634790(300)	2P–9SD (1 + 4 + 5)	(Boltzmann population of fine structure)	[162][i]
$2S_{1/2}$–$10D_{5/2}$	789359610.238(38)	$2S_{F=3/2}$–$10D_{5/2\,F=3/2\,5/2\,7/2}$	(−0.004357)(3) = −7/45 hfs($10D_{5/2\,F=3/2-5/2}$) −2/9 hfs($10D_{5/2\,F=5/2-7/2}$)	[118][e]
$2S_{1/2}$–$12D_{3/2}$	799409168.038(9)	$2S_{F=3/2}$–$12D_{3/2\,F=1/2\,3/2\,5/2}$	(0.002524)(5) = 1/12 hfs($8D_{3/2\,F=1/2-3/2}$) + 3/20 hfs($8D_{3/2\,F=3/2-5/2}$)	[103]
$2S_{1/2}$–$12D_{5/2}$	799409184.967(7)	$2S_{F=3/2}$–$12D_{5/2\,F=3/2\,5/2\,7/2}$	(−0.002521)(5) = −7/45 hfs($12D_{5/2\,F=3/2-5/2}$) −2/9 hfs($12D_{5/2\,F=5/2-7/2}$)	[103]
$3P_{1/2}$–$3S_{1/2}$	315.3(4)	$3P_{1/2\,F=1/2\,3/2}$–$3S_{F=1/2\,3/2}$	hfs($3S_{1/2}$,$3P_{1/2}$)[f]	[159][h]
$3P_{1/2}$–$3P_{3/2}$	3250.7(10)	$3P_{1/2\,F=1/2\,3/2}$–$3P_{3/2\,F=1/2\,3/2\,5/2}$	hfs($3P_{1/2}$,$3P_{3/2}$)[f]	[159][h]
$3S_{1/2}$–$3P_{3/2}$	2934.5(50)	$3P_{1/2\,F=3/2}$–$3P_{3/2\,F=5/2}$	hfs($3P_{1/2}$,$3P_{3/2}$)[f]	[160][h]
$3D_{3/2}$–$3P_{3/2}$	5 (5)	$3D_{3/2\,F=1/2\,3/2\,5/2}$–$3P_{3/2\,F=1/2\,3/2\,5/2}$	hfs($3D_{3/2}$,$3P_{3/2}$)[f]	[159][h]
$4P_{1/2}$–$4S_{1/2}$	133(5)	$4P_{1/2\,F=1/2\,3/2}$–$4S_{F=1/2\,3/2}$	hfs($4S_{1/2}$,$4P_{3/2}$)[f]	[159][h]
$4P_{1/2}$–$4P_{3/2}$	1371.8(3)	$4P_{1/2\,F=1/2\,3/2}$–$4P_{3/2\,F=1/2\,3/2\,5/2}$	(0.00)(4) hfs($4P_{1/2}$,$4P_{3/2}$)	[161]

[a] The uncertainty given for corrections accounting for non-statistical excitation of hfs sub-levels includes only the uncertainty of the adopted hfs splittings. Numerical values of fractions of hfs intervals in the corrections were calculated using formulas (23), (24).
[b] The $2P_{1/2}$–2S transition frequency was measured by Cosens [153] and subsequently corrected by Vorburger and Cosens [155].
[c] Derived by adding the measured isotope shift to the hydrogen Ritz frequency (see text).
[d] The 2S–$2P_{3/2}$ frequency was measured by Dayhoff et al. [47] and subsequently corrected by Taylor et al. [156].
[e] Corrected for the revised frequency of the reference laser (see text).
[f] In the measurements of Series [158], Tate et al. [157], and Wilcox and Lamb [159], theoretical values of hfs were used along with some assumptions about populations of hfs components. It is not possible to estimate the numerical values of corrections resulting from these assumptions.
[g] Derived by subtracting the measured ($2P_{3/2}$–$3S_{1/2}$)–($2P_{3/2}$–$3D_{5/2}$) separation, 0.1326(10) cm^{-1} (or 3975 ± 30 MHz) [158], from the $2P_{3/2}$–$3D_{5/2}$ interval measured by Tate et al. [157].
[h] Uncertainties given by Wilcox and Lamb [159] and Lamb and Sanders [160] were divided by two (see Beyer [2]).
[i] Relibrated (see text).

6. Fine structure measurements in deuterium

Table G lists the most precise measurements of fine structure intervals in deuterium selected for the present compilation.

Huber et al. [152] measured the hydrogen–deuterium isotope shift of the 1S–2S two-photon transition frequency using a laser frequency comb and a CH_4-stabilized He-Ne reference laser. The measured value was the frequency difference between the $1S_{F=1}$–$2S_{F=1}$ hyperfine transition in hydrogen and the $1S_{F=3/2}$–$2S_{F=3/2}$ hyperfine transition in deuterium. The value for the 1S–2S isotope shift was obtained from the measured hfs interval by adding the hfs correction, Δf_{hfs} = [hfs(H,1S) – hfs(H,2S)]/4 – [hfs(D,1S) – hfs(D,2S)]/3 = 215225585(14) Hz. Some of the hfs splittings used to derive this value were subsequently re-measured with greater precision [14,83]. Using the currently adopted values for these hfs intervals, we obtain Δf_{hfs} = 215225590(5) Hz. The uncertainty of the latter value is determined by the measurement uncertainty of the 2S hfs splittings [83]. The difference from the value used by Huber et al. [152] is negligibly small compared to the uncertainty of their frequency measurement, ±150 Hz, which was dominated by the frequency fluctuations of the CH_4-stabilized He-Ne standard. Therefore, their final result for the fine-structure isotope shift, 670994334.64(15) kHz, remains unchanged. To obtain the deuterium 1S–2S transition frequency, we added to this value the hydrogen 1S–2S frequency precisely measured by Hänsch et al. [115].

The $2P_{1/2}$–$2S_{1/2}$ transition frequency was measured by Cosens [153] by a radio-frequency (rf) method similar to that of Lamb et al. [42–47]. In this type of experiment, an atomic beam is excited to the metastable 2S state by an electron beam. After that, the beam passes through a known magnetic field, where transitions between Zeeman sublevels of the hyperfine structure are resonantly induced by a superposed radio-frequency electric field. The actual measured quantities are profiles of the quenching signal of a selected hyperfine transition. The observed quenching line shapes are fitted to the theoretical one, derived from the Bethe–Lamb theory for the lifetime of the 2S state perturbed by an rf electric field. The line shape must be averaged over the velocity distribution of the atoms in the beam. Cosens [153], following Lamb et al. [42–47], assumed that the distribution had a $U^2\exp(-U^2)$ dependence, where U is a dimensionless velocity variable, $U^2 = (Mv^2/2kT)$ (M is the mass of the atom, v is its velocity, T is the temperature, and k is Boltzmann's constant). However, measurements by Robiscoe and Shyn [154] showed that the actual velocity distribution was close to $U^4\exp(-U^2)$. They noted that the velocity distribution depends on the apparatus geometry. However, their apparatus was virtually identical to the one used by Cosens [153]. Vorburger and Cosens [155] corrected the original measurement of Cosens [153] according to the changed theoretical line shape. Their result is 1 059.28(6) MHz.

The previous measurement of the $2P_{1/2}$–$2S_{1/2}$ interval was made by Triebwasser et al. [46]. Their result (slightly corrected by Taylor et al. [156]) was 1 059.00(6) MHz. Robiscoe and Shyn [154] noted that the apparatus geometry used by Triebwasser et al. [46] was rather different from theirs. Thus, the velocity distributions in their experiments cannot be assumed to be the same. In the experiment of Triebwasser et al. [46] the velocity distribution was distorted by motional field quenching. This effect was used to measure indirectly the effective power n in an assumed $U^n\exp(-U^2)$ distribution by determining the inverse first moment of the distribution (see Appendix VII in Triebwasser et al. [46]). It was found that the value of n is close to 2 in both hydrogen and deuterium. In this procedure Triebwasser et al. [46] assumed that the velocity distribution function $U^n\exp(-U^2)$ is valid for the entire range of velocities $0 \leqslant U \leqslant \infty$. However, Robiscoe and Shyn [154] showed that kinematic effects in a collimated atomic beam result in the existence of the lower and upper cut-off velocities in the distribution.

For example, if the interval $0.5 \leqslant U \leqslant \infty$ is assumed, the apparent value of n would be close to 1.8, while if $0 \leqslant U \leqslant 1.5$, $n \approx 3.6$. The cut-off velocities depend strongly on the apparatus geometry, especially on the beam alignment and collimation. As noted by Robiscoe and Shyn [154], the possible differences in the beam geometry in the Triebwasser et al. [46] experiments on the $2P_{1/2}$–$2S_{1/2}$ measurement and on the measurement of the velocity distribution could have resulted in the differences between the velocity distributions in those two experiments. This would result in an error in the estimate of the motional Stark effect correction, which is proportional to $\langle U^2\rangle$. Thus, we conclude that the measurement of Triebwasser et al. [46] contains an unaccounted for systematic uncertainty and adopt the result of Vorburger and Cosens [155] instead. This conclusion is corroborated by an indirect measurement by van Wijngaarden and Drake [23] who deduced the value of the Lamb shift from the measurement of the anisotropy in the angular distribution of the Lyman-α radiation induced by quenching of the $2S_{1/2}$ state by an electric field. Their result, 1 059.36(16) MHz, is in good agreement with that of Vorburger and Cosens, 1 059.28(6) MHz.

The $2S_{1/2}$–$2P_{3/2}$ transition frequency was measured by Dayhoff et al. [47] in a similar experiment with rf excitation of an atomic beam. Taylor et al. [156] reviewed those measurements and introduced a small correction resulting from improved values of some auxiliary constants used in the magnetic field measurements. They also re-evaluated the measurement uncertainties. The value given by Taylor et al. [156] is 9 912.607(56) MHz (see their Eq. 150). However, in the derivation of the uncertainty they did not account for the possible systematic errors due to the uncertainty of the velocity distribution in the atomic beam (see the above discussion of the Triebwasser et al. experiment [46]). It is not possible to evaluate this uncertainty exactly. However, from comparison of the results of Triebwasser et al. [46] and Vorburger and Cosens [155] for the $2P_{1/2}$–$2S_{1/2}$ interval it is seen that this uncertainty can be as large as 0.3 MHz. Thus, our adopted value for the $2S_{1/2}$–$2P_{3/2}$ frequency is 9 912.61(30) MHz.

Tate et al. [157] measured the hydrogen-deuterium isotope shifts of the $2P_{1/2}$–$3D_{3/2}$ and $2P_{3/2}$–$3D_{5/2}$ transitions by means of saturated absorption spectroscopy. Their results are 124 261.2(18) MHz and 124 259.2(16) MHz, respectively. We add the Ritz frequencies of these transitions in hydrogen (see Table 5) to the measured isotope shifts to obtain the frequencies of these transitions in deuterium, 456 810 113.8(19) MHz and 456 800 225.9(16) MHz, respectively. Since the hydrogen values possess an unknown systematic shift x (resulting from the hydrogen measurements of Zhao et al. [111]; see the previous section), we retain this additive quantity in the deuterium values, too. However, since the absolute value of x is probably much smaller than the uncertainty of the isotope shift measurements [157], it has a relatively small effect on the determination of the energy levels.

Zhao et al. [111] measured the frequencies of the $2S_{F=3/2}$–$3P_{1/2}$ and $2S_{F=3/2}$–$3P_{3/2}$ transitions using a set-up of crossed laser and atomic beams. Similar to their hydrogen measurements (see Section 4), they gave preference to the results obtained with the virtual mirrors method. After applying corrections for the 2S and 3P hyperfine structure, optical pumping, second-order Doppler effect, and photon recoil, they obtained the fine-structure 2S–$3P_{1/2}$ and 2S–$3P_{3/2}$ frequencies 456 805 812.0(3) MHz and 456 809 062.9(3) MHz, respectively (corresponding to wave numbers 15 237.401 736(10) cm^{-1} and 15 237.510 173(10) cm^{-1}). We applied the same re-calibration procedure to those measurements as we did for hydrogen, based on the comparison of the hydrogen 2S–4P measurements of Zhao et al. [121] with the measurements of Berkeland et al. [117] (see Section 4). Thus, we decreased the measured frequencies by 6.6 parts in 10^{10}, resulting in the values 456 805 811.7(3) MHz and 456 809 062.6(3) MHz. The change of the hyperfine corrections due to the more precise currently adopted values of the hfs intervals proved to be insignificant. Since there are doubts about the identity of the handling of the reference laser in the hydrogen and deuterium measurements of Zhao et al. [111], we assume that their deuterium measurements could have a systematic shift different from that in their hydrogen measurements. To trace the effects of this possible shift, we introduce an additive quantity "+y" in Table G. It may have a value similar to the quantity x.

In a later experiment, Zhao et al. [121] measured the frequencies of the $2S_{F=3/2}$–$4P_{1/2}$ and $2S_{F=3/2}$–$4P_{3/2}$ transitions using a similar method. After applying corrections for the 2S and 4P hyperfine structure, light shifts, Stark and second-order Doppler effects, photon recoil, and other minor corrections, they obtained the fine-structure 2S–$4P_{1/2}$ and 2S–$4P_{3/2}$ frequencies 616 687 770.38(19) MHz and 616 689 142.15(17) MHz, respectively. To determine the uncertainties, we combined the statistical uncertainties given in Table II of Zhao et al. in Ref. [121] with the uncertainties of the corrections due to the 4P hyperfine structure, light shifts, Stark and second-order Doppler effects, photon recoil, and other minor corrections given in their Table I. The re-calibration procedure based on the comparison of the hydrogen 2S–4P measurements of Zhao et al. [121] with the measurements of Berkeland et al. [117] (see Section 4) results in decreasing the measured frequencies by 6.6 parts in 10^{10}. We note that the correction for the hfs of $4P_{1/2}$ and $4P_{3/2}$ was estimated by Zhao et al. [121] (see their Table I) to be +0.17 MHz and −0.17 MHz, respectively. From our calculations of the hfs intervals (see Section 3.1.3), these corrections are $(1/9) \times \mathrm{hfs}(4P_{1/2}) = 0.19$ MHz and $(-5/36) \times \mathrm{hfs}(4P_{3/2,F=1/2-3/2}) - (1/4) \times \mathrm{hfs}(4P_{3/2,F=3/2-5/2}) = -0.19$ MHz, respectively. Taking account of the revised hfs correction, we obtain 616 687 769.99(19) MHz and 616 689 141.73(17) MHz for 2S–$4P_{1/2}$ and 2S–$4P_{3/2}$, respectively.

Series [158] interferometrically measured the separation of the weak $2P_{3/2}$–$3S_{1/2}$ transition from $2P_{3/2}$–$3D_{5/2}$. He obtained 0.132 6(10) cm^{-1} for this separation, which corresponds to

3 975(30) MHz. By subtracting this separation from the $2P_{3/2}$–$3D_{5/2}$ frequency measured by Tate et al. [157], we obtain 456 796 251(30) MHz for the frequency of the $2P_{3/2}$–$3S_{1/2}$ transition.

Weitz et al. [101] measured the frequency differences between the ($2S_{F=3/2}$–$4S_{1/2,F=3/2}$, $4D_{5/2}$) and one quarter of the ($1S_{F=3/2}$–$2S_{F=3/2}$) two-photon transitions using counterpropagating laser beams. In this type of experiment, there is no photon recoil and no first-order Doppler shift. There are only very small theoretical corrections to the measured intervals, especially in the case of the 2S–4S transition, where the large hfs intervals allowed excitation of selected $F = 3/2$ hyperfine sublevels. In the case of $2S_{F=3/2}$–$4D_{5/2}$, however, the intensities of the unresolved hfs components were assumed to be proportional to their calculated oscillator strengths. The smallness of the resulting correction (−68 kHz) and the increased estimated uncertainty of the final result (±41 kHz, as compared to the ±20 kHz for 2S–4S), seem to justify the use of this assumption.

The measurements of the 2S–8S,D and 2S–12S,D two-photon transitions by Biraben et al. [103] were described in Section 4.

The measurement of the $2S$–$10D_{5/2}$ transition by Garreau et al. [118] was corrected by Biraben [134] for the revised frequency of the reference laser (see the corresponding comments for hydrogen in Section 4).

The $3P_{1/2}$–$3S_{1/2}$, $3P_{1/2}$–$3P_{3/2}$, $3D_{3/2}$–$3P_{3/2}$, and $4P_{1/2}$–$4S_{1/2}$ intervals were measured by Wilcox and Lamb [159], and $3S_{1/2}$–$3P_{3/2}$ by Lamb and Sanders [160] by an rf method similar to the one used by Lamb et al. [42–45] for the $n = 2$ intervals. The uncertainties given in Table G for those measurements have been converted from those originally given by the authors, so that they correspond to one standard deviation (see Beyer [2]).

The $4P_{1/2}$–$4P_{3/2}$ fine structure interval was measured by Zhao et al. [161]. Although this team later measured this interval to a greater precision [121], the later measurement cannot be considered statistically independent from their 2S–4P measurements. Thus, we use their earlier less precise result instead.

Csillag [162] measured the wave numbers of the Balmer series lines 2P–nD (n = 4–9) using a Fabry–Perot interferometer with photographic recording. As a light source, he used a high-frequency ring discharge cooled by liquid nitrogen. With this technique, the Balmer series lines were observed as distinct doublets. The "red" component of each doublet was a blend of the fine-structure components 1, 4, and 5 (see Fig. 1), and the "violet" one was a blend of components 2a, 2b, 3a, and 3b. The measured intensity ratios of the "red" and "violet" components matched the theoretical ones (assuming Boltzmann populations) within the 5% measurement uncertainty. This allowed Csillag to calculate the shifts of the most intense fine-structure transitions in each of the two components ($1 = 2P_{3/2}$–$nD_{5/2}$ in "red" and $2b = 2P_{1/2}$–$nD_{3/2}$ in "violet") from the observed maxima and determine their positions. The statistical uncertainties of the measured wave numbers varied between 0.0013 cm^{-1} and 0.010 cm^{-1} (corresponding to 40 MHz and 300 MHz) depending on the signal-to-noise ratio. According to Csillag [162], the main source of a systematic error was the uncertainty of the ^{198}Hg wavelength standards, ±0.0001 Å (18 MHz). Csillag [162] discussed several other sources of possible systematic errors. Among those, he mentioned the use of non-standard air. According to his estimates, it could produce an error in the derived value of the Rydberg constant smaller than 0.002 cm^{-1}, which corresponds to an error in the individual line measurements about 0.005 cm^{-1} (or 14 MHz).

However, our estimates lead to quite different conclusions. By using Ciddor's formulas for the refractive index of air [163], and assuming reasonable possible differences of the air parameters from standard conditions (temperature 20 °C, humidity 50%, and pressure 90 kPa, vs. the standard conditions of temperature 15 °C, zero humidity, and pressure 101.325 kPa), we estimated that the use of non-standard air could lead to uncertainties in the measured frequencies as large as 100 MHz. Since the exact conditions of Csillag's experiment are not known, we recalibrated his measurements in the following way. We determined the correction factor for the measured frequencies as a mean of the ratios of the frequencies resulting from the second step of the level-optimization procedure (see the following section) to those measured by Csillag [162], weighted by inverse squares of their statistical uncertainties. The resulting correction factor is $\Delta\sigma/\sigma - 1 = +(9.2 \pm 2.6) \times 10^{-8}$. This corresponds to an increase in the measured frequencies of approximately 60 MHz. The frequency values given in Table G account for this correction factor. The uncertainties are computed as a square root of the sum of the statistical and systematic uncertainties.

7. Level optimization for deuterium

The level optimization procedure for deuterium, similar to that employed for hydrogen, was carried out in several steps. In the first step, the measured fine-structure intervals from Table G (excluding Csillag's measurements of the Balmer series [162]) were used to derive the energy levels using the LOPT code [139]. Then the Ritz series formulas were fitted for the $nD_{5/2}$ and $nS_{1/2}$ series as was done for H. Since there are fewer available data than for hydrogen (in particular, there are no precise measurements of the $n = 6$ levels), the number of terms used in the Ritz formulas was reduced by one compared to hydrogen, and the uncertainties of the derived values are considerably greater. The resulting parameters of the fittings are given in Tables H and I.

The uncertainties of the fitted energy levels and of the ionization energy were estimated in a manner similar to that for hydrogen, namely, by comparing the fitted energy values from different fittings with each other and with experimental values, where available. The resulting value of the ionization energy, 3 288 981 521.1(23) MHz, agrees well with the QED calculations of Jentschura et al. [10], which yield 3 288 981 522.062(3) MHz. The fitted energy values of the $nD_{5/2}$ and $nS_{1/2}$ levels also agree with the QED calculations within the estimated uncertainties, except for $5D_{5/2}$ which deviates from the theory by −4.0(23) MHz. This is less than two standard deviations, which is statistically acceptable.

Table H
Ionization energy and quantum defect expansion coefficients for the $nD_{5/2}$ series of D obtained by fitting different versions of the extended Ritz formula (25).

	3-term Ritz formula	4-term Ritz formula (exact fit)
E_I	3 288 981 519.6(4) MHz	3 288 981 521.1 MHz
$c_0 \times 10^6$	7.50(7)	8.13
$c_1 \times 10^6$	−68.0(16)	−113.8
$c_2 \times 10^6$	193(9)	1041
$c_3 \times 10^6$	–	−4 380
Std. dev. of levels	0.3 MHz	–

Table I
Quantum defect expansion coefficients for the $nS_{1/2}$ series of D obtained by fitting different versions of the extended Ritz formula (25).

	4-term Ritz formula	5-term Ritz formula (exact fit)
E_I (fixed)	3 288 981 521.1 MHz	3 288 981 521.1 MHz
$c_0 \times 10^6$	23.23(25)	24.21
$c_1 \times 10^6$	−48(4)	−79
$c_2 \times 10^6$	82(13)	369
$c_3 \times 10^6$	−51(10)	−923
$c_4 \times 10^6$	–	615
Std. dev. of levels	8 MHz	–

The fine-structure splittings $n\mathrm{D}_{3/2}$–$n\mathrm{D}_{5/2}$, $n\mathrm{P}_{1/2}$–$n\mathrm{S}_{1/2}$, and $n\mathrm{P}_{1/2}$–$n\mathrm{P}_{3/2}$ were also derived from interpolations and extrapolations similar to those used for hydrogen. In total, we obtained 35 interpolated and extrapolated fine-structure intervals. These intervals were added to the input of the least-squares level optimization program on equal footing with 23 measured values (not including Csillag's measurements [162]) to make the second step of the level optimization procedure. From this step, we obtained the Ritz values of the 2P–nD fine-structure intervals with uncertainties smaller than 2.3 MHz. These Ritz values were used to determine the correction factor for Csillag's measurements [162] (see the previous section). The twelve corrected measurements of Csillag were included in the third and final step of the level optimization procedure. Since their uncertainties are relatively large, they changed the optimized level values by very small amounts. The resulting optimized energy levels are listed in Table 6.

Table 7 lists the input frequency intervals (measured and derived semi-empirically from Ritz-formula fittings, interpolations, and extrapolations, as described above), along with the Ritz values of these intervals (differences between the corresponding energy levels) resulting from the least-squares level optimization procedure. In this table, we also give a comparison with theoretical values of these intervals resulting from precise QED calculations of Jentschura et al. [10]. We included in Table 7 several transitions that were not part of the level optimization procedure either because their energies were not measured or the measurement uncertainty was too large to have a noticeable effect on the optimized energies. In particular, the table includes the 1S–$2\mathrm{P}_{3/2}$ transition measured by Herzberg [164] who used lines of ^{198}Hg I between 1 203 Å and 1 236 Å for calibration. The currently adopted Ritz wave numbers of those lines [165] are systematically lower by 0.006 cm^{-1} (180 MHz) than those used by Herzberg. Therefore, we decreased his measured value of the 1S–$2\mathrm{P}_{3/2}$ interval by this amount.

Transitions that were not measured are denoted in Table 7 with a symbol "P" (meaning "predicted interval") in the column of references.

In Tables 6 and 7 we also include the calculated shifts of the fine-structure levels and transitions caused by the electric quadrupole and off-diagonal magnetic hyperfine interactions. These shifts are small compared to experimental uncertainties and thus are only of theoretical interest at present. The magnetic hfs in deuterium is much smaller than in hydrogen. However, there is an electric quadrupole hfs, which is absent in hydrogen. So, the behavior of the hfs shifts along the series is different from hydrogen insofar as it decreases fast with increasing n. From Table 6, it can be seen that for the 2P and 3P levels the magnitudes of the hfs shifts exceed the uncertainties of the high-precision QED calculations [10], which are done in an approximation that does not describe the hfs interactions. Table 7 shows that, similar to the hydrogen case, the effect of the hfs is more pronounced for fine-structure intervals between highly excited states. For example, the hfs correction for the $4\mathrm{D}_{3/2}$–$4\mathrm{D}_{5/2}$ interval is 265 Hz, which should be compared with the uncertainty 15 Hz of the fine-structure QED calculation [10]. Thus, in high-precision fine-structure calculations for deuterium, as well as for hydrogen, the hfs should not be neglected.

An overall agreement of the optimized experimental energy level values with the QED calculations of Jentschura et al. [10] is good. Out of 54 experimental level values in Table 6, only seven deviate from the QED calculations by more than one standard deviation, and one level ($8\mathrm{P}_{1/2}$) deviates by more than two standard deviations. The $3\mathrm{D}_{3/2}$ and $3\mathrm{D}_{5/2}$ levels that are based on the hydrogen $n = 3$ measurements of Zhao et al. [111], are both lower than the calculated values by approximately 2 MHz. This deviation is in the same direction as for the corresponding levels in hydrogen and is partially due to the same systematic shift. However, most of it is probably due to the relatively large uncertainties of the isotope shift measurements of Tate et al. [157], which are somewhat smaller than 2 MHz. The other $n = 3$ levels, which are based on the 2S–3P frequencies measured in deuterium by Zhao et al. [111] (recalibrated as described in Section 4), agree with the calculations very well. This indicates that their systematic error (denoted as y in Tables 6 and 7) is probably smaller than 0.1 MHz. The 4P levels are based mainly on the recalibrated measurements of Zhao et al. [121]. They agree with the calculations reasonably well.

The optimized value of the $2\mathrm{P}_{3/2}$ level is based on the $2\mathrm{S}_{1/2}$–$2\mathrm{P}_{3/2}$ measurement by Dayhoff et al. [47]. It is lower than the theoretical value by 0.2 MHz, which is within our increased estimate of the experimental uncertainty (see the previous section). However, the technique used by Dayhoff et al. [47] is potentially capable of a much better measurement precision. A new measurement of the interval by this method would be most welcome.

8. Fine structure measurements and derived energy levels in tritium

Tritium is considerably more difficult to handle in experiments than hydrogen and deuterium, because of its radioactivity. Although there are only a few reported measurements of the tritium spectrum, these data are important for diagnostics of tritium in thermonuclear reactors (see, for example, Skinner et al. [166]).

The first measurement of the tritium spectrum was made by Pomerance and Terranova [167]. Those authors observed the first three members of the Balmer series and measured the tritium-hydrogen isotope shift of the Balmer-α line to be 2.36(5) Å, corresponding to 5.48(12) cm^{-1} or 164(4) GHz. With the low spectral resolution of this experiment, the fine structure of the series lines was not resolved.

Kireyev [168] investigated the fine structure of the Balmer-α line using a capillary discharge tube and a Fabry-Perot interferometer. He was able to resolve partially the fine-structure components of the line, which appeared as three broadened peaks comprising the components (1, 4, 5), (3a, 3b), and (2a, 2b) (see Fig. 1). The measured mean separation from the corresponding peaks in hydrogen was 5.530 5(25) cm^{-1}, or 165 800(75) MHz.

A somewhat better spectral resolution was achieved in the experiment of Kibble et al. [39] who were able to resolve the weak component 5 from the merged profile of components 1 and 4. In that experiment, the light source was an rf discharge lamp filled with a mixture of deuterium or tritium and helium. The reduction of the Doppler-broadened line widths was achieved by immersing the lamp in liquid helium. Unfortunately, Kibble et al. [39] did not give the measured transition energies. They included only a scatter plot of deviations of different measurements of the Rydberg constant from a median value (see Fig. 4 of Kibble et al. [39]). We converted this plot into a set of transition energy measurements. Each of the six measured transitions has several data points on the plot, and from each data point we obtained a separate value of the transition energy. Since the measurement uncertainties for each of these data points are not known, we took an un-weighted mean as the final value for each transition. For the most intense transitions, there are many (up to 11) individual measurements, so the root of mean square of deviation from the mean (combined with a systematic uncertainty of ±5 parts in 10^8 in the Rydberg) gives a good estimate of the measurement uncertainty. However, for the weakest transition, $2\mathrm{P}_{3/2}$–$3\mathrm{S}_{1/2}$ (designated with label 5) there are only two data points. For this transition, based on the relative intensities given in Fig. 3 of Ref. [39], we estimated the statistical uncertainty to be twice as large as for the $2\mathrm{P}_{3/2}$–$3\mathrm{D}_{3/2}$ transition (designated with label 4). The values of transition frequencies and their derived uncertainties are given in Table J.

Table J
Tritium Balmer-α fine-structure transition frequencies derived from the measurements of Kibble et al. [39].

Transition	Frequency (MHz)
$2P_{3/2}$–$3D_{5/2}$	456 841 590(60)
$2P_{3/2}$–$3D_{3/2}$	456 840 480(110)
$2P_{3/2}$–$3S_{1/2}$	456 837 830(210)
$2P_{1/2}$–$3D_{3/2}$	456 851 440(70)
$2S_{1/2}$–$3P_{3/2}$	456 850 380(70)
$2S_{1/2}$–$3P_{1/2}$	456 847 300(70)

Tate et al. [157] measured the isotope shifts (IS) of the $2S_{1/2}$–$3P_{1/2}$, $2S_{1/2}$–$3P_{3/2}$, $2P_{1/2}$–$3D_{3/2}$, and $2P_{3/2}$–$3D_{5/2}$ transitions between H, D, and T by means of saturated absorption spectroscopy. They also gave the measured separations of the $2S_{1/2}$–$3P_{1/2}$, $2S_{1/2}$–$3P_{3/2}$, and $2P_{1/2}$–$3D_{3/2}$ transitions from $2P_{3/2}$–$3D_{5/2}$ in tritium. Since the transition frequencies in H and D are now known to a high precision, the data on isotope shifts and line separations in tritium given by Tate et al. [157] provide two or three separate values of frequency for each transition. We determined the frequencies of fine-structure transitions in tritium as weighted averages of those separate values. This derivation and the final values of transition frequencies are summarized in Table K. In the first two columns, the two uncertainty values given in parentheses have the following meaning. The first one is the statistical uncertainty of the measured IS, and the second one is the systematic uncertainty, which is a combination of the ±1 MHz frequency calibration uncertainty and the ±1 MHz uncertainty due to unknown pressure shifts [157]. The inverse squares of the statistical uncertainties were used as weights in averaging the values in the first two columns of the table. The resulting statistical uncertainties were combined in quadrature with the systematic uncertainty. The inverse square of this combined uncertainty was used as a weight in averaging the first two columns with the value in the third column weighted by the inverse square of its total uncertainty (given in parentheses). The last column has the resulting mean frequency and its total uncertainty.

In total, there are six fine-structure transitions for which the frequencies were measured. Four of them, resulting from the measurements of Tate et al. [157], are listed in Table K, and two ($2P_{3/2}$–$3D_{3/2}$ and $2P_{3/2}$–$3S_{1/2}$) are taken from the measurements of Kibble et al. [39], listed in Table J. To derive the eight energy levels with $n = 2$ and 3, involved in these transitions, we need at least two additional transition frequencies. They can be obtained from theoretical calculations.

Calculated values of energy levels of tritium were given by Garcia and Mack [31] and Erickson [37]. However, those calculations have limited precision due to omission of high-order QED corrections. Erickson's calculations are, in principle, more accurate, since they include higher-order corrections. To determine the excitation energies from Erickson's calculated values of the total energies [37], we adjusted Erickson's values in the following way. First, we scaled the energy values according to the currently adopted values of the Rydberg constant and the masses of the electron and the triton [24]. To do so, it was necessary to correct the triton mass misprinted as 3.015 007 743 74 u in Table A of Erickson [37] to the value actually used in his calculations, 3.015 500 743 74 u [169]. Second, for the nS levels we accounted for the correction of the electron self-energy term proportional to $(Z\alpha)^6$ that was found by Mohr [52], Sapirstein [53], and other investigators to be incorrectly given by Erickson. Jentschura et al. [170] calculated the coefficients $A_{60}(n\text{S})$ corresponding to this term for $n \leqslant 8$. The correction to Erickson's values for the total energies due to this term amounts to −222 kHz for $1S_{1/2}$, −45 kHz for $2S_{1/2}$, and −14 kHz for $3S_{1/2}$.

To verify the accuracy of Erickson's calculations (scaled to the adopted values of the fundamental constants and corrected for the above-mentioned error in the self-energy expression), we made similar calculations of the total energies of levels in hydrogen and deuterium and compared them with the calculations of Jentschura et al. [10]. We found that differences between the two calculations in many cases are significantly greater than the uncertainties given by Erickson [37]. They are the greatest for the $1S_{1/2}$ level. The differences are obviously due to the higher-order terms not included by Erickson and his less accurate account for the nuclear effects. Thus, to calculate the excitation energies, we used the value of the ionization potential from a more accurate calculation given by Mohr et al. [6] as 109 718.5439 cm^{-1} with a reference to a private communication from S. Kotochigova. This value, corresponding to 3 289 279 196.4 MHz, was obtained using the same theoretical formulas as those used by Jentschura et al. [10] for hydrogen and deuterium. Presumably, the uncertainty of the QED calculation for this value is much smaller than the uncertainty due to the imprecise knowledge of the fundamental constants, which is ±3.8 MHz, and the rounding error, which is ±0.7 MHz. The magnitude of the differences between Erickson's calculations and those of Jentschura et al. [10] for excited states is much smaller for deuterium than for hydrogen. This indicates that the large fraction of Erickson's errors is due to higher-order mass corrections that rapidly decrease with increasing nuclear mass. Thus, to estimate the uncertainties of the scaling and correction of Erickson's calculations for tritium, we combined in quadrature the uncertainties given by Erickson [37] and the differences of his (scaled and corrected) total energies for deuterium from those of Jentschura et al. [10]. These combined uncertainties range from 0.5 kHz for the total energy of $3D_{5/2}$ to 30 kHz for $2S_{1/2}$.

We used the $1S_{1/2}$–$2P_{1/2}$ transition frequency, 2 466 954 603.3(39) MHz, determined as described above, as the basis for the derivation of the $n = 2$ and 3 levels (the uncertainty of this value is mainly due to the uncertainties of the Rydberg constant and triton mass). One more transition used in this derivation is $3D_{3/2}$–$3P_{3/2}$. Its frequency in deuterium was measured to be 5 ± 5 MHz [159]. Assuming that the isotope shift is mainly determined by the reduced-mass factor, we accepted this frequency in tritium to be the same as in deuterium within the uncertainty of the Wilcox and Lamb measurement [159]. The energy levels of tritium, as derived from the six observed and two additionally accepted transition frequencies, are listed in Table 8.

In the last column of Table 8 we give the differences of energy levels calculated with the data given by Garcia and Mack [31] from those calculated with Erickson's data [37]. In these calculations we adjusted the original values from both sources in a similar manner.

Table 8 shows that almost all theoretical data for transition energies agree well with experimental data and with each other.

Table K
Tritium Balmer-α fine-structure transition frequencies derived from the measurements of Tate et al. [157].

Transition	Frequency from IS(H-T) (MHz)	Frequency from IS(D-T) (MHz)	Frequency from line separation in T (MHz)	Mean measured frequency (MHz)
$2P_{3/2}$–$3D_{5/2}$	456 841 568.8(8)(14)	456 841 568.7(17)(14)	–	456 841 568.8(16)
$2P_{1/2}$–$3D_{3/2}$	456 851 457.2(19)(14)	456 851 457.4(21)(14)	456 851 457.1(18)	456 851 457.2(13)
$2S_{1/2}$–$3P_{3/2}$	456 850 406.2(11)(14)	456 850 406.2(12)(14)	456 850 404.9(24)	456 850 405.8(14)
$2S_{1/2}$–$3P_{1/2}$	456 847 154.3(18)(14)	456 847 154.2(18)(14)	456 847 152.9(29)	456 847 153.8(16)

The only substantial deviation is for the $2P_{1/2}$–$3D_{3/2}$ interval. Its experimental value, based on the measurements of Tate et al. [157], is 456 851 457.2(13) MHz, while the theory gives 456 851 460.866(5) MHz. The difference is −3.7(13) MHz. In this regard, we note that, in deriving the fine-structure intervals from the experimental data of Tate et al. [157], we assumed that their measurements are free of systematic errors. If systematic errors are present, it could lead to accumulation of errors in summation of the H-D and D-T isotope shifts and transition separations. Thus, the observed deviation from theory in tritium probably indicates that systematic errors on the order of 1 MHz were indeed present. Since the determination of the $n = 3$ levels is ultimately based on the measurements of Zhao et al. [111], their systematic error (denoted as x) propagates from hydrogen to tritium. However, the probable value of x is considerably smaller than the observed deviation from theory.

9. Comparison with QED calculations

It is not easy to compare the experimental energy levels and transition frequencies of H, D, and T with the QED calculations because the latter are in fact adjusted to fit the experimental transition frequencies by adjusting the fundamental constants entering into the QED equations (see Mohr et al. [6,24]). In this adjustment procedure, there is no easy way to trace individual contributions of a particular measurement to the resulting values of the fundamental constants. Such tracing was done by the authors of Refs. [6,24] case-by-case by excluding a particular measurement from the adjustment procedure and comparing the resulting values and uncertainties of the fundamental constants with the original optimized values [171]. In this way, it was found that the contributions of the measured H and D frequency intervals to the adjusted values of the fundamental constants are significant for only three constants: the proton charge radius R_p, the deuteron charge radius R_d, and the Rydberg constant R_∞. In the least-squares determination of these three constants [6,24] there are three degrees of freedom to fit 25 input values (23 frequency measurements in H and D and two direct measurements of R_p and R_d using electron-proton and electron–deuteron elastic scattering [172,173]). The two 1S–2S measurements by Hänsch et al. [115] (1S–2S frequency in H) and Huber et al. [152] (H–D isotope shift of the 1S–2S frequency) having sub-kilohertz uncertainties exactly determine two of these three degrees of freedom. It is difficult to determine which particular combinations of R_p, R_d, and R_∞ correspond to these two degrees of freedom. However, one thing can be stated with certainty: the exact agreement of those two ultra-precise 1S–2S measurements with the QED calculations cannot be considered as a confirmation of the QED theory, because it is the result of the fitting of the fundamental constants based on these (and other) transitions.[6]

The remaining degree of freedom is determined by the remaining 23 input values, of which 21 are the H and D frequency measurements. All other physical constants besides R_p, R_d, and R_∞, entering in the QED calculations, such as the fine-structure constant, Planck's constant, and the mass ratios of electron, proton, deuteron, and triton, are mainly determined by other independent (non-spectroscopic) measurements (see Mohr et al. [6,24] and references therein for more details). Thus, we may say that the comparisons given in Table 4 through Table 8 are meaningful (for estimation of the validity of the QED theory), except for the special case of the above-mentioned 1S–2S frequencies in H and D. In these comparisons, the highest weight should be given to the measurements having the smallest ratios of the measurement uncertainties to the QED calculation uncertainties (see below). In Tables C and G (input values for the level optimization in H and D), there are ten measurements having this ratio smaller than or equal to 3. They are the 2S–8S, 2S–8D, and 2S–12D measurements in H and D by de Beauvoir et al. [103]. All of them were included in the CODATA 2006 adjustment of the fundamental constants. They all agree with the QED calculations within the measurement uncertainties.

It should be noted that uncertainties of measurements and QED calculations are of a different nature. The measurement uncertainties consist of two parts, random and systematic. Everywhere in this paper, where experimental uncertainties are given, they are meant to be on the level of one standard deviation, where the random and systematic parts are combined in quadrature, unless otherwise stated. In cases when several measured values contain the same systematic uncertainty, the latter is given separately as an additive systematic shift. The uncertainties of the QED calculations have been explained in detail by Mohr et al. [6]. The uncertainty of a calculated quantity (energy level or interval) consists of two parts: the uncertainty of a theoretical expression for this quantity and the uncertainty due to variances and covariances of the values of the fundamental constants entering that expression. The second part has statistical nature, as the values of the fundamental constants are obtained together with their variances and covariances by a least-squares adjustment procedure from a set of measured values, which have well-defined statistical uncertainties. The first part is not strictly statistical. Theoretical expressions for energy levels and intervals are expanded in Taylor series truncated at certain points. The sums of omitted (uncalculated) high-order terms are estimated based on character and convergence trends of each of those expansions. These estimates are treated as variances, or standard uncertainties, of approximate theoretical expressions. Uncertainties of different values are in general correlated with each other to some extent. This correlation is described by covariances of different values. Expressions for variances and covariances of calculated energy levels are given by Eq. (58) through (60) of Ref. [6]. Neglecting here the complicated nature of the total theoretical uncertainties, we compare them directly with statistical standard deviations of the measured values.

In Table 5 (output of the level optimization for Table H), in addition to the ten measured frequency intervals discussed above, there are eight more intervals having ratios of Ritz uncertainties to QED calculation uncertainties that are smaller than or equal to 3 while being determined by transition frequencies other than 1S–2S. They are the $2P_{1/2,3/2}$–$8S_{1/2}$, $2P_{1/2,3/2}$–$8D_{3/2}$, $2P_{1/2,3/2}$–$12D_{3/2}$, $2P_{3/2}$–$8D_{5/2}$, and $2P_{3/2}$–$12D_{5/2}$ intervals in H. These intervals were determined by combining the measurements of de Beauvoir et al. [103] with the $n = 2$ fine-structure intervals, $2P_{1/2}$–$2S_{1/2}$ measured by Lundeen and Pipkin [88], and $2S_{1/2}$–$2P_{3/2}$ measured by Hagley and Pipkin [98]. These frequency intervals also agree with the QED calculations within their uncertainties.

Thus, we can state that 18 frequency intervals in H and D, determined with high precision either by direct measurements or by combinations of such, are effectively fitted by the QED calculations with only one free parameter (some combination of R_∞, R_p, and R_d) to within the uncertainties of their determination. This can be considered as a convincing confirmation of the validity of the QED theory.

[6] The relative uncertainties of the final values of R_∞, R_p, and R_d resulting from the CODATA 2006 least-squares adjustment [6,24] are 6.6×10^{-12}, 7.8×10^{-3}, and 1.3×10^{-3}, respectively. If the two 1S–2S frequency measurements [115,152] are deleted from the input of the adjustment procedure, the resulting uncertainties of R_∞, R_p, and R_d become 6.9×10^{-12}, 1.2×10^{-2}, and 2.8×10^{-3}, respectively [171]. The uncertainty of R_∞ increases only slightly, while those of R_p, and R_d increase significantly. Without the high-precision 1S–2S data, these two charge radii are determined largely by the direct measurements using electron–proton and electron–deuteron elastic scattering having relative uncertainties 2.0×10^{-2} and 4.7×10^{-3}, respectively. Numerous other spectroscopic data also contribute to the reduction of the uncertainties of the adjusted values of R_p, and R_d.

However, there remain some outstanding discrepancies in the lower-accuracy measurements. The most significant of them are the following.

(1) The $n = 3$ levels in hydrogen, based on the $2S_{1/2}$–$3P_{1/2,3/2}$ measurements of Zhao et al. [111], all deviate from the QED calculations by −0.7 MHz, which is four statistical standard deviations. The original estimate of the systematic uncertainty x given by Zhao et al. [111] (±0.07 MHz) is too small to explain this deviation.
(2) The $n = 6$ levels in hydrogen deviate from the QED calculations by −0.17 MHz, which is also four statistical standard deviations. These levels are based on the measurement of the frequency intervals ν(2S–6S,6D)–1/4 ν(1S–3S) by de Beauvoir et al. [103]. Therefore, they contain one quarter of the systematic uncertainty x of the n=3 levels. The deviation of the $n = 6$ levels from the QED calculations is exactly one quarter of that for $n = 3$ levels, which is consistent with the assumption of $x = -0.7$ MHz, meaning that the (re-calibrated) measurements of Zhao et al. [111] have this large systematic error. An independent direct measurement of the 1S–3S frequency, which is currently underway [128–130], should resolve this discrepancy.
(3) The $2P_{3/2}$ level in deuterium, based on the $2S_{1/2}$–$2P_{3/2}$ measurement by Dayhoff et al. [47], disagrees with the QED calculations by four standard deviations if the originally reported measurement uncertainty (±0.06 MHz) is correct. We assumed a five times greater uncertainty for those measurements, which eliminates the discrepancy. However, this assumption can only be verified by repeating those measurements.

10. Line series of hydrogen

Bohr [26] wrote: "If the theory is right, we may therefore never expect to be able in experiments with vacuum tubes to observe the lines corresponding to high members of the Balmer series of the emission spectrum of hydrogen; it might, however, be possible to observe the lines by investigation of the absorption spectrum of this gas..." He pointed out that observation of emission lines from highly excited states of hydrogen requires extremely low particle densities in the emitting medium. This condition is easily fulfilled in many astrophysical objects but is very difficult to implement in laboratory, as low density results in low line intensities. At higher densities the sizes of the orbits of excited electrons become comparable with the mean distances between the atoms, and the excited states are destroyed by collisions. Those considerations were confirmed by the history of observations of the hydrogen spectrum. They indicate that highly excited states of hydrogen are mostly important for astrophysical applications.

The width of the fine structure (distance between the outermost components of the fine structure of levels with the same principal quantum number n) decreases with increasing n roughly as n^{-3}. Therefore, with increasing n, the widths of the series lines of the higher members of the series quickly become too small to be resolved even by means of elaborate spectroscopic methods such as Fourier-transform spectroscopy (FTS). In addition, emission lines are broadened by the thermal Doppler and other effects. Thus, the series lines observed in emission spectra of celestial objects are usually unresolved blends of the fine-structure components. The exact positions of the line centers in such spectra depend on the population kinetics in the emitting medium. So, to derive these positions from the precisely known fine-structure intervals, one would need additional information on the excitation cross sections and the plasma parameters such as particle densities and velocity distributions. These parameters are very difficult to obtain. Thus, it would be useful to obtain empirical information on positions of line centers pertinent to typical conditions of astrophysical observations. The goal of this section is to obtain this empirical information. For completeness, we also review the laboratory observations.

10.1. Laboratory observations

As predicted by Bohr [26], observation of high members of the hydrogen series lines in the laboratory has been very difficult to accomplish.

The first member of the Lyman series was observed by Lyman [27] at approximately 1216 Å (121.6 nm, or 2465000 GHz; no estimate of uncertainty was given). Wilkinson [150] measured the Lyman-α, β, and δ lines in a hollow cathode discharge using a grating spectrograph. The wavelengths were 1215.662(5) Å, 1025.728(3) Å, and 949.742(4) Å (frequencies 2466084(10) GHz, 2922729(9) GHz, and 3156567(13) GHz, respectively). Sansonetti et al. [149] measured the Lyman-α wavelength in a Pt-Ne hollow cathode discharge containing a small amount of hydrogen as an impurity. They obtained the wavelength 1215.6701(20) Å (frequency 2466068(4) GHz) for this line.

Curtis [143,144] measured the wavelengths of the first six members of the Balmer series ($n = 2 \to n = 3, 4, \ldots$) using a capillary discharge excited by an inductance coil in conjunction with a grating spectrograph. His measurements had uncertainties ranging from 0.0006 Å to 0.0017 Å (100 to 220 MHz).

Wood [174] using a specially designed gas discharge tube succeeded in observation and measurement of the Balmer series up to $n = 22$. Although he gave only a qualitative description of his measurement uncertainties, stating that they are of the same order of magnitude as those of Curtis [143,144], the uncertainties can be estimated from deviations of his measured wavelengths from Curtis's calculations based on the Rydberg formula. From these deviations, it is seen that the measured wavelengths are systematically too high by approximately 0.005 Å (too low by ≈1.0 GHz) and have a random scatter of about ±0.007 Å (1.4 GHz). Similar conclusions can also be obtained based on purely experimental data (using deviations of the measured wavelengths from the Ritz values, as described further in this article). With this approach, it was possible to determine that the systematic shift of Wood's measurements linearly depends on the wavelength, varying from +0.002 Å at 3700 Å to −0.007 Å at 4300 Å, while the statistical uncertainty is ±0.006 Å (1.2 GHz).

Paschen [175] identified and measured the first two members of the series named after him ($n = 3 \to n = 4, 5, \ldots$). He obtained air wavelengths of 18751.3 ± 1.0 Å and 12817.6 ± 1.5 Å for these two lines, corresponding to frequencies of 159835(9) and 233827(27) GHz. His result for the first member of the series still remains the only published measurement of this line.

Poetker [176] used a gas-discharge tube similar in design to Wood's, of about 2 m length, to measure the wavelengths of five lines of the Paschen series $n = 3 \to n = 7 - 11$ with uncertainties of ±1 Å (30 to 40 GHz).

Brackett [28] was the first to observe the first two members of the series bearing his name ($n = 4 \to n = 5, 6, \ldots$). His measurements had relatively large uncertainties. The measured wavelengths were 4.05(3) and 2.63(2) μm, corresponding to frequencies of 159(4) and 234(9) THz. He also observed the third member of the Paschen series ($n = 2 \to n = 6$) with similar low resolution.

Pfund [29] and Humphreys [30] observed the leading members of the two series named after them ($n = 5 \to n = 6, 7, \ldots$ and $n = 6 \to n = 7, 8, \ldots$) at approximately 7.40 μm and 12.4 μm (frequencies 40.5 and 24.2 THz), respectively. Humphreys [30] also observed the second member of the Pfund series (at 4.65 μm or 64.4 THz) and three additional members of the Brackett series

($n = 4 \rightarrow n = 7$, 8, and 9, at 2.17, 1.94, and 1.82 μm, corresponding to 138, 154, and 165 THz, respectively).

The first member of the seventh series of hydrogen was observed by Hansen and Strong [177] using a water-cooled discharge tube and an echelle grating spectrometer with a bolometer cooled by liquid helium. Using known absorption lines of water as references, they measured the wave numbers of the $n = 7 \rightarrow n = 8$ and Humphreys-α ($n = 6 \rightarrow n = 7$) lines to be 524.60 cm^{-1} and 808.28 cm^{-1} (vacuum wavelengths 190 621 Å and 123 719.5 Å, or frequencies 15 727.1 GHz and 24 231.6 GHz, respectively). We estimate the uncertainties of those measurements as 1/10 of the resolving power, which corresponds to ±0.04 cm^{-1} and ±0.02 cm^{-1}, or ±1.2 GHz and ±0.6 GHz, respectively.

10.2. Solar observations

The vast majority of the hydrogen series measurements were made on solar spectra. Although the first observations of Wollaston [178] and Fraunhofer [179] were made with an absorption spectrum, in subsequent years it was found that absorption lines were very broad in the solar spectrum, which precluded their precise measurement. However, the spectra of the solar limb and prominences were found to contain much sharper emission lines. These lines are best seen during solar eclipses or with equipment capable of analyzing the light coming from outside the solar disk.

The Lyman series was observed in spectra of the solar corona up to $n = 21$ (see Feldman et al. [180], Curdt et al. [151], and Parenti et al. [181]). All those observations were made using data recorded by the extreme UV spectrometer SUMER on the satellite-borne Solar and Heliospheric Observatory.

The Balmer series was measured in solar eclipse spectra by Evershed [182] and Dyson [183] up to $n = 31$ and by Mitchell [184,185] up to $n = 37$. We corrected the wavelengths given in the early publications [182–184] according to the revised wavelength scale [186]. This correction amounts to −0.178 Å at 4860 Å and −0.140 Å for wavelengths between 3890 Å and 3 650 Å. If only the hydrogen lines are taken into account, it appears that Dyson's measurements [183] are the most accurate, deviating from calculated values by ±0.02 Å on average. However, when species other than hydrogen are considered, a much larger uncertainty is revealed. We compared Dyson's wavelengths of 318 lines belonging to He I, Mg I, Al I, Si I-II, Ca I-II, Sc I-II, Ti I-II, V I-II, Cr I-II, Mn I, Fe I-II, Co I, Ni I, Zn I-II, Sr II, Y II, Zr I-II, Ba II, La II, Ce II, Eu II, Gd II, Er II, and Pb I with more precise experimental values [187,188]. Only the lines for which Dyson did not indicate any blending were used in this comparison. The comparison clearly shows that the uncertainty of Dyson's measurements [183] is ±0.05 Å (corresponding to ±6 to ±11 GHz) with no noticeable systematic shift. Evershed's measurements [182] are slightly less accurate, but they agree with Dyson's within ±0.05 Å. Similar comparisons were done for Mitchell's wavelength measurements [184,185]. In the region 3600–4910 Å, containing the Balmer series (except for the Balmer-α line), 482 lines of He I, Ti I-II, Cr II, V I, Fe I-II, Ni I, and Y II, for which Mitchell [185] indicated no blending, were compared with more accurate wavelengths from the NIST databases [187,188]. This comparison showed that Mitchell's wavelengths [185] are accurate to ±0.034 Å (corresponding to ±4 to ±8 GHz) with a nearly constant systematic shift equal to +0.0092(16) Å. However, the hydrogen lines in the region 3800–4900 Å systematically deviate from the calculated values by +0.15 Å. To understand the reason for these deviations, it is necessary to consider the nature of the data. The wavelengths given in the 1947 paper [185] are averaged over observations of several different eclipses. In this averaging, a greater weight had been given to the 1905 measurement [184], as it had a much better overall quality due to exceptionally good observational conditions.

However, as noted by Mitchell [184], the lines having intensities greater than 25 on his adopted scale were poorly measured due to the difficulty in visually finding a center of the line appearing as a long arc in the spectrum of a slitless spectrograph. In Mitchell's line list [184] there are very few lines of such great intensity, and most of them belong to hydrogen. Comparing those of He, Mg, Ca, Ti, and Sr with the reference wavelengths [187,188] shows that the wavelengths measured by Mitchell [184] are systematically shifted towards longer wavelengths, the shift being greater for more intense lines. The shift is as large as +0.31 Å for the two intense Ca II lines at 3933.66 Å and 3 968.47 Å. This shift can be approximated as linear in intensity. Removing this intensity-dependent shift from the hydrogen wavelengths brings them into fair agreement with the measurements of Dyson [183] and Evershed [182], as well as with the calculated values. However, this procedure cannot provide for uncertainties smaller than 0.08 Å. Therefore, we conclude that Mitchell's wavelengths [184,185] for the hydrogen lines in the region 3703–4862 Å should be disregarded, and Dyson's measurements [183] should be used instead. In the shorter-wavelength region containing the less intense, higher members of the Balmer series ($n = 17$–37), the uncertainty of Mitchell's wavelengths [185] (after removing the systematic shift of 0.009 Å) can be safely assumed to be the same as for the other species (i.e., ±0.034 Å) except for blended lines, for which a more reasonable uncertainty estimate is ±0.10 Å. The wavelength region 6200–8900 Å, containing the Balmer-α line and the Paschen series, was investigated by Mitchell [185] during a different eclipse, and those measurements do not exhibit such a measurement flaw. In particular, his wavelength of the Balmer-α line, 6562.80 Å, is in fairly good agreement with the laboratory measurement of Curtis [143], 6562.793 0(17) Å.

As noted above, the Paschen series was observed by Mitchell [185] for $n = 11$–40 in a spectrum of a solar eclipse. By comparing his wavelengths of other species (He I, O I, Mg I, Al I, K I, Ca I-II, Sc II, Ti I, Cr I, Fe I-II, Ni I, and Ba II) with those known now with higher accuracy [187] we determined that Mitchell's wavelengths in this spectral region possess a systematic shift equal to +0.012(3) Å (in terms of frequency, it varies between −0.46 GHz at 8863 Å and −0.87 GHz at 6439 Å). One hundred reference lines were used in this comparison, which provided sufficient statistics. The uncertainties of Mitchell's measurements [185] in the wavelength region 6439 Å to 8863 Å were determined as a standard deviation of his wavelengths from the reference values and are equal to ±0.03 Å (±1.2 GHz at 6439 Å and ±2.2 GHz at 8 863 Å). For the wavelengths given with only one digit after the decimal point, we adopted the uncertainty of ±0.1 Å.

High-precision measurements of solar emission lines became possible after the advent of Fourier-transform (FT) spectroscopy. Such measurements were pioneered by Brault and Noyes [189] who, using the FT spectrometer of the National Solar Observatory at Kitt Peak, measured the wave number of the Humphreys line ($n = 6 \rightarrow n = 7$) to be 808.283(3) cm^{-1}, corresponding to vacuum wavelength 123719.0(5) Å or frequency 24231.71(9) GHz. They also measured a few members of the next three series ($n = 7 \rightarrow n = 8$–10, $n = 8 \rightarrow n = 11$, 12, 14, and $n = 9 \rightarrow n = 14$, 15). The spectral resolution was typically 0.005 cm^{-1} or 150 MHz ($\Delta\lambda/\lambda = 160000$).

For the Paschen-β line ($n = 3 \rightarrow n = 5$), Mohler's solar wavelength table [190] lists the wavelength 12818.2(3) Å corresponding to frequency 233816(5) GHz. A more precise measurement was accomplished by Chang and Deming [191]. They used the FT spectrograph of the National Solar Observatory at Kitt Peak to observe infrared emission spectra of solar prominences. The spectral resolution was 0.02 cm^{-1}, which corresponds to a frequency interval of 600 MHz and a wavelength interval of 0.4 Å at 2150 cm^{-1} and 0.024 Å at 9140 cm^{-1}. The wavelength scale was calibrated using telluric N_2O lines. The observed line positions were corrected for the Earth's motion, solar rotation, and the gravitational redshift. The resulting wave number of the Paschen-β line was 7 799.352(5) cm^{-1}, corresponding to a frequency of 233 818.69(15) GHz and an air wavelength of 12 818.072(8) Å. Chang and Deming [191] also measured the Paschen-γ line, seven members of the Brackett series ($n = 4 \rightarrow n = 5$, 7, 9–12, and 16), and two members of the Pfund series ($n = 5 \rightarrow n = 7$ and $n = 5 \rightarrow n = 8$). We inferred the measurement uncertainties from their Fig. 3.

Farmer et al. [192] measured the wave numbers of the $n = 7 \rightarrow n = 8$ and $n = 8 \rightarrow n = 9$ lines in the solar spectrum using an FT spectrometer at the International Scientific Station of Jungfraujoch in the Bernese Alps, Switzerland. This location is particularly favorable for infrared solar spectroscopy because of its high altitude (3.58 km) and exceptionally low zenith water vapor column densities during the winter months. The spectral resolution of the instrument was 0.006 cm^{-1} (or 180 MHz). The spectrum was calibrated using telluric N_2O and CO_2 lines. The measured wave numbers of the solar lines were corrected for the Doppler shift. The uncertainties of the resulting wave numbers of the $n = 7 \rightarrow n = 8$ and $n = 8 \rightarrow n = 9$ lines, 524.605 cm^{-1} and 359.667 cm^{-1}, respectively, were conservatively estimated as ±0.001 cm^{-1}. The corresponding frequency values are 15 727.26(3) GHz and 10 782.55(3) GHz, respectively.

Chang and Deming [193] reported their FTS measurements of solar prominence spectra in the region 520–1 230 cm^{-1}. They observed all the lines investigated earlier by Brault and Noyes [189] and the additional $n = 8 \rightarrow n = 13$ and $n = 9 \rightarrow n = 16$ lines. To calibrate the frequency scale, they used calculated wave numbers of the line centers of the hydrogen lines. To decouple their measurements from theoretical calculations, we re-calibrated their frequency scale separately for each of the three investigated prominences using the well-measured hydrogen wave numbers from Brault and Noyes [189] and Farmer et al. [192]. The frequency scales were slightly different for different prominences because of the different Doppler shifts caused by the movement of plasma in the prominences. We derived the wave numbers of the hydrogen lines as weighted means of the three measurements. (The systematic shifts due to solar gravity and rotation and due to Earth's rotational and orbital motion were already accounted for by Chang and Deming [193]). The uncertainties were calculated as the root of the sum of the squares of statistical uncertainties given by Chang and Deming [193] and the systematic uncertainties of the frequency-scale factors. The total uncertainties of the mean wave numbers varied between 0.0014 cm^{-1} and 0.008 cm^{-1} (40 and 240 MHz) for different lines. For example, the wave number of the Humphreys-α line ($n = 6 \rightarrow n = 7$) is 808.282 5(13) cm^{-1} (frequency 24 231.70(4) GHz).

FTS measurements of several other members of the Brackett, Pfund, and Humphreys series are given by Geller [194]. Those measurements were made using the data obtained in the ATMOS experiment on the Space Shuttle. The hydrogen lines were observed as broad absorption peaks and were measured with uncertainties 0.01 cm^{-1} to 0.03 cm^{-1} (300–900 MHz), depending on the noise level and the degree of blending with other lines. We estimated the uncertainties for each line by looking at the spectrograms given by Farmer and Norton [195].

Boreiko and Clark [196] measured three far-infrared lines arising from high-n transitions ($n = 13 \rightarrow n = 14$, $n = 14 \rightarrow n = 15$, and $n = 15 \rightarrow n = 16$) in high-resolution solar spectra obtained with a Michelson interferometer on a balloon-based telescope.

Boreiko et al. [197] observed the $n = 11 \rightarrow n = 12$, $n = 12 \rightarrow n = 13$, and $n = 13 \rightarrow n = 14$ transitions in the solar spectrum using a balloon-borne FT spectrometer. Although they did not give the measured wave numbers, they can be obtained from the spectrograms presented in their Fig. 1 with uncertainties of 0.005 to 0.010 cm^{-1} (150 to 300 MHz). Recently, Clark [198] re-processed those balloon observations and obtained the wave numbers of these three lines with uncertainties of 0.003–0.006 cm^{-1} (90–180 MHz).

Clark et al. [199] observed solar emission lines from high-n transitions ($n = 19 \rightarrow n = 20$ and $n = 21 \rightarrow n = 22$) in the submillimeter wavelength region with a dual-beam polarizing FT spectrometer at the James Clerk Maxwell Telescope in Mauna Kea, Hawaii, at an altitude of 4.092 km. Their wave numbers are 29.621(5) cm^{-1} and 22.096(3) cm^{-1} (frequencies 888.02(15) and 662.42(9) GHz).

10.3. Radio recombination lines

Nebulas surrounding supernovae contain large amounts of ionized hydrogen at very low particle densities. Astronomers call such objects H II regions, which is to say that most of the hydrogen is ionized in them. Under such conditions, recombination of protons with electrons creates highly excited hydrogen atoms. These excited atoms decay radiatively into lower states and can be detected by analyzing the emitted radiation. Transitions between closely located highly excited levels with high principal quantum numbers $n \sim 100$, $\Delta\ n = 1, 2, \ldots$ have frequencies in the radio range that can be detected and measured using large radio telescopes. The radio-frequency lines corresponding to these transitions are called radio recombination lines (RRLs). Such lines of hydrogen were first convincingly detected in 1964 by Sorochenko and Borodzich [200] and Dravskikh et al. [201] at 8 872.5 MHz and 5 762.9 MHz, respectively. Since then, many tens of hydrogen RRLs with principle quantum numbers from 10 to about 250 have been detected, and the observations have been extended to both the lower and higher frequency regions corresponding to wavelength regions from meters to micrometers. A comprehensive review of observations and theory of RRLs was given recently by Gordon and Sorochenko [202]. The RRLs have important applications for diagnostics of astrophysical objects and for understanding collisional processes and line-profile formation in Rydberg atoms. However, since their fine structure is smeared out by Doppler broadening, and their observed frequencies depend on the macroscopic velocities of emitting media, which cannot be measured independently, the RRLs provide no additional information about the atomic structure of hydrogen. Consequently, we did not make use of them in this compilation.

10.4. The concept of distinctive energies

In atomic spectroscopy, the fine structure of atomic energy levels is often unresolved. The energy levels of atomic spectra are usually grouped in spectroscopic terms according to a certain coupling scheme (e.g., *LS* or *jj*). We say that the fine structure of a term is unresolved if all observed transitions originating from its fine-structure components or terminating on them are merged together in unresolved line profiles. In such cases, the term is usually represented by a single energy value corresponding to its center of gravity. This center-of-gravity energy is theoretically calculated as an average of the energies of the fine structure components weighted by their statistical weights ($2j + 1$). To find experimental energy values of the terms, it is common to use unresolved spectral lines as if the term is represented by a single energy value. The hydrogen spectrum, as originally noted by Rydberg [203], uniquely differs from the spectra of all other elements insofar as it does not exhibit a discernible term structure. This is, of course, because of the near degeneracy of its energy levels with respect to the quantum numbers l and j.

Since most of the observations of the hydrogen series lines do not resolve the fine structure, it is of interest to investigate how well the levels with the same principle quantum number n can be represented by a single energy. Towle et al. [204] calculated the distributions of intensities of the fine-structure components of transitions between high members of hydrogen series in the local thermodynamic equilibrium (LTE) approximation. They showed that these distributions are highly asymmetric, and their shapes vary strongly depending on the difference between the principal quantum numbers Δn. The distribution of intensities is the sharpest for $\Delta n = 1$, where its maximum is located very close to the lowest-frequency fine-structure component. With increasing Δn, the distribution of intensities widens, and its maximum shifts towards higher frequencies. Therefore, the accuracy of the single-energy approximation will depend on the participating transitions. It will also depend on the way the energy of the level complex is determined. For example, one can use the statistical weights of the fine-structure components to find the center of gravity of the level complex. Another way is to use the observed relative intensities of individual fine-structure transitions as weights. In this case, the energy will depend on the population kinetics in the medium. Since most of the observed hydrogen series lines have a completely unresolved fine structure, the only feasible way is to use the observed wavelengths to find the corresponding energies of the level complexes. Here we will investigate the effect of using different transition sets on empirical energies of the level complexes.

If the hydrogen energy levels were truly degenerate in l and j, then the energy levels could easily be found from the observed spectral lines by using a least-squares level optimization procedure. This procedure would not depend on the particular set of spectral lines and would require only that every level participates in at least one observed transition. The uncertainties of the optimized energy levels would then depend only on the measurement uncertainties of the lines. The fine structure, however small, distorts this clear picture. If we used only the Lyman series in the level optimization, then we would find the centers of gravity of the $n\mathrm{P}_{1/2}$ and $n\mathrm{P}_{3/2}$ levels (assuming populations proportional to statistical weights). Adding the Balmer and other series would shift the optimized energies, since the higher-L transitions would tend to decrease the spacing between the excited levels. We will call the least-squares-optimized energy levels obtained with a particular set of transitions the *distinctive energies* corresponding to this transition set. Here, the transition set is defined not only by the scope of participating transitions, but also by their energy uncertainties.

To investigate the effect of varying transition sets on the distinctive energies of the merged levels, we made numerical experiments using the transition data calculated by Garcia and Mack [31]. In addition to the QED energies of the fine-structure energy levels, they calculated the wave numbers of the line centers obtained by weighting of the wave numbers of individual fine-structure components by their line strengths. These line centers correspond to those expected to be observed in emission under LTE conditions. Besides the line centers, Garcia and Mack also give the minimum and maximum wave numbers of the fine structure transitions for each $n_1 \to n_2$ complex (n_1 = 1–12, n_2 = 2–20). Our numerical experiments consisted in the selection of four different sets of line centers calculated by Garcia and Mack [31], assigning uncertainties to these wave numbers, obtaining least-squares optimized distinctive energies from these sets of line centers, and comparing the results obtained with different sets. The transition wave numbers calculated by Garcia and Mack [31] were scaled to the currently adopted values of the fundamental constants [6].

The first test set of data included the full scope of line centers calculated by Garcia and Mack [31]. The uncertainties of the line center positions were assumed to be equal to a fraction f of the total transition width (difference between the minimum and maximum wave number of the $n = i \to n = k$ transition complex). The value of f was initially set to one half and then gradually decreased in order to obtain deviations of the precise positions of line centers from the least-squares optimized values ($\Delta\nu_{ik}$) statistically consistent with the assumed uncertainties ($\delta\nu_{ik}$). The criterion of statistical consistency of uncertainties is that the mean square of the relative deviations of transition wave numbers from the optimized values ($\Delta\nu_{ik}/\delta\nu_{ik}$) is equal to one. The resulting value of f was close to 1/8. However, the optimized values of the wave numbers of the Lyman series ($n = 1 \to n = k$, $k = 2, 3, 4, \ldots$) were systematically higher than the calculated transition centers [31] by about 4 values of assigned uncertainties $\delta\nu_{1k}$ for $k \geqslant 4$. This clearly indicates that the Lyman series should not be included in such level optimization procedure.

The second test set was made from the first set by deleting all members of the Lyman series except $n = 1 \to n = 12$. The latter was chosen because it is the highest principal quantum number for which the fine structure is precisely known from experiments (see Sections 4 and 5). Its wave number was set to be equal to the calculated center of gravity of the $n = 12$ complex [31]. Fixing one member of the Lyman series is necessary in order to tie all other energies to the ground level $n = 1$. For this test set, the criterion of statistical consistency was satisfied with a much smaller uncertainty fraction $f = 1/65$ (for the definition of f, see above). For example, the uncertainty of the Balmer-α line was set to 0.007 cm^{-1}. The deviations of the optimized values of transition wave numbers now exhibit a distribution close to normal, the largest relative deviation being for the $\Delta n = 2$ members of the Brackett, Pfund, and higher series ($n = i \to n = i + \Delta n$, $i = 4, 5, \ldots$), $\Delta\nu_{4,6}/\delta\nu_{4,6} = -2.7$, $\Delta\nu_{5,7}/\delta\nu_{5,7} = -2.6$, $\Delta\nu_{6,8}/\delta\nu_{6,8} = -2.3$, and $\Delta\nu_{7,9}/\delta\nu_{7,9} = -2.2$. However, if one looks at the behavior of the deviations of the optimized wave numbers from the precise ones along any particular line series, a clearly systematic behavior is observed, the deviations being negative for the lower members of the series and positive for the higher ones.

We derived the third test data set from the second one by deleting all transitions that were not experimentally observed (except for the $n = 1 \to n = 12$ transition, which is necessary to fix all energies relative to the ground state). This leaves us with the following series members: $n = 2 \to n = k$ ($k = 2 - 20$), $n = 3 \to n = k$ ($k = 3 - 20$), $n = 4 \to n = k$ ($k = 5 - 7$, 9–11), $n = 5 \to n = k$ ($k = 6 - 9$, 9–11), $n = 6 \to n = k$ (k = 7, 8, 10), $n = 7 \to n = k$ (k = 8–10), $n = 8 \to n = k$ (k = 9, 11–15), $n = 9 \to n = k$ (k = 14–16), $n = 11 \to n = 12$, and $n = 12 \to n = 13$. For this test set, the criterion of statistical consistency was satisfied with a somewhat smaller uncertainty fraction $f = 1/78$. The systematic behavior of the deviations of the optimized wave numbers from the precise ones along the line series is qualitatively similar to that of the second data set: $\Delta\nu_{4,6}/\delta\nu_{4,6} = -2.3$, $\Delta\nu_{5,7}/\delta\nu_{5,7} = -1.3$, $\Delta\nu_{6,8}/\delta\nu_{6,8} = -0.7$, and $\Delta\nu_{7,9}/\delta\nu_{7,9} = -0.7$.

The purpose of the fourth data set was to examine the effect of experimental measurement uncertainties on the distinctive energies. To this end, the set of lines was left the same as in the third set, but the uncertainties of the wave numbers were set to the values equal to those that will be used in the level optimization of the experimental measurements (described in Section 10.5). Those experimental uncertainties are generally much greater than the fractions of the fine-structure width used in the sets 1–3. The corresponding values of f (uncertainty relative to the fine-structure width) are in the range 1/30 through 8 (2.2 on average). Since the squares of line uncertainties are used as weights in the level optimization procedure (see the work of Kramida and Nave [139]), changing these uncertainties shifts the resulting optimized energy levels. The questions are: (1) Do the line uncertainties used introduce significant systematic errors in the optimized energy levels? (2) Do these uncertainties result in noticeable systematic errors in the optimized Ritz wave numbers of the series lines? The least-squares level optimization of the fourth test data set provides the following answers to these questions:

(1) The resulting optimized distinctive energies are a good approximation to the centers of gravity of the fine-structure level complexes. The root mean square of the ratios of the differences between the former and the latter to the uncertainties of the optimized level values is equal to 0.15. The maximum difference amounts to about 1/3 of the uncertainty of the optimized level value for $n = 3$, 5, and 20, and 1/5 for $n = 2$ and 18. Thus we can conclude that a similar set of experimental data in the absence of systematic shifts in the measured wave numbers should result in the distinctive energies that are very close (within the level-optimization uncertainties) to the centers of gravity of the fine-structure level complexes.

(2) The Ritz values of the optimized transition wave numbers have uncertainties much smaller than the input values of the observed transition uncertainties. This is, of course, due to the fact that each of the distinctive energy levels is determined by several transitions, which reduces the resulting statistical uncertainty. However, it is also evident that some of these Ritz wave numbers still bear the signature of the same systematic trends that were observed with the test data sets 2 and 3. Most noticeably, the optimized Ritz wave numbers of the lower members of the Balmer series ($n = 4$, 5, and 6) are shifted downwards from the (virtually exact) input values by 0.008 cm^{-1}, 0.005 cm^{-1}, and 0.004 cm^{-1}, while the uncertainties of the corresponding optimized distinctive energy values are equal to 0.011 cm^{-1}. At the same time, the optimized Ritz wave numbers of several members of the Paschen series ($n = 8$ through 16) are shifted upwards from the input values by (0.004–0.007) cm^{-1}, while the uncertainties of the corresponding optimized distinctive energy values are in the range (0.006–0.008) cm^{-1}.

The values of the distinctive energy levels obtained in these numerical experiments are compared in Table L with the theoretical centers of gravity of the fine-structure level complexes from Garcia and Mack [31].

10.5. Level optimization for experimental data on line series

The findings of the previous section provide the basis for the optimization of the actual experimental data for the hydrogen series. In particular, it can be expected that, by optimizing the set of observed transitions, we can verify whether the assumption of Boltzmann population of levels is valid for the hydrogen series lines in the solar atmosphere. If it is valid, then the resulting distinctive energy levels should be close to the centers of gravity of the fine-structure level complexes.

The level optimization was performed in several steps. In the first step, the $n = 12$ energy was fixed at the center-of gravity value calculated by Garcia and Mack [31] (adjusted according to the changed values of the fundamental constants). For the series lines, we used the following sets of measurements:

$n = 2 \rightarrow n = k$ ($k = 4$–16)	Dyson [183];
$n = 2 \rightarrow n = k$ ($k = 3$, 17–37)	Mitchell [185] (corrected for systematic shift, see Section 9.2);
$n = 3 \rightarrow n = 4$	Paschen [175];
$n = 3 \rightarrow n = 5, 6$	Chang and Deming [191];
$n = 3 \rightarrow n = k$ ($k = 7$–10)	Poetker [176];
$n = 3 \rightarrow n = k$ ($k = 11$–40)	Mitchell [185] (corrected for systematic shift, see Section 9.2);
$n = 4 \rightarrow n = k$ ($k = 5$, 7, 10, 11)	Chang and Deming [191];
$n = 4 \rightarrow n = 6$	Geller [194];
$n = 5 \rightarrow n = 6, 9$	Geller [194];
$n = 5 \rightarrow n = 7, 8$	Chang and Deming [191];
$n = 6 \rightarrow n = 7$	Chang and Deming [193] (re-calibrated;

Table L
Distinctive energy levels obtained from numerical experiments with various theoretical transition data sets compared with the theoretical centers of gravity of the level complexes [31].

n	$E_{c.g.}$ [31][a] (cm^{-1})	$E_{opt} - E_{c.g.}$[b] (cm^{-1})			
		Set 1	Set 2	Set 3	Set 4
2	(82 259.110 89)	(0.002)(11)	(−0.000 1)(14)	(0.001 0)(12)	(0.002)(10)
3	(97 492.308 66)	(−0.006)(4)	(−0.003 0)(5)	(−0.002 0)(5)	(0.002)(6)
4	(102 823.900 24)	(−0.005 4)(20)	(−0.002 0)(3)	(−0.000 4)(4)	(0.000)(4)
5	(105 291.658 68)	(−0.004 1)(11)	(−0.001 12)(14)	(0.000 09)(24)	(−0.001)(5)
6	(106 632.167 36)	(−0.003 0)(7)	(−0.000 64)(9)	(0.000 02)(18)	(0.000 3)(21)
7	(107 440.451 04)	(−0.002 4)(4)	(−0.000 38)(6)	(0.000 03)(13)	(0.000 0)(16)
8	(107 965.057 90)	(−0.001 9)(3)	(−0.000 22)(4)	(−0.000 06)(7)	(−0.000 1)(14)
9	(108 324.726 49)	(−0.001 62)(22)	(−0.000 12)(3)	(−0.000 01)(10)	(−0.000 2)(16)
10	(108 581.995 24)	(−0.001 40)(16)	(−0.000 06)(3)	(0.000 06)(19)	(0.000 0)(24)
11	(108 772.344 97)	(−0.001 23)(13)	(−0.000 02)(21)	(0.000 01)(5)	(−0.000 1)(16)
12	(108 917.121 53)	(−0.001 10)(10)	–	–	–
13	(109 029.791 71)	(−0.001 35)(14)	(−0.000 03)(3)	(0.000 01)(4)	(0.000 1)(20)
14	(109 119.192 04)	(−0.001 16)(11)	(−0.000 01)(3)	(0.000 10)(11)	(0.000 1)(22)
15	(109 191.315 66)	(−0.000 98)(10)	(0.000 02)(3)	(0.000 17)(11)	(0.000)(3)
16	(109 250.343 56)	(−0.000 83)(8)	(0.000 04)(3)	(0.000 17)(13)	(0.000)(6)
17	(109 299.264 43)	(−0.000 70)(7)	(0.000 06)(3)	(0.003 4)(18)	(0.00)(3)
18	(109 340.260 60)	(−0.000 59)(6)	(0.000 09)(3)	(0.003 5)(18)	(0.01)(3)
19	(109 374.955 65)	(−0.000 50)(5)	(0.000 11)(3)	(0.003 6)(18)	(0.00)(3)
20	(109 404.577 72)	(−0.000 43)(4)	(0.000 13)(3)	(0.003 7)(18)	(0.01)(3)
f[c]	–	1/8	1/65	1/78	1/30–8

[a] Theoretical center-of-gravity energies of level complexes from Garcia and Mack [31], scaled to the current values of the fundamental constants [6].
[b] The optimized energies E_{opt} were determined using the LOPT code [139] using the four test sets of transitions described in the text. The uncertainties given in parentheses after the values of the energy differences were determined from the elements of the covariance matrix of the level-optimization procedure [139].
[c] The uncertainty fractions, meaning the ratios of uncertainties assumed for the transition wave numbers in the input of the level-optimization procedure to the widths of the corresponding fine structures (see text).

	see Section 9.2);
$n = 6 \rightarrow n = 8, 10$	Geller [194];
$n = 7 \rightarrow n = 8$	Farmer et al. [192];
$n = 7 \rightarrow n = 9, 10$	Chang and Deming [193] (re-calibrated; see Section 9.2);
$n = 8 \rightarrow n = 9$	Farmer et al. [192];
$n = 8 \rightarrow n = k$ ($k = 11$–15)	Chang and Deming [193] (re-calibrated; see Section 9.2);
$n = 9 \rightarrow n = 14, 16$	Chang and Deming [193] (re-calibrated; see Section 9.2);
$n = 9 \rightarrow n = 15$	Brault and Noyes [189];
$n = 11 \rightarrow n = 12$	Clark [198];
$n = 12 \rightarrow n = 13$	Clark [198];
$n = 13 \rightarrow n = 14$	Clark [198];
$n = 14 \rightarrow n = 15$	Boreiko and Clark [196];
$n = 15 \rightarrow n = 16$	Boreiko and Clark [196];
$n = 19 \rightarrow n = 20$	Clark et al. [199];
$n = 21 \rightarrow n = 22$	Clark et al. [199].

The vast majority of the measured lines were taken from solar observations. The only exceptions are a few members of the Paschen series, for which laboratory measurements were used. The uncertainties of those measurements are relatively large, so they had virtually no influence on the optimized energy values.

As a second step, we included the re-calibrated laboratory measurements of Wood [174]. In order to preserve the general character of the data as pertinent to solar observations, Wood's measurements were corrected using the Ritz wave numbers of the Balmer series obtained in the first step of the level optimization. The deviations of Wood's wavelength measurements [174] from the Ritz values are given in Fig. 6.

We approximated these deviations by a linear dependence on wavelength, which we subtracted from Wood's original wavelengths [174]. The uncertainties of the corrected values, as determined by the standard deviation of the data points in Fig. 6 from the linear fit, are ±0.006 Å. Since the Ritz wavelengths used in this correction were based on the solar observations, Wood's corrected measurements are tied to the solar measurements and can be considered on equal footing with them.

In the second step of the level optimization, Dyson's measurements of the Balmer series for $n = 5$–22 were replaced with Wood's corrected values having much smaller uncertainties.

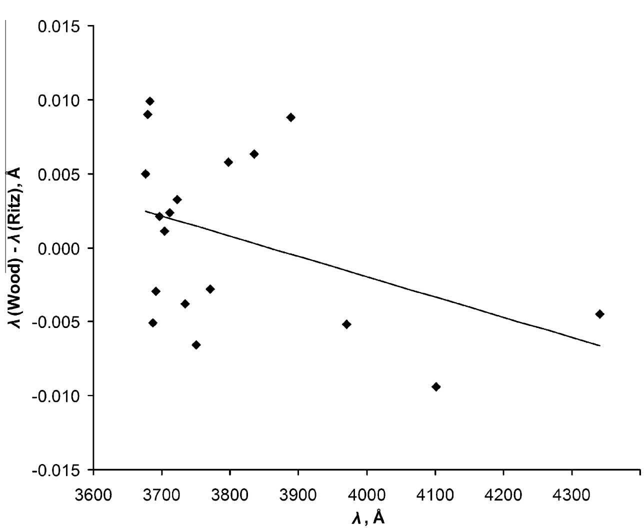

Fig. 6. Deviations of Wood's wavelengths of the Balmer series [174] from the Ritz values obtained in the first step of the level optimization. The solid line is a linear fit.

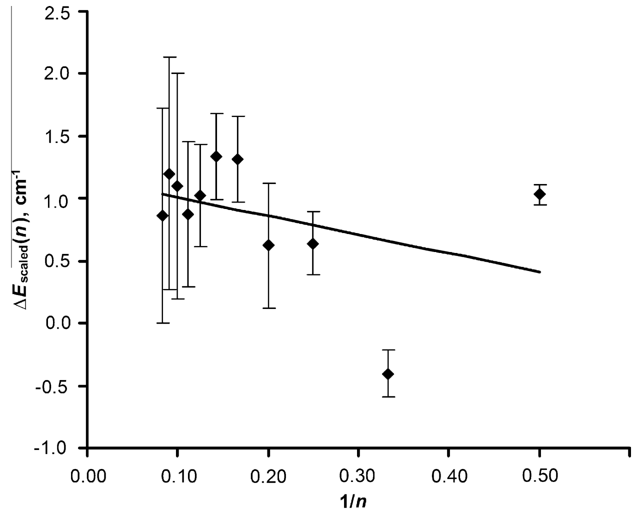

Fig. 7. Scaled energy differences $\Delta E_{scaled}(n) = [E(n) - E(nP_{3/2})] \times n^3$ as a function of $1/n$ for $n = 2$–12. The straight line is a linear fit. The interpolated values for $n = 9$–12 (the four leftmost data points) were used in the third step of the level optimization to fix the positions of excited levels relative to the ground level (see text). The error bars are experimental uncertainties.

In the third step of the level optimization, we investigated the behavior of the energy differences $E(n)$–$E(nP_{3/2})$ along the series. The values $E(nP_{3/2})$ were determined in Section 5 from high-precision experimental data on the hydrogen fine structure. As with the fine-structure intervals presented in Figs. 3–5, it was found that the differences $E(n) - E(nP_{3/2})$ scale nearly as n^3. Thus, the dependence of the scaled quantity $\Delta E_{scaled}(n) = [E(n) - E(nP_{3/2})] \times n^3$ on $1/n$ was approximated with a linear function, and the interpolated values of the energy intervals $E(n) - E(nP_{3/2})$ for $n = 9$ to 12 were inserted into the input data set on equal footing with the experimental transition wave numbers. The level optimization and the described interpolation of $\Delta E_{\rm scaled}(n)$ were iterated until convergence of $\Delta E_{\rm scaled}(n)$ was achieved. This allowed us to decouple the optimized energy levels from the theoretical $n = 12$ value used in the initial steps. The final plot of $\Delta E_{\rm scaled}(n)$ is presented in Fig. 7.

In the fourth step of the level optimization, in order to facilitate a direct comparison with the numerical experiments described in the previous section, we removed all $n = i \rightarrow n = k$ transitions with $i \geqslant 13$ and $k \geqslant 21$.

The levels with $n \geqslant 13$ found in the third step of the level optimization had relatively large uncertainties, ranging from 0.002 cm^{-1} for $n = 14$–0.15 cm^{-1} for $n = 38$–40. To determine them more precisely, we performed the fifth, and final, step of level optimization. We fitted the distinctive energy levels obtained in the third step to the 4-term extended Ritz formula (25) with a fixed value of the ionization energy obtained in Section 5. The parameters of the fitting are given in Table M. In this fitting, the experimental energy level values were weighted by inverse squares of their uncertainties. The standard deviation of experimental level values from the fit was 0.0010 cm^{-1}. In the final optimization step, we used the fitted level values for levels with $n \geqslant 13$.

The final optimized distinctive energy levels are compared in Table 9 with the values obtained in the preliminary steps of the level optimization and in the numerical experiment described in the previous section (based on theoretical data from Garcia and Mack [31]).

Comparison with the level values from the third step of the level optimization procedure is given in order to show that fixing the $n = 13$–40 levels at their interpolated values does not introduce significant shifts in the level values for lower n. These shifts are indeed

Table M
Quantum defect expansion coefficients for the distinctive energy levels of hydrogen series obtained by fitting the 4-term extended Ritz formula (25).

	4-term Ritz formula
E_l (fixed)	109678.771732 cm^{-1}
$c_0 \times 10^6$	6.7(11)
$c_1 \times 10^6$	−100(50)
$c_2 \times 10^6$	1500(500)
$c_3 \times 10^6$	−4500(1500)
Std. dev. of levels	0.0010 cm^{-1}

well below the indicated uncertainties. This fixing dramatically decreases the uncertainties of the high-n levels. The largest deviation between the levels found in the third step of the optimization from the interpolated energies is observed for the $n = 14$ level, where it amounts to one combined uncertainty. This level was based to a large extent on the 13 → 14 transition observed by Clark [198] at 89.395(7) cm^{-1}. Clark [198] noted that his values of the wave numbers of the 11 → 12, 12 → 13, and 13 → 14 transitions were derived by decomposition of the blended line complexes containing components of Mg I. He also noted that the spectra were relatively noisy (see the raw spectra given by Boreiko et al. [197]).

Comparison with the level values from the fourth step of the level optimization procedure shows that adding the observed $n = i \to n = k$ transitions with $i \geqslant 13$ and $k \geqslant 21$ does not introduce any noticeable systematic shifts in the levels with $n \leqslant 20$.

Comparison of the optimized experimental level values with those from the third set of data obtained in the numerical experiment (see the previous section) indicates good agreement (within the combined uncertainties), except for $n = 2$. For $n = 2$, the experimental value is higher than the model one by several standard deviations. This deviation is even more pronounced in comparison of the optimized $n = 2$ level value with the theoretical center of gravity of this level (in the last column of Table 9). This clearly indicates that for the $n = 2$ level the conditions for a Boltzmann population (for the fine-structure components of the level complex) are not fulfilled in the solar corona, while for the higher levels it is a fairly good approximation.

The calculations of Garcia and Mack [31], used in our comparisons have, of course, been superseded by several more recent, far more accurate, QED calculations (e.g., Ref. [10]). However, the differences between the energies calculated by Garcia and Mack [31] and those by Jentschura et al. [10] are smaller than 2 MHz, corresponding to 7×10^{-5} cm^{-1}. These differences are much smaller than the intrinsic uncertainties of the distinctive energy levels and can be neglected.

The observed and Ritz wavelengths and wave numbers of the hydrogen series lines are presented in Table 10. To convert vacuum wave numbers to air wavelengths and vice versa, we used the five-parameter formula for the index of refraction of air from Peck and Reeder [205].

The uncertainties of the Ritz wave numbers given in Table 10 were initially calculated by the LOPT code [139] with the assumption of no systematic errors. However, as shown in the numerical experiment with test data set 4 (see Section 9.4), there are systematic errors caused by the intrinsic inaccuracy of the concept of distinctive energy levels. These errors are approximately of the same size as the random uncertainties obtained in the level optimization procedure. Therefore, we increased the uncertainties of the Ritz wave numbers by a factor of $\sqrt{2}$.

Comparison of the measured wave numbers with the Ritz values reveals three outlying measurements for which the differences $\Delta\sigma_{\mathrm{exp-Ritz}}$ exceed two standard deviations. These are the $n = 4 \to n = 9$, $n = 4 \to n = 12$, and $n = 4 \to n = 16$ transitions at 550.790 cm^{-1}, 6093.34 cm^{-1}, and 6426.545 cm^{-1}, respectively [191]. These deviations are most probably caused by an unidentified blending. We excluded those measurements from the level-optimization procedure.

It should be emphasized that the optimized distinctive energy levels and Ritz wave numbers given in Tables 9 and 10 are based entirely on experimental observations and not on theoretical calculations. They are pertinent to the conditions in the solar corona and should be used to interpret solar spectra.

10.6. Intensities of the series lines

Emission intensities of the series lines depend on many parameters of the emitting plasma, such as particle densities, velocity distributions, and plasma dimensions. In each of the large number of solar observations typically several members of one or two series were observed. Different observations involve greatly different amounts of emitting plasma having different parameters. Since most or all of these parameters are not known, it is difficult to establish a uniform intensity scale for all those observations. To establish such a scale, we assumed that the intensities of the series lines in the solar corona can be approximated by a simple formula

$$I_{\mathrm{obs}}(n) \propto \frac{e^{-E_{\mathrm{up}}/kT}}{\lambda n^3}, \tag{27}$$

where E_{up} is the energy of the upper level and T is the electron temperature. This formula is applicable if the levels are excited only from the ground state by electron collisions.

If this relation is valid, the plot of the quantity $\ln[I_{\mathrm{obs}}(n)\ \lambda n^3]$ versus E_{up} must be a straight line with a negative slope equal to the inverse of the electron temperature. The actual data based on several different observations are presented in Fig. 8.

Intensities of the lowest members of the Lyman and Balmer series, as measured by Feldman et al. [180] and Fontenla [206], respectively, are lowered by self-absorption (Lyman-α is not included in Fig. 8). In some observations of high series lines (e.g., Refs. [196,197]) the plot has a positive slope instead of a negative one. This indicates that in the conditions of those particular observations there was a significant contribution of recombination to the populations of the highly excited levels. For the rest of the observations, we deduced effective temperatures from the slopes of the plots, and then re-scaled the observed intensities to a uniform scale corresponding to an arbitrarily chosen $kT = 2$ eV (the actual values of kT obtained for different observations varied between 0.1 eV and 2.5 eV). The relative line intensities given by Mitchell [185] are visual estimates on the scale from 0 to 200. We derived separate conversion curves for Mitchell's intensities of the Balmer and Paschen series, as these two series were

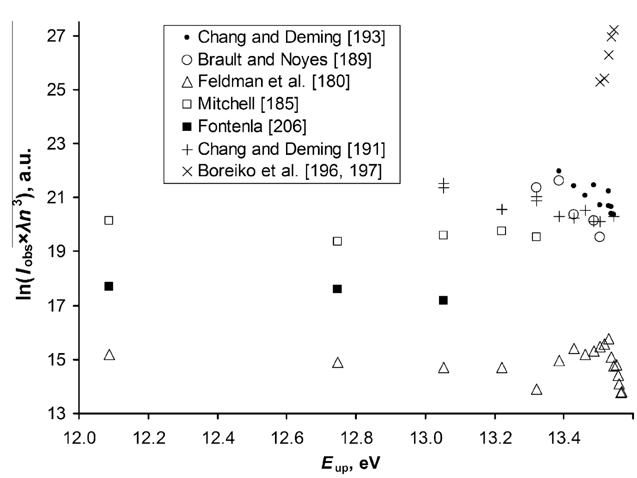

Fig. 8. Relative intensities of the hydrogen series observed by various authors.

measured in different experiments with different instruments. For the Balmer series, Mitchell's intensity scale appears to be logarithmic in the entire intensity range, while for the Paschen series intensities smaller than 20 are linear, and greater than 20 are logarithmic. Thus, for the Paschen series from Mitchell [185] we used two separate scales, one for the weak lines and another for the strong ones. In Fig. 8, only a few members of the Balmer series from Mitchell [185] are shown. If two or more measurements are available for the same line, we rescaled each measured intensity and then took an average. The rescaled observed relative intensities are given in Table 10 along with predicted ones (given in parentheses). They can be used as a qualitative guide to expected relative intensities in emission of the solar corona, provided that possible temperature variations are taken into account.

In the laboratory observations, the intensities of the series lines are very different. In accordance with Bohr's prediction [26], with increasing principal quantum number they decrease much faster than in the Sun. For example, the intensity ratio of the Balmer-α and Balmer-τ ($n = 2 \rightarrow n = 22$) lines in Wood's experiments [174] was 900 000, while in the coronal spectra it is only a few hundred.

10.7. Use of the hydrogen and deuterium series lines for calibration

The hydrogen and deuterium series lines are often used for calibration of spectra emitted by laboratory light sources. With traditional spectroscopic equipment such as grating spectrometers, the fine structure of these lines is often completely or at least partially unresolved. In such blended spectra, the positions of the observed peaks depend on the relative intensities of the individual components of the blends, which in general are determined by detailed balance between various excitation and de-excitation processes. In low-density plasmas such as in the solar corona, the main excitation process is usually electron-impact excitation from the ground state, and line intensities are proportional to its rate (although in the previous section we saw some examples of coronal spectra in which recombination was more important). In dense plasmas, where relative populations of closely separated excited levels approach Boltzmann's distribution, line intensities are approximately proportional to radiative rates multiplied by statistical weights of upper levels. Such dependence is a characteristic of LTE. The conditions in laboratory light sources are usually intermediate between coronal equilibrium and LTE and strongly vary between different types of sources. Therefore, it is impossible to give precise estimates of the peak positions applicable to all experimental arrangements. Even if the plasma conditions correspond to LTE, and the plasma is optically thin, the peak positions depend on the degree of blending, which is determined by interplay between the fine-structure intervals, spectral resolution, and line broadening. Self-absorption, if present, can also shift the observed peaks. For these reasons, the reference wavelengths of unresolved multiplets were given in the previous compilations of hydrogen spectra [207–210] as averages of calculated fine-structure components weighted by calculated line strengths (without any plasma-kinetics modeling). Little guidance was given on possible deviations of such average values from those observed in experiments. The purpose of this section is to provide such guidance.

Reader [210] published a set of reference data for hydrogen and deuterium that includes calculated center-of-gravity wavelengths of several blended lines of the Lyman and Balmer series. Similar but less precise values for the Lyman series were given earlier by Kaufman and Edlén [209]. Because of the good agreement between the present experimental energy levels and high-precision QED calculations, the mean wave numbers calculated using the Ritz values of the fine-structure intervals instead of theoretical ones agree with those given by Reader [210] to within the number of digits given by him. Therefore, Reader's data can be used as a guide in determining the expected positions of the series lines observed at moderate spectral resolution. We here extend his reference data set and analyze its applicability to laboratory measurements by comparing the calculated peak positions with various observations.

Kaufman and Edlén [209] noted that for the Lyman series observed in emission the position of the unresolved doublets could be shifted by self-absorption (if present) toward longer wavelengths by up to one-sixth of the doublet interval. It should be noted that an unfavorable combination of the line broadening with the apparatus resolution can result in greater shifts. An extreme example is provided by Herzberg's measurement of the deuterium Lyman-α line [164]. He observed this line in absorption using atomic deuterium gas cooled by liquid air and a 3 m concave grating spectrograph with a dispersion of 0.50 Å/mm in the fifth order. The resulting line width (full width at half maximum) was approximately equal to the doublet separation. Under such conditions, the $1S_{1/2}$–$2P_{1/2,3/2}$ doublet looked like an asymmetric triangle with a slight hump on its longer-wavelength side, corresponding to the weaker $1S_{1/2}$–$2P_{1/2}$ transition. The maximum of the composite line profile was measured to be at 82 281.644(17) cm^{-1} (we corrected Herzberg's original measurement for the revised ^{198}Hg I standard as described in Section 7). This differs from the calculated center of gravity of the blended line by 0.099 cm^{-1}, which is 27% of the doublet separation.

We calculated the wave numbers corresponding to the centers of gravity of the Lyman series lines in a way similar to Reader [210], except that we used the Ritz wave numbers of the fine structure components instead of theoretical ones, and the line strengths calculated by Jitrik and Bunge [211] instead of those from older calculations used by Reader [210]. Despite slight differences in the line strengths (line strengths from Jitrik and Bunge are considerably more precise) and in the fine structure intervals, the resulting positions of the line centers for $n \leqslant 7$ exactly coincide with Reader's to the number of given decimal figures. Per the discussion above, giving more than three figures after the decimal point is not justified for $n \leqslant 6$. Since the data of Jitrik and Bunge extend to higher principal quantum numbers, in Table 11 we extend Reader's tabulation up to $n = 12$. For $n \geqslant 7$, we give one more decimal place, as the fine structure splittings decrease as n^{-3}.

Using the same approach, we extended Reader's tabulation [210] for the Balmer series to $n = 12$. The width of the fine structure of the Balmer series lines is dominated by the $n = 2$ fine-structure splitting. Therefore, it varies only slightly along the series. As will be shown below, the variation of conditions of excitation and observation in different experiments validates giving the center-of-gravity wave numbers of various groups of fine-structure components to four digits after the decimal point. Our values for $n \leqslant 7$ agree with Reader's [210] to within one in the last digit. The differences in all cases are due to the more precise line strengths of Jitrik and Bunge [211] as compared to those used by Reader [210]. For deuterium, following the recommendations of Jitrik and Bunge [211], we used the hydrogen line strengths. The differences between the theoretical fine-structure intervals used by Reader (following Erickson [37]) and our Ritz values are negligible compared to the precision of the derived values of the centers of gravity of various transition groups.

In Table 12 we give a key to the designations of various fine-structure components and their groups used by different authors for the Balmer-α lines. The right part of this table contains relative intensities of the fine-structure components and their groups according to the different models explained below. The same designations are used for components of the other members of the Balmer series.

The five intensity models listed in Table 12 were used to estimate the variation of the positions of the centers of gravity of Balmer-α component groups under different excitation conditions.

The first intensity model, designated as "Model 1", is purely theoretical and assumes that the relative intensities of the fine-structure components are exactly equal to the ratios of the theoretical line strengths (taken from Ref. [211]). The other models are based on various experiments in which the relative intensities of the three groups of the Balmer-α components designated as "red", "violet1", and "violet2", as well as separations between them, were measured. All models have the following common features: 1) the intensity of the strongest ("red") component was arbitrarily set to 10, and all other intensities are given relative to it; 2) intensity ratios 3b:5 and 2b:4 for components having a common upper level ($3S_{1/2}$ or $3D_{3/2}$, respectively) were assumed to be exactly equal to the ratios of the line strengths calculated by Jitrik and Bunge [211] (same for hydrogen and deuterium); 3) the wave numbers of the fine-structure components were set to be equal to the Ritz values from Tables 5 and 7 for hydrogen and deuterium, respectively; and 4) the position of a center of gravity of any group of fine-structure components was assumed to be a weighted mean of the wave numbers of the components with weights equal to their relative intensities.

In the experiments of Williams [38], Drinkwater et al. [212], Kuhn and Series [213], and Kireyev [168], the intensity ratios violet1:red and violet2:red, as well as wave number separations violet1–red and violet1–violet2, were measured. These four measured quantities together with the relations implied by the models uniquely define all relative intensities of the fine-structure components. We used this direct approach to derive models 2, 3, 4, and 5. Kireyev [168] gave the measured ratios and separations only for tritium. However, the difference between the isotope shifts of the fine-structure components is negligibly small compared to his experimental uncertainties, and there is no reason to expect a significant variation in the relative intensities between deuterium and tritium. Therefore, we applied his intensity measurements to deuterium. Similarly, we applied the deuterium measurements of Kuhn and Series [213] to hydrogen as well. Kopfermann [214] gave only the two separations (violet1–red and violet1–violet2) for hydrogen; he did not measure the relative intensities. Therefore, to build Model 2 for hydrogen from his measurements, we assumed theoretical intensity ratios violet1:red and violet2:red. Drinkwater et al. [212] noted that the large enhancement (by a factor of 2.2 compared to theory) of the violet2:violet1 intensity ratio observed in deuterium by Williams [38] can be partially explained by a possible hydrogen contamination in the deuterium discharge. However, Williams's measured value for this ratio in hydrogen is also enhanced relative to theory (Model 1) by a factor of 1.7. In the experiment of Drinkwater et al. [212], this enhancement factor was 1.3 for both hydrogen and deuterium, while Kireyev [168] observed this enhancement in tritium by a factor of 1.8. For the purpose of the present investigation we assumed that all variations of the relative intensities in the different experiments described above are real, and those variations result in shifts of the observed centers of blended components.

Table 13 presents a comparison of wave numbers of various component groups and intervals between them calculated from the Ritz fine-structure intervals using different models of relative intensities of the fine-structure transitions described above. Besides the measurements quoted above, we also included in this table the separations between the most intense Balmer-α components measured by Williams and Gibbs [215] both in hydrogen and deuterium. Table 13 shows that the center of gravity of the Balmer-α line can be shifted from Model 1 (based on theoretical line ratios) by as much as 0.02 cm^{-1}. The same applies to the position of the "violet" group. The smallest shifts (up to 0.007 cm^{-1}) can be expected for the most intense and compact "red" group.

For the higher members of the Balmer series, there are only a few measurements of relative intensities and separations of the "red" and "violet" groups [40,162]. Therefore, we give only the Model 1 calculations (based on the theoretical line strengths [211]) for $n \geqslant 4$. Although the deuterium measurements of Csillag [162] agree very well with this model, the somewhat more precise violet–red separation measurements of Houston and Hsieh [40] in hydrogen deviate from this model by as much as +0.0062(6) cm^{-1} (186 ± 18 MHz) for $n = 7$. (Houston and Hsieh [40] noted that their method of measurement used for the $n = 4$–7 lines could not be applied to the Balmer-α line. However, they gave the lower and upper bounds for the value of the Balmer-α separation. The value we give in Table 13, 0.3098(20) cm^{-1}, is a straight mean of these bounds, to which we assigned the same uncertainty as they gave for the uncorrected measurement of this line). The deviation for the $n = 6$ "doublet" separation, +0.0026(6) cm^{-1} (78 ± 18 MHz) [40], is smaller than for $n = 7$, but still substantial. If these deviations are real, they probably indicate that the relative intensities of the components within the "red" and "violet" groups strongly deviate from the ratios of the line strengths, which would mean a deviation from Boltzmann populations. In any case, the conclusion is that the observed center-of-gravity wave numbers of the $n \geqslant 4$ Balmer lines can deviate from the theoretical values assuming intensities proportional to the line strengths by as much as 0.006 cm^{-1} (180 MHz), unless the correspondence of the relative intensities to the theoretical line strengths is experimentally verified. For $n = 3$ the deviation can be as large as 0.02 cm^{-1} (600 MHz).

11. Conclusions

In this work, we have critically compiled the published results of experimental measurements of the fine and hyperfine structure of hydrogen, deuterium, and tritium. From these data, we derived reference sets of essentially experimental energy levels and transition frequencies in these spectra. These experimental data provide a convincing confirmation of the validity of the modern QED theory of the hydrogen fine structure. The largest deviations of experiments from the theory are observed for the $n = 3$ levels in hydrogen. The discrepant measurements are those by Zhao et al. [111] calibrated using laser interferometry. The most likely reason for the discrepancy is related to the operating conditions of the reference laser standard. In deuterium, the discrepant measurement is that of Dayhoff et al. [47] using a radio-frequency method with an atomic beam. The probable cause of the discrepancy is insufficient knowledge of the velocity distribution of the atoms in the beam.

The main purely theoretical corrections to the measured fine structure intervals are related to the hyperfine structure. Although they are small compared to the measured quantities, the accuracy with which they can be predicted only marginally exceeds the experimental uncertainties. In hydrogen, the size of the fine-structure shifts due to the off-diagonal magnetic hfs exceeds the uncertainties of the QED calculations of the fine structure of the lower excited levels. In deuterium, the electric quadrupole hfs shifts the lower fine-structure energy levels by amounts far exceeding the uncertainties of the QED calculations of the fine structure. These effects might become measurable as the precision of experiments improves. This indicates a need for further theoretical and experimental work on the hfs.

We have critically compiled experimental data on the hydrogen series lines obtained in laboratory and solar observations. In several data sets, systematic errors were revealed and corrected. From these data, we derived a set of distinctive energy levels describing the series of lines with unresolved fine structure and a set of Ritz wave numbers of the corresponding transitions. These sets can be used as reference data to interpret solar spectra.

Note added in Proof

While this paper was already in print, Kolachevsky et al. [216] published the result of their improved measurement of the 1S hfs interval in hydrogen. Their new experimental value, 177 556 834(7) Hz, is in excellent agreement with the semiempirical value, 177 556 838.2(3) Hz [17]. None of the numerical results given in the present paper are affected by this new measurement.

Acknowledgments

The author is grateful to P.J. Mohr for fruitful discussions on various aspects of the QED theory of fine and hyperfine structure and the determination of the fundamental constants. The correspondence with S.R. Lundeen, D.A. Van Baak, F. Nez, F. Biraben, and J.C. Bergquist regarding the details of their experiments on the fine structure measurements and hyperfine corrections is gratefully acknowledged. The author also thanks T.A. Clark for the help with solar observations of high members of the hydrogen series and for providing his latest data prior to publication. Fruitful discussions with J. Reader and W.C. Martin, as well as generous help provided by C.J. Sansonetti on the interpretation of interferometric measurements, are greatly appreciated. The author expresses his gratitude to one of the referees for providing multiple important corrections and stylistic improvements. This work is supported in part by the Office of Fusion Energy Sciences of the U.S. Department of Energy and by the National Aeronautics and Space Administration.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.adt.2010.05.001.

References

[1] G.W. Series (Ed.), The Spectrum of Atomic Hydrogen – Advances, World Scientific, Singapore, 1988.
[2] H.-J. Beyer, Lamb-shift and fine-structure measurements on one-electron systems, in: P.G. Burke, H. Kleinpoppen (Eds.), Progress in Atomic Spectroscopy, Plenum Press, New York, NY, 1978, pp. 529–606 (Part A, Chapter 12).
[3] F.M. Pipkin, Lamb shift measurements, in: T. Kinoshita (Ed.), Quantum Electrodynamics, World Scientific, Singapore, 1990. p. 696 (Chapter 14).
[4] P.J. Mohr, B.N. Taylor, Rev. Mod. Phys. 72 (2000) 351.
[5] P.J. Mohr, B.N. Taylor, Rev. Mod. Phys. 77 (2005) 1.
[6] P.J. Mohr, B.N. Taylor, D.B. Newell, Rev. Mod. Phys. 80 (2008) 633.
[7] G.K. Woodgate, Elementary Atomic Structure, Clarendon Press, Oxford, 1980.
[8] P.A.M. Dirac, The Principles of Quantum Mechanics, fourth ed., Clarendon Press, Oxford, 1958 (originally published in 1930).
[9] W.E. Lamb Jr., R.C. Retherford, Phys. Rev. 72 (1947) 241.
[10] U.D. Jentschura, S. Kotochigova, E.O. LeBigot, P.J. Mohr, B.N. Taylor, The Energy Levels of Hydrogen and Deuterium (version 2.1). National Institute of Standards and Technology, Gaithersburg, MD, 2005. Available from: <http://physics.nist.gov/HDEL> (accessed 22 January 2008).
[11] J. Vanier, R. Larouche, Metrologia 14 (1978) 31.
[12] P. Petit, M. Desaintfuscien, C. Audoin, Metrologia 16 (1980) 7.
[13] Y.M. Cheng, Y.L. Hua, C.B. Chen, J.H. Gao, W. Shen, IEEE Trans. Instrum. Meas. 29 (1980) 316.
[14] S.G. Karshenboim, Phys. Rep. 422 (1-2) (2005) 1.
[15] D.J. Wineland, N.F. Ramsey, Phys. Rev. A 5 (1972) 821.
[16] M.S. Mathur, S.B. Crampton, D. Kleppner, N.F. Ramsey, Phys. Rev. 158 (1967) 14.
[17] U.D. Jentschura, V.A. Yerokhin, Phys. Rev. A 73 (2006) 062503.
[18] C.E. Carlson, V. Nazaryan, K. Griffioen, Phys. Rev. A 78 (2008) 022517.
[19] H.A. Bethe, E.E. Salpeter, Quantum Mechanics of One- and Two-Electron Atoms, Springer-Verlag, Berlin, 1957 (reprinted by Plenum Publishing Corporation, NY, 1977).
[20] S. Brodsky, R.G. Parsons, Phys. Rev. 163 (1967) 134;
Erratum: Phys. Rev. 176 (1968) 423.
[21] I.I. Sobel'man, Introduction to the Theory of Atomic Spectra, Pergamon Press, NY, 1972.
[22] G.W.F. Drake, P.S. Farago, A. van Wijngaarden, Phys. Rev. A 11 (1975) 1621.
[23] A. van Wijngaarden, G.W.F. Drake, Phys. Rev. A 17 (1978) 1366.
[24] P.J. Mohr, B.N. Taylor, D.B. Newell, The 2006 CODATA Recommended Values of the Fundamental Physical Constants, web version 5.0. National Institute of Standards and Technology, Gaithersburg, MD 20899, USA, 2007. This database was developed by J. Baker, M. Douma, S. Kotochigova. Available from: <http://physics.nist.gov/constants> (accessed January 18 2008).
[25] W.V. Houston, Phys. Rev. 51 (1937) 446.
[26] N. Bohr, Philos. Mag. 26 (1913) 1.
[27] T. Lyman, Phys. Rev. 33 (1914) 504.
[28] F.S. Brackett, Astrophys. J. 56 (1922) 154.
[29] A.H. Pfund, J. Opt. Soc. Am. 9 (1924) 193.
[30] C.J. Humphreys, J. Res. Natl. Bur. Stand. (US) 50 (1) (1953) 1.
[31] J.D. Garcia, J.E. Mack, J. Opt. Soc. Am. 55 (1965) 654.
[32] H. Grotch, D.R. Yennie, Z. Phys. A 202 (1967) 425.
[33] H. Grotch, D.R. Yennie, Rev. Mod. Phys. 41 (1969) 350.
[34] J. Sapirstein, D.R. Yennie, Theory of hydrogenic bound states, in: T. Kinoshita (Ed.), Quantum Electrodynamics, World Scientific, Singapore, 1990. p. 560 (Chapter 12).
[35] M.I. Eides, H. Grotch, V.A. Shelyuto, Phys. Rep. 342 (2001) 63.
[36] W.A. Barker, F.N. Glover, Phys. Rev. 99 (1955) 317.
[37] G.W. Erickson, J. Phys. Chem. Ref. Data 6 (1977) 831.
[38] R.C. Williams, Phys. Rev. 54 (1938) 558.
[39] B.P. Kibble, W.R.C. Rowley, R.E. Shawyer, G.W. Series, J. Phys. B 6 (1973) 1079.
[40] W.V. Houston, Y.M. Hsieh, Phys. Rev. 45 (1934) 263.
[41] S. Pasternack, Phys. Rev. 54 (1938) 1113.
[42] W.E. Lamb Jr., R.C. Retherford, Phys. Rev. 79 (1950) 549.
[43] W.E. Lamb Jr., R.C. Retherford, Phys. Rev. 81 (1951) 222.
[44] W.E. Lamb Jr., Phys. Rev. 85 (1952) 259.
[45] W.E. Lamb Jr., R.C. Retherford, Phys. Rev. 86 (1952) 1014.
[46] S. Triebwasser, E.S. Dayhoff, W.E. Lamb Jr., Phys. Rev. 89 (1953) 98.
[47] E.S. Dayhoff, S. Triebwasser, W.E. Lamb Jr., Phys. Rev. 89 (1953) 106.
[48] H.A. Bethe, Phys. Rev. 72 (1947) 339.
[49] H.A. Bethe, L.M. Brown, J.R. Stehn, Phys. Rev. 77 (1950) 370.
[50] W.R. Johnson, G. Soff, At. Data Nucl. Data Tables 33 (1985) 405.
[51] G.W. Erickson, Phys. Rev. Lett. 27 (1971) 780.
[52] P.J. Mohr, Phys. Rev. Lett. 34 (1975) 1050;
P.J. Mohr, At. Data Nucl. Data Tables 29 (1983) 453.
[53] J. Sapirstein, Phys. Rev. Lett. 47 (1981) 1723.
[54] S. Kotochigova, P.J. Mohr, B.N. Taylor, Can. J. Phys. 80 (2002) 1373.
[55] U.D. Jentschura, S. Kotochigova, E.-O. Le Bigot, P.J. Mohr, B.N. Taylor, Phys. Rev. Lett. 95 (2005) 163003.
[56] H.E. White, A.Y. Eliason, Phys. Rev. 44 (1933) 753.
[57] B.W. Petley, K. Morris, R.E. Shawyer, J. Phys. B 13 (1980) 3099.
[58] S.R. Lundeen, P.E. Jessop, F.M. Pipkin, Phys. Rev. Lett. 34 (1975) 377.
[59] C.W. Fabjan, F.M. Pipkin, Phys. Rev. A 6 (1972) 556.
[60] V.W. Hughes, in: C.H. Townes (Ed.), Quantum Electronics, Columbia University Press, New York, 1960. p. 582.
[61] J.E. Nafe, E.B. Nelson, I.I. Rabi, Phys. Rev. 71 (1947) 914.
[62] J.E. Nafe, E.B. Nelson, Phys. Rev. 73 (1948) 718.
[63] E.B. Nelson, J.E. Nafe, Phys. Rev. 75 (1949) 1194.
[64] A.G. Prodell, P. Kusch, Phys. Rev. 79 (1950) 1009.
[65] A.G. Prodell, P. Kusch, Phys. Rev. 88 (1952) 184.
[66] A.G. Prodell, P. Kusch, Phys. Rev. 106 (1957) 87.
[67] P. Kusch, Phys. Rev. 100 (1955) 1188.
[68] J.P. Wittke, R.H. Dicke, Phys. Rev. 103 (1956) 620.
[69] L.W. Anderson, F.M. Pipkin, J.C. Baird Jr., Phys. Rev. Lett. 4 (1960) 69.
[70] F.M. Pipkin, R.H. Lambert, Phys. Rev. 127 (1962) 787.
[71] D. Kleppner, H.M. Goldenberg, N.F. Ramsey, Appl. Opt. 1 (1962) 55.
[72] S.B. Crampton, D. Kleppner, N.F. Ramsey, Phys. Rev. Lett. 11 (1963) 338.
[73] S.B. Crampton, H.G. Robinson, D. Kleppner, N.F. Ramsey, Phys. Rev. 141 (1966) 55.
[74] R. Vessot, H. Peters, J. Vanier, R. Beehler, D. Halford, R. Harrach, D. Allan, D. Glaze, C. Snider, J. Barnes, L. Cutler, L. Bodily, IEEE Trans. Instrum. Meas. 15 (1966) 165.
[75] H. Hellwig, R.F.C. Vessot, M.W. Levine, P.W. Zitzewitz, D.W. Allan, D.J. Glaze, IEEE Trans. Instrum. Meas. 19 (1970) 200.
[76] D. Kleppner, The proton moment in Bohr magnetons – and all that, in: D.N. Langenberg, B.N. Taylor (Eds.), Natl. Bur. Stand. (US), Spec. Publ. No. 343, U.S. Government Printing Office, Washington, DC, 1971, p. 411.
[77] L. Essen, R.W. Donaldson, M.J. Bangham, E.G. Hope, Nature (London) 229 (1971) 110.
[78] P.F. Winkler, D. Kleppner, T. Myint, F.G. Walther, Phys. Rev. A 5 (1972) 83.
[79] D.J. Larson, P.A. Valberg, N.F. Ramsey, Phys. Rev. Lett. 23 (1969) 1369.
[80] J.W. Heberle, H.A. Reich, P. Kusch, Phys. Rev. 101 (1956) 612.
[81] H.A. Reich, J.W. Heberle, P. Kusch, Phys. Rev. 104 (1956) 1585.
[82] N.E. Rothery, E.A. Hessels, Phys. Rev. A 61 (2000) 044501.
[83] N. Kolachevsky, M. Fischer, S.G. Karshenboim, T.W. Hänsch, Phys. Rev. Lett. 92 (2004) 033003;
N. Kolachevsky, P. Fendel, S.G. Karshenboim, T.W. Hänsch, Phys. Rev. A 70 (2004) 062503.
[84] H. Kopfermann, Nuclear Moments, Academic Press, NY, 1958.
[85] G. Breit, Phys. Rev. 35 (1930) 1447.
[86] V.M. Shabaev, J. Phys. B 24 (1991) 4479.
[87] A.V. Nefiodov, G. Plunien, G. Soff, Phys. Lett. B 552 (2003) 35.
[88] S.R. Lundeen, F.M. Pipkin, Metrologia 22 (1986) 9.
[89] E. Latvamaa, L. Kurittu, P. Pyykkö, L. Tataru, J. Phys. B 6 (1973) 591.
[90] S.R. Lundeen, Separated Oscillatory Field Measurement of the Lambshift in H, $n = 2$, Ph.D. Thesis, Harvard University, 1975.
[91] J.P. Santos, F. Parente, P. Indelicato, Eur. Phys. J. D 3 (1998) 43.

[92] S.R. Lundeen, F.M. Pipkin, Phys. Rev. Lett. 34 (1975) 1368.
[93] G. Newton, P.J. Unsworth, D.A. Andrews, J. Phys. B 8 (1975) 2928.
[94] D.A. Andrews, G. Newton, Phys. Rev. Lett. 37 (1976) 1254.
[95] G. Newton, D.A. Andrews, P.J. Unsworth, Philos. Trans. R. Soc. London Ser. A 290 (1979) 373.
[96] V.G. Pal'chikov, Yu.L. Sokolov, V.P. Yakovlev, Metrologia 21 (1985) 99.
[97] K.A. Safinya, K.K. Chan, S.R. Lundeen, F.M. Pipkin, Phys. Rev. Lett. 45 (1980) 1934.
[98] E.W. Hagley, F.M. Pipkin, Phys. Rev. Lett. 72 (1994) 1172.
[99] C.W. Fabjan, F.M. Pipkin, Phys. Rev. Lett. 25 (1970) 421.
[100] D.A. Van Baak, B.O. Clark, F.M. Pipkin, Phys. Rev. A 19 (1979) 787.
[101] M. Weitz, A. Huber, F. Schmidt-Kaler, D. Leibfried, W. Vassen, C. Zimmermann, K. Pachucki, T.W. Hänsch, L. Julien, F. Biraben, Phys. Rev. A 52 (1995) 2664.
[102] B. de Beauvoir, F. Nez, L. Julien, B. Cagnac, F. Biraben, D. Touahri, L. Hilico, O. Acef, A. Clairon, J.J. Zondy, Phys. Rev. Lett. 78 (1997) 440.
[103] B. de Beauvoir, C. Schwob, O. Acef, J.-J. Zondy, L. Jozefowski, L. Hilico, F. Nez, L. Julien, A. Clairon, F. Biraben, Eur. Phys. J. D 12 (2000) 61;
Erratum: Eur. Phys. J. D 14 (2001) 398.
[104] F. Nez, private communication, 2008.
[105] R.V. Reid Jr., M.L. Vaida, Phys. Rev. Lett. 29 (1972) 494;
Erratum: Phys. Rev. Lett. 34 (1975) 1064.
[106] C.W. Fabjan, F.M. Pipkin, M. Silverman, Phys. Rev. Lett. 26 (1971) 347.
[107] R.A. Brown, F.M. Pipkin, Ann. Phys. (NY) 80 (1973) 479.
[108] J.C. Baird, J. Brandenberger, K.I. Gondaira, H. Metcalf, Phys. Rev. A 5 (1972) 564.
[109] M. Glass-Maujean, Opt. Commun. 8 (1973) 260.
[110] M. Glass-Maujean, L. Julien, T. Dohnalik, J. Phys. B 11 (1978) 421.
[111] P. Zhao, W. Lichten, H. Layer, J.C. Bergquist, Phys. Rev. A 34 (1986) 5138.
[112] B. Cagnac, G. Grynberg, F. Biraben, J. Phys. (Paris) 34 (1973) 845.
[113] G. Grynberg, Spectroscopie d'absorption à deux photons sans élargissement Doppler. Application à l'étude du sodium et du néon, Thèse de doctorat d'état, Laboratoire de Physique de l'École Normale Supérieure, Université de Paris VI, Paris, France, Centre National de la Recherche Scientifique Report No. CNRS AO 12497, 1976, 256 pp.
[114] G. Grynberg, B. Cagnac, Rep. Prog. Phys. 40 (1977) 791.
[115] T.W. Hänsch, J. Alnis, P. Fendel, M. Fischer, C. Gohle, M. Herrmann, R. Holzwarth, N. Kolachevsky, Th. Udem, M. Zimmermann, Philos. Trans. R. Soc. London Ser. A 363 (2005) 2155.
[116] T.W. Hänsch, M.H. Nayfeh, S.A. Lee, S.M. Curry, I.S. Shahin, Phys. Rev. Lett. 32 (1974) 1336.
[117] D.J. Berkeland, E.A. Hinds, M.G. Boshier, Phys. Rev. Lett. 75 (1995) 2470.
[118] J.C. Garreau, M. Allegrini, L. Julien, F. Biraben, J. Phys. (Paris) 51 (1990) 2293.
[119] D.A. Van Baak, B.O. Clark, S.R. Lundeen, F.M. Pipkin, Phys. Rev. A 22 (1980) 591;
Erratum: Phys. Rev. A 24 (1981) 1162.
[120] E.W. Weber, J.E.M. Goldsmith, Phys. Rev. Lett. 41 (1978) 940.
[121] P. Zhao, W. Lichten, Z.-X. Zhou, H.P. Layer, J.C. Bergquist, Phys. Rev. A 39 (1989) 2888.
[122] M. Fischer, N. Kolachevsky, M. Zimmermann, R. Holzwarth, Th. Udem, T.W. Hänsch, M. Abgrall, J. Grünert, I. Maksimovic, S. Bize, H. Marion, F. Pereira Dos Santos, P. Lemonde, G. Santarelli, P. Laurent, A. Clairon, C. Salomon, M. Haas, U.D. Jentschura, C.H. Keitel, Phys. Rev. Lett. 92 (2004) 230802.
[123] M. Niering, R. Holzwarth, J. Reichert, P. Pokasov, Th. Udem, M. Weitz, T.W. Hänsch, P. Lemonde, G. Santarelli, M. Abgrall, P. Laurent, C. Salomon, A. Clairon, Phys. Rev. Lett. 84 (2000) 5496.
[124] C. Schwob, L. Jozefowski, B. de Beauvoir, L. Hilico, F. Nez, L. Julien, F. Biraben, O. Acef, J.-J. Zondy, A. Clairon, Phys. Rev. Lett. 82 (1999) 4960;
Erratum: Phys. Rev. Lett. 86 (2001) 4193.
[125] S. Bourzeix, B. de Beauvoir, F. Nez, M.D. Plimmer, F. de Tomasi, L. Julien, F. Biraben, D.N. Stacey, Phys. Rev. Lett. 76 (1996) 384.
[126] F. Nez, M.D. Plimmer, S. Bourzeix, L. Julien, F. Biraben, R. Felder, O. Acef, J.J. Zondy, P. Laurent, A. Clairon, M. Abed, Y. Millerioux, P. Juncar, Phys. Rev. Lett. 69 (1992) 2326.
[127] F. Biraben, J.C. Garreau, L. Julien, M. Allegrini, Phys. Rev. Lett. 62 (1989) 621.
[128] G. Hagel, C. Schwob, L. Jozefowski, B. de Beauvoir, L. Hilico, F. Nez, L. Julien, F. Biraben, O. Acef, A. Clairon, Laser Phys. 11 (10) (2001) 1076.
[129] O. Arnoult, F. Nez, C. Schwob, L. Julien, F. Biraben, Can. J. Phys. 83 (2005) 273.
[130] E. Peters, S. Reinhardt, S. Diddams, T. Udem, T.W. Hänsch, in: Book of Abstracts of the International Conference on Precision Physics of Simple Atomic Systems, Windsor, ON, Canada, July 21–26, 2008, p. 54.
[131] S.L. Kaufman, W.E. Lamb Jr., K.R. Lea, M. Leventhal, Phys. Rev. A 4 (1971) 2128.
[132] B.L. Cosens, T.V. Vorburger, Phys. Rev. A 2 (1970) 16.
[133] T.W. Shyn, T. Rebane, R.T. Robiscoe, W.L. Williams, Phys. Rev. A 3 (1971) 116.
[134] F. Biraben, private communication, 2008.
[135] D.A. Jennings, C.R. Pollock, F.R. Petersen, R.E. Drullinger, K.M. Evenson, J.S. Wells, J.L. Hall, H.P. Layer, Opt. Lett. 8 (1983) 136.
[136] T.J. Quinn, Metrologia 40 (2003) 103.
[137] S.N. Lea, W.R.C. Rowley, H.S. Margolis, G.P. Barwood, G. Huang, P. Gill, J.-M. Chartier, R.S. Windeler, Metrologia 40 (2003) 84.
[138] A.A. Madej, J.E. Bernard, L. Robertsson, L.-S. Ma, M. Zucco, R.S. Windeler, Metrologia 41 (2004) 152.
[139] A.E. Kramida, G. Nave, Eur. Phys. J. D 37 (2006) 1.
[140] W. Ritz, Phys. Z. 9 (16) (1908) 521.
[141] A. Fowler, Mon. Not. Roy. Astron. Soc. 73 (1912) 62.
[142] A. Fowler, Report on Series in Line Spectra, Fleetway Press, London, 1922, 184 pp.
[143] W.E. Curtis, Proc. R. Soc. London Ser. A 90 (1914) 605.
[144] W.E. Curtis, Proc. R. Soc. London Ser. A 96 (1919) 147.
[145] D. Hartree, Proc. Cambridge Philos. Soc. 24 (1927) 426.
[146] G.W.F. Drake, Quantum defect theory and analysis of high-precision helium term energies, in: B. Bederson, A. Dalgarno (Eds.), Advances in Atomic Molecular and Optical Physics, vol. 32, Academic Press, New York, 1994, pp. 93–116.
[147] C. Sansonetti, Fortran Computer Code RITZPL, private communication, 2005.
[148] K.-H. Weber, C.J. Sansonetti, Phys. Rev. A 35 (1987) 4650.
[149] C.J. Sansonetti, F. Kerber, J. Reader, M.R. Rosa, Astrophys. J. Suppl. Ser. 153 (2004) 555.
[150] P.G. Wilkinson, J. Opt. Soc. Am. 45 (1955) 862.
[151] W. Curdt, P. Brekke, U. Feldman, K. Wilhelm, B.N. Dwivedi, U. Schühle, P. Lemaire, Astron. Astrophys. 375 (2001) 591.
[152] A. Huber, Th. Udem, B. Gross, J. Reichert, M. Kourogi, K. Pachucki, M. Weitz, T.W. Hänsch, Phys. Rev. Lett. 80 (1998) 468.
[153] B.L. Cosens, Phys. Rev. 173 (1968) 49.
[154] R.T. Robiscoe, T.W. Shyn, Phys. Rev. Lett. 24 (1970) 559.
[155] T.V. Vorburger, B.L. Cosens, in: D.N. Langenberg, B.N. Taylor (Eds.), Natl. Bur. Stand. (US) Spec. Publ. No. 343, U.S. Government Printing Office, Washington, DC, 1971, p. 361.
[156] B.N. Taylor, W.H. Parker, D.N. Langenberg, Rev. Mod. Phys. 41 (1969) 375;
Erratum: Rev. Mod. Phys. 45 (1973) 109.
[157] D.A. Tate, P.E.G. Baird, M.G. Boshier, E.A. Hinds, D.N. Stacey, G.K. Woodgate, J. Phys. B 21 (1988) 421.
[158] G.W. Series, Proc. R. Soc. London Ser. A 208 (1951) 277.
[159] L.R. Wilcox, W.E. Lamb Jr., Phys. Rev. 119 (1960) 1915.
[160] W.E. Lamb Jr., T.M. Sanders Jr., Phys. Rev. 119 (1960) 1901.
[161] P. Zhao, W. Lichten, H. Layer, J.C. Bergquist, Phys. Rev. Lett. 58 (1987) 1293.
[162] L. Csillag, Acta Phys. Acad. Sci. Hung. 24 (1968) 1;
L. Csillag, Results and possibilities in the determination of the Rydberg constant, in: J.H. Sanders, A.H. Wapstra (Eds.), Proceedings of the Fourth International Conference on Atomic Masses and Fundamental Constants, Teddlington, England, 1971, Plenum Press, New York, NY, 1972, p. 411.
[163] P.E. Ciddor, Appl. Opt. 35 (1996) 1566.
[164] G. Herzberg, Proc. R. Soc. London Ser. A 234 (1956) 516.
[165] E.B. Saloman, J. Phys. Chem. Ref. Data 35 (2006) 1519.
[166] C.H. Skinner, H. Adler, R.V. Budny, J.H. Kamperschroer, L.C. Johnson, A.T. Ramsey, D.P. Stotler, Nucl. Fusion 35 (1995) 143.
[167] H. Pomerance, D. Terranova, Am. J. Phys. 18 (1950) 466.
[168] P.S. Kireyev, Sov. Phys. Dokl. (Engl. Transl.) 112 (1957) 8.
[169] G.W. Erickson, private communication, 2008. Although the original data files used in Erickson's calculations [37] were lost, it was possible to guess the correct value of the triton mass based on the values of the tritium atomic mass available at that time.
[170] U.D. Jentschura, A. Czarnecki, K. Pachucki, Phys. Rev. A 72 (2005) 062102.
[171] P. Mohr, private communication, 2008.
[172] I. Sick, Phys. Lett. B 576 (2003) 62.
[173] I. Sick, Can. J. Phys. 85 (2007) 409.
[174] R.W. Wood, Proc. R. Soc. London Ser. A 97 (1920) 455;
R.W. Wood, Philos. Mag. 42 (1921) 729;
R.W. Wood, Philos. Mag. 44 (1922) 538.
[175] F. Paschen, Ann. Phys. (Leipzig) 332 (13) (1908) 537.
[176] A.H. Poetker, Phys. Rev. 30 (1927) 418.
[177] P. Hansen, J. Strong, Appl. Opt. 12 (1973) 429.
[178] W.H. Wollaston, Philos. Trans. R. Soc. London Ser. A 92 (1802) 365.
[179] J. von Fraunhofer, Denkschriften der Königlichen Akademie der Wissenschaften zu München 5 (1814–1815) 193.;
J. Fraunhofer, Ann. Phys. (Leipzig) 74 (1823) 337.
[180] U. Feldman, W.E. Behring, W. Curdt, U. Schühle, K. Wilhelm, P. Lemaire, T.M. Moran, Astrophys. J. Suppl. Ser. 113 (1997) 195.
[181] S. Parenti, J.-C. Vial, P. Lemaire, Astron. Astrophys. 443 (2005) 679.
[182] J. Evershed, Philos. Trans. R. Soc. London Ser. A 201 (1903) 457.
[183] F.W. Dyson, Philos. Trans. R. Soc. London Ser. A 206 (1906) 403.
[184] S.A. Mitchell, Astrophys. J. 38 (1913) 407.
[185] S.A. Mitchell, Astrophys. J. 105 (1947) 1.
[186] K. Burns, W.F. Meggers, Standard Solar Wave Lengths (4073–4754 Å), Publ. Allegheny Obs. Univ. Pittsburgh 6 (7) (1929) 105;
K. Burns, Standard Solar Wave Lengths (3592–4107 Å and 4761–5892 Å), Publ. Allegheny Obs. Univ. Pittsburgh 6 (9) (1929) 141.
[187] Yu. Ralchenko, A.E. Kramida, J. Reader, NIST ASD Team, NIST Atomic Spectra Database, version 3.1.5, National Institute of Standards and Technology, Gaithersburg, MD, 2008. Available from: <http://physics.nist.gov/asd3> (accessed April 28 2008).
[188] J.E. Sansonetti, W.C. Martin, J. Phys. Chem. Ref. Data 34 (2005) 1559. Available from: <http://physics.nist.gov/PhysRefData/Handbook/index.html>.
[189] J. Brault, R. Noyes, Astrophys. J. 269 (1983) L61.
[190] O.C. Mohler, A Table of Solar Spectrum Wave Lengths, 11,984 Å to 25,578 Å, Univ. Michigan, Ann Arbor, MI, 1955, 83 pp.
[191] E.S. Chang, D. Deming, Sol. Phys. 165 (1996) 257.
[192] C.B. Farmer, L. Delbouille, G. Roland, C. Servais, The Solar Spectrum between 16 and 40 Microns, San Juan Capistrano Research Institute Technical Report 94-2, San Juan Capistrano, CA 92675, USA, 1994, p. 1.
[193] E.S. Chang, D. Deming, Sol. Phys. 179 (1998) 89.

[194] M. Geller, A High-Resolution Atlas of the Infrared Spectrum of the Sun and the Earth Atmosphere from Space, vol. III, NASA Reference Publication 1224, 1992, 456 pp.
[195] C.B. Farmer, R.H. Norton, A High-Resolution Atlas of the Infrared Spectrum of the Sun and the Earth Atmosphere from Space, Vols. I and II, NASA Reference Publication 1224, 1989.
[196] R.T. Boreiko, T.A. Clark, Astron. Astrophys. 157 (1986) 353.
[197] R.T. Boreiko, T.A. Clark, D.A. Naylor, J.R. Busler, High-*n* hydrogen lines in solar infrared spectra from balloon-borne, Mauna Kea, and ATMOS observations, in D.M. Rabin, J.T. Jefferies, C. Lindsey (Eds.). Infrared Solar Physics, Proceedings of the 154th Symposium of the IAU, Tucson, AZ, March 2–6, 1992, Kluwer Academic Publishers, Dordrecht, Holland, 1994, p. 365.
[198] T.A. Clark, private communication, 2009.
[199] T.A. Clark, D.A. Naylor, G.R. Davis, Astron. Astrophys. 357 (2000) 757.
[200] R.L. Sorochenko, E.V. Borodzich, Dokl. Akad. Nauk SSSR 163 (1965) 603 (in Russian);
R.L. Sorochenko, E.V. Borodzich, Sov. Phys. Dokl. (Engl. Transl.) 10 (1966) 588.
[201] A.F. Dravskikh, Z.V. Dravskikh, V.A. Kolbasov, G.S. Misezhnikov, D.E. Nikulin, V.B. Shteinshleiger, Dokl. Akad. Nauk. SSSR 163 (1965) 332 (in Russian);
A.F. Dravskikh, Z.V. Dravskikh, V.A. Kolbasov, G.S. Misezhnikov, D.E. Nikulin, V.B. Shteinshleiger, Sov. Phys. Dokl. (Engl. Trans.) 10 (1966) 672.
[202] M.A. Gordon, R.L. Sorochenko, Radio recombination lines: their physics and astronomical applications, Astrophysics and Space Science Library, vol. 282, Kluwer Academic Publishers, Dordrecht, 2002. 380 pp.
[203] J.R. Rydberg, Recherches sur la Constitution des Spectres d'Emission des Elements Chimique, Kung. Sven. Vetenskapsakad. Handl. 23 (11) (1890) 1 (in French);
Shortened English translation J.R. Rydberg, Philos. Mag. Ser. 5 (29) (1890) 331.
[204] J.P. Towle, P.A. Feldman, J.K.G. Watson, Astrophys. J. Suppl. Ser. 107 (1996) 747.
[205] E.R. Peck, K. Reeder, J. Opt. Soc. Am. 62 (1972) 958.
[206] J.M. Fontenla, Sol. Phys. 64 (1979) 177.
[207] C.E. Moore, A multiplet table of astrophysical interest. Revised Edition. Part I – Table of multiplets, Contrib. Princeton Univ. Obs. 20 (Part I) (1945) 110pp.
[208] C.E. Moore, Tables of spectra of hydrogen, carbon, nitrogen, and oxygen atoms and ions, in: J.W. Gallagher (Ed.), CRC Series in Evaluated Data in Atomic Physics, CRC Press, Boca Raton, FL, 1993. 339pp.
[209] V. Kaufman, B. Edlén, J. Phys. Chem. Ref. Data 3 (1974) 825.
[210] J. Reader, Appl. Spectrosc. 58 (2004) 1469.
[211] O. Jitrik, C.F. Bunge, J. Phys. Chem. Ref. Data 33 (2004) 1059. Numerical results available online at: <http://www.fisica.unam.mx/research/tables/spectra/1el>.
[212] J.W. Drinkwater, O. Richardson, W.E. Williams, Proc. R. Soc. London Ser. A 174 (1940) 164.
[213] H. Kuhn, G.W. Series, Proc. R. Soc. London Ser. A 202 (1950) 127.
[214] H. Kopfermann, Naturwissenschaften 22 (1934) 218.
[215] R.C. Williams, R.C. Gibbs, Phys. Rev. 45 (1934) 475.
[216] N. Kolachevsky, A. Matveev, J. Alnis, C.G. Parthey, S.G. Karshenboim, T.W. Hänsch, Phys. Rev. Lett. 102 (2009) 213002.

Explanation of Tables

In all Tables, purely theoretical values are given in parentheses. Values obtained using semi-empirical scaling methods are given in square brackets. Uncertainty in the units of the last decimal place is given in parentheses next to the value.

Table 1. Hfs splittings and off-diagonal shifts in hydrogen (Hz)

n	Principal quantum number
$\Delta E_{c.g.}$	Shifts of centers of gravity of the hyperfine doublets caused by off-diagonal magnetic hfs
Ref.	Source reference. Values without a reference are calculated in this work (see Sections 3.1.1 and 3.1.2)

Table 2. Hfs splittings and shifts in deuterium (Hz)

n	Principal quantum number
$\Delta E_{c.g.}$	Shifts of centers of gravity of the hyperfine multiplets caused by off-diagonal magnetic hfs and electric quadrupole hfs

Table 3. Hfs splittings and off-diagonal shifts in tritium (Hz)

n	Principal quantum number
$\Delta E_{c.g.}$	Shifts of centers of gravity of the hyperfine doublets caused by off-diagonal magnetic hfs
Ref.	Source reference. Values without a reference are calculated in this work (see Section 3.1.4)

Table 4. Optimized energy levels of hydrogen derived from available experimental transition frequencies compared with QED calculations of Jentschura et al. [10]

LS term	*LS* term designation (principal quantum number followed by a letter symbol for the total orbital momentum L)
J	Total angular momentum ($\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{S}$).
E_{exp}	Energy levels (MHz) derived from measured transition frequencies using the LOPT program [139]. The values enclosed in square brackets were determined from a single interpolated or extrapolated transition frequency (see Section 5). The additive quantity x for the n = 3 and 6 levels is defined in Table C
Unc.	Uncertainty of the excitation energy from the ground level (MHz)
Num. trans.	Number of transitions connecting the level with other levels contributing to the level-optimization procedure
E_{th}	Sum of the values from Jentschura et al. [10] (calculated in an approximation that does not describe hfs interactions) and the hyperfine shifts caused by the off-diagonal magnetic hfs (MHz). The uncertainties include contributions from off-diagonal hfs shifts and the fundamental constants, as well as covariances between the parameters involved in determination of the latter [6,24]
$\Delta E_{off\text{-}d}$	Hyperfine shifts (MHz) caused by off-diagonal magnetic hfs (see Table 1 and text in Section 3.1.2)
ΔE_{exp-th}	Deviation of experiment from theory (MHz)

Table 5. Frequencies of fine-structure transitions in hydrogen derived from the level-optimization procedure compared with QED calculations of Jentschura et al. [10]

E_{exp}	Transition frequencies (MHz) as measured or derived from experiments. Interpolated and extrapolated values are enclosed in square brackets
Ref.	Source reference. Semi-empirical values determined in the present paper are referred to as follows: Ritz fit1 – Ritz formula fitting for the nS series; Ritz fit2 – Ritz formula fitting for the $n\mathrm{D}_{5/2}$ series; Extrap. 1 – extrapolation of the $n\mathrm{P}_{1/2}$–$n\mathrm{S}_{1/2}$ intervals; Interp. 2 – interpolation of the $n\mathrm{D}_{3/2}$–$n\mathrm{D}_{5/2}$ intervals; Interp. 3 and Extrap. 3 – interpolation and extrapolation of the $n\mathrm{P}_{1/2}$–$n\mathrm{P}_{3/2}$ intervals. These semi-empirical calculations are described in Section 5. "P" in this column means "predicted interval"
E_{Ritz}	Ritz frequencies (MHz) and uncertainties determined by level optimization (see Section 5). For transitions that alone determine one of the levels involved, this column is empty. The additive quantity x determining the systematic shifts of the n = 3 and 6 levels is defined in Table C
$\Delta E_{exp-Ritz}$	Deviation of measured frequency from Ritz value (MHz)
E_{th}	Sum of values from Jentschura et al. [10] (calculated in an approximation that does not describe hfs interactions) and hyperfine shifts caused by off-diagonal magnetic hfs (MHz). The uncertainties include contributions from off-diagonal hfs shifts and the fundamental constants, as well as covariances between the parameters involved in determination of the latter [6,24]
$\Delta E_{off\text{-}d}$	Hyperfine shifts (MHz) caused by off-diagonal magnetic hfs (see Table 1 and text in Section 3.1.2)
$\Delta E_{Ritz-th}$	Deviation of Ritz frequency from theory (MHz)

Table 6. Optimized energy levels of deuterium derived from available experimental transition frequencies compared with QED calculations of Jentschura et al. [10]

LS term	*LS* term designation (principal quantum number followed by a letter symbol for the total orbital momentum *L*)
J	Total angular momentum ($\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{S}$)
E_{exp}	Energy levels (MHz) derived from measured transition frequencies using the LOPT program [139]. The values enclosed in square brackets were determined from a single interpolated or extrapolated transition frequency (see Section 7). The additive quantities *x* and *y* denote possible systematic shifts in the measurements of Zhao et al. [111] for hydrogen and deuterium, respectively (see Section 6)
Unc.	Uncertainty of the excitation energy from the ground level (MHz)
Num. trans.	Number of transitions connecting the level with other levels contributing to level optimization
E_{th}	Sum of values from Jentschura et al. [10] (calculated in an approximation that does not describe hfs interactions) and hyperfine shifts caused by electric quadrupole hfs and off-diagonal magnetic hfs (MHz). The uncertainty values are the combined uncertainties of the QED fine-structure calculations (including covariances between the parameters involved in their determination [6,24]) and uncertainties of the hfs calculations
ΔE_{hfs}	Hyperfine shifts (MHz) caused by electric quadrupole hfs and off-diagonal magnetic hfs (see Table 2 and text in Section 3.1.3)
$\Delta E_{\mathrm{exp-th}}$	Deviation of experiment from theory (MHz)

Table 7. Frequencies of fine-structure transitions in deuterium derived from the level-optimization procedure compared with QED calculations of Jentschura et al. [10]

E_{exp}	Transition frequencies (MHz) as measured or derived from experiments. Interpolated and extrapolated values are enclosed in square brackets
Ref.	Source reference. Semi-empirical values determined in the present paper are referred to as follows: Ritz fit1 – Ritz formula fitting for the *n*S series; Ritz fit2 – Ritz formula fitting for the $n\mathrm{D}_{5/2}$ series; Extrap. 1 – extrapolation of the $n\mathrm{P}_{1/2}$–$n\mathrm{S}_{1/2}$ intervals; Interp. 2 – interpolation of the $n\mathrm{D}_{3/2}$–$n\mathrm{D}_{5/2}$ intervals; Interp. 3 and Extrap. 3 – interpolation and extrapolation of the $n\mathrm{P}_{1/2}$–$n\mathrm{P}_{3/2}$ intervals. These semi-empirical calculations are described in Sections 5 and 7. "P" in this column means "predicted interval"
E_{Ritz}	Ritz frequencies (MHz) and uncertainties determined by level optimization (see Section 7). For transitions that alone determine one of the levels involved, this column is empty. The additive quantities *x* and *y* denote possible systematic shifts in the measurements of Zhao et al. [111] for hydrogen and deuterium, respectively (see Section 6)
$\Delta E_{\mathrm{exp-Ritz}}$	Deviation of measured frequency from Ritz value (MHz)
E_{th}	Sum of values from Jentschura et al. [10] (calculated in an approximation that does not describe hfs interactions) and hyperfine shifts caused by electric quadrupole hfs and off-diagonal magnetic hfs. The uncertainty values are the combined uncertainties of the QED fine-structure calculations (including covariances between the parameters involved in their determination [6,24]) and uncertainties of the hfs calculations
ΔE_{hfs}	Hyperfine shifts caused by electric quadrupole hfs and off-diagonal magnetic hfs (see Table 2 and text in Section 3.1.3)
$\Delta E_{\mathrm{Ritz-th}}$	Deviation of Ritz frequency from theory (MHz)

Table 8. Experimental and theoretical energy levels of tritium

E_{exp}	Transition frequencies (MHz) as measured or derived from experiments. Interpolated and extrapolated values are enclosed in square brackets. The additive quantities after the energy values have the following meaning: *x* is the systematic uncertainty of the hydrogen $n = 3$ energies resulting from the measurements of Zhao et al. [111] (see Section 4; this is the same quantity as in Tables 4–7 for Tables H and D); *z* describes the uncertainty of the energy of the $2\mathrm{P}_{1/2}$ level, which was fixed at its theoretical value; *w* represents the uncertainty of the $3\mathrm{P}_{3/2}$ level, which is based on the $3\mathrm{D}_{3/2}$–$3\mathrm{P}_{3/2}$ separation, 5 ± 5 MHz, extrapolated from the measurement of this separation in deuterium [159]; *v* represents the uncertainty of the $2\mathrm{P}_{1/2}$–$2\mathrm{P}_{3/2}$ interval, ±110 MHz
Unc.	Uncertainty of the energy separation from the $2\mathrm{P}_{1/2}$ level (MHz)
E_{th}	Theoretical energy (MHz). The ionization limit (ionization potential, IP) is quoted from Mohr et al. [6] (see Section 7). Its uncertainty combines the uncertainties of the Rydberg constant and triton mass with the rounding error of the IP value given by Mohr et al. [6]. For all other levels, the energies were obtained from scaled and corrected total energies calculated by Erickson [37] by adding the IP from Mohr et al. [6]. Their uncertainties correspond to those of the total energies (i.e., they are given relative to the IP)
$\Delta E_{\mathrm{exp-th}}$	Deviation of experiment from theory (MHz)
$\Delta E_{[31]\mathrm{-th}}$	Difference between calculations of Garcia and Mack [31] and theoretical value in column "E_{th}" (MHz)

Table 9. Optimized distinctive energy levels of the hydrogen series (see Section 10.4 for the definition of distinctive energy levels)

n	Principal quantum number
E_{opt}	Optimized level value (cm^{-1}). The values in square brackets are interpolated by fitting the 4-term extended Ritz formula (25) (see text and Table M)
ν_{opt}	Optimized value of the frequency of the transition to the ground state (MHz)
Num. lines	Number of observed connecting lines
$E_{\mathrm{opt}} - E_{\mathrm{step3}}$	Difference between the optimized level value and the one obtained in Step 3 of the level optimization procedure (see Section 10.5) (cm^{-1})
$E_{\mathrm{opt}} - E_{\mathrm{step4}}$	Difference between the optimized level value and the one obtained in Step 4 of the level optimization procedure (see Section 10.5) (cm^{-1})
$E_{\mathrm{opt}} - E_{\mathrm{set3}}$	Difference between the optimized level value and the one obtained from Set 3 of the numerical experiment described in Section 10.4 (cm^{-1})
$E_{\mathrm{opt}} - E_{\mathrm{c.g.}}$	Difference between the optimized level value and theoretical center-of-gravity level value (cm^{-1})

Table 10. Observed and Ritz wavelengths and wave numbers of the hydrogen series lines

Transition	Principal quantum numbers of the lower and upper levels
λ_{obs}	Observed wavelength (Å). Wavelengths below 20 000 Å are given in standard air, above that in vacuum. Conversion from vacuum wave numbers to air wavelengths and vice versa was done using the five-parameter formula for the index of refraction of air from Peck and Reeder [205]
σ_{obs}	Observed vacuum wave number (cm^{-1})
Ref.	Source reference
I_{obs}	Observed relative intensity (arbitrary units) (see Section 10.6)
σ_{Ritz}	Ritz wave number (cm^{-1})
$\Delta\sigma_{\mathrm{obs-Ritz}}$	Deviation of the observed wave number from Ritz value (cm^{-1})
$\Delta\sigma_{\mathrm{fs}}$	Width of the fine structure (separation between the outermost components), from Garcia and Mack [31]
ν_{Ritz}	Ritz transition frequency (GHz)

Table 11. Wave numbers of the Lyman series lines of hydrogen and deuterium derived from experimental fine-structure intervals weighted by calculated line strengths

n_{upper}	Principal quantum number of the upper level
σ	Wave number derived from experimental fine-structure intervals weighted by calculated line strengths (cm^{-1})
σ_{obs}	Observed wave number (cm^{-1}) and its measurement uncertainty (in parentheses) in the units of the last significant figure of the value, followed by a reference to the source
$\Delta\sigma$	Maximum possible deviation in the wave number under varying conditions of excitation and observation (estimated as 27% of the fine-structure interval; see Section 10.7)
ν	Transition frequency derived from experimental fine-structure intervals weighted by calculated line strengths (GHz). The quantities in parentheses, given next to the values, are maximum possible deviations in transition frequencies under varying conditions of excitation and observation (see description of the previous column), in the units of the last given decimal figure of the value

Table 12. Designations of fine-structure components and their groups, used by different authors for the Balmer-α lines, and their relative intensities according to different models

Models of the relative intensities (1–5) are described in Section 10.7.

Table 13. Wave numbers of component groups and their separations for the Balmer series lines of hydrogen and deuterium derived from experimental fine-structure intervals weighted by different models

n_{upper}	Principal quantum number of the upper level
Group or interval	Designations of transition groups and separations between them are defined in Table 12
Model	Models of the relative intensities (1–5) are described in Section 10.7
σ	Wave number derived from experimental fine-structure intervals weighted by model intensities of the fine-structure components (cm^{-1})
σ_{obs}	Observed wave number (cm^{-1})
$\Delta\sigma$	Differences of wave numbers calculated with different intensity models from Model 1 (cm^{-1})

Table 1
Hfs splittings and off-diagonal shifts in hydrogen (Hz). See page 621 for Explanation of Tables.

n	$S_{1/2}$	Ref.	$P_{1/2}$	$\Delta E_{c.g.}$	$P_{3/2}$	$\Delta E_{c.g.}$
1	1 420 405 751.768(1)	[14]				
2	[177 556 838.2](3)	[17]	(59 169 500)(600)	(−1 870)(19)	(23 651 600)(240)	(935)(9)
3	[52 609 473.2](3)	[17]	(17 531 800)(180)	(−554)(6)	(7 007 920)(70)	(277)(3)
4	[22 194 585.2](2)	[17]	(7 396 230)(70)	(−234)(2)	(2 956 470)(30)	(116.9)(12)
5	[11 363 609.1](1)	[17]	(3 785 860)(40)	(−119.7)(12)	(1 513 715)(15)	(59.8)(6)
6	[6 576 153.79](6)	[17]	(2 191 470)(22)	(−69.3)(7)	(875 992)(9)	(34.6)(3)
7	[4 141 246.81](4)	[17]	(1 380 050)(14)	(−43.6)(4)	(551 645)(6)	(21.8)(2)
8	[2 774 309.35](3)	[17]	(924 525)(9)	(−29.2)(3)	(369 559)(4)	(14.6)(2)
9	[1 948 484.71](19)		(649 323)(6)	(−20.5)(2)	(259 553)(3)	(10.3)(1)
10	[1 420 444.48](14)		(473 356)(5)	(−15.0)(2)	(189 214)(2)	(7.5)(1)
11	[1 067 200.41](11)		(355 639)(4)	(−11.2)(1)	(142 159.3)(14)	(5.6)(1)
12	[822 015.69](8)		(273 933)(3)	(−8.7)(1)	(109 498.9)(11)	(4.30)(5)
n	$D_{3/2}$	$\Delta E_{c.g.}$	$D_{5/2}$	$\Delta E_{c.g.}$	$F_{5/2}$	$\Delta E_{c.g.}$
3	(4 205 000)(4000)	(−1 496)(15)	(2 700 900)(2 700)	(998)(10)		
4	(1 774 000)(1800)	(−631)(6)	(1 139 400)(1 100)	(421)(4)	(813 000)(800)	(−1202)(12)
5	(908 300)(900)	(−323)(3)	(583 400)(600)	(216)(2)	(416 200)(400)	(−616)(6)
6	(525 600)(500)	(−187)(2)	(337 600)(300)	(124.7)(12)	(240 890)(240)	(−356)(4)
7	(331 000)(300)	(−117.8)(12)	(212 610)(210)	(78.5)(8)	(151 690)(150)	(−224)(2)
8	(221 750)(220)	(−78.9)(8)	(142 430)(140)	(52.6)(5)	(101 620)(100)	(−150)(2)
9	(155 740)(160)	(−55.4)(6)	(100 030)(100)	(36.9)(4)	(71 370)(70)	(−105.5)(11)
10	(113 540)(110)	(−40.4)(4)	(72 920)(70)	(26.9)(3)	(52 030)(50)	(−77.0)(8)
11	(85 300)(90)	(−30.4)(3)	(54 790)(50)	(20.2)(2)	(39 090)(40)	(−57.9)(6)
12	(65 700)(70)	(−23.4)(2)	(42 200)(40)	(15.6)(2)	(30 110)(30)	(−44.5)(4)
n	$F_{7/2}$	$\Delta E_{c.g.}$				
4	(601 400)(600)	(902)(9)				
5	(307 900)(300)	(462)(5)				
6	(178 180)(180)	(267)(3)				
7	(112 210)(110)	(168)(2)				
8	(75 170)(80)	(112.7)(11)				
9	(52 790)(50)	(79.2)(8)				
10	(38 490)(40)	(57.7)(6)				
11	(28 920)(30)	(43.4)(4)				
12	(22 272)(22)	(33.4)(3)				

Table 2
Hfs splittings and shifts in deuterium (Hz). See page 621 for Explanation of Tables. Values without a reference are calculated in this work (see Section 3.1.3).

n	Hfs splitting $F = 1/2$–$F = 3/2$	Hfs splitting $F = 1/2$–$F = 3/2$	$\Delta E_{c.g}$	Hfs splitting $F = \|j\text{-}1\|$–$F = j$	Hfs splitting $F = j\|$–$F = j + 1$	$\Delta E_{c.g.}$
	$n\mathrm{S}_{1/2}$	$n\mathrm{P}_{1/2}$		$n\mathrm{P}_{3/2}$		
1	327384352.5222(17)[a]					
2	40 924 454(7)[b]	(13 633 390)(140)	(13 530)(70)	(2 699 520)(130)	(4 554 030)(90)	(14 240)(80)
3	[12 125 772](3)	(4 039 550)(40)	(4 009)(21)	(799 860)(40)	(1 349 350)(30)	(4 218)(22)
4	[5 115 551.2](12)	(1 704 183)(17)	(1 691)(9)	(337 443)(17)	(569 259)(11)	(1 779)(9)
5	[2 619 157.8](6)	(872 541)(9)	(866)(5)	(172 771)(9)	(291 461)(5)	(911)(5)
6	[1 515 714.2](4)	(504 942)(5)	(501)(3)	(99 983)(5)	(168 669)(3)	(527)(3)
7	[954 501.2](2)	(317 981)(3)	(315.6)(17)	(62 963)(3)	(106 217)(2)	(332)(2)
8	[639 440.63](15)	(213 022.1)(21)	(211.4)(11)	(42 180.4)(21)	(71 157.3)(13)	(222.4)(12)
9	[449 099.25](11)	(149 612.1)(15)	(148.5)(8)	(29 624.6)(15)	(49 976.0)(9)	(156.2)(8)
10	[327 393.15](8)	(109 067.2)(11)	(108.2)(6)	(21 596.4)(11)	(36 432.5)(7)	(113.9)(6)
11	[245 975.19](6)	(81 943.7)(8)	(81.3)(4)	(16 225.7)(8)	(27 372.3)(5)	(85.6)(4)
12	[189 463.44](5)	(63 117.5)(6)	(62.6)(3)	(12 497.9)(6)	(21 083.6)(4)	(65.9)(3)
n	Hfs splitting $F = \|j\text{-}1\|$–$F = j$	Hfs splitting $F = j$–$F = j + 1$	$\Delta E_{c.g.}$	Hfs splitting $F = \|j\text{-}1\|$–$F = j$	Hfs splitting $F = j$–$F = j + 1$	$\Delta E_{c.g.}$
	$n\mathrm{D}_{3/2}$			$n\mathrm{D}_{5/2}$		
3	(482 900)(500)	(808 500)(800)	(453.5)(24)	(344 700)(400)	(484 800)(500)	(1 080)(6)
4	(203 720)(200)	(341 100)(300)	(191.3)(10)	(145 430)(150)	(204 520)(210)	(456)(2)
5	(104 300)(100)	(174 640)(180)	(98.0)(5)	(74 460)(70)	(104 720)(110)	(233.4)(12)
6	(60 360)(60)	(101 060)(100)	(56.7)(3)	(43 090)(40)	(60 600)(60)	(135.1)(7)
7	(38 010)(40)	(63 640)(60)	(35.7)(2)	(27 140)(30)	(38 160)(40)	(85.0)(4)
8	(25 460)(30)	(42 640)(40)	(23.92)(13)	(18 178)(18)	(25 570)(30)	(57.0)(3)
9	(17 885)(18)	(29 940)(30)	(16.80)(9)	(12 767)(13)	(17 956)(18)	(40.0)(2)
10	(13 038)(13)	(21 830)(22)	(12.25)(6)	(9 307)(9)	(13 090)(13)	(29.17)(15)
11	(9 796)(10)	(16 401)(16)	(9.20)(5)	(6 993)(7)	(9 834)(10)	(21.92)(11)
12	(7 545)(8)	(12 633)(13)	(7.09)(4)	(5 386)(5)	(7 575)(8)	(16.88)(9)
n	Hfs splitting $F = \|j\text{-}1\|$–$F = j$	Hfs splitting $F = j$–$F = j + 1$	$\Delta E_{c.g.}$	Hfs splitting $F = \|j\text{-}1\|$–$F = j$	Hfs splitting $F = j$–$F = j + 1$	$\Delta E_{c.g.}$
	$n\mathrm{F}_{5/2}$			$n\mathrm{F}_{7/2}$		
4	(103 790)(100)	(146 100)(150)	(−177.2)(9)	(80 990)(80)	(103 970)(100)	(351.7)(18)
5	(53 140)(50)	(74 800)(80)	(−90.7)(5)	(41 470)(40)	(53 230)(50)	(180.1)(9)
6	(30 750)(30)	(43 290)(40)	(−52.5)(3)	(23 996)(24)	(30 810)(30)	(104.2)(5)
7	(19 366)(19)	(27 260)(30)	(−33.06)(17)	(15 111)(15)	(19 400)(19)	(65.6)(3)
8	(12 974)(13)	(18 262)(18)	(−22.14)(12)	(10 123)(10)	(12 996)(13)	(44.0)(2)
9	(9 112)(9)	(12 826)(13)	(−15.56)(8)	(7 110)(7)	(9 128)(9)	(30.87)(16)
10	(6 643)(7)	(9 350)(9)	(−11.35)(6)	(5 183)(5)	(6 654)(7)	(22.51)(12)
11	(4 991)(5)	(7 025)(7)	(−8.53)(4)	(3 894)(4)	(4 999)(5)	(16.92)(9)
12	(3 844)(4)	(5 411)(5)	(−6.56)(3)	(3 000)(3)	(3 851)(4)	(13.02)(7)

[a] Measured by Wineland and Ramsey [15].
[b] Measured by Kolachevsky et al. [83].

Table 3
Hfs splittings and off-diagonal shifts in tritium (Hz). See page 621 for Explanation of Tables.

n	$S_{1/2}$	Ref.	$P_{1/2}$	$\Delta E_{c.g.}$	$P_{3/2}$	$\Delta E_{c.g.}$
1	1 516 701 470.775(7)	[16][a]				
2	[189 594 000](400)		(63 172 400)(600)	(−2129)(21)	(25 246 300)(300)	(1 065)(11)
3	[56 176 010](110)		(18 717 840)(190)	(−631)(6)	(7 480 440)(80)	(316)(3)
4	[23 699 210](50)		(7 896 590)(80)	(−266)(3)	(3 155 820)(30)	(133.1)(13)
5	[12 133 974](24)		(4 043 050)(40)	(−136.3)(14)	(1 615 778)(16)	(68.1)(7)
6	[7 021 965](14)		(2 339 725)(23)	(−78.9)(8)	(935 057)(9)	(39.4)(4)
7	[4 421 990](9)		(1 473 411)(15)	(−49.7)(5)	(588 840)(6)	(24.8)(2)
8	[2 962 385](6)		(987 070)(10)	(−33.3)(3)	(394 477)(4)	(16.6)(2)
9	[2 080 576](4)		(693 250)(7)	(−23.4)(2)	(277 054)(3)	(11.7)(2)
10	[1 516 739](3)		(505 379)(5)	(−17.0)(2)	(201 972)(2)	(8.5)(1)
11	[1 139 548](2)		(379 699)(4)	(−12.8)(1)	(151 745)(2)	(6.4)(1)
12	[877 742](2)		(292 464)(3)	(−9.9)(1)	(116 881.9)(12)	(4.9)(1)
n	$D_{3/2}$	$\Delta E_{c.g.}$	$D_{5/2}$	$\Delta E_{c.g.}$	$F_{5/2}$	$\Delta E_{c.g.}$
3	(4 489 000)(5 000)	(−1704)(17)	(2 883 000)(3 000)	(1 136)(11)		
4	(1 894 000)(1 900)	(−719)(7)	(1 216 200)(1 200)	(479)(5)	(867 800)(900)	(−1369)(14)
5	(969 700)(1 000)	(−368)(4)	(622 700)(600)	(245)(2)	(444 300)(500)	(−701)(7)
6	(561 200)(600)	(−213)(2)	(360 400)(400)	(142.0)(14)	(257 100)(300)	(−406)(4)
7	(353 400)(400)	(−134.1)(13)	(226 930)(230)	(89.4)(9)	(161 930)(160)	(−255)(3)
8	(236 740)(240)	(−89.8)(9)	(152 030)(150)	(59.9)(6)	(108 480)(110)	(−171)(2)
9	(166 270)(170)	(−63.1)(6)	(106 770)(110)	(42.1)(4)	(76 190)(80)	(−120.2)(12)
10	(121 210)(120)	(−46.0)(5)	(77 840)(80)	(30.7)(3)	(55 540)(60)	(−87.6)(9)
11	(91 070)(90)	(−34.6)(3)	(58 480)(60)	(23.0)(2)	(41 730)(40)	(−65.9)(7)
12	(70 150)(70)	(−26.6)(3)	(45 040)(50)	(17.7)(2)	(32 140)(30)	(−50.7)(5)
n	$F_{7/2}$	$\Delta E_{c.g.}$				
4	(641 800)(700)	(1027)(10)				
5	(328 600)(300)	(526)(5)				
6	(190 150)(190)	(304)(3)				
7	(119 740)(120)	(192)(2)				
8	(80 220)(80)	(128.3)(13)				
9	(56 340)(60)	(90.1)(9)				
10	(41 070)(40)	(65.7)(7)				
11	(30 860)(30)	(49.4)(5)				
12	(23 769)(24)	(38.0)(4)				

[a] Rescaled according to the adopted value of hfs(1S) for hydrogen (see introduction to Section 3).

Table 4
Optimized energy levels of hydrogen derived from available experimental transition frequencies compared with QED calculations of Jentschura *et al.* [10]. See page 621 for Explanation of Tables.

LS term	*J*	E_{exp} (MHz)		Unc. (MHz)	Num. trans.	E_{th} (MHz)	ΔE_{off-d} (MHz)	ΔE_{exp-th} (MHz)
1S	1/2	0.000 00		–	3	0.000 00	–	–
2P	1/2	2 466 060 355.339		0.009	3	(2 466 060 355.341 2)(24)	(−0.0019)	−0.002
2S	1/2	2 466 061 413.187 07		0.000 03	9	(2 466 061 413.187 10)(5)	–	−0.000 03
2P	3/2	2 466 071 324.389		0.012	3	(2 466 071 324.385 6)(24)	(0.000 9)	0.003
3P	1/2	2 922 742 963. 15	+*x*	0.21	4	(2 922 742 963.793 3)(7)	(−0.0006)	−0.64
3S	1/2	2 922 743 277.97	+*x*	0.22	4	(2 922 743 278.671 6)(14)	–	−0.70
3D	3/2	2 922 746 207.9	+*x*	0.6	5	(2 922 746 208.549 9)(7)	(−0.0015)	−0.6
3P	3/2	2 922 746 213.24	+*x*	0.21	5	(2 922 746 213.883 4)(7)	(0.000 3)	−0.64
3D	5/2	2 922 747 291.13	+*x*	0.22	4	(2 922 747 291.889 6)(7)	(0.001 0)	−0.76
4P	1/2	3 082 581 430.756		0.015	4	(3 082 581 430.738 0)(17)	(−0.0002)	0.018
4S	1/2	3 082 581 563.823		0.010	5	(3 082 581 563.815 6)(20)	–	0.007
4D	3/2	3 082 582 799.84		0.23	3	(3 082 582 799.579 8)(17)	(−0.0006)	0.26
4P	3/2	3 082 582 801.858		0.010	4	(3 082 582 801.868 1)(17)	(0.000 1)	−0.010
4F	5/2	3 082 583 256.6		1.6	1	(3 082 583 255.803 3)(17)	(−0.0012)	0.8
4D	5/2	3 082 583 256.630		0.024	4	(3 082 583 256.614 2)(17)	(0.000 4)	0.016
4F	7/2	3 082 583 484.6		0.4	1	(3 082 583 484.320 7)(17)	(0.000 9)	0.3
5P	1/2	3 156 563 616.570		1.1	3	(3 156 563 616.462 9)(22)	(−0.0001)	0.1
5S	1/2	[3 156 563 684.771]		1.1	3	(3 156 563 684.658 4)(24)	–	0.1
5D	3/2	3 156 564 315.79		0.7	2	(3 156 564 317.298 4)(22)	(−0.0003)	−1.5
5P	3/2	3 156 564 318.589		1.1	3	(3 156 564 318.480 7)(22)	(0.000 1)	0.1
5D	5/2	[3 156 564 549.71]		0.7	3	(3 156 564 551.300 1)(22)	(0.000 2)	−1.6
5F	7/2	3 156 564 666.9		1.6	1	(3 156 564 667.882 5)(22)	(0.000 5)	−1.0
6P	1/2	[3 196 751 390.790]	+1/4 *x*	0.03	1	(3 196 751 390.971)(3)	(−0.0001)	−0.18
6S	1/2	3 196 751 430.284	+1/4 *x*	0.021	1	(3 196 751 430.455)(3)	–	−0.17
6D	3/2	[3 196 751 796.39]	+1/4 *x*	0.04	1	(3 196 751 796.543)(3)	(−0.0002)	−0.15
6P	3/2	[3 196 751 797.051]	+1/4 *x*	0.03	1	(3 196 751 797.231)(3)	(0.000 0)	−0.18
6D	5/2	3 196 751 931.779	+1/4 *x*	0.011	1	(3 196 751 931.960)(3)	(0.000 1)	−0.18
7P	1/2	[3 220 983 314.518]		1.2	1	(3 220 983 314.718)(3)	(0.000 0)	−0.2
7S	1/2	[3 220 983 339.400]		1.2	1	(3 220 983 339.590)(3)	–	−0.2
7D	3/2	[3 220 983 570.129]		0.7	1	(3 220 983 570.120)(3)	(−0.0001)	0.0
7P	3/2	[3 220 983 570.356]		1.2	1	(3 220 983 570.554)(3)	(0.000 0)	−0.2
7D	5/2	[3 220 983 655.4]		0.7	1	(3 220 983 655.397)(3)	(0.000 1)	0.0
8P	1/2	[3 236 710 746.525]		0.018	1	(3 236 710 746.537)(3)	(0.000 0)	−0.012
8S	1/2	3 236 710 763.199		0.009	1	(3 236 710 763.203)(3)	–	−0.004
8D	3/2	3 236 710 917.637		0.008	1	(3 236 710 917.636)(3)	(−0.0001)	0.001
8P	3/2	[3 236 710 917.916]		0.019	1	(3 236 710 917.928)(3)	(0.000 0)	−0.012
8D	5/2	3 236 710 974.771		0.006	1	(3 236 710 974.765)(3)	(0.000 1)	0.006
9P	1/2	[3 247 493 411.886]		1.2	1	(3 247 493 411.755)(3)	(0.000 0)	0.1
9S	1/2	[3 247 493 423.600]		1.2	1	(3 247 493 423.461)(3)	–	0.1
9D	3/2	[3 247 493 531.869]		0.7	1	(3 247 493 531.922)(3)	(−0.000 1)	−0.1
9P	3/2	[3 247 493 532.260]		1.2	1	(3 247 493 532.128)(3)	(0.000 0)	0.1
9D	5/2	[3 247 493 572.0]		0.7	1	(3 247 493 572.046)(3)	(0.000 0)	0.0
10P	1/2	[3 255 206 183.059 0]		1.2	1	(3 255 206 182.761)(3)	(0.000 0)	0.3
10S	1/2	[3 255 206 191.600]		1.2	1	(3 255 206 191.296)(3)	–	0.3
10D	3/2	[3 255 206 270.342]		0.04	1	(3 255 206 270.363)(3)	(0.000 0)	−0.02
10P	3/2	[3 255 206 270.811 4]		1.2	1	(3 255 206 270.512)(3)	(0.000 0)	0.3
10D	5/2	3 255 206 299.60		0.04	1	(3 255 206 299.613)(3)	(0.000 0)	−0.01
11P	1/2	[3 260 912 757.682 0]		1.2	1	(3 260 912 757.359)(3)	(0.000 0)	0.3
11S	1/2	[3 260 912 764.100]		1.2	1	(3 260 912 763.772)(3)	–	0.3
11D	3/2	[3 260 912 823.116]		0.7	1	(3 260 912 823.176)(3)	(0.000 0)	−0.1
11P	3/2	[3 260 912 823.611 7]		1.2	1	(3 260 912 823.288)(3)	(0.000 0)	0.3
11D	5/2	[3 260 912 845.1]		0.7	1	(3 260 912 845.152)(3)	(0.000 0)	−0.1
12P	1/2	[3 265 253 073.256 0]		1.2	1	(3 265 253 072.974)(3)	(0.000 0)	0.3
12S	1/2	[3 265 253 078.200]		1.2	1	(3 265 253 077.913)(3)	–	0.3
12D	3/2	3 265 253 123.660		0.009	1	(3 265 253 123.669)(3)	(0.000 0)	−0.009
12P	3/2	[3 265 253 124.038 6]		1.2	1	(3 265 253 123.756)(3)	(0.000 0)	0.3
12D	5/2	3 265 253 140.591		0.007	1	(3 265 253 140.596)(3)	(0.000 0)	−0.005
Limit		[3 288 086 856.8]		0.7		(3 288 086 857.128)(3)	–	−0.3

Table 5
Frequencies of fine-structure transitions in hydrogen derived from the level-optimization procedure compared with QED calculations of Jentschura et al. [10]. See page 621 for Explanation of Tables.

Transition	E_{exp} (MHz)	Ref.	E_{Ritz} (MHz)		$\Delta E_{exp-Ritz}$ (MHz)	E_{th} (MHz)	$\Delta E_{off\text{-}d}$ (MHz)	$\Delta E_{Ritz\text{-}th}$ (MHz)
$1S_{1/2}-2P_{1/2}$	2 466 068 000(4 000)[a]	[149]	2 466 060 355.339(9)		8 000	(2 466 060 355.341 2)(24)	(−0.001 9)	−0.002
$1S_{1/2}-2S_{1/2}$	2 466 061 413.187 07(3)	[115]	2 466 061 413.187 07(3)		0.000 00	(2 466 061 413.187 10)(5)	–	−0.000 03
$1S_{1/2}-2P_{3/2}$	2 466 068 000(4 000)[a]	[149]	2 466 071 324.389(12)		−3 000	(2 466 071 324.385 6)(24)	(0.000 9)	0.003
$1S_{1/2}-3P_{1/2}$	2 922 729 000(9 000)[a]	[150]	2 922 742 963.15(21)	+x	−14 000	(2 922 742 963.793 3)(7)	(−0.000 6)	−0.64
$1S_{1/2}-3S_{1/2}$		P	2 922 743 277.97(22)	+x		(2 922 743 278.671 6)(14)	–	−0.70
$1S_{1/2}-3P_{3/2}$	2 922 729 000(9 000)[a]	[150]	2 922 746 213.24(21)	+x	−17 000	(2922746213.8834)(7)	(0.000 3)	−0.64
$1S_{1/2}-4P_{1/2}$	3 082 570 000(60 000)[a]	[151]	3 082 581 430.756(15)		−10 000	(3 082 581 430.738 0)(17)	(−0.000 2)	0.018
$1S_{1/2}-4S_{1/2}$		P	3 082 581 563.823(10)			(3 082 581 563.815 6)(20)	–	0.007
$1S_{1/2}-4P_{3/2}$	3 082 570 000(60 000)[a]	[151]	3 082 582 801.858(10)		−10 000	(3 082 582 801.868 1)(17)	(0.000 1)	−0.010
$1S_{1/2}-5P_{1/2}$	3 156 567 000(13 000)[a]	[150]	3 156 563 616.6(11)		3 000	(3 156 563 616.462 9)(22)	(−0.000 1)	0.1
$1S_{1/2}-5S_{1/2}$	[3 156 563 685.1](12)	Ritz fit1	3 156 563 684.8(11)		0.3	(3 156 563 684.658 4)(24)	–	0.1
$1S_{1/2}-5P_{3/2}$	3 156 567 000(13 000)[a]	[150]	3 156 564 318.6(11)		3 000	(3 156 564 318.480 7)(22)	(0.000 1)	0.1
$1S_{1/2}-5D_{5/2}$	[3 156 564 549.6](7)	Ritz fit2	3 156 564 549.7(7)		−0.1	(3 156 564 551.300 1)(22)	(0.000 2)	−1.6
$1S_{1/2}-6P_{1/2}$	3 196 760 000(70 000)[a]	[151]	3 196 751 390.79(3)	+x/4	10 000	(3 196 751 390.971)(3)	(0.000)	−0.18
$1S_{1/2}-6S_{1/2}$		P	3 196 751 430.284(21)	+x/4		(3 196 751 430.455)(3)	–	−0.17
$1S_{1/2}-6P_{3/2}$	3 196 760 000(70 000)[a]	[151]	3 196 751 797.05(3)	+x/4	10 000	(3 196 751 797.231)(3)	(0.000)	−0.18
$1S_{1/2}-7P_{1/2}$	3 220 980 000(70 000)[a]	[151]	3 220 983 314.5(12)		−3000	(3 220 983 314.718)(3)	(0.000)	−0.2
$1S_{1/2}-7S_{1/2}$	[3 220 983 339.4](12)[b]	Ritz fit1				(3 220 983 339.590)(3)	–	−0.2
$1S_{1/2}-7P_{3/2}$	3 220 980 000(70 000)[a]	[151]	3 220 983 570.4(12)		−3 000	(3 220 983 570.554)(3)	(0.000)	−0.2
$1S_{1/2}-7D_{5/2}$	[3 220 983 655.4](7)[b]	Ritz fit2				(3 220 983 655.397)(3)	(0.000)	0.0
$1S_{1/2}-8P_{1/2}$	3 236 700 000(70 000)[a]	[151]	3 236 710 746.525(18)		−10 000	(3 236 710 746.537)(3)	(0.000)	−0.012
$1S_{1/2}-8S_{1/2}$		P	3 236 710 763.199(9)			(3 236 710 763.203)(3)	–	−0.004
$1S_{1/2}-8P_{3/2}$	3 236 700 000(70 000)[a]	[151]	3 236 710 917.916(19)		−10 000	(3 236 710 917.928)(3)	(0.000)	−0.012
$1S_{1/2}-9P_{1/2}$	3 247 490 000(70 000)[a]	[151]	3 247 493 411.9(12)		0	(3 247 493 411.755)(3)	(0.000)	0.1
$1S_{1/2}-9S_{1/2}$	[3 247 493 423.6](12)[b]	Ritz fit1				(3 247 493 423.461)(3)	–	0.1
$1S_{1/2}-9P_{3/2}$	3 247 490 000(70 000)[a]	[151]	3 247 493 532.3(12)		0	(3 247 493 532.128)(3)	(0.000)	0.2
$1S_{1/2}-9D_{5/2}$	[3 247 493 572.0](7)[b]	Ritz fit2				(3 247 493 572.046)(3)	(0.000)	0.0
$1S_{1/2}-10P_{1/2}$	3 255 180 000(70 000)[a]	[151]	3 255 206 183.1(12)		−30 000	(3 255 206 182.761)(3)	(0.000)	0.3
$1S_{1/2}-10S_{1/2}$	[3 255 206 191.6](12)[b]	Ritz fit1				(3 255 206 191.296)(3)	–	0.3
$1S_{1/2}-10P_{3/2}$	3 255 180 000(70 000)[a]	[151]	3 255 206 270.8(12)		−30 000	(3 255 206 270.512)(3)	(0.000)	0.3
$1S_{1/2}-11P_{1/2}$	3 260 920 000(70 000)[a]	[151]	3 260 912 757.7(12)		10 000	(3 260 912 757.359)(3)	(0.000)	0.3
$1S_{1/2}-11S_{1/2}$	[3 260 912 764.1](12)[b]	Ritz fit1				(3 260 912 763.772)(3)	–	0.3
$1S_{1/2}-11P_{3/2}$	3 260 920 000(70 000)[a]	[151]	3 260 912 823.6(12)		10 000	(3 260 912 823.288)(3)	(0.000)	0.3
$1S_{1/2}-11D_{5/2}$	[3 260 912 845.1](7)[b]	Ritz fit2				(3 260 912 845.152)(3)	(0.000)	−0.1
$1S_{1/2}-12P_{1/2}$	3 265 250 000(70 000)[a]	[151]	3 265 253 073.3(12)		0	(3 265 253 072.974)(3)	(0.000)	0.3
$1S_{1/2}-12S_{1/2}$	[3 265 253 078.2](12)[b]	Ritz fit1				(3 265 253 077.913)(3)	–	0.3
$1S_{1/2}-12P_{3/2}$	3 265 250 000(70 000)[a]	[151]	3 265 253 124.0(12)		0	(3 265 253 123.756)(3)	(0.000)	0.2
$2P_{1/2}-2S_{1/2}$	1 057.847(9)	[88]	1 057.848(9)		−0.001	(1 057.845 9)(24)	(0.001 9)	0.002
$2P_{1/2}-2P_{3/2}$	10 969.13(10)	[108]	10 969.050(15)		0.08	(10 969.044 4)(9)	(0.002 8)	0.006
$2P_{1/2}-3S_{1/2}$		P	456 682 922.63(22)	+x		(456 682 923.331)(4)	(0.002)	−0.70
$2P_{1/2}-3D_{3/2}$	456 685 852.8(17) + x	[116]	456 685 852.6(6)	+x	0.2	(456 685 853.208)(3)	(0.000)	−0.6
$2P_{1/2}-4S_{1/2}$		P	616 521 208.484(13)			(616 521 208.475)(4)	(0.002)	0.009
$2P_{1/2}-4D_{3/2}$		P	616 522 444.50(23)			(616 522 444.239)(4)	(0.001)	0.26
$2P_{1/2}-5S_{1/2}$		P	690 503 329.4(11)			(690 503 329.317)(5)	(0.002)	0.1
$2P_{1/2}-5D_{3/2}$		P	690 503 960.5(7)			(690 503 961.957)(5)	(0.002)	−1.5
$2P_{1/2}-6S_{1/2}$		P	730 691 074.945(23)	+x/4		(730 691 075.114)(5)	(0.002)	−0.17
$2P_{1/2}-6D_{3/2}$		P	730 691 441.05(4)	+x/4		(730 691 441.202)(5)	(0.002)	−0.15
$2P_{1/2}-7S_{1/2}$		P	754 922 984.1(12)			(754 922 984.249)(5)	(0.002)	−0.1
$2P_{1/2}-7D_{3/2}$		P	754 923 214.8(7)			(754 923 214.779)(5)	(0.002)	0.0
$2P_{1/2}-8S_{1/2}$		P	770 650 407.860(12)			(770 650 407.862)(5)	(0.002)	−0.002
$2P_{1/2}-8D_{3/2}$		P	770 650 562.298(12)			(770 650 562.295)(5)	(0.002)	0.003
$2P_{1/2}-9S_{1/2}$		P	781 433 068.3(12)			(781 433 068.120)(5)	(0.002)	0.2
$2P_{1/2}-9D_{3/2}$		P	781 433 176.5(7)			(781 433 176.581)(5)	(0.002)	−0.1
$2P_{1/2}-10S_{1/2}$		P	789 145 836.3(12)			(789 145 835.954)(5)	(0.002)	0.3
$2P_{1/2}-10D_{3/2}$		P	789 145 915.00(5)			(789 145 915.021)(5)	(0.002)	−0.02
$2P_{1/2}-11S_{1/2}$		P	794 852 408.8(12)			(794 852 408.431)(5)	(0.002)	0.4
$2P_{1/2}-11D_{3/2}$		P	794 852 467.8(7)			(794 852 467.834)(5)	(0.002)	0.0
$2P_{1/2}-12S_{1/2}$		P	799 192 722.9(12)			(799 192 722.572)(5)	(0.002)	0.3
$2P_{1/2}-12D_{3/2}$		P	799192768.321(13)			(799 192 768.328)(5)	(0.002)	−0.007
$2S_{1/2}-2P_{3/2}$	9 911.201(12)	[98]	9 911.202(12)		−0.001	(9 911.198 5)(24)	(0.000 9)	0.003
$2S_{1/2}-3P_{1/2}$	456 681 549.9(3) + x	[111]	456 681 549.96(21)	+x	−0.1	(456 681 550.606 1)(7)	(−0.000 6)	−0.65
$2S_{1/2}-3S_{1/2}$		P	456 681 864.78(22)	+x		(456 681 865.484 5)(14)	–	−0.70
$2S_{1/2}-3P_{3/2}$	456 684 800.1(3) + x	[111]	456 684 800.05(21)	+x	0.1	(456 684 800.696 3)(7)	(0.000 3)	−0.65
$2S_{1/2}-4P_{1/2}$	616 520 017.568(15)	[117]	616 520 017.569(15)		−0.001	(616 520 017.550 9)(17)	(−0.000 2)	0.018
$2S_{1/2}-4S_{1/2}$	616 520 150.636(10)	[101]	616 520 150.636(10)		0.000	(616 520 150.628 5)(20)	–	0.008
$2S_{1/2}-4P_{3/2}$	616 521 388.672(10)	[117]	616 521 388.671(10)		0.001	(616 521 388.681 0)(17)	(0.000 1)	−0.010
$2S_{1/2}-4D_{5/2}$	616521843.441(24)	[101]	616 521 843.443(24)		−0.002	(616 521 843.427 1)(17)	(0.000 4)	0.016
$2S_{1/2}-5P_{1/2}$		P	690 502 203.4(11)			(690 502 203.275 8)(22)	(−0.000 1)	0.1
$2S_{1/2}-5S_{1/2}$		P	690502271.6(11)			(690 502 271.471 3)(24)	–	0.1
$2S_{1/2}-5P_{3/2}$		P	690502905.4(11)			(690 502 905.293 6)(22)	(0.000 1)	0.1
$2S_{1/2}-6P_{1/2}$		P	730 689 977.60(3)	+x/4		(730 689 977.784)(3)	(0.000)	−0.18
$2S_{1/2}-6S_{1/2}$	730 690 017.097(21)[b] +x/4	[103]				(730 690 017.268)(3)	–	−0.17
$2S_{1/2}-6P_{3/2}$		P	730 690 383.86(4)	+x/4		(730 690 384.044)(22)	(0.000)	−0.18

Table 5 (*continued*)

Transition	E_{exp} (MHz)	Ref.	E_{Ritz} (MHz)	$\Delta E_{exp-Ritz}$ (MHz)	E_{th} (MHz)	$\Delta E_{off\text{-}d}$ (MHz)	$\Delta E_{Ritz\text{-}th}$ (MHz)
$2S_{1/2}$–$6D_{5/2}$	730 690 518.592(11)[b] +x/4	[103]			(730 690 518.773)(3)	(0.000)	−0.18
$2S_{1/2}$–$7P_{1/2}$		P	754921901.3(12)		(754 921 901.530)(3)	(0.000)	−0.2
$2S_{1/2}$–$7S_{1/2}$		P	754 921 926.2(12)		(754 921 926.403)(3)	–	−0.2
$2S_{1/2}$–$7P_{3/2}$		P	754 922 157.2(12)		(754 922 157.367)(3)	(0.000)	−0.2
$2S_{1/2}$–$8P_{1/2}$		P	770 649 333.338(18)		(770 649 333.350)(3)	(0.000)	−0.012
$2S_{1/2}$–$8S_{1/2}$	770 649 350.012(9)[b]	[103]			(770 649 350.016)(3)	–	−0.004
$2S_{1/2}$–$8D_{3/2}$	770 649 504.450(8)[b]	[103]			(770 649 504.449)(3)	(0.000)	0.001
$2S_{1/2}$–$8P_{3/2}$		P	770 649 504.729(19)		(770 649 504.741)(3)	(0.000)	−0.012
$2S_{1/2}$–$8D_{5/2}$	770 649 561.584(7)[b]	[103]			(770 649 561.578)(3)	(0.000)	0.006
$2S_{1/2}$–$9P_{1/2}$		P	781 431 998.7(12)		(781 431 998.567)(3)	(0.000)	0.1
$2S_{1/2}$–$9S_{1/2}$		P	781 432 010.4(12)		(781 432 010.274)(3)	–	0.1
$2S_{1/2}$–$9P_{3/2}$		P	781 432 119.1(12)		(781 432 118.941)(3)	(0.000)	0.2
$2S_{1/2}$–$10P_{1/2}$		P	789 144 769.9(12)		(789 144 769.573)(3)	(0.000)	0.3
$2S_{1/2}$–$10S_{1/2}$		P	789 144 778.4(12)		(789 144 778.108)(3)	–	0.3
$2S_{1/2}$–$10P_{3/2}$		P	789 144 857.6(12)		(789 144 857.325)(3)	(0.000)	0.3
$2S_{1/2}$–$10D_{5/2}$	789 144 886.41(4)[b]	[118]			(789 144 886.426)(3)	(0.000)	−0.02
$2S_{1/2}$–$11P_{1/2}$		P	794 851 344.5(12)		(794 851 344.172)(3)	(0.000)	0.3
$2S_{1/2}$–$11S_{1/2}$		P	794 851 350.9(12)		(794 851 350.585)(3)	–	0.3
$2S_{1/2}$–$11P_{3/2}$		P	794 851 410.4(12)		(794 851 410.101)(3)	(0.000)	0.3
$2S_{1/2}$–$12P_{1/2}$		P	799 191 660.1(12)		(799 191 659.786)(3)	(0.000)	0.3
$2S_{1/2}$–$12S_{1/2}$		P	799 191 665.0(12)		(799 191 664.726)(3)	–	0.3
$2S_{1/2}$–$12D_{3/2}$	799 191 710.473(10)[b]	[103]			(799 191 710.482)(3)	(0.000)	−0.009
$2S_{1/2}$–$12P_{3/2}$		P	799 191 710.9(12)		(799 191 710.569)(3)	(0.000)	0.3
$2S_{1/2}$–$12D_{5/2}$	799 191 727.404(7)[b]	[103]			(799 191 727.409)(3)	(0.000)	−0.005
$2P_{3/2}$–$3S_{1/2}$		P	456 671 953.58(22) +x		(456 671 954.286)(4)	(−0.001)	−0.71
$2P_{3/2}$–$3D_{3/2}$		P	456 674 883.5(6) +x		(456 674 884.165)(3)	(−0.002)	−0.7
$2P_{3/2}$–$3D_{5/2}$	456 675 968(4) + x	[116]	456 675 966.74(22) +x	1	(456 675 967.504)(3)	(0.000)	−0.76
$2P_{3/2}$–$4S_{1/2}$		P	616 510 239.434(16)		(616 510 239.430)(4)	(−0.001)	0.004
$2P_{3/2}$–$4D_{3/2}$		P	616 511 475.45(23)		(616 511 475.194)(4)	(−0.002)	0.26
$2P_{3/2}$–$4D_{5/2}$		P	616 511 932.24(3)		(616 511 932.228)(4)	(−0.001)	0.01
$2P_{3/2}$–$5S_{1/2}$		P	690 492 360.4(11)		(690 492 360.273)(5)	(−0.001)	0.1
$2P_{3/2}$–$5D_{3/2}$		P	690 492 991.4(7)		(690 492 992.913)(5)	(−0.001)	−1.5
$2P_{3/2}$–$5D_{5/2}$		P	690 493 225.3(7)		(690 493 226.914)(5)	(−0.001)	−1.6
$2P_{3/2}$–$6S_{1/2}$		P	730 680 105.895(24) +x/4		(730 680 106.070)(5)	(−0.001)	−0.18
$2P_{3/2}$–$6D_{3/2}$		P	730 680 472.00(4) +x/4		(730 680 472.157)(5)	(−0.001)	−0.16
$2P_{3/2}$–$6D_{5/2}$		P	730 680 607.390(16) +x/4		(730 680 607.575)(5)	(−0.001)	−0.19
$2P_{3/2}$–$7S_{1/2}$		P	754 912 015.0(12)		(754 912 015.204)(5)	(−0.001)	−0.2
$2P_{3/2}$–$7D_{3/2}$		P	754 912 245.7(7)		(754 912 245.734)(5)	(−0.001)	0.0
$2P_{3/2}$–$7D_{5/2}$		P	754 912 331.0(7)		(754 912 331.012)(5)	(−0.001)	0.0
$2P_{3/2}$–$8S_{1/2}$		P	770 639 438.810(15)		(770 639 438.818)(5)	(−0.001)	−0.008
$2P_{3/2}$–$8D_{3/2}$		P	770 639 593.248(15)		(770 639 593.250)(5)	(−0.001)	−0.002
$2P_{3/2}$–$8D_{5/2}$		P	770 639 650.382(14)		(770 639 650.380)(5)	(−0.001)	0.002
$2P_{3/2}$–$9S_{1/2}$		P	781 422 099.2(12)		(781 422 099.075)(5)	(−0.001)	0.1
$2P_{3/2}$–$9D_{3/2}$		P	781422207.5(7)		(781 422 207.537)(5)	(−0.001)	0.0
$2P_{3/2}$–$9D_{5/2}$		P	781 422 247.6(7)		(781 422 247.660)(5)	(−0.001)	−0.1
$2P_{3/2}$–$10S_{1/2}$		P	789 134 867.2(12)		(789 134 866.910)(5)	(−0.001)	0.3
$2P_{3/2}$–$10D_{3/2}$		P	789 134 945.95(5)		(789 134 945.977)(5)	(−0.001)	−0.03
$2P_{3/2}$–$10D_{5/2}$		P	789 134 975.21(4)		(789 134 975.227)(5)	(−0.001)	−0.02
$2P_{3/2}$–$11S_{1/2}$		P	794 841 439.7(12)		(794 841 439.386)(5)	(−0.001)	0.3
$2P_{3/2}$–$11D_{3/2}$		P	794 841 498.7(7)		(794 841 498.790)(5)	(−0.001)	−0.1
$2P_{3/2}$–$11D_{5/2}$		P	794 841 520.7(7)		(794 841 520.766)(5)	(−0.001)	−0.1
$2P_{3/2}$–$12S_{1/2}$		P	799 181 753.8(12)		(799 181 753.528)(5)	(−0.001)	0.3
$2P_{3/2}$–$12D_{3/2}$		P	799 181 799.271(15)		(799 181 799.283)(5)	(−0.001)	−0.012
$2P_{3/2}$–$12D_{5/2}$		P	799 181 816.202(14)		(799 181 816.211)(5)	(−0.001)	−0.009
$3P_{1/2}$–$3S_{1/2}$	314.82(5)	[59]	314.82(5)	0.00	(314.878 5)(7)	(0.0006)	−0.06
$3P_{1/2}$–$3D_{3/2}$	3 245(3)	[110]	3 244.8(6)	0	(3 244.756 69)(19)	(−0.000 94)	0.0
$3P_{1/2}$–$3P_{3/2}$	3 250.3(4)[a]	[111] Interp. 3	3 250.09(3)	0.2	(3 250.090 2)(3)	(0.000 8)	0.00
	[3 250.09](3)			0.00			
$3S_{1/2}$–$3D_{3/2}$	2 929.9(8)	[109]	2 929.9(6)	0.0	(2 929.878 3)(7)	(−0.001 5)	0.0
$3S_{1/2}$–$3P_{3/2}$	2 933.5(12)	[106]	2 935.27(6)	−1.8	(2 935.211 8)(7)	(0.000 3)	0.06
$3S_{1/2}$–$3D_{5/2}$	4 013.16(5)	[119]	4 013.16(5)	0.00	(4 013.218 0)(7)	(0.001 0)	−0.06
$3D_{3/2}$–$3P_{3/2}$	5.5(9)	[120]	5.3(6)	0.2	(5.333 49)(19)	(0.001 77)	0.0
$3D_{3/2}$–$3D_{5/2}$	1 083(3)	[110]	1 083.2(6)	0	(1 083.339 70)(5)	(0.002 49)	−0.1
$3P_{3/2}$–$3D_{5/2}$	1 078.0(11)	[110]	1 077.89(7)	0.1	(1 078.006 21)(19)	(0.000 72)	−0.12
$4P_{1/2}$–$4S_{1/2}$	133.2(6)	[106]	133.067(18)	0.1	(133.077 6)(3)	(0.000 2)	−0.011
$4P_{1/2}$–$4P_{3/2}$	1 370.85(22)	[121]	1371.102(18)	−0.25	(1 371.130 18)(11)	(0.000 35)	−0.028
$4P_{1/2}$–$4D_{3/2}$	1 371.1(12)	[106]	1369.08(23)	2.0	(1 368.841 81)(8)	(−0.000 40)	0.24
$4S_{1/2}$–$4D_{3/2}$	1 235.0(21)	[110]	1 236.02(23)	−1.0	(1 235.764 2)(3)	(−0.000 6)	0.26
$4S_{1/2}$–$4P_{3/2}$	1 237.8(3)	[107]	1 238.035(14)	−0.2	(1 238.052 5)(3)	(0.000 1)	−0.018
$4S_{1/2}$–$4D_{5/2}$	1 693.0(4)	[110]	1 692.81(3)	0.2	(1 692.798 6)(3)	(0.000 4)	0.01
$4D_{3/2}$–$4F_{5/2}$	456.8(16)[b]	[106]			(456.223 544)(25)	(−0.000 571)	0.6
$4D_{3/2}$–$4D_{5/2}$	[456.85](23)	Interp. 2 [110]	456.79(22)	0.06	(457.034 445)(22)	(0.001 052)	−0.24
	458.0(22)[a]			1.2			
$4P_{3/2}$–$4D_{5/2}$	455.7(16)	[106]	454.77(3)	0.9	(454.746 08)(8)	(0.000 30)	0.02

(*continued on next page*)

Table 5 (*continued*)

Transition	E_{exp} (MHz)	Ref.	E_{Ritz} (MHz)	$\Delta E_{exp-Ritz}$ (MHz)	E_{th} (MHz)	$\Delta E_{off\text{-}d}$ (MHz)	$\Delta E_{Ritz\text{-}th}$ (MHz)
$4D_{5/2}$–$4F_{7/2}$	228.0(4)[b]	[106]			(227.706 503)(22)	(0.000481)	0.3
$5P_{1/2}$–$5S_{1/2}$	[68.201](19)	Extrap. 1 [106]	68.201(19)	0.000	(68.19555)(16)	(0.00012)	0.005
	65(5)[a]			−3			
$5P_{1/2}$–$5D_{3/2}$	704(7)	[106]	699.2(13)	5	(700.835 55)(4)	(−0.00020)	−1.6
$5P_{1/2}$–$5P_{3/2}$	[702.019](12)	Extrap. 3	702.019(12)	0.000	(702.01779)(6)	(0.00018)	0.001
$5S_{1/2}$–$5P_{3/2}$	622(11)	[106]	633.818(22)	−12	(633.82224)(16)	(0.00006)	−0.004
$5D_{3/2}$–$5D_{5/2}$	[233.92](8)	Interp. 2	233.92(8)	0.00	(234.001668)(12)	(0.000539)	−0.08
$5P_{3/2}$–$5D_{5/2}$	232(3)	[106]	231.1(13)	1	(232.819 43)(4)	(0.00016)	−1.7
$5D_{5/2}$–$5F_{7/2}$	117.2(15)[b]	[106]			(116.582 406)(11)	(0.000246)	0.6
$6P_{1/2}$–$6S_{1/2}$	[39.494](23)[b]	Extrap. 1			(39.48442)(9)	(0.00007)	0.010
$6P_{1/2}$–$6P_{3/2}$	[406.261](8)[b]	Extrap. 3			(406.25987)(3)	(0.00010)	0.001
$6D_{3/2}$–$6D_{5/2}$	[135.39](4)[b]	Interp. 2			(135.417614)(6)	(0.000312)	−0.03
$7P_{1/2}$–$7S_{1/2}$	[24.882](20)[b]	Extrap. 1			(24.87235)(6)	(0.00004)	0.010
$7P_{1/2}$–$7P_{3/2}$	[255.838](5)[b]	Extrap. 3			(255.836904)(21)	(0.000065)	0.001
$7D_{3/2}$–$7D_{5/2}$	[85.271](20)[b]	Interp. 2			(85.277547)(4)	(0.000196)	−0.007
$8P_{1/2}$–$8S_{1/2}$	[16.674](16)[b]	Extrap. 1			(16.66585)(4)	(0.00003)	0.008
$8P_{1/2}$–$8P_{3/2}$	[171.391](4)[b]	Extrap. 3			(171.390621)(15)	(0.000044)	0.000
$8D_{3/2}$–$8D_{5/2}$		P	57.134(10)		(57.129283)(3)	(0.000 132)	0.005
$9P_{1/2}$–$9S_{1/2}$	[11.714](13)[b]	Extrap. 1			(11.70657)(3)	(0.00002)	0.007
$9P_{1/2}$–$9P_{3/2}$	[120.374](3)[b]	Extrap. 3			(120.373042)(10)	(0.000031)	0.001
$9D_{3/2}$–$9D_{5/2}$	[40.131](11)[b]	Interp. 2			(40.1237126)(19)	(0.0000923)	0.007
$10P_{1/2}$–$10S_{1/2}$	[8.541](11)[b]	Extrap. 1			(8.534937)(21)	(0.000015)	0.006
$10P_{1/2}$–$10P_{3/2}$	[87.7524](20)[b]	Extrap. 3			(87.751906)(8)	(0.000023)	0.0005
$10D_{3/2}$–$10D_{5/2}$	[29.258](13)[b]	Interp. 2			(29.2501812)(14)	(0.0000673)	0.008
$11P_{1/2}$–$11S_{1/2}$	[6.418](9)[b]	Extrap. 1			(6.412897)(16)	(0.000011)	0.005
$11P_{1/2}$–$11P_{3/2}$	[65.9297](16)[b]	Extrap. 3			(65.929279)(6)	(0.000017)	0.0004
$11D_{3/2}$–$11D_{5/2}$	[21.984](14)[b]	Interp. 2			(21.9760905)(11)	(0.0000506)	0.008
$12P_{1/2}$–$12S_{1/2}$	[4.944](7)[b]	Extrap. 1			(4.939843)(12)	(0.000009)	0.004
$12P_{1/2}$–$12P_{3/2}$	[50.7826](12)[b]	Extrap. 3			(50.782314)(4)	(0.000013)	0.0003
$12D_{3/2}$–$12D_{5/2}$		P	16.931(12)		(16.9271830)(7)	(0.0000390)	0.004

[a] These values were not included in the least-squares level optimization procedure.
[b] This transition alone determines one of the levels involved.

Table 6
Optimized energy levels of deuterium derived from available experimental transition frequencies compared with QED calculations of Jentschura et al. [10]. See page 622 for Explanation of Tables.

LS term	*J*	E_{exp} (MHz)		Unc. (MHz)	Num. trans.	E_{th} (MHz)	ΔE_{hfs} (MHz)	$\Delta E_{exp\text{-}th}$ (MHz)
1S	1/2	0.000 00		–	1	0.000 00	–	–
2P	1/2	2 466 731 348.24		0.06	1	(2 466 731 348.302 2)(24)	0.013 5	−0.06
2S	1/2	2 466 732 407.521 71		0.000 15	7	(2 466 732 407.521 74)(16)	–	−0.000 03
2P	3/2	2 466 742 320.1		0.3	2	(2 466 742 320.338 4)(24)	0.014 2	−0.2
3P	1/2	2 923 538 219.24	+*y*	0.3	3	(2 923 538 219.106 4)(7)	0.004 0	0.1
3S	1/2	2 923 538 534.6	+*y*	0.5	2	(2 923 538 534.391 8)(14)	–	0.2
3D	3/2	2 923 541 462.4	+*x*	1.8	1	(2 923 541 464.742 2)(7)	0.000 5	−2.3
3P	3/2	2 923 541 470.1	+*y*	0.3	3	(2 923 541 470.083 1)(7)	0.004 2	0.0
3D	5/2	2 923 542 546.0	+*x*	1.6	1	(2 923 542 548.375 7)(7)	0.001 1	−2.4
4P	1/2	3 083 420 177.53		0.17	3	(3 083 420 177.863 0)(18)	0.001 7	−0.33
4S	1/2	3 083 420 311.095		0.020	2	(3 083 420 311.112 4)(20)	–	−0.017
4D	3/2	[3 083 421 547.07]		0.23	1	(3 083 421 547.075 7)(18)	0.000 2	−0.01
4P	3/2	3 083 421 549.28		0.15	2	(3 083 421 549.367 2)(18)	0.001 8	−0.09
4D	5/2	3 083 422 004.24		0.04	1	(3 083 422 004.234 1)(18)	0.000 5	0.01
5P	1/2	[3 157 422 490.52]		6	1	(3 157 422 494.051 3)(22)	0.000 9	−4
5S	1/2	[3 157 422 559]		6	1	(3 157 422 562.334 8)(24)	–	−3
5P	3/2	[3 157 423 192.83]		6	1	(3 157 423 196.260 6)(22)	0.000 9	−3
5D	3/2	[3 157 423 199.03]		2.3	1	(3 157 423 195.076 7)(22)	0.000 1	4.0
5D	5/2	[3 157 423 433.10]		2.3	1	(3 157 423 429.141 8)(22)	0.000 2	4.0
6P	1/2	[3 197 621 201.33]		6	1	(3 197 621 203.609)(3)	0.000 5	−2
6S	1/2	[3 197 621 241]		6	1	(3 197 621 243.145)(3)	–	−2
6P	3/2	[3 197 621 607.77]		6	1	(3 197 621 609.980)(3)	0.000 5	−2
6D	3/2	[3 197 621 610.971]		2.3	1	(3 197 621 609.291)(3)	0.000 1	1.7
6D	5/2	[3 197 621 746.429]		2.3	1	(3 197 621 744.746)(3)	0.000 1	1.7
7P	1/2	[3 221 859 720.00]		6	1	(3 221 859 720.838)(3)	0.000 3	−1
7S	1/2	[3 221 859 745]		6	1	(3 221 859 745.743)(3)	–	−1
7P	3/2	[3 221 859 975.95]		6	1	(3 221 859 976.745)(3)	0.000 3	−1
7D	3/2	[3 221 859 976.488]		2.3	1	(3 221 859 976.310)(3)	0.000 0	0.2
7D	5/2	[3 221 860 061.791]		2.3	1	(3 221 860 061.610)(3)	0.000 1	0.2
8P	1/2	[3 237 591 432.01]		0.03	1	(3 237 591 432.077)(3)	0.000 2	−0.07
8S	1/2	3 237 591 448.768		0.007	1	(3 237 591 448.764)(3)	–	0.004
8D	3/2	3 237 591 603.224		0.006	1	(3 237 591 603.222)(3)	0.000 0	0.002
8P	3/2	[3 237 591 603.48]		0.04	1	(3 237 591 603.514)(3)	0.000 2	−0.03
8D	5/2	3 237 591 660.371		0.006	1	(3 237 591 660.367)(3)	0.000 1	0.004
9P	1/2	[3 248 377 032.22]		6	1	(3 248 377 031.247)(3)	0.000 1	1
9S	1/2	[3 248 377 044]		6	1	(3 248 377 042.969)(3)	–	1
9D	3/2	[3 248 377 151.356]		2.3	1	(3 248 377 151.448)(3)	0.000 0	−0.1
9P	3/2	[3 248 377 152.650]		6	1	(3 248 377 151.653)(3)	0.000 2	1
9D	5/2	[3 248 377 191.492]		2.3	1	(3 248 377 191.582)(3)	0.000 0	−0.1
10P	1/2	[3 256 091 901.411]		6	1	(3 256 091 900.891)(3)	0.000 1	1
10S	1/2	[3 256 091 910]		6	1	(3 256 091 909.437)(3)	–	1
10D	3/2	[3 256 091 988.501]		0.04	1	(3 256 091 988.517)(3)	0.000 0	−0.02
10P	3/2	[3 256 091 989.205]		6	1	(3 256 091 988.667)(3)	0.000 1	1
10D	5/2	3 256 092 017.76		0.04	1	(3 256 092 017.775)(3)	0.000 0	−0.01
11P	1/2	[3 261 800 028.545]		6	1	(3 261 800 028.244)(3)	0.000 1	0
11S	1/2	[3 261 800 035]		6	1	(3 261 800 034.665)(3)	–	0
11D	3/2	[3 261 800 094.117]		2.3	1	(3 261 800 094.078)(3)	0.000 0	0.0
11P	3/2	[3 261 800 094.506]		6	1	(3 261 800 094.191)(3)	0.000 1	0
11D	5/2	[3 261 800 116.1]		2.3	1	(3 261 800 116.060)(3)	0.000 0	0.0
12P	1/2	[3 266 141 526.027]		6	1	(3 266 141 524.854)(3)	0.000 1	1
12S	1/2	[3 266 141 531]		6	1	(3 266 141 529.800)(3)	–	1
12D	3/2	3 266 141 575.560		0.009	1	(3 266 141 575.563)(3)	0.000 0	−0.003
12P	3/2	[3 266 141 576.834]		6	1	(3 266 141 575.650)(3)	0.000 1	1
12D	5/2	3 266 141 592.489		0.007	1	(3 266 141 592.495)(3)	0.000 0	−0.006
Limit		[3 288 981 521.1]		2.3	–	(3 288 981 522.062)(3)	–	−1.0

Table 7
Frequencies of fine-structure transitions in deuterium derived from the level-optimization procedure compared with QED calculations of Jentschura et al. [10]. See page 622 for Explanation of Tables.

Transition	E_{exp} (MHz)	Ref.	E_{Ritz} (MHz)		$\Delta E_{exp-Ritz}$(MHz)	E_{th} (MHz)	ΔE_{hfs} (MHz)	$\Delta E_{Ritz-th}$ (MHz)
$1S_{1/2}$–$2P_{1/2}$		P	2 466 731 348.24(6)			(2 466 731 348.302 2)(24)	(0.013 5)	−0.06
$1S_{1/2}$–$2S_{1/2}$	2 466 732 407.521 71(15)	[152]	2 466 732 407.521 71(15)		0.000 00	(2 466 732 407.521 74)(16)	–	−0.000 03
$1S_{1/2}$–$2P_{3/2}$	2 466 742 200(1 000)[a]	[164][a]	2 466 742 320.1(3)		−120	(2 466 742 320.338 4)(24)	(0.014 2)	−0.2
$1S_{1/2}$–$3P_{1/2}$		P	2 923 538 219.2(3)	+y		(2 923 538 219.106 4)(7)	(0.004 0)	0.1
$1S_{1/2}$–$3S_{1/2}$		P	2 923 538 534.6(5)	+y		(2 923 538 534.391 8)(14)	–	0.2
$1S_{1/2}$–$3P_{3/2}$		P	2 923 541 470.1(3)	+y		(2 923 541 470.083 1)(7)	(0.004 2)	0.0
$1S_{1/2}$–$4P_{1/2}$		P	3 083 420 177.53(17)			(3 083 420 177.863 0)(18)	(0.001 7)	−0.33
$1S_{1/2}$–$4S_{1/2}$		P	3 083 420 311.095(20)			(3 083 420 311.112 4)(20)	–	−0.017
$1S_{1/2}$–$4P_{3/2}$		P	3 083 421 549.28(15)			(3 083 421 549.367 2)(18)	(0.001 8)	−0.09
$1S_{1/2}$–$5P_{1/2}$		P	3 157 422 491(7)			(3 157 422 494.051 3)(22)	(0.000 9)	−3
$1S_{1/2}$–$5S_{1/2}$	[3 157 422 559](6)[b]	Ritz fit1				(3 157 422 562.334 8)(24)	–	−3
$1S_{1/2}$–$5P_{3/2}$		P	3 157 423 193(7)			(3 157 423 196.260 6)(22)	(0.000 9)	−3
$1S_{1/2}$–$5D_{5/2}$	[3 157 423 433.2](23)	Ritz fit2	3 157 423 433.1(23)		0.1	(3 157 423 429.141 8)(22)	(0.000 2)	4.0
$1S_{1/2}$–$6P_{1/2}$		P	3 197 621 201(7)			(3 197 621 203.609)(3)	(0.001)	−3
$1S_{1/2}$–$6S_{1/2}$	[3 197 621 241](6)[b]	Ritz fit1				(3 197 621 243.145)(3)	–	−2
$1S_{1/2}$–$6P_{3/2}$		P	3 197 621 608(7)			(3 197 621 609.980)(3)	(0.001)	−2
$1S_{1/2}$–$6D_{5/2}$	[3 197 621 746.3](23)	Ritz fit2	3 197 621 746.4(23)		−0.1	(3 197 621 744.746)(3)	(0.000)	1.7
$1S_{1/2}$–$7P_{1/2}$		P	3 221 859 720(6)			(3 221 859 720.838)(3)	(0.000)	−1
$1S_{1/2}$–$7S_{1/2}$	[3 221 859 745](6)[b]	Ritz fit1				(3 221 859 745.743)(3)	–	−1
$1S_{1/2}$–$7P_{3/2}$		P	3 221 859 976(6)			(3 221 859 976.745)(3)	(0.000)	−1
$1S_{1/2}$–$7D_{5/2}$	[3 221 860 061.9](23)	Ritz fit2	3 221 860 061.8(23)		0.1	(3221860061.610)(3)	(0.000)	0.2
$1S_{1/2}$–$8P_{1/2}$		P	3 237 591 432.01(3)			(3 237 591 432.077)(3)	(0.000)	−0.07
$1S_{1/2}$–$8S_{1/2}$		P	3 237 591 448.768(7)			(3 237 591 448.764)(3)	–	0.004
$1S_{1/2}$–$8P_{3/2}$		P	3 237 591 603.48(4)			(3 237 591 603.514)(3)	(0.000)	−0.03
$1S_{1/2}$–$9P_{1/2}$		P	3 248 377 032(7)			(3 248 377 031.247)(3)	(0.000)	1
$1S_{1/2}$–$9S_{1/2}$	[3 248 377 044](6)[b]	Ritz fit1				(3 248 377 042.969)(3)	–	1
$1S_{1/2}$–$9P_{3/2}$		P	3 248 377 153(7)			(3 248 377 151.653)(3)	(0.000)	1
$1S_{1/2}$–$9D_{5/2}$	[3 248 377 191.5](23)	Ritz fit2	3 248 377 191.5(23)		0.0	(3 248 377 191.582)(3)	(0.000)	−0.1
$1S_{1/2}$–$10P_{1/2}$		P	3 256 091 901(7)			(3 256 091 900.891)(3)	(0.000)	0
$1S_{1/2}$–$10S_{1/2}$	[3 256 091 910](6)[b]	Ritz fit1				(3 256 091 909.437)(3)	–	1
$1S_{1/2}$–$10P_{3/2}$		P	3 256 091 989(7)			(3 256 091 988.667)(3)	(0.000)	0
$1S_{1/2}$–$11P_{1/2}$		P	3 261 800 029(7)			(3 261 800 028.244)(3)	(0.000)	1
$1S_{1/2}$–$11S_{1/2}$	[3 261 800 035](6)[b]	Ritz fit1				(3 261 800 034.665)(3)	–	0
$1S_{1/2}$–$11P_{3/2}$		P	3 261 800 095(7)			(3 261 800 094.191)(3)	(0.000)	1
$1S_{1/2}$–$11D_{5/2}$	[3 261 800 116.1](23)[b]	Ritz fit2				(3 261 800 116.060)(3)	(0.000)	0.0
$1S_{1/2}$–$12P_{1/2}$		P	3 266 141 526(6)			(3 266 141 524.854)(3)	(0.000)	1
$1S_{1/2}$–$12S_{1/2}$	[3 266 141 531](6)[b]	Ritz fit1				(3 266 141 529.800)(3)	–	1
$1S_{1/2}$–$12P_{3/2}$		P	3 266 141 577(7)			(3 266 141 575.650)(3)	(0.000)	1
$2P_{1/2}$–$2S_{1/2}$	1059.28(6)	[153,155]	1059.28(6)		0.00	(1059.219 5)(24)	(−0.013 5)	0.06
$2P_{1/2}$–$2P_{3/2}$		P	10 971.9(4)			(10 972.036 2)(9)	(0.000 7)	−0.1
$2P_{1/2}$–$3S_{1/2}$		P	456 807 186.4(5)	+y		(456 807 186.090)(4)	(−0.014)	0.3
$2P_{1/2}$–$3D_{3/2}$	456 810 113.8(19) + x	[157]	456 810 114.2(18)	+x	−0.4	(456 810 116.440)(3)	(−0.013)	−2.2
$2P_{1/2}$–$4S_{1/2}$		P	616 688 962.86(7)			(616 688 962.810)(4)	(−0.014)	0.06
$2P_{1/2}$–$4D_{3/2}$	616 690 180(40)	[162][c]	616 690 198.83(24)		−19	(616 690 198.774)(4)	(−0.013)	0.06
$2P_{1/2}$–$5S_{1/2}$		P	690 691 211(7)			(690 691 214.032)(5)	(−0.014)	−3
$2P_{1/2}$–$5D_{3/2}$	690 691 810(50)	[162][c]	690 691 850.8(23)		−40	(690 691 846.774)(5)	(−0.013)	4.0
$2P_{1/2}$–$6S_{1/2}$		P	730 889 893(7)			(730889894.842)(5)	(−0.014)	−2
$2P_{1/2}$–$6D_{3/2}$	730 890 320(60)	[162][c]	730 890 262.7(23)		60	(730 890 260.989)(5)	(−0.013)	1.7
$2P_{1/2}$–$7S_{1/2}$		P	755 128 397(7)			(755 128 397.440)(5)	(−0.014)	0
$2P_{1/2}$–$7D_{3/2}$	755 128 600(60)	[162][c]	755 128 628.2(23)		−30	(755 128 628.007)(5)	(−0.013)	0.2
$2P_{1/2}$–$8S_{1/2}$		P	770 860 100.53(6)			(770 860 100.462)(5)	(−0.014)	0.07
$2P_{1/2}$–$8D_{3/2}$	770 860 360(210)	[162][c]	770 860 254.98(6)		110	(770860254.920)(5)	(−0.014)	0.06
$2P_{1/2}$–$9S_{1/2}$		P	781 645 696(7)			(781 645 694.667)(5)	(−0.014)	1
$2P_{1/2}$–$9D_{3/2}$	781 645 760(300)	[162][c]	781 645 803.1(23)		−40	(781 645 803.145)(5)	(−0.014)	0.0
$2P_{1/2}$–$10S_{1/2}$		P	789 360 562(7)			(789 360 561.135)(5)	(−0.014)	1
$2P_{1/2}$–$10D_{3/2}$		P	789 360 640.26(7)			(789 360 640.215)(5)	(−0.014)	0.05
$2P_{1/2}$–$11S_{1/2}$		P	795 068 687(7)			(795 068 686.362)(5)	(−0.014)	1
$2P_{1/2}$–$11D_{3/2}$		P	795 068 745.9(23)			(795 068 745.776)(5)	(−0.014)	0.1
$2P_{1/2}$–$12S_{1/2}$		P	799 410 183(7)			(799 410 181.498)(5)	(−0.014)	2
$2P_{1/2}$–$12D_{3/2}$		P	799 410 227.32(6)			(799 410 227.261)(5)	(−0.014)	0.06
$2S_{1/2}$–$2P_{3/2}$	9912.6(3)	[156,47][d]	9 912.6(3)		0.0	(9 912.816 7)(24)	(0.014 2)	−0.2
$2S_{1/2}$–$3P_{1/2}$	456 805 811.7(3) + y	[111]	456 805 811.7(3)	+y	0.0	(456 805 811.584 7)(7)	(0.004 0)	0.1
$2S_{1/2}$–$3S_{1/2}$		P	456 806 127.1(5)	+y		(456 806 126.870 1)(14)	–	0.2
$2S_{1/2}$–$3P_{3/2}$	456 809 062.6(3) + y	[111]	456 809 062.6(3)	+y	0.0	(456 809 062.561 4)(7)	(0.004 2)	0.0
$2S_{1/2}$–$4P_{1/2}$	616 687 769.99(19)	[121]	616 687 770.01(17)		0.01	(616 687 770.341 3)(17)	(0.001 7)	−0.33
$2S_{1/2}$–$4S_{1/2}$	616 687 903.573(20)	[101]	616 687 903.573(20)		0.000	(616 687 903.590 7)(20)	–	−0.018
$2S_{1/2}$–$4P_{3/2}$	616 689 141.73(17)	[121]	616 689 141.76(16)		−0.01	(616 689 141.845 5)(17)	(0.001 8)	−0.09
$2S_{1/2}$–$4D_{5/2}$	616 689 596.72(4)	[101]	616 689 596.72(4)		0.00	(616 689 596.712 4)(17)	(0.000 5)	0.01
$2S_{1/2}$–$5P_{1/2}$		P	690 690 083(6)			(690 690 086.529 6)(22)	(0.000 9)	−4
$2S_{1/2}$–$5P_{3/2}$		P	690 690 785(7)			(690 690 788.738 8)(22)	(0.000 9)	−4
$2S_{1/2}$–$6P_{1/2}$		P	730 888 794(7)			(730 888 796.088)(3)	(0.001)	−2
$2S_{1/2}$–$6P_{3/2}$		P	730 889 200(7)			(730 889 202.458)(3)	(0.001)	−2

Table 7 (*continued*)

Transition	E_{exp} (MHz)	Ref.	E_{Ritz} (MHz)		$\Delta E_{exp-Ritz}$(MHz)	E_{th} (MHz)	ΔE_{hfs} (MHz)	$\Delta E_{Ritz\text{-}th}$ (MHz)
$2S_{1/2}-7P_{1/2}$		P	755 127 312(7)			(755 127 313.316)(3)	(0.000)	−1
$2S_{1/2}-7P_{3/2}$		P	755 127 568(7)			(755 127 569.223)(3)	(0.000)	−1
$2S_{1/2}-8P_{1/2}$		P	770 859 024.49(3)			(770 859 024.555)(3)	(0.000)	−0.07
$2S_{1/2}-8S_{1/2}$	770 859 041.246(7)[b]	[103]				(770 859 041.243)(3)	–	0.003
$2S_{1/2}-8D_{3/2}$	770 859 195.702(6)	[103]	770 859 195.702(7)		0.000	(770 859 195.700)(3)	(0.000)	0.002
$2S_{1/2}-8P_{3/2}$		P	770 859 195.96(4)			(770 859 195.993)(3)	(0.000)	−0.03
$2S_{1/2}-8D_{5/2}$	770 859 252.850(6)	[103]	770 859 252.849(6)		0.001	(770 859 252.845)(3)	(0.000)	0.004
$2S_{1/2}-9P_{1/2}$		P	781 644 625(7)			(781 644 623.726)(3)	(0.000)	1
$2S_{1/2}-9P_{3/2}$		P	781 644 745(7)			(781 644 744.132)(3)	(0.000)	1
$2S_{1/2}-10P_{1/2}$		P	789 359 494(7)			(789 359 493.370)(3)	(0.000)	1
$2S_{1/2}-10P_{3/2}$		P	789 359 582(7)			(789 359 581.145)(3)	(0.000)	1
$2S_{1/2}-10D_{5/2}$	789 359 610.24(4)[b]	[118]				(789 359 610.253)(3)	(0.000)	−0.01
$2S_{1/2}-11P_{1/2}$		P	795 067 621(6)			(795 067 620.722)(3)	(0.000)	0
$2S_{1/2}-11P_{3/2}$		P	795 067 687(6)			(795 067 686.669)(3)	(0.000)	0
$2S_{1/2}-12P_{1/2}$		P	799 409 119(7)			(799 409 117.332)(3)	(0.000)	2
$2S_{1/2}-12D_{3/2}$	799 409 168.038(9)[b]	[103]				(799 409 168.042)(3)	(0.000)	−0.004
$2S_{1/2}-12P_{3/2}$		P	799 409 169(7)			(799 409 168.129)(3)	(0.000)	1
$2S_{1/2}-12D_{5/2}$	799 409 184.967(7)[b]	[103]				(799 409 184.973)(3)	(0.000)	−0.006
$2P_{3/2}-3S_{1/2}$	456 796 251(30)	[158]	456 796 214.5(6)	+y	40	(456 796 214.053)(4)	(−0.014)	0.4
$2P_{3/2}-3D_{3/2}$		P	456 799 142.3(18)	+x		(456 799 144.404)(3)	(−0.014)	−2.1
$2P_{3/2}-3D_{5/2}$	456 800 225.9(16) + x	[157]				(456 800 228.037)(3)	(−0.013)	−2.1
$2P_{3/2}-4S_{1/2}$		P	616 677 991.0(3)			(616 677 990.774)(4)	(−0.014)	0.2
$2P_{3/2}-4D_{3/2}$		P	616 679 227.0(4)			(616 679 226.737)(4)	(−0.014)	0.3
$2P_{3/2}-4D_{5/2}$	616 679 760(50)	[162][c]	616 679 684.1(3)		80	(616 679 683.896)(4)	(−0.014)	0.2
$2P_{3/2}-5S_{1/2}$		P	690 680 239(7)			(690680241.996)(5)	(−0.014)	−3
$2P_{3/2}-5D_{3/2}$		P	690 680 878.9(23)			(690680874.738)(5)	(−0.014)	4.2
$2P_{3/2}-5D_{5/2}$	690 681 100(70)	[162][c]	690 681 113.0(23)		−10	(690 681 108.803)(5)	(−0.014)	4.2
$2P_{3/2}-6S_{1/2}$		P	730 878 921(7)			(730 878 922.806)(5)	(−0.014)	−2
$2P_{3/2}-6D_{3/2}$		P	730 879 290.9(23)			(730 879 288.953)(5)	(−0.014)	1.9
$2P_{3/2}-6D_{5/2}$	730 879 480(80)	[162][c]	730 879 426.3(23)		50	(730879424.407)(5)	(−0.014)	1.9
$2P_{3/2}-7S_{1/2}$		P	755 117 425(7)			(755 117 425.404)(5)	(−0.014)	0
$2P_{3/2}-7D_{3/2}$		P	755117 656.4(23)			(755 117 655.971)(5)	(−0.014)	0.4
$2P_{3/2}-7D_{5/2}$	755 117 710(50)	[162][c]	755 117 741.7(23)		−30	(755 117 741.272)(5)	(−0.014)	0.4
$2P_{3/2}-8S_{1/2}$		P	770 849 128.7(3)			(770 849 128.426)(5)	(−0.014)	0.3
$2P_{3/2}-8D_{3/2}$		P	770 849 283.1(3)			(770 849 282.884)(5)	(−0.014)	0.2
$2P_{3/2}-8D_{5/2}$	770 849 570(210)	[162][c]	770 849 340.3(3)		230	(770 849 340.028)(5)	(−0.014)	0.3
$2P_{3/2}-9S_{1/2}$		P	781 634 724(7)			(781 634 722.631)(5)	(−0.014)	1
$2P_{3/2}-9D_{3/2}$		P	781634831.3(24)			(781 634 831.109)(5)	(−0.014)	0.2
$2P_{3/2}-9D_{5/2}$	781 634 790(300)	[162][c]	781 634 871.4(23)		−100	(781 634 871.244)(5)	(−0.014)	0.2
$2P_{3/2}-10S_{1/2}$		P	789 349 590(7)			(789 349 589.099)(5)	(−0.014)	1
$2P_{3/2}-10D_{3/2}$		P	789 349 668.4(3)			(789 349 668.179)(5)	(−0.014)	0.2
$2P_{3/2}-10D_{5/2}$		P	789 349 697.7(3)			(789 349 697.437)(5)	(−0.014)	0.3
$2P_{3/2}-11S_{1/2}$		P	795 057 715(7)			(795,057,714.326)(5)	(−0.014)	1
$2P_{3/2}-11D_{3/2}$		P	795 057 774.0(23)			(795 057 773.740)(5)	(−0.014)	0.3
$2P_{3/2}-11D_{5/2}$		P	795 057 796.0(23)			(795 057 795.722)(5)	(−0.014)	0.3
$2P_{3/2}-12S_{1/2}$		P	799 399 211(7)			(799 399 209.462)(5)	(−0.014)	2
$2P_{3/2}-12D_{3/2}$		P	799 399 255.5(3)			(799 399 255.225)(5)	(−0.014)	0.3
$2P_{3/2}-12D_{5/2}$		P	799 399 272.4(3)			(799 399 272.157)(5)	(−0.014)	0.2
$3P_{1/2}-3S_{1/2}$	315.3(4)	[159]	315.4(4)		−0.1	(315.2854)(7)	(−0.004 0)	0.1
$3P_{1/2}-3D_{3/2}$		P	3 243.2(18)	+x − y		(3 245.635 76)(19)	(−0.003 56)	−2.4
$3P_{1/2}-3P_{3/2}$	3 250.7(10)	[159]	3 250.9(4)	− y	−0.2	(3 250.976 7)(3)	(0.000 2)	−0.1
$3P_{1/2}-4D_{3/2}$		P	159 883 327.8(4)	− y		(159 883 327.969 3)(11)	(−0.003 8)	−0.2
$3P_{1/2}-5D_{3/2}$		P	233 884 979.8(23)	− y		(233 884 975.970 3)(16)	(−0.003 9)	3.8
$3P_{1/2}-6D_{3/2}$		P	274 083 391.7(23)	− y		(274 083 390.184 8)(18)	(−0.004 0)	1.5
$3P_{1/2}-7D_{3/2}$		P	298 321 757.2(24)	− y		(298 321 757.203 1)(20)	(−0.004 0)	0.0
$3P_{1/2}-8D_{3/2}$		P	314 053 384.0(3)	− y		(314 053 384.115 7)(21)	(−0.004 0)	−0.1
$3P_{1/2}-9D_{3/2}$		P	324 838 932.1(23)	− y		(324 838 932.341 2)(22)	(−0.004 0)	−0.2
$3P_{1/2}-10D_{3/2}$		P	332 553 769.3(3)	− y		(332 553 769.410 0)(22)	(−0.004 0)	−0.1
$3P_{1/2}-11D_{3/2}$		P	338 26 1874.9(23)	− y		(338 261 874.971 6)(22)	(−0.004 0)	−0.1
$3P_{1/2}-12D_{3/2}$		P	342 603 356.3(3)	− y		(342 603 356.457 0)(23)	(−0.004 0)	−0.2
$3S_{1/2}-3P_{3/2}$	2 935(5)	[160]	2 935.5(6)	− y	−1	(2 935.691 3)(7)	(0.004 2)	−0.2
$3S_{1/2}-4P_{1/2}$		P	159 881 642.9(5)	− y		(159 881 643.471 2)(4)	(0.001 7)	−0.6
$3S_{1/2}-4P_{3/2}$		P	159 883 014.7(5)	− y		(159 883 014.975 4)(4)	(0.001 8)	−0.3
$3S_{1/2}-5P_{1/2}$		P	233 883 956(7)	− y		(233 883 959.659 5)(9)	(0.000 9)	−4
$3S_{1/2}-5P_{3/2}$		P	233 884 658(7)	− y		(233 884 661.868 8)(9)	(0.000 9)	−4
$3S_{1/2}-6P_{1/2}$		P	274 082 667(7)	− y		(274 082 669.217 6)(11)	(0.000 5)	−2
$3S_{1/2}-6P_{3/2}$		P	274 083 073(7)	− y		(274 083 075.588 2)(11)	(0.000 5)	−3
$3S_{1/2}-7P_{1/2}$		P	298 321 185(7)	− y		(298 321 186.446 3)(13)	(0.000 3)	−1
$3S_{1/2}-7P_{3/2}$		P	298 321 441(7)	− y		(298 321 442.352 9)(13)	(0.000 3)	−1
$3S_{1/2}-8P_{1/2}$		P	314052897.4(5)	− y		(314 052 897.685 2)(14)	(0.000 2)	−0.3
$3S_{1/2}-8P_{3/2}$		P	314053068.9(5)	− y		(314 053 069.122 5)(14)	(0.000 2)	−0.2
$3S_{1/2}-9P_{1/2}$		P	324 838 498(7)	− y		(324 838 496.855 5)(15)	(0.000 1)	1
$3S_{1/2}-9P_{3/2}$		P	324 838 618(6)	− y		(324 838 617.261 5)(15)	(0.000 2)	1
$3S_{1/2}-10P_{1/2}$		P	332 553 367(7)	− y		(332 553 366.499 5)(15)	(0.000 1)	1

(*continued on next page*)

Table 7 (*continued*)

Transition	E_{exp} (MHz)	Ref.	E_{Ritz} (MHz)		$\Delta E_{exp-Ritz}$(MHz)	E_{th} (MHz)	ΔE_{hfs} (MHz)	$\Delta E_{Ritz-th}$ (MHz)
$3S_{1/2}$–$10P_{3/2}$		P	332 553 455(7)	− y		(332 553 454.275 3)(15)	(0.000 1)	1
$3S_{1/2}$–$11P_{1/2}$		P	338 261 494(6)	− y		(338 261 493.851 8)(15)	(0.000 1)	0
$3S_{1/2}$–$11P_{3/2}$		P	338 261 560(7)	− y		(338 261 559.799 0)(15)	(0.000 1)	0
$3S_{1/2}$–$12P_{1/2}$		P	342 602 991(7)	− y		(342 602 990.462 5)(16)	(0.000 1)	1
$3S_{1/2}$–$12P_{3/2}$		P	342 603 042(7)	− y		(342 603 041.258 6)(16)	(0.000 1)	1
$3D_{3/2}$–$3P_{3/2}$	5(5)	[159]	7.7(18)	+y − x	−3	(5.340 89)(19)	(0.003 76)	2.4
$3D_{3/2}$–$3D_{5/2}$		P	1 083.6(24)			(1 083.633 53)(4)	(0.000 63)	0.0
$3P_{3/2}$–$3D_{5/2}$		P	1 075.9(17)	+x − y		(1 078.292 63)(19)	(−0.003 14)	−2.4
$3P_{3/2}$–$4D_{3/2}$		P	159 880 077.0(4)	-y		(159 880 076.992 6)(11)	(−0.004 0)	0.0
$3P_{3/2}$–$4D_{5/2}$		P	159 880 534.1(3)	-y		(159 880 534.151 0)(11)	(−0.003 8)	−0.1
$3P_{3/2}$–$5D_{3/2}$		P	233 881 728.9(23)	-y		(233 881 724.993 6)(16)	(−0.004 1)	3.9
$3P_{3/2}$–$5D_{5/2}$		P	233 881 963.0(23)	-y		(233 881 959.058 7)(16)	(−0.004 0)	3.9
$3P_{3/2}$–$6D_{3/2}$		P	274 080 140.9(23)	-y		(274 080 139.208 2)(18)	(−0.004 2)	1.7
$3P_{3/2}$–$6D_{5/2}$		P	274 080 276.3(23)	-y		(274 080 274.662 5)(18)	(−0.004 1)	1.6
$3P_{3/2}$–$7D_{3/2}$		P	298 318 506.4(23)	-y		(298 318 506.226 4)(20)	(−0.004 2)	0.2
$3P_{3/2}$–$7D_{5/2}$		P	298 318 591.7(23)	-y		(298 318 591.527 2)(20)	(−0.004 1)	0.2
$3P_{3/2}$–$8D_{3/2}$		P	314 050 133.1(3)	-y		(314 050 133.139 0)(21)	(−0.004 2)	0.0
$3P_{3/2}$–$8D_{5/2}$		P	314 050 190.3(3)	-y		(314 050 190.283 7)(21)	(−0.004 2)	0.0
$3P_{3/2}$–$9D_{3/2}$		P	324 835 681.3(24)	-y		(324 835 681.364 6)(22)	(−0.004 2)	−0.1
$3P_{3/2}$–$9D_{5/2}$		P	324 835 721.4(23)	-y		(324 835 721.499 1)(22)	(−0.004 2)	−0.1
$3P_{3/2}$–$10D_{3/2}$		P	332 550 518.4(3)	-y		(332 550 518.434 0)(22)	(−0.004 2)	0.0
$3P_{3/2}$–$10D_{5/2}$		P	332 550 547.7(3)	-y		(332 550 547.692 1)(22)	(−0.004 2)	0.0
$3P_{3/2}$–$11D_{3/2}$		P	338 258 624.0(23)	-y		(338 258 623.994 9)(22)	(−0.004 2)	0.0
$3P_{3/2}$–$11D_{5/2}$		P	338 258 646.0(23)	-y		(338 258 645.977 0)(22)	(−0.004 2)	0.0
$3P_{3/2}$–$12D_{3/2}$		P	342 600 105.5(3)	-y		(342 600 105.480 3)(23)	(−0.004 2)	0.0
$3P_{3/2}$–$12D_{5/2}$		P	342 600 122.4(3)	-y		(342 600 122.412 1)(23)	(−0.004 2)	0.0
$4P_{1/2}$–$4S_{1/2}$	133(6)	[159]	133.57(17)		−1	(133.249 4)(3)	(−0.001 7)	0.32
$4P_{1/2}$–$4D_{3/2}$		P	1 369.5(3)			(1 369.212 67)(8)	(−0.001 50)	0.3
$4P_{1/2}$–$4P_{3/2}$	1371.8(3)	[161]	1 371.75(19)		0.0	(1 371.504 17)(11)	(0.000 09)	0.25
$4P_{1/2}$–$5D_{3/2}$		P	74 003 021.5(23)			(74 003 017.213 7)(5)	(−0.001 6)	4.3
$4P_{1/2}$–$6D_{3/2}$		P	114 201 433.4(23)			(114 201 431.428 3)(8)	(−0.001 6)	2.0
$4P_{1/2}$–$7D_{3/2}$		P	138 439 799.0(23)			(138 439 798.446 5)(9)	(−0.001 7)	0.6
$4P_{1/2}$–$8D_{3/2}$		P	154 171 425.69(17)			(154 171 425.359 0)(10)	(−0.001 7)	0.33
$4P_{1/2}$–$9D_{3/2}$		P	164 956 973.8(23)			(164 956 973.584 6)(11)	(−0.001 7)	0.2
$4P_{1/2}$–$10D_{3/2}$		P	172 671 810.97(17)			(172 671 810.654 0)(11)	(−0.001 7)	0.32
$4P_{1/2}$–$11D_{3/2}$		P	178 379 916.6(23)			(178 379 916.215 0)(12)	(−0.001 7)	0.4
$4P_{1/2}$–$12D_{3/2}$		P	182 721 398.03(17)			(182 721 397.700 4)(12)	(−0.001 7)	0.33
$4S_{1/2}$–$4P_{3/2}$		P	1 238.18(16)			(1 238.254 8)(3)	(0.001 8)	−0.07
$4S_{1/2}$–$5P_{1/2}$		P	74 002 179(7)			(74 002 82.938 91)(20)	(0.000 87)	−4
$4S_{1/2}$–$5P_{3/2}$		P	74 002 882(7)			(74 002 885.148 17)(20)	(0.000 91)	−3
$4S_{1/2}$–$6P_{1/2}$		P	114 200 890(7)			(114 200 892.497 0)(5)	(0.000 5)	−2
$4S_{1/2}$–$6P_{3/2}$		P	114 201 297(7)			(114 201 298.867 7)(5)	(0.000 5)	−2
$4S_{1/2}$–$7P_{1/2}$		P	138 439 409(6)			(138 439 409.725 7)(6)	(0.000 3)	−1
$4S_{1/2}$–$7P_{3/2}$		P	138 439 665(7)			(138 439 665.632 4)(6)	(0.000 3)	−1
$4S_{1/2}$–$8P_{1/2}$		P	154171120.92(4)			(154 171 120.964 6)(7)	(0.000 2)	−0.04
$4S_{1/2}$–$8P_{3/2}$		P	154171292.39(5)			(154 171 292.402 0)(7)	(0.000 2)	−0.01
$4S_{1/2}$–$9P_{1/2}$		P	164 956 721(7)			(164 956 720.135 0)(8)	(0.000 1)	1
$4S_{1/2}$–$9P_{3/2}$		P	164 956 842(7)			(164 956 840.540 9)(8)	(0.000 2)	1
$4S_{1/2}$–$10P_{1/2}$		P	172 671 590(7)			(172 671 589.778 9)(9)	(0.000 1)	0
$4S_{1/2}$–$10P_{3/2}$		P	172 671 678(7)			(172 671 677.554 8)(9)	(0.000 1)	0
$4S_{1/2}$–$11P_{1/2}$		P	178 379 717(7)			(178 379 717.131 2)(9)	(0.000 1)	0
$4S_{1/2}$–$11P_{3/2}$		P	178 379 783(7)			(178 379 783.078 5)(9)	(0.000 1)	0
$4S_{1/2}$–$12P_{1/2}$		P	182 721 215(6)			(182 721 213.741 9)(9)	(0.000 1)	1
$4S_{1/2}$–$12P_{3/2}$		P	182 721 266(7)			(182 721 264.538 0)(9)	(0.000 1)	1
$4D_{3/2}$–$4P_{3/2}$		P	2.2(3)			(2.291 50)(8)	(0.001 59)	−0.1
$4D_{3/2}$–$4D_{5/2}$	[457.17](23)	Interp. 2	457.17(23)		0.00	(457.158 405)(15)	(0.000 265)	0.01
$4P_{3/2}$–$4D_{5/2}$		P	454.96(16)			(454.866 90)(8)	(−0.001 32)	0.09
$4P_{3/2}$–$5D_{3/2}$		P	74 001 649.8(24)			(74 001 645.709 5)(5)	(−0.001 7)	4.1
$4P_{3/2}$–$5D_{5/2}$		P	74 001 883.8(23)			(74 001 879.774 7)(5)	(−0.001 5)	4.0
$4P_{3/2}$–$6D_{3/2}$		P	114 200 061.7(23)			(114 200 059.924 1)(8)	(−0.001 7)	1.8
$4P_{3/2}$–$6D_{5/2}$		P	114 200 197.1(24)			(114 200 195.378 4)(8)	(−0.001 6)	1.7
$4P_{3/2}$–$7D_{3/2}$		P	138 438 427.2(23)			(138 438 426.942 4)(9)	(−0.001 7)	0.3
$4P_{3/2}$–$7D_{5/2}$		P	138 438 512.5(23)			(138 438 512.243 0)(9)	(−0.001 7)	0.3
$4P_{3/2}$–$8D_{3/2}$		P	154 170 053.94(16)			(154 170 053.854 9)(10)	(−0.001 8)	0.09
$4P_{3/2}$–$8D_{5/2}$		P	154 170 111.09(15)			(154 170 110.999 7)(10)	(−0.001 7)	0.09
$4P_{3/2}$–$9D_{3/2}$		P	164 955 602.1(23)			(164 955 602.080 4)(11)	(−0.001 8)	0.0
$4P_{3/2}$–$9D_{5/2}$		P	164 955 642.2(23)			(164 955 642.215 1)(11)	(−0.001 7)	0.0
$4P_{3/2}$–$10D_{3/2}$		P	172 670 439.22(16)			(172 670 439.149 9)(11)	(−0.001 8)	0.07
$4P_{3/2}$–$10D_{5/2}$		P	172 670 468.48(16)			(172 670 468.408 1)(11)	(−0.001 7)	0.07
$4P_{3/2}$–$11D_{3/2}$		P	178 378 544.8(23)			(178 378 544.710 8)(12)	(−0.001 8)	0.1
$4P_{3/2}$–$11D_{5/2}$		P	178 378 566.8(23)			(178 378 566.692 8)(12)	(−0.001 8)	0.1
$4P_{3/2}$–$12D_{3/2}$		P	182 720 026.28(15)			(182 720 026.196 2)(12)	(−0.001 8)	0.08
$4P_{3/2}$–$12D_{5/2}$		P	182 720 043.21(15)			(182 720 043.128 0)(12)	(−0.001 8)	0.08
$5P_{1/2}$–$5S_{1/2}$	[68.48](12)[b]	Extrap. 1				(68.283 45)(16)	(−0.000 87)	0.20

Table 7 (*continued*)

Transition	E_{exp} (MHz)	Ref.	E_{Ritz} (MHz)	$\Delta E_{exp-Ritz}$(MHz)	E_{th} (MHz)	ΔE_{hfs} (MHz)	$\Delta E_{Ritz\text{-}th}$ (MHz)
$5P_{1/2}$–$5P_{3/2}$	[702.31](10)[b]	Extrap. 3			(702.209 27)(6)	(0.000 05)	0.10
$5D_{3/2}$–$5D_{5/2}$	[234.07](4)	Interp. 2	234.07(4)	0.00	(234.065 135)(8)	(0.000 135)	0.00
$6P_{1/2}$–$6S_{1/2}$	[39.67](7)[b]	Extrap. 1			(39.535 29)(9)	(−0.000 50)	0.13
$6P_{1/2}$–$6P_{3/2}$	[406.44](6)[b]	Extrap. 3			(406.370 68)(3)	(0.000 03)	0.07
$6D_{3/2}$–$6D_{5/2}$	[135.458](24)	Interp. 2	135.458(24)	0.000	(135.454 343)(4)	(0.000 078)	0.004
$7P_{1/2}$–$7S_{1/2}$	[25.00](5)[b]	Extrap. 1			(24.904 39)(6)	(−0.000 32)	0.10
$7P_{1/2}$–$7P_{3/2}$	[255.95](4)[b]	Extrap. 3			(255.906 685)(21)	(0.000 016)	0.04
$7D_{3/2}$–$7D_{5/2}$	[85.303](15)	Interp. 2	85.303(15)	0.000	(85.300 676)(3)	(0.000 049)	0.002
$8P_{1/2}$–$8S_{1/2}$	[16.76](3)[b]	Extrap. 1			(16.687 31)(4)	(−0.000 21)	0.07
$8P_{1/2}$–$8P_{3/2}$	[171.47](3)[b]	Extrap. 3			(171.437 369)(15)	(0.000 011)	0.03
$8D_{3/2}$–$8D_{5/2}$		P	57.147(9)		(57.144 777)(2)	(0.000 033)	0.002
$9P_{1/2}$–$9S_{1/2}$	[11.78](3)[b]	Extrap. 1			(11.721 65)(3)	(−0.000 15)	0.06
$9P_{1/2}$–$9P_{3/2}$	[120.430](21)[b]	Extrap. 3			(120.405 875)(10)	(0.000 008)	0.024
$9D_{3/2}$–$9D_{5/2}$	[40.136](13)	Interp. 2	40.136(13)	0.000	(40.134 595 3)(13)	(0.000 023 2)	0.001
$10P_{1/2}$–$10S_{1/2}$	[8.589](19)[b]	Extrap. 1			(8.545 926)(21)	(−0.000 108)	0.043
$10P_{1/2}$–$10P_{3/2}$	[87.794](16)[b]	Extrap. 3			(87.775 840)(8)	(0.000 006)	0.018
$10D_{3/2}$–$10D_{5/2}$	[29.259](15)[b]	Interp. 2			(29.258 114 7)(10)	(0.000 016 9)	0.001
$11P_{1/2}$–$11S_{1/2}$	[6.455](15)[b]	Extrap. 1			(6.421 154)(16)	(−0.000 081)	0.034
$11P_{1/2}$–$11P_{3/2}$	[65.961](12)[b]	Extrap. 3			(65.947 261)(6)	(0.000 004)	0.014
$11D_{3/2}$–$11D_{5/2}$	[21.983](12)[b]	Interp. 2			(21.982 051 0)(7)	(0.000 012 7)	0.001
$12P_{1/2}$–$12S_{1/2}$	[4.973](11)[b]	Extrap. 1			(4.946 201)(12)	(−0.000 063)	0.027
$12P_{1/2}$–$12P_{3/2}$	[50.807](10)[b]	Extrap. 3			(50.796 165)(4)	(0.000 003)	0.011
$12D_{3/2}$–$12D_{5/2}$		P	16.929(11)		(16.931 774 0)(6)	(0.000 009 8)	−0.003

[a] Herzberg's measurement of the $1S_{1/2}$–$2P_{3/2}$ transition [164] was recalibrated according to the revised reference wave numbers of ^{198}Hg I lines [165]. This value was not included in the least-squares level optimization procedure.
[b] This transition alone determines one of the levels involved.
[c] Recalibrated (see text in Section 6).
[d] Uncertainty of the Dayhoff et al. [47] measurement has been increased to ±0.3 MHz (see Section 6).

Table 8
Experimental and theoretical energy levels of tritium. See page 622 for Explanation of Tables.

Level	E_{exp} (MHz)		Unc. (MHz)	E_{th} (MHz)	$\Delta E_{\mathrm{exp\text{-}th}}$ (MHz)	ΔE[31]$_{-\mathrm{th}}$ (MHz)
$1S_{1/2}$	0		3.9	0	–	–
$2P_{1/2}$	(2466954603.3)	$+z$	–	(2466954603.290)(5)	–	0.6
$2S_{1/2}$	2466955659.7	$+w$	6	(2466955662.45)(3)	−3	0.4
$2P_{3/2}$	2466965580.5	$+z+v$	110	(2466965576.312)(4)	4	0.12
$3P_{1/2}$	2923802813.5	$+x+w$	6	(2923802818.2257)(12)	−5	0.19
$3S_{1/2}$	2923803410	$+x+z+v$	240	(2923803133.489)(9)	277	0.16
$3D_{3/2}$	2923806060.5	$+x+z$	1.3	(2923806064.1555)(10)	−4	0.06
$3P_{3/2}$	2923806065.5	$+x+w$	6	(2923806069.4940)(9)	−4	0.05
$3D_{5/2}$	2923807149.3	$+x+z+v$	110	(2923807147.8860)(5)	1	0.02
Limit	–			(**3289279196.4**)(**39**)	–	−1.2

Table 9
Optimized distinctive energy levels of the hydrogen series. See page 623 for Explanation of Tables.

n	E_{opt} (cm^{-1})	ν_{opt} (MHz)	Num. lines	$E_{opt} - E_{step3}$ (cm^{-1})	$E_{opt} - E_{step4}$ (cm^{-1})	$E_{opt} - E_{set3}$ (cm^{-1})	$E_{opt} - E_{c.g.}$ (cm^{-1})
1	0.000 0	0					
2	82 259.158(10)	2 466 067 500(300)	35	0.002(14)	0.006(15)	0.047(14)	0.047(10)
3	97 492.304(5)	2 922 745 750(150)	38	−0.001(8)	−0.002(8)	−0.005(8)	−0.005(5)
4	102 823.904(3)	3 082 583 090(90)	8	0.000(4)	0.000(4)	0.006(5)	0.004(3)
5	105 291.657(4)	3 156 564 470(120)	7	0.000(6)	0.000(6)	0.001(6)	−0.002(4)
6	106 632.168 1(16)	3 196 751 980(50)	7	−0.000 1(23)	0.000 3(23)	0.002 2(26)	0.000 7(16)
7	107 440.450 8(9)	3 220 983 680(30)	8	−0.000 1(13)	0.000 3(13)	0.001 4(18)	−0.000 2(9)
8	107 965.056 8(7)	3 236 710 976(21)	11	−0.000 1(11)	0.000 4(11)	0.000 7(16)	−0.001 1(7)
9	108 324.725 3(7)	3 247 493 566(21)	10	−0.000 1(11)	0.000 3(11)	0.000 7(17)	−0.001 2(7)
10	108 581.994 5(8)	3 255 206 303(24)	6	0.000 0(11)	0.000 0(11)	0.000 9(25)	−0.000 7(8)
11	108 772.344 5(6)	3 260 912 852(18)	6	0.000 0(8)	0.000 1(8)	0.001 3(17)	−0.000 5(6)
12	108 917.120 9(5)	3 265 253 139(15)	6	0.000 0(7)	0.000 3(7)	0.001 1(5)	−0.000 6(5)
13	[109 029.791 3](9)	[3 268 630 910](30)	5	0.000 3(31)	0.006 6(24)	0.001 2(22)	−0.000 4(9)
14	[109 119.191 7](9)	[3 271 311 070](30)	6	−0.002 0(20)	−0.001 0(22)	0.001 3(24)	−0.000 3(9)
15	[109 191.315 4](9)	[3 273 473 280](30)	6	0.001(3)	0.002(3)	0.002(3)	−0.000 3(9)
16	[109 250.343 3](10)	[3 275 242 900](30)	4	0.003(4)	0.004(4)	0.001(6)	−0.000 3(10)
17	[109 299.264 2](10)	[3 276 709 510](30)	2	0.00(3)	0.00(3)	−0.002(30)	−0.000 2(10)
18	[109 340.260 4](10)	[3 277 938 540](30)	2	−0.04(3)	−0.04(3)	−0.005(30)	−0.000 2(10)
19	[109 374.955 5](10)	[3 278 978 680](30)	3	0.005(23)	0.004(23)	0.000(30)	−0.000 2(10)
20	[109 404.577 6](10)	[3 279 866 720](30)	3	0.008(23)	0.005(23)	−0.008(30)	−0.000 1(10)
21	[109 430.069 6](10)	[3 280 630 950](30)	3	0.034(23)			−0.000 1(10)
22	[109 452.165 0](10)	[3 281 293 360](30)	3	0.033(23)			−0.000 1(10)
23	[109 471.441 6](10)	[3 281 871 260](30)	2	−0.01(4)			−0.000 1(10)
24	[109 488.359 2](10)	[3 282 378 430](30)	2	−0.01(4)			0.000 0(10)
25	[109 503.287 5](10)	[3 282 825 970](30)	2	−0.01(4)			−0.000 1(10)
26	[109 516.526 7](10)	[3 283 222 870](30)	2	0.01(4)			−0.000 1(10)
27	[109 528.322 3](10)	[3 283 576 500](30)	2	−0.03(4)			−0.000 1(10)
28	[109 538.876 8](10)	[3 283 892 910](30)	2	−0.01(4)			0.000 0(10)
29	[109 548.358 4](10)	[3 284 177 160](30)	2	−0.01(4)			0.000 0(10)
30	[109 556.907 7](10)	[3 284 433 470](30)	2	−0.02(4)			0.000 0(10)
31	[109 564.643 1](10)	[3 284 665 370](30)	2	0.02(4)			0.000 0(10)
32	[109 571.664 7](10)	[3 284 875 870](30)	2	0.01(4)			0.000 0(10)
33	[109 578.057 7](10)	[3 285 067 530](30)	2	0.01(4)			0.000 0(10)
34	[109 583.894 9](10)	[3 285 242 520](30)	2	0.00(4)			0.000 0(10)
35	[109 589.239 0](10)	[3 285 402 730](30)	2	0.00(4)			0.000 0(10)
36	[109 594.143 9](10)	[3 285 549 780](30)	2	−0.10(12)			−0.000 1(10)
37	[109 598.656 6](10)	[3 285 685 070](30)	2	−0.01(12)			0.000 0(10)
38	[109 602.817 7](10)	[3 285 809 810](30)	1	−0.04(15)			0.000 0(10)
39	[109 606.662 8](10)	[3 285 925 090](30)	1	−0.16(15)			−0.000 1(10)
40	[109 610.223 2](10)	[3 286 031 820](30)	1	0.02(15)			0.000 0(10)

Table 10
Observed and Ritz wavelengths and wave numbers of the hydrogen series lines. See page 623 for Explanation of Tables.

Transition	λ_{obs} (Å)	σ_{obs} (cm^{-1})	Ref.	I_{obs} (a.u.)	σ_{Ritz} (cm^{-1})	$\Delta\sigma_{obs\text{-}Ritz}$ (cm^{-1})	$\Delta\sigma_{fs}$ (cm^{-1})	ν_{Ritz} (GHz)
2–3	6562.79(3)	15233.21(7)	[185][a]	500000	15233.146(16)	0.07	(0.464)	456678.2(5)
2–4	4861.35(5)	20564.67(21)	[183][b]	180000	20564.746(14)	−0.07	(0.407)	616515.6(4)
2–5	4340.472(6)	23032.49(3)	[174][a]	90000	23032.499(14)	−0.01	(0.387)	690496.9(4)
2–6	4101.734(6)	24373.05(4)	[174][a]	70000	24373.010(14)	0.04	(0.378)	730684.5(4)
2–7	3970.075(6)	25181.32(4)	[174][a]	30000	25181.293(14)	0.02	(0.374)	754916.2(4)
2–8	3889.064(6)	25705.84(4)	[174][a]	70000	25705.899(14)	−0.06	(0.371)	770643.5(4)
2–9	3835.397(6)	26065.53(4)	[174][a]	30000	26065.567(14)	−0.04	(0.369)	781426.0(4)
2–10	3797.909(6)	26322.80(4)	[174][a]	17000	26322.837(14)	−0.03	(0.369)	789138.8(4)
2–11	3770.633(6)	26513.21(4)	[174][a]	9000	26513.187(14)	0.03	(0.368)	794845.4(4)
2–12	3750.151(6)	26658.02(4)	[174][a]	8000	26657.963(14)	0.05	(0.367)	799185.6(4)
2–13	3734.369(6)	26770.67(4)	[174][a]	6000	26770.633(14)	0.04	(0.367)	802563.4(4)
2–14	3721.946(6)	26860.03(4)	[174][a]	5000	26860.034(14)	−0.01	(0.367)	805243.6(4)
2–15	3711.978(6)	26932.15(4)	[174][a]	3300	26932.157(14)	0.00	(0.367)	807405.8(4)
2–16	3703.859(6)	26991.19(4)	[174][a]	2800	26991.185(14)	0.00	(0.367)	809175.4(4)
2–17	3697.157(6)	27040.11(4)	[174][a]	2300	27040.106(14)	0.01	(0.366)	810642.0(4)
2–18	3691.551(6)	27081.18(4)	[174][a]	2300	27081.102(14)	0.07	(0.366)	811871.0(4)
2–19	3686.831(6)	27115.85(4)	[174][a]	2000	27115.798(14)	0.05	(0.366)	812911.2(4)
2–20	3682.823(6)	27145.36(4)	[174][a]	1700	27145.420(14)	−0.06	(0.366)	813799.2(4)
2–21	3679.370(6)	27170.84(4)	[174][a]	1700	27170.912(14)	−0.08	(0.366)	814563.4(4)
2–22	3676.376(6)	27192.96(4)	[174][a]	1400	27193.007(14)	−0.05	(0.366)	815225.8(4)
2–23	3673.81(3)	27211.95(22)	[185][a]	1300	27212.284(14)	−0.33	(0.366)	815803.8(4)
2–24	3671.32(10)	27230.4(7)	[185][a]	1200*bl*	27229.201(14)	1.2	(0.366)	816310.9(4)
2–25	3669.45(3)	27244.28(22)	[185][a]	1300	27244.130(14)	0.15	(0.366)	816758.5(4)
2–26	3667.73(3)	27257.06(22)	[185][a]	1000	27257.369(14)	−0.31	(0.366)	817155.4(4)
2–27	3666.08(3)	27269.32(22)	[185][a]	900	27269.164(14)	0.16	(0.366)	817509.0(4)
2–28	3664.65(10)	27280.0(7)	[185][a]	1000*bl*	27279.719(14)	0.2	(0.366)	817825.4(4)
2–29	3663.41(10)	27289.2(7)	[185][a]	900*bl*	27289.200(14)	0.0	(0.366)	818109.6(4)
2–30	3662.22(10)	27298.1(7)	[185][a]	1100*bl*	27297.750(14)	0.3	(0.366)	818366.0(4)
2–31	3661.27(3)	27305.15(22)	[185][a]	800	27305.485(14)	−0.34	(0.366)	818597.8(4)
2–32	3660.32(10)	27312.2(7)	[185][a]	700*bl*	27312.507(14)	−0.3	(0.366)	818808.4(4)
2–33	3659.41(3)	27319.03(22)	[185][a]	700	27318.900(14)	0.13	(0.366)	819000.0(4)
2–34	3658.65(3)	27324.70(22)	[185][a]	700	27324.737(14)	−0.03	(0.366)	819175.0(4)
2–35	3658.04(10)	27329.3(7)	[185][a]	700*bl*	27330.081(14)	−0.8	(0.366)	819335.2(4)
2–36	3657.25(3)	27335.16(22)	[185][a]	700	27334.986(14)	0.18	(0.366)	819482.3(4)
2–37	3656.65(3)	27339.65(22)	[185][a]	700	27339.499(14)	0.15	(0.366)	819617.6(4)
3–4	18751.3(10)	5331.5(3)	[175]	(51000)	5331.600(7)	−0.1	(0.144)	159837.35(21)
3–5	12818.072(8)	7799.352(5)	[191]	32000	7799.353(6)	−0.001	(0.144)	233818.72(18)
3–6	10938.17(13)	9139.79(11)	[191]	14000	9139.864(8)	−0.07	(0.144)	274006.23(24)
3–7	10049.8(10)	9947.7(10)	[176]	(13000)	9948.147(7)	−0.4	(0.144)	298237.94(21)
3–8	9546.2(10)	10472.5(11)	[176]	(9000)	10472.753(7)	−0.3	(0.144)	313965.24(21)
3–9	9229.7(10)	10831.6(12)	[176]	(6500)	10832.421(8)	−0.8	(0.144)	324747.81(24)
3–10	9015.3(10)	11089.2(12)	[176]	(4800)	11089.691(8)	−0.5	(0.144)	332460.57(24)
3–11	8862.89(10)	11279.90(13)	[185][a]	(3600)	11280.041(8)	−0.14	(0.144)	338167.12(24)
3–12	8750.46(3)	11424.83(4)	[185][a]	2200	11424.817(7)	0.02	(0.144)	342507.40(21)
3–13	8665.02(3)	11537.49(4)	[185][a]	(2200)	11537.487(8)	0.00	(0.144)	345885.16(24)
3–14	8598.39(3)	11626.89(4)	[185][a]	2300	11626.888(8)	0.00	(0.144)	348565.33(24)
3–15	8545.38(3)	11699.02(4)	[185][a]	(1400)	11699.011(8)	0.01	(0.144)	350727.53(24)
3–16	8502.49(3)	11758.03(4)	[185][a]	(1200)	11758.039(8)	−0.01	(0.144)	352497.14(24)
3–17	8467.26(3)	11806.95(4)	[185][a]	700	11806.960(8)	−0.01	(0.144)	353963.76(24)
3–18	8437.95(3)	11847.96(4)	[185][a]	700	11847.956(8)	0.01	(0.144)	355192.79(24)
3–19	8413.32(3)	11882.65(4)	[185][a]	700	11882.652(8)	0.00	(0.144)	356232.95(24)
3–20	8392.40(3)	11912.27(4)	[185][a]	700	11912.274(8)	0.00	(0.144)	357120.99(24)
3–21	8374.48(3)	11937.76(4)	[185][a]	600	11937.766(8)	−0.01	(0.144)	357885.22(24)
3–22	8359.00(3)	11959.87(4)	[185][a]	500	11959.861(8)	0.01	(0.144)	358547.61(24)
3–23	8345.54(3)	11979.16(4)	[185][a]	500	11979.138(8)	0.02	(0.144)	359125.52(24)
3–24	8333.78(3)	11996.06(4)	[185][a]	370	11996.055(8)	0.00	(0.144)	359632.68(24)
3–25	8323.42(3)	12010.99(4)	[185][a]	290	12010.984(8)	0.01	(0.144)	360080.24(24)
3–26	8314.26(3)	12024.22(4)	[185][a]	290	12024.223(8)	0.00	(0.144)	360477.14(24)
3–27	8306.10(3)	12036.04(4)	[185][a]	250	12036.018(8)	0.02	(0.144)	360830.74(24)
3–28	8298.83(3)	12046.58(4)	[185][a]	210	12046.573(8)	0.01	(0.144)	361147.17(24)
3–29	8292.30(3)	12056.07(4)	[185][a]	170	12056.054(8)	0.01	(0.144)	361431.41(24)
3–30	8286.42(3)	12064.62(4)	[185][a]	140	12064.604(8)	0.02	(0.144)	361687.73(24)
3–31	8281.13(3)	12072.33(4)	[185][a]	140	12072.339(8)	−0.01	(0.144)	361919.62(24)
3–32	8276.32(3)	12079.34(4)	[185][a]	140	12079.361(8)	−0.02	(0.144)	362130.13(24)
3–33	8271.94(3)	12085.74(4)	[185][a]	140	12085.754(8)	−0.01	(0.144)	362321.79(24)
3–34	8267.94(3)	12091.59(4)	[185][a]	110	12091.591(8)	0.00	(0.144)	362496.78(24)
3–35	8264.28(3)	12096.94(4)	[185][a]	110	12096.935(8)	0.01	(0.144)	362656.99(24)
3–36	8260.89(10)	12101.91(15)	[185][a]	110	12101.840(8)	0.07	(0.144)	362804.04(24)
3–37	8257.89(10)	12106.30(15)	[185][a]	110	12106.353(8)	−0.05	(0.144)	362939.33(24)
3–38	8254.99(10)	12110.56(15)	[185][a]	90	12110.514(8)	0.04	(0.144)	363064.08(24)
3–39	8252.29(10)	12114.52(15)	[185][a]	90	12114.359(8)	0.16	(0.144)	363179.35(24)
3–40	8249.99(10)	12117.90(15)	[185][a]	90	12117.919(8)	−0.02	(0.144)	363286.07(24)
4–5	40522.79(8)	2467.747(5)	[191]	11000	2467.753(6)	−0.006	(0.069)	73981.37(18)

Table 10 (*continued*)

Transition	λ_{obs} (Å)	σ_{obs} (cm^{-1})	Ref.	I_{obs} (a.u.)	σ_{Ritz} (cm^{-1})	$\Delta\sigma_{obs\text{-}Ritz}$ (cm^{-1})	$\Delta\sigma_{fs}$ (cm^{-1})	ν_{Ritz} (GHz)
4–6	26258.71(21)	3808.26(3)	[194]	(9000)	3808.264(6)	0.00	(0.064)	114168.88(18)
4–7	21661.178(23)	4616.554(5)	[191]	8000	4616.547(6)	0.007	(0.066)	138400.60(18)
4–8				(4500)	5141.153(6)		(0.067)	154127.89(18)
4–9	18179.21(4)	5500.790(11)[c]	[191]	2800	5500.821(6)	−0.031	(0.067)	164910.46(18)
4–10	17366.885(21)	5758.085(7)	[191]	3100	5758.091(6)	−0.006	(0.068)	172623.23(18)
4–11	16811.10(3)	5948.450(10)	[191]	1600	5948.440(6)	0.010	(0.068)	178329.74(18)
4–12	16411.36(13)	6093.34(5)[c]	[191]	1400	6093.217(6)	0.12	(0.068)	182670.05(18)
4–13				(1200)	6205.887(5)		(0.068)	186047.81(15)
4–14				(1000)	6295.288(5)		(0.068)	188727.99(15)
4–15				(800)	6367.411(6)		(0.068)	190890.18(18)
4–16	15560.46(6)	6426.545(24)[c]	[191]	800	6426.439(6)	0.106	(0.068)	192659.79(18)
5–6	74599.0(6)	1340.500(10)	[194]	(3000)	1340.511(6)	−0.011	(0.037)	40187.51(18)
5–7	46537.8(3)	2148.789(12)	[191]	4200	2148.794(6)	−0.005	(0.033)	64419.22(18)
5–8	37405.76(17)	2673.385(12)	[191]	2500	2673.400(6)	−0.015	(0.035)	80146.52(18)
5–9	32969.8(22)	3033.08(20)	[194]	(1800)	3033.068(6)	0.01	(0.035)	90929.09(18)
5–10				(1400)	3290.337(7)		(0.036)	98641.82(21)
5–11				(1100)	3480.688(7)		(0.036)	104348.40(21)
5–12				(900)	3625.464(6)		(0.037)	108688.68(18)
5–13				(700)	3738.134(6)		(0.037)	112066.44(18)
5–14				(580)	3827.535(6)		(0.037)	114746.61(18)
5–15				(480)	3899.658(7)		(0.037)	116908.81(21)
5–16				(400)	3958.686(6)		(0.037)	118678.42(18)
6–7	123719.12(20)	808.2825(13)	[193][d]	410	808.2827(18)	−0.0002	(0.022)	24231.71(5)
6–8	75024.4(17)	1332.90(3)	[194]	620	1332.8887(21)	0.01	(0.019)	39959.00(6)
6–9				(540)	1692.5572(23)		(0.020)	50741.59(7)
6–10	51286.5(8)	1949.83(3)	[194]	(450)	1949.8264(24)	0.00	(0.021)	58454.32(7)
6–11				(370)	2140.1764(23)		(0.021)	64160.87(7)
6–12				(300)	2284.9528(23)		(0.022)	68501.16(7)
6–13				(240)	2397.623(3)		(0.022)	71878.93(9)
6–14				(200)	2487.0236(25)		(0.022)	74559.09(7)
6–15				(170)	2559.147(3)		(0.022)	76721.30(9)
6–16				(140)	2618.175(3)		(0.022)	78490.91(9)
7–8	190619.6(4)	524.6050(10)	[192]	540	524.6060(11)	−0.0010	(0.014)	15727.29(3)
7–9	113086.81(18)	884.2764(14)	[193][d]	420	884.2745(13)	0.0019	(0.013)	26509.88(4)
7–10	87600.64(14)	1141.5442(18)	[193][d]	420	1141.5437(16)	0.0005	(0.013)	34222.62(5)
7–11				(360)	1331.8937(15)		(0.013)	39929.17(4)
7–12				(300)	1476.6701(14)		(0.013)	44269.46(4)
7–13				(250)	1589.3405(18)		(0.014)	47647.23(5)
7–14				(210)	1678.7409(17)		(0.014)	50327.39(5)
7–15				(180)	1750.8646(18)		(0.014)	52489.60(5)
7–16				(150)	1809.8925(19)		(0.014)	54259.21(6)
8–9	278035.0(8)	359.6670(10)	[192]	(220)	359.6685(10)	−0.0015	(0.0097)	10782.59(3)
8–10				(270)	616.9377(14)		(0.0086)	18495.33(4)
8–11	123871.53(18)	807.2880(12)	[193][d]	340	807.2877(11)	0.0003	(0.0084)	24201.88(3)
8–12	105035.07(18)	952.0630(16)	[193][d]	190	952.0641(11)	−0.0011	(0.0088)	28542.16(3)
8–13	93920.3(3)	1064.733(3)	[193][d]	60	1064.7345(16)	−0.002	(0.0090)	31919.94(5)
8–14	86644.60(23)	1154.140(3)	[193][d]	180	1154.1349(14)	0.005	(0.0092)	34600.09(4)
8–15	81548.4(5)	1226.265(7)	[193][d]	120	1226.2586(17)	0.006	(0.0094)	36762.31(5)
8–16				(130)	1285.2865(17)		(0.0095)	38531.92(5)
9–10				(110)	257.2692(14)		(0.0069)	7712.74(4)
9–11				(140)	447.6192(13)		(0.0062)	13419.29(4)
9–12				(140)	592.3956(11)		(0.0059)	17759.57(3)
9–13				(130)	705.0660(17)		(0.0061)	21137.35(5)
9–14	125870.5(3)	794.4674(20)	[193][d]	210	794.4664(14)	0.0010	(0.0063)	23817.50(4)
9–15	115395.4(4)	866.586(3)	[189]	110	866.5901(16)	−0.004	(0.0065)	25979.72(5)
9–16	108035.9(7)	925.618(6)	[193][d]	80	925.6180(17)	0.000	(0.0066)	27749.33(5)
10–11				(60)	190.3500(14)		(0.0051)	5706.55(4)
10–12				(80)	335.1264(13)		(0.0046)	10046.84(4)
10–13				(90)	447.7968(18)		(0.0043)	13424.61(5)
10–14				(80)	537.1972(17)		(0.0044)	16104.77(5)
10–15				(70)	609.3209(17)		(0.0046)	18266.98(5)
10–16				(70)	668.3488(18)		(0.0047)	20036.59(5)
11–12	690717(24)	144.777(5)	[198]	14*bl*	144.7764(11)	0.001	(0.0039)	4340.29(3)
11–13				(20)	257.4468(16)		(0.0035)	7718.06(5)
11–14				(21)	346.8472(15)		(0.0033)	10398.22(4)
11–15				(21)	418.9709(16)		(0.0033)	12560.43(5)
11–16				(19)	477.9988(16)		(0.0034)	14330.04(5)
12–13	887610(80)	112.662(10)	[198]	9*bl*	112.6704(14)	−0.008	(0.0030)	3377.77(4)
12–14				(13)	202.0708(14)		(0.0028)	6057.93(4)
12–15				(14)	274.1945(15)		(0.0026)	8220.14(4)
12–16				(14)	333.2224(15)		(0.0025)	9989.76(4)
13–14	1118630(90)	89.395(7)	[198]	14*bl*	89.4004(18)	−0.005	(0.0024)	2680.16(5)
13–15				(20)	161.5241(19)		(0.0022)	4842.37(6)
13–16				(22)	220.5520(19)		(0.0019)	6611.98(6)
14–15	1387500(800)	72.07(4)	[196]	19	72.1237(18)	−0.05	(0.0019)	2162.21(5)

(*continued on next page*)

Table 10 (*continued*)

Transition	λ_{obs} (Å)	σ_{obs} (cm^{-1})	Ref.	I_{obs} (a.u.)	σ_{Ritz} (cm^{-1})	$\Delta\sigma_{obs\text{-}Ritz}$ (cm^{-1})	$\Delta\sigma_{fs}$ (cm^{-1})	ν_{Ritz} (GHz)
14–16				(28)	131.1516(18)		(0.0018)	3931.83(5)
15–16	1694230(140)	59.024(5)	[196]	16	59.0278(18)	−0.004	(0.0016)	1769.61(5)
16–17				(11)	48.9209(20)		(0.0013)	1466.61(6)
17–18				(8)	40.9962(20)		(0.0011)	1229.04(6)
18–19				(6)	34.6951(20)		(0.0009)	1040.13(6)
19–20	3376000(600)	29.621(5)	[199]	(4)	29.6221(20)	−0.001	(0.0008)	888.05(6)
20–21				(3)	25.4920(20)		(0.0007)	764.23(6)
21–22	4525700(600)	22.096(3)	[199]	(2)	22.0954(18)	0.001	(0.0006)	662.40(5)
22–23				(2)	19.2766(20)		(0.0005)	577.90(6)
23–24				(1)	16.9176(20)		(0.0005)	507.18(6)
24–25				(1)	14.9283(20)		(0.0004)	447.54(6)
25–26				(1)	13.2392(20)		(0.0004)	396.90(6)
26–27				(1)	11.7956(20)		(0.0003)	353.62(6)

[a] Corrected for the systematic shift (see text in Sections 10.2 and 10.5).
[b] Corrected for the revised wavelength scale (see text in Section 10.2).
[c] These lines were excluded from the level-optimization procedure.
[d] Recalibrated (see text in Section 10.2).

Table 11
Wave numbers of the Lyman series lines of hydrogen and deuterium derived from experimental fine-structure intervals weighted by calculated line strengths. See page 623 for Explanation of Tables.

	Hydrogen					Deuterium			
n_{upper}	σ (cm^{-1})	σ_{obs} (cm^{-1})		$\Delta\sigma$[a] (cm^{-1})	ν (GHz)	σ (cm^{-1})	σ_{obs} (cm^{-1})	$\Delta\sigma$[a] (cm^{-1})	ν (GHz)
2	82 259.163	82 259.16(14)	[149]	0.099	2 466 068(3)	82 281.545	82 281.644(17) [164]	0.099	2 466 739(3)
3	97 492.283	97 491.7(3)	[150]	0.029	2 922 745.1(9)	97 518.810		0.029	2 923 540.4(9)
4	102 823.879	102 826.0(15)	[181]	0.012	3 082 582.3(4)	102 851.857		0.012	3 083 421.1(4)
5	105 291.644	105 291.8(4)	[150]	0.006	3 156 564.08(18)	105 320.293		0.006	3 157 422.95(18)
6	106 632.158	106 631.0(16)	[181]	0.004	3 196 751.67(12)	106 661.171		0.004	3 197 621.46(12)
7	107 440.444 2	107 440.1(16)	[181]	0.002 3	3 220 983.49(7)	107 469.677 9		0.002 3	3 221 859.89(7)
8	107 965.053 0	107 962.3(16)	[181]	0.001 5	3 236 710.86(4)	107 994.429 5		0.001 5	3 237 591.53(4)
9	108 324.722 8	108 325.0(16)	[181]	0.001 1	3 247 493.49(3)	108 354.197 2		0.001 1	3 248 377.11(3)
10	108 581.992 5	108 583.9(16)	[181]	0.000 8	3 255 206.243(24)	108 611.536 8		0.000 8	3 256 091.964(24)
11	108 772.342 8	108 773.4(16)	[181]	0.000 6	3 260 912.801(18)	108 801.939 0		0.000 6	3 261 800.073(18)
12	108 917.119 8	108 917.6(16)	[181]	0.000 5	3 265 253.106(15)	108 946.755 4		0.000 5	3 266 141.547(15)

[a] Maximum possible deviation under varying conditions of excitation and observation (estimated as 27% of the fine-structure interval; see Section 10.7).

Table 12
Designations of fine-structure components and their groups, used by different authors for the Balmer-α lines, and their relative intensities according to different models. See page 623 for Explanation of Tables.

Component	This paper	Reader [210]	Kireyev [168]	Relative intensities				
				Model 1 H, D	Model 2 H / D	Model 3 H, D	Model 4 H / D	Model 5 H / D
$2P_{1/2}–3D_{3/2}$	2b	E	g	(4.9040)	[3.75]/[5.05]	[5.40]	[5.15]/[4.48]	[4.08]/[5.40]
$2S_{1/2}–3P_{3/2}$	2a	B	f	(2.0433)	[3.20]/[2.67]	[2.20]	[2.63]/[2.42]	[5.09]/[3.19]
$2P_{1/2}–3S_{1/2}$	3b	C	d	(0.0958)	[0.30]/[0.11]	[0.09]	[0.05]/[0.38]	[0.06]/[0.03]
$2S_{1/2}–3P_{1/2}$	3a	A	e	(1.0217)	[0.82]/[2.07]	[1.61]	[1.61]/[1.06]	[2.46]/[3.00]
$2P_{3/2}–3D_{5/2}$	1	G	a	(8.8280)	[8.65]/[8.77]	[8.74]	[8.86]/[8.34]	[9.07]/[8.85]
$2P_{3/2}–3D_{3/2}$	4	F	b	(0.9808)	[0.75]/[1.01]	[1.08]	[1.03]/[0.90]	[0.82]/[1.08]
$2P_{3/2}–3S_{1/2}$	5	D	c	(0.1916)	[0.60]/[0.23]	[0.18]	[0.11]/[0.77]	[0.11]/[0.07]
Component group				[211]	[214]/[168]	[213]	[212]	[38]
2a + 2b + 3a + 3b	violet	A + B + C + E	d + e + f + g					
1 + 4 + 5	red	D + F + G	a + b + c	10.0000	10.00/10.00	10.00	10.00/10.00	10.00/10.00
1 + 4	red1	F + G	a + b					
2a + 2b	violet1	B + E	f + g	(6.9472)	(6.947)/7.72	7.60	7.78/6.90	9.17/8.59
3a + 3b	violet2	A + C	d + e	(1.1175)	(1.117)/2.18	1.70	1.66/1.44	2.51/3.03
1 + 2a + 2b + 3a + 3b + 4 + 5	center	A + B + C + D + E + F + G	a + b + c + d + e + f + g					

Table 13
Wave numbers of component groups and their separations for the Balmer series lines of hydrogen and deuterium derived from experimental fine-structure intervals weighted by different models. See page 623 for Explanation of Tables.

			Hydrogen				Deuterium			
n_{upper}	Group or interval	Model[a]	σ (cm^{-1})	σ_{obs} (cm^{-1})		$\Delta\sigma^{b}$ (cm^{-1})	σ (cm^{-1})	σ_{obs} (cm^{-1})		$\Delta\sigma^{b}$ (cm^{-1})
3	center	1	15 233.2018				15 237.3467			
		2	15 233.1975			−0.0043	15 237.3560			0.0093
		3	15 233.2098			0.0080	15 237.3546			0.0080
		4	15 233.2122			0.0094	15 237.3431			−0.0036
		5	15 233.2214			0.0196	15 237.3652			0.0186
	violet	1	15 233.3721				15 237.5170			
		2	15 233.3682			−0.0039	15 237.5047			−0.0123
		3	15 233.3662			−0.0059	15 237.5112			−0.0059
		4	15 233.3656			−0.0065	15 237.5124			−0.0047
		5	15 233.3544			−0.0177	15 237.4983			−0.0187
	red	1	15 233.0645				15 237.2093			
		2	15 233.0598			−0.0046	15 237.2087			−0.0006
		3	15 233.0642			−0.0002	15 237.2091			−0.0002
		4	15 233.0654			−0.0009	15 237.2019			−0.0074
		5	15 233.0661			−0.0016	15 237.2106			0.0013
	red1	1	15 233.0670				15 237.2118			
		2	15 233.0677			−0.0007	15 237.2117			−0.0001
		3	15 233.0666			−0.0004	15 237.2114			−0.0004
		4	15 233.0668	15 233.0670(14)	[212]	−0.0001	15 237.2119			0.0001
		5	15 233.0676			0.0006	15 237.2115			−0.0003
	violet–red	1	0.3076	0.3098(20)	[40]		0.3077			
				0.308	[215]					
		2	0.3084			0.0008	0.2960			−0.0118
		3	0.3020			−0.0056	0.3021			−0.0056
		4	0.3002			−0.0075	0.3105			−0.0027
		5	0.2883	0.313(3)	[38]	−0.0193	0.2877			−0.0200
	violet1–red1	1	0.3230				0.3231	0.321	[215]	
		2	0.3165	0.326	[214]	−0.0066	0.3214	0.321(3)	[168]	−0.0017
		3	0.3236			0.0005	0.3237	0.3213(30)	[213]	0.0005
		4	0.3216	0.3198(10)	[212]	−0.0014	0.3210	0.3205(10)	[212]	−0.0021
		5	0.3132	0.313(3)	[38]	−0.0098	0.3207	0.3212(21)	[38]	−0.0024
	violet1–violet2	1	0.1293				0.1293			
		2	0.1151	0.105	[214]	−0.0142	0.1290	0.129(3)	[168]	−0.0003
		3	0.1309			0.0016	0.1309	0.133(3)	[213]	0.0017
		4	0.1302	0.1317(10)	[212]	0.0009	0.1190	0.1190(10)	[212]	−0.0102
		5	0.1230			−0.0063	0.1300	0.130(3)	[38]	0.0007
4	center	1	20 564.7791				20 570.3747	20 570.380(4)	[162]	
	violet	1	20 564.9518				20 570.5475	20 570.5471(13)	[162]	
	red	1	20 564.6216				20 570.2172	20 570.2196(17)	[162]	
	violet–red	1	0.3302	0.3298(4)	[40]		0.3302	0.3275(21)	[162]	
5	Venter	1	23 032.5392	23 032.49(3)	[174]		23 038.8064	23 038.803(4)	[162]	
	violet	1	23 032.7120				23 038.9792	23 038.9779(16)	[162]	
	red	1	23 032.3733				23 038.6405	23 038.6401(21)	[162]	
	violet–red	1	0.3387	0.3388(4)	[40]		0.3386	0.3378(26)	[162]	
6	center	1	24 373.0507	24 373.05(4)	[174]		24 379.6826	24 379.684(5)	[162]	
	violet	1	24 373.2232				24 379.8550	24 379.8563(18)	[162]	
	red	1	24 372.8807				24 379.5125	24 379.5144(24)	[162]	
	violet–red	1	0.3425	0.3451(6)	[40]		0.3425	0.342(3)	[162]	
7	center	1	25 181.3367	25 181.32(4)	[174]		25 188.1884	25 188.186(4)	[162]	
	violet	1	25 181.5087				25 188.3605	25 188.3591(19)	[162]	
	red	1	25 181.1643				25 188.0161	25 188.0149(13)	[162]	
	violet–red	1	0.3444	0.3506(6)	[40]		0.3445	0.3442(23)	[162]	
8	center	1	25 705.9452	25 705.84(4)	[174]		25 712.9397	25 712.943(6)	[162]	
	violet	1	25 706.1169				25 713.1114	25 713.114(7)	[162]	
	red	1	25 705.7714				25 712.7659	25 712.773(7)	[162]	
	violet–red	1	0.3455				0.3456	0.341(9)	[162]	
9	center	1	26 065.6149	26 065.53(4)	[174]		26 072.7073	26 072.701(8)	[162]	
	violet	1	26 065.7863				26 072.8788	26 072.876(10)	[162]	
	red	1	26 065.4402				26 072.5326	26 072.530(10)	[162]	
	violet–red	1	0.3461				0.3462	0.347(14)	[162]	
10	center	1	26 322.8845	26 322.80(4)	[174]		26 330.0469			
	violet	1	26 323.0557				26 330.2182			
	red	1	26 322.7092				26 329.8716			
	violet–red	1	0.3465				0.3466			
11	center	1	26 513.2349	26 513.21(4)	[174]		26 520.4491			
	violet	1	26 513.4059				26 520.6201			

(continued on next page)

Table 13 (*continued*)

n_{upper}	Group or interval	Model[a]	Hydrogen σ (cm^{-1})	σ_{obs} (cm^{-1})		$\Delta\sigma$[b] (cm^{-1})	Deuterium σ (cm^{-1})	σ_{obs} (cm^{-1})	$\Delta\sigma$[b] (cm^{-1})
	red	1	26513.0591				26520.2733		
	violet–red	1	0.3468				0.3469		
12	center	1	26658.0119	26658.02(4)	[174]		26665.2655		
	violet	1	26658.1828				26665.4364		
	red	1	26657.8358				26665.0894		
	violet–red	1	0.3470				0.3470		

[a] Models of relative intensities (1 through 5) are described in Section 10.7.
[b] Differences of wave numbers calculated with different intensity models from Model 1.

Update

Atomic Data and Nuclear Data Tables

Volume 126, Issue , March 2019, Page 295–298

DOI: https://doi.org/10.1016/j.adt.2018.11.003

Atomic Data and Nuclear Data Tables 126 (2019) 295–298

Contents lists available at ScienceDirect

Atomic Data and Nuclear Data Tables

journal homepage: www.elsevier.com/locate/adt

Corrigendum

Erratum to "A critical compilation of experimental data on spectral lines and energy levels of hydrogen, deuterium, and tritium" [At. Data Nucl. Data Tables 96 (2010) 586–644]

A. Kramida

National Institute of Standards and Technology, 100 Bureau Dr., MS 8422, Gaithersburg, MD 20899, USA

ARTICLE INFO

Article history:
Received 19 October 2018
Accepted 26 November 2018
Available online 13 December 2018

ABSTRACT

In the critical compilation of experimental and theoretical data on the H, D, and T spectra (Kramida, 2010), there were errors in some of the hyperfine structure parameters for deuterium. These errors are corrected here.

DOI of original article: https://doi.org/10.1016/j.adt.2010.05.001.
E-mail address: alexander.kramida@nist.gov.

https://doi.org/10.1016/j.adt.2018.11.003
0092-640X/

Contents

References 296
Explanation of Tables 297
Table 1. Corrected hfs-induced shifts $\Delta E_{\text{c.g.}}$ in nL_j ($n > 1$, $L > 0$) levels of deuterium (Hz) 297

It was recently brought to my attention by Dr. Thomas Udem of Max-Planck-Institut für Quantenoptik, Germany, that there is an error in the description of deuterium quadrupole hyperfine structure (hfs) shifts in my article [1]. In Section 3.1.3 of that article, I have incorrectly stated that those shifts are non-zero for the $2P_{1/2}$ levels of deuterium (in fact, for all $nP_{1/2}$ levels). In my derivation, I used Eq. (27.45) from Chapter 7 of Sobel'man [2] describing the hyperfine splittings of atomic energy levels, corresponding to Eq. (19) of Ref. [1]. However, I overlooked the fact that Sobel'man [2] gave in Chapter 6 equations (23.10) through (23.15) for the quadrupole hfs-induced shifts of energy levels. According to these equations, an additional shift Δ (independent of F) must be added to the right side of Eq. (20) of Ref. [1]:

$$\Delta E_{nljF}^{\text{quad}} = B_{nlj} C\,(C+1) + \Delta, \tag{1}$$

where $C = F(F+1) - j(j+1) - I(I+1)$, F, j, and I are the total, orbital, and nuclear angular momentum quantum numbers, respectively, and B_{nlj} is the magnetic dipole hfs constant correctly described by Eq. (19) of Ref. [1]. Neglecting the fact that Sobelman's Eqs. (23.14) and (23.15) both contain $(2j-1)^{1/2}$ in the denominator (which is zero for $j = 1/2$), the additional term Δ can be obtained from these equations in the following form:

$$\Delta = -\frac{4}{3} B_{nlj} I(I+1) j(j+1), \tag{2}$$

where the above-mentioned term $(2j-1)^{1/2}$ cancels out.

It can easily be shown that Eqs. (1) and (2) lead to a zero quadrupole shift for $j = 1/2$ levels with any value of I.

Although Sobelman's equations leading to this result are mathematically inapplicable in the case $j = 1/2$, the result is correct: it was already well known as early as in 1940 [3] that quadrupole hyperfine shift is zero for $j = 0$ and $j = 1/2$ levels because their wavefunctions are spherically symmetric. Thus, Eqs. (1) and (2) are valid for all $j \neq 0$, which is probably the reason for Sobelman's not mentioning the exception for $j = 1/2$ in his book [2].

The omission of the independent of F term Δ in Ref. [1] resulted in incorrect values for hfs-induced shifts of centers of gravity of all fine-structure levels of deuterium (except $nS_{1/2}$) in Table 2 of Ref. [1]. The corrected values are given here in Table 1.

All data derived from experiments reported in Ref. [1] were not affected by this error, because the hfs-induced shifts were much smaller than experimental uncertainties. However, the hfs-induced shifts given in columns "ΔE_{hfs}" of Tables 6 and 7 of Ref. [1] are incorrect for the non-S levels, as well as the theoretical energies E_{th} to which those shifts were added. The correct values of ΔE_{hfs} and E_{th} are easy to obtain by adding the differences of the shift values given here in Table 1 from those given in Table 2 of Ref. [1].

The author is grateful to Drs. Thomas Udem of Max-Planck-Institut für Quantenoptik, Germany, Charlotte Froese Fischer of the University of British Columbia, Canada, Michel Godefroid of Université libre de Bruxelles, Belgium, and Alexander Petrov of the St. Petersburg Nuclear Physics Institute, Russia for helpful discussions.

References

[1] A.E. Kramida, At. Data Nucl. Data Tables 96 (2010) 586.
[2] I.I. Sobel'man, Introduction to the Theory of Atomic Spectra, Pergamon Press, NY, 1972.
[3] H. Kopfermann, Kernmomente, Edwards Brothers, Michigan, 1940.

Explanation of Tables

Table 1	**Corrected hfs-induced shifts $\triangle E_{\mathrm{c.g.}}$ in nL_j ($n > 1$, $L > 0$) levels of deuterium (Hz)**	
	n	Principal quantum number

Table 1
Corrected hfs-induced shifts $\Delta E_{c.g.}$ in nL_j ($n > 1$, $L > 0$) levels of deuterium (Hz).

n	$n\mathrm{P}_{1/2}$	$n\mathrm{P}_{3/2}$	$n\mathrm{D}_{3/2}$	$n\mathrm{D}_{5/2}$	$n\mathrm{F}_{5/2}$	$n\mathrm{F}_{7/2}$
2	(−470.1)(25)	(235.1)(12)				
3	(−139.3)(7)	(69.6)(4)	(−376.1)(20)	(250.7)(13)		
4	(−58.8)(3)	(29.38)(15)	(−158.7)(8)	(105.8)(6)	(−302.2)(16)	(226.7)(12)
5	(−30.09)(16)	(15.04)(8)	(−81.2)(4)	(54.2)(3)	(−154.7)(8)	(116.1)(6)
6	(−17.41)(9)	(8.71)(5)	(−47.01)(25)	(31.34)(16)	(−89.5)(5)	(67.2)(4)
7	(−10.96)(6)	(5.48)(3)	(−29.60)(16)	(19.74)(10)	(−56.4)(3)	(42.29)(22)
8	(−7.35)(4)	(3.673)(19)	(−19.83)(10)	(13.22)(7)	(−37.78)(20)	(28.33)(15)
9	(−5.16)(3)	(2.580)(14)	(−13.93)(7)	(9.29)(5)	(−26.53)(14)	(19.90)(10)
10	(−3.761)(20)	(1.880)(10)	(−10.15)(5)	(6.77)(4)	(−19.34)(10)	(14.51)(8)
11	(−2.826)(15)	(1.413)(7)	(−7.63)(4)	(5.09)(3)	(−14.53)(8)	(10.90)(6)
12	(−2.177)(11)	(1.088)(6)	(−5.88)(3)	(3.918)(21)	(−11.19)(6)	(8.39)(4)

All purely theoretical values are given in parentheses. Uncertainty in the units of the last decimal place is given in parentheses next to the value. For example, (−470.1)(25) means a purely theoretical value −470.1± 2.5.